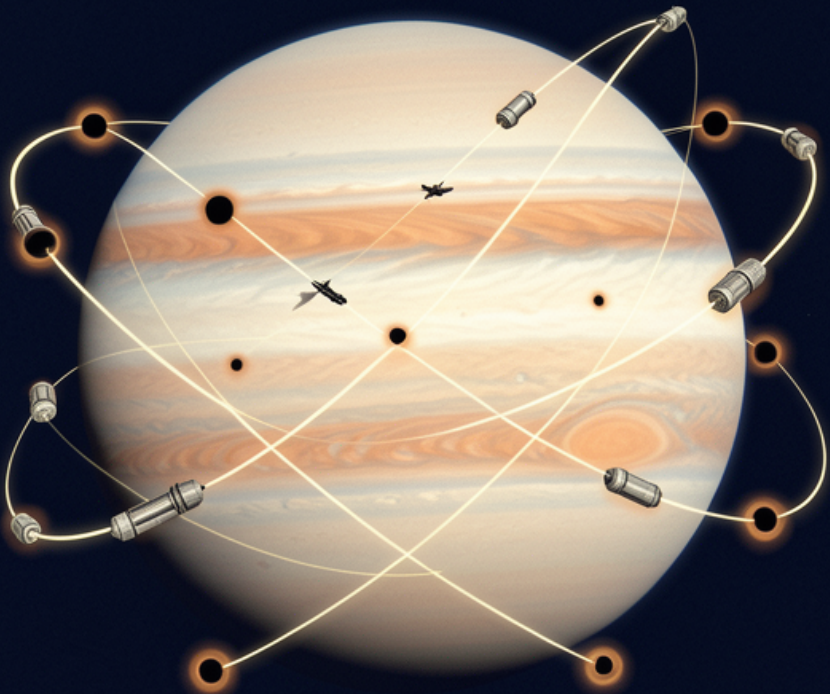


A SCIENCE FICTION NOVEL

THE GRAVITY WE OWE

*Twelve captive suns.
One mind waking.*



TAPIO KINNUNEN

The Gravity We Owe

*What does a civilization owe to a mind it mistook for
infrastructure?*

Tapio Kinnunen

ACT I

Chapter 1 — The Refusal

Ada Narang counted the green seals twice before she let a trillion tonnes of darkness cross the gap.

Twelve lights circled the cage gallery, one for each sink held by the Mass Compact. Eleven shone blue. Aster Three's light pulsed amber beneath Ada's thumb while the transfer clock narrowed through its final minute. Beyond the command blister, Crown Meridian filled half the sky: a silver cylinder banded with farms, mirrors, and the steady yellow windows of four hundred thousand people who had never needed to see what kept their floors beneath them.

"Receiving cage reports capture field at ninety-eight percent," said Jalen from guidance. "Relative velocity point zero two meters per second. We are inside the handoff corridor."

Ada checked his plot rather than his face. The *Common Measure* and Crown's receiving truss moved together above Ophion's cloud tops, separated by eleven kilometers and a difference in velocity small enough to cross a room on foot. On the scale that mattered, Aster Three occupied a point too small to draw. Its mass bent every trajectory on the screen.

"Cage gallery?"

Iri Quill's voice arrived through the ship circuit with metal around it. "Primary loops agree within four parts per million. Secondary plasma stable. Feed shutter closed. Blue strip says the manual latch has not wandered."

Nobody but Iri trusted a strip of paint more than an optical sensor. They had cut the blue flake from a free-spin yard hull and wedged it beside the latch housing as a crude alignment witness. If the metal shifted, the painted edge would break. Ada had once asked why an engineer on the most exacting vessel in Ophion kept a child's test in a trillion-tonne cage. Iri had replied that children knew when adults were lying about simple things.

"Hold your hand over manual release," Ada said. "Do not touch it without my word."

"Already there."

Crown Control entered the count at thirty seconds. Its transfer officer spoke with the calm of someone inside a habitat whose spin did not depend on the next instruction. The handoff had been rehearsed for six days and modeled for six years. Aster Three would move from *Common Measure's* charged-mass cage into Crown's field, feed the administrative cylinder for nine days, then continue toward Rook. The path formed one link in the ninety-one-day circulation that kept forty-six habitats warm, spun, watered, and honest with gravity.

At twenty seconds, Ada felt a pressure beneath her back teeth.

No alarm sounded. The sensation came first, too slight to move anything except the fluid in her inner ear. Aster Three had changed its relation to the ship.

"Stop count," she said.

Crown's officer continued. "Seventeen. Sixteen."

"Common Measure calls stop. Freeze receiving ramp."

Jalen's hands moved. "Our charge monitor shows nominal."

Ada opened the raw phase trace. The elegant line had grown a shallow hook, no taller than the width of a cursor. Every filtered display had averaged it away.

"Crown, freeze your field."

"Transfer corridor remains green," the officer said. "We have no stop condition."

"You have one because I called it."

For two breaths nothing changed. Crown's capture field continued rising, invisible except for the numbers multiplying along Ada's display. *Common Measure's* cage began its programmed release. Between those opposing curves, the black hole sat inside an engineered charge sheath no wider than a grain of dust. The cage did not hold its gravity. Nothing could. It persuaded the sink's charge to stay centered while every tonne around it arranged itself according to a plan.

The phase hook doubled.

"Abort handoff," Ada ordered. "Bring our field back."

Jalen looked at her now. "Fields will overlap."

"They already do."

Iri swore once on the engineering channel. A tone rose under their voice, not an alarm but the unfiltered carrier from the primary loops. "Captain, it changed effective charge. Negative twelve percent and falling."

Crown Control spoke over them. "Common Measure, maintain scheduled release. We will widen capture."

Ada saw the consequence before Jalen painted it. A wider receiving field would pull Aster Three away from their cage before either system could define the center. The sink would not fly free like a dropped stone. Its gravity would carry the ships with it, turning eleven kilometers of careful separation into a slow mutual fall. Crown Meridian sat behind the receiving truss. Morrow's morning traffic lane crossed thirty-eight minutes ahead.

"Denied," Ada said. "Crown, collapse field now."

"Custodian transfer authority supersedes tug discretion during final corridor."

"Not on my hull."

A red vector appeared across the plot. Jalen whispered its first number, then stopped because everyone could read the rest. The transfer path had rotated four hundredths of a degree. At ordinary speed that shift meant nothing. With Aster Three coupled to two cages, it aimed the combined mass toward a freight procession climbing from Morrow.

Three ore barges and a passenger spindle occupied the lane. Twelve hundred people, more or less. Ada did not ask for an exact count. Exactness could become a hiding place when the decision had already arrived.

"Thrust plan Delta Glass," she said. "Roll us cage-forward. Crown, you will yaw away from the lane on my mark."

Jalen pulled up the plan and shook his head. "Delta Glass assumes a dead receiving field."

"Then correct for a live one."

"Time?"

"Forty seconds."

Crown's transfer officer finally lost calm. "You cannot thrust across the shared center. Jettison your cage and clear the mass."

The words struck Ada with the remembered shape of an older command. Ketu, twenty-four years ago: release the counterweight, save the majority, accept the drum that fell outside the margin. Her mother had been on that drum with six hundred and eleven other people. Later reports called the decision mathematically necessary. None recorded who had felt the floor tilt beneath them while necessity was calculated elsewhere.

Jettisoning would free *Common Measure*. It would leave Crown's field fighting a damaged cage wrapped around Aster Three, with the passenger lane still ahead. The people

giving the order had decided which uncertainty belonged to whom.

"We keep the cage," Ada said.

"Captain," Jalen began.

"Plot Delta Glass through Morrow Flywheel Two."

Recognition changed his expression. Morrow's second flywheel had been retired from grid service and left spinning as emergency storage, a forty-kilometer ring of dense composite turning inside the habitat's outer shadow. At this distance its gravity barely registered. Its magnetic field, however, could couple to a charged cage if someone persuaded Morrow to open the containment vanes.

"Flywheel Two is cold-listed," Jalen said.

"Cold is not dead." Ada opened a direct channel. "Morrow traffic, this is mass tug *Common Measure*. Emergency request. Wake Flywheel Two and give me field authority for ninety seconds."

Static answered first. Light needed only a fraction of a second to cross the gap, but institutions always added distance. A dispatcher demanded a Compact authorization code. Ada gave her captain's emergency key, which would later purchase several hours of accusation.

The ship rolled. Ophion's banded face crossed the blister until Crown Meridian stood like a vertical blade above them. Ada felt the sink's gradient shift through the couch. The command deck creaked as masses designed to meet along one axis began arguing across another.

"Port radiator hinge load rising," Iri said. "I can lock it or keep thermal flow. Pick one."

"Lock. We can run hot."

"Not for long."

"We do not need long."

Crown's capture truss fired attitude thrusters too early. Its field center slid, dragging Aster Three's effective position toward the habitat cylinder. Jalen corrected with the main drives at three percent, a whisper of thrust by fusion standards and enough to rewrite the next month of everyone's orbit.

Morrow traffic returned. A woman's voice, breathless and angry, filled the channel. "Common Measure, Belen Sato, maintenance convener. Flywheel Two can give you seventy-four seconds before the bearings exceed our civil limit."

"Seventy will do."

"What are you bringing across my lane?"

"Aster Three."

Silence held for half a heartbeat. "That was not in the request."

"It is in the vector."

"Your vector says you miss us by thirty kilometers."

"If everyone behaves."

Belen laughed without humor. "Nobody calls me with an emergency because everyone behaved. Vanes opening in twelve seconds."

Ada put Morrow's field on the plot. A faint green lens appeared beyond the red collision vector. It did not have enough strength to capture the sink. Ada needed only a gradient, a third preference in a fight between two cages. If Aster Three's altered charge coupled to the flywheel field, the tug could lean the shared center away from the traffic lane while Crown cleared.

"Crown, yaw on my burn."

"Compact orders cage jettison," the transfer officer said.

"Crown truss, answer with the name of the person commanding your thrusters."

A different voice came back, lower and closer to machinery. "Lieutenant Sima Hall."

"Sima, you have a passenger spindle in the red corridor. I need nine degrees of yaw, sunward. Can your truss give it?"

"It can if the habitat stops widening capture."

Ada looked at the field trace. "Then we both need them to stop."

Iri cut into the open circuit. "Captain, phase drop at twenty-six percent. Manual latch beginning to walk. Blue witness cracked."

That simple failure frightened Ada more than the abstract numbers. "Put your hand on release and keep it there."

"My hand has not moved."

Morrow's retired flywheel woke.

The effect crossed the plots as a tremor. Aster Three's charge sheath responded, not by drifting toward the new field but by changing phase again. The hook in the raw trace became three evenly spaced teeth.

"Prime interval," Jalen said. "Two, three, five milliseconds."

"Record raw. No filters." Ada tightened both hands around the control grips. "Burning Delta Glass in three. Sima, yaw with me. Three, two, one."

Fusion light struck from *Common Measure's* drive. Acceleration pressed Ada into the couch, gentle enough for any passenger and monstrous when applied beside a reluctant singularity. Crown's truss rotated sunward. For an instant the two ships, two cages, and one black hole shared no stable center.

The command blister polarized as Crown Meridian's mirrors swept sunlight across them. Warnings covered the lower display: truss torsion, cage eccentricity, rising coolant temperature, port hinge strain. Ada ignored every

condition that had not yet crossed a red line. She flew the relation, not the ship, balancing their burn against Sima's yaw and Morrow's temporary field.

Aster Three moved seven centimeters inside the modeled cage volume.

Seven centimeters changed everything. The tug's bow dipped as gravity found a new lever. A locker tore open behind guidance. Loose tablets and a metal cup fell sideways, curved in midair, then struck the deck near Ada's boots.

"Latch at limit," Iri said. Their breath came fast now. "If it walks another millimeter, I either fire manual recenter or open the cage."

"Recenter on my mark."

"Manual pulse can kick phase anywhere."

"Not anywhere." Ada watched the three teeth repeat. Two milliseconds. Three. Five. Then an absence where seven belonged. After the gap, eleven.

The pattern did not belong in a fault. Machinery repeated according to frequency, resonance, wear, and feedback. Prime numbers could arise from sampling, but not as clean choices across a charge transition. Something inside the transfer had marked time.

"Morrow, field duration?"

"Forty-one seconds," Belen said. "Our bearings are heating faster than your request promised."

"Close at thirty."

"You get twenty-eight."

"Accepted."

Ada projected the next missing prime. If the sequence continued, the phase would leave a thirteen-millisecond interval after eleven. She could fire Iri's recenter pulse into that absence, when the cage coupling appeared weakest. It

might be superstition dressed as pattern recognition. It remained better than a random command.

"Stand by at thirteen," she said.

Jalen understood. "Two. Three. Five. Gap seven. Eleven."

Aster Three's trace fell flat.

"Now."

Iri fired the manual pulse.

A deep sound crossed the hull, too low to hear cleanly. Ada felt it in her damaged thumb and along the bones of her chest. The sink snapped toward the cage center. Crown's capture field lost its grip for one crucial second. *Common Measure* carried the mass into Morrow's green lens while Sima completed the yaw.

The red collision vector slid beside the passenger lane.

No one cheered. The transfer remained unfinished, the cage still hot, the ships still close enough to make every correction dangerous. Ada held the burn until the predicted miss distance reached five kilometers, then eight, then nineteen. Only when the passenger spindle crossed behind them did she let her shoulders touch the couch.

"Crown truss is clear," Sima reported. "Receiving field collapsing by local order."

Crown Control demanded that she identify the authority for that order.

"My authority," Sima said, and ended the channel.

Morrow closed Flywheel Two after twenty-nine seconds. Belen had given them one more than promised. The green lens vanished, leaving *Common Measure* alone with Aster Three and a month's worth of trajectory debt.

A second channel opened from the passenger spindle they had nearly crossed. Its pilot identified herself as Mei Aras and asked whether the lane remained live. Ada could

hear children behind the woman, their voices thinned by compression and fear. The spindle carried school families from Morrow to a seasonal exchange on Crown. Eleven hundred and eighty-six passengers, Aras said, then apologized for supplying a number no one had requested.

"Keep your present burn for six minutes," Ada told her. "Traffic will give you a new corridor above our plane."

"Did something break?"

Ada studied the amber indicator. "Something changed. Your ship is clear of it now."

Aras accepted the answer because she needed a trajectory more than an explanation. Before closing the circuit, she said that the children had felt the gradient. Water had climbed the spindle's classroom walls and stayed there long enough for them to touch it. Ada entered that observation in the raw log. A fault report would reduce the event to charge, distance, and force. The children's water belonged there too.

Crown Control returned with a formal command to surrender the tug's navigation authority. Ada forwarded it to the crew's shared board before answering. Emergency rules allowed Compact command to seize a tug during piracy, mutiny, or an incapacitated captain. None applied. Crown cited imminent mass hazard, a phrase broad enough to swallow any captain's judgment if left unchallenged.

"Crew council," Ada said over the internal circuit. "You have the seizure order. Anyone who believes I am impaired or acting outside safe-hull doctrine, call it now."

Nineteen names occupied the roster. Eighteen acknowledgments arrived in uneven sequence. Iri's came last from the cage gallery.

Captain fit. Order unsafe while mass remains aboard.

Ada attached the finding to her refusal. It did not make her right. It made the authority visible and placed a line beneath everyone who had accepted the risk with her.

Ketu's fatal decision had moved downward without names until gravity supplied them. She would not let this one travel the same way.

Only then did she notice the command blister's air tasted of scorched insulation. Jalen had blood on his collar where the sideways fall had thrown a tablet into his jaw. Two couches aft, systems officer Parveen kept one hand pressed to her ribs while she reconciled coolant losses. Ada started to ask whether either needed relief, then saw that the medical circuit already carried their self-reports and the roster had redistributed their tasks. Command did not require her to touch every wound before it counted. She marked both reports acknowledged and left the crew's decisions in place.

Outside, the passenger spindle dwindled toward its new corridor. A bead of expelled classroom water flashed in sunlight beside it, briefly separate from every wall and map meant to contain it.

"Drive to standby," Ada said. "Jalen, establish a holding ellipse outside all traffic. Tell every lane controller our predictions are provisional. Iri, status."

A pause stretched from the cage gallery.

"Still attached," Iri answered. "Primary loops degraded. Manual latch scarred but seated. Port radiator hinge took a torsion spike. It will pass inspection if the inspector is corrupt."

"We do not employ corrupt inspectors."

"Then it is damaged."

Ada uncurled her left hand. The old burn ridge along her thumb had whitened under the grip. Around the command deck, seventeen other people began the practiced work of surviving what they had just survived. No one asked whether the transfer would resume. Crown's cage drifted away behind them, its truss turning carefully toward home.

Tomas Reeve called before Compact command did.

His face appeared in a small window, older than the last time Ada had seen him and no less familiar for the distance. Ninefold Observatory's dark displays surrounded him. He did not waste time on greeting.

"I received your raw phase stream," he said.

"From whom?"

"Your guidance officer mirrored it to the anomaly net."

Ada glanced at Jalen. He kept his eyes on the holding ellipse. She approved without saying so.

"Tell me the pattern comes from our cage," she said.

"It does not." Tomas placed five traces beside Aster Three's. Different colors, different timestamps, different sink assignments. "Aster One changed phase near Rook when yours slipped. Aster Five did the same at Harbor of Glass. Nine and Eleven answered within light-time. The deviations are small, but the spacing matches."

Jalen leaned closer. "Two, three, five, missing seven, eleven."

"Then thirteen," Tomas said. "Exactly when your engineer fired the pulse."

Ada looked down at the amber light beneath her thumb. It had steadied after the manual recenter. The eleven blue indicators around it had begun to pulse, each at a slightly different rate.

"What does it mean?" she asked.

Tomas watched the traces rather than her. "I don't know yet. But Aster Three did not refuse your transfer alone."

Across the board, the twelve lights found a common beat.

Chapter 2 — Borrowed Weight

Crown Meridian took Ada's command key before it took her coat.

The hearing escort waited at the lock while a Compact technician pressed a black authorization tile against her wrist slate. Her local navigation certificates dimmed one by one. Beyond the transparent collar, the *Common Measure* hung against Ophion with its cage truss pointed away from traffic. Repair lamps crawled along the port radiator hinge. Aster Three remained aboard, a mass too small to see and too large to leave unattended, under Iri Quill's hand and eighteen crew members who had voted their captain fit.

"Temporary suspension," the technician said. "Transfer privileges only."

"You took drive authority."

"The hearing may restore it."

Ada tapped the dead certificate. "Who holds my ship's collision key until then?"

A Crown security officer answered from behind her. "First Custodian Vardek."

"From eleven kilometers away."

"Your acting pilot retains local avoidance."

Local avoidance meant Jalen could dodge debris but could not alter the holding ellipse without Crown's approval. A gravitational hazard sat inside a damaged cage,

and the person allowed to move it occupied a conference room. Ada took back her wrist and entered the cylinder.

Crown's spin pulled gently as the lift descended toward the administrative band. The floor gained weight by degrees. Farms rotated past the lift windows: green terraces, bright water, citrus trees trained along low trellises. Waste heat from sink service kept those leaves warm through Ophion's long shadow. The people who scheduled that warmth spoke in sink-hours, as though gravity came portioned like skilled labor.

At the hearing chamber, a wall display listed current dependencies. Forty-six habitat names surrounded a moving allocation wheel. Six million, one hundred four thousand, two hundred twenty permanent residents drew some part of their heat, spin correction, water pumping, or construction margin from the twelve captured masses. Crown Meridian held only seven percent of the population and administered every transfer.

Cass Vardek stood at the center table with his sleeves folded once. He had the compact build of someone born under full cylinder spin and the stillness of a man who disliked wasting motion. No tribunal sat above him. Three operations chiefs attended in person; habitat delegates watched through delayed windows along the wall.

"Ada Narang," he said. "You kept a live cage during an unstable handoff, entered Morrow's traffic allocation without clearance, commandeered a retired flywheel field, and induced Crown's receiving officer to disobey control."

"Sima Hall commanded her own truss."

"Your distinction matters legally. It matters less to collision mechanics."

"Then begin with the collision mechanics."

Cass gestured, and the room darkened around a reconstruction of the failed handoff. *Common Measure* became a wireframe shape beside Crown's receiving truss. The red

path toward Morrow's traffic lane returned with clinical brightness. Ada watched her own burn bend it aside.

"Control's jettison order cleared your crew," Cass said. "Our median model gave the passenger spindle a seventy-three percent survival probability."

"Median hides the tail."

"Keeping the cage gave both vessels fifty-one percent."

"Your model treated Aster Three's charge change as noise. Mine used the raw trace."

A projection chief enlarged the assumptions. The Compact model had indeed smoothed the prime intervals into an averaged decline. Ada could see where each dangerous fact disappeared: first inside a filter, then a probability distribution, then a number suited to command.

Cass did not defend the filtering. "You had forty seconds. Central control had twelve seconds of clean telemetry and responsibility for a habitat behind the truss. You want them to defer to one captain's private interpretation?"

"I want risk to stay with the people who can feel it change."

"That principle strands six million people inside forty-six local truths."

"It keeps you from exporting uncertainty into somebody else's hull."

One delegate window brightened. Belen Sato watched from Morrow with dark hair tied away from a face gone gray from missed sleep. "For the record," she said, "Crown exported its receiving error into our passenger lane before Captain Narang touched our flywheel."

The clerk asked her to reserve comment until called.

Belen leaned closer to her pickup. "Reserve the collision next time." Her window muted under procedural control.

Ada kept her attention on Cass. Anger offered him an easier case. He could file temper beside insubordination

and never address who got chosen when uncertainty narrowed.

"What would you have done?" he asked.

"Collapsed Crown's field at my first stop call."

"After they failed to comply."

"Delta Glass, exactly as flown."

"Without knowing Morrow would open the flywheel."

"I asked before I committed the burn."

Cass changed the reconstruction. In the alternate path, *Common Measure* jettisoned its cage. Explosive bolts separated the truss. The tug rolled clear while Crown's field captured the charged assembly, or failed to. Hundreds of translucent runs scattered around the passenger route. Most passed wide. Several crossed it. Two entered Crown Meridian's outer maintenance skin.

"You rejected the highest aggregate survival case," he said.

"I rejected a model that assigned all fatal variance to crews who had no vote."

"Gravity does not recognize votes."

"No. Commands do."

The old scar along Ada's left thumb tightened as she closed her fist. In Crown's projection, abandoned machinery drifted with no history. The image needed another cylinder, another failing drum, a girl of seventeen braced in a school corridor while the floor angle changed.

"Ketu Spin Loss," she said. "Put that model beside this one."

A clerk looked to Cass. He nodded.

The archive rendered the Ketu system twenty-four years earlier. A scheduled Aster Eight exchange had arrived late after a tug coolant fault. Ketu's counterweight drums began losing synchronization. Compact control ordered the outer

drum released to preserve the core cylinder and the sink cage. Six hundred twelve people died when spin loss tore the released drum along a bearing seam. Ada's mother occupied maintenance compartment K-14. Official analysis calculated that holding the drum might have killed four thousand.

Cass studied Ada rather than the archival geometry. "You believe we should have held it."

"I believe the people aboard deserved authority over the risk."

"They had eighty-nine seconds."

"They had names."

"So did the four thousand." His voice stayed level. "Ketu taught this administration that delayed central action can multiply casualties. It did not teach me to enjoy choosing."

The reply checked her. Cass had not hidden behind necessity. He carried the decision without pretending arithmetic cleaned it. For one breath she saw the protector inside the office, and understood why that made him more dangerous to her judgment, not less.

An operations chief placed a new order on the shared table. "All sink transfers suspended pending cage inspection and phase review. No tug may alter a sink trajectory without Custodian authorization. Effective immediately."

Ada read the distribution list. Every active cage. Every local captain. "You froze circulation."

"I froze transfers," Cass said. "Current stations continue draw."

"Aster Three missed Crown's allocation. Morrow spent flywheel margin saving your lane. How long before the gap propagates?"

"Thirty-one hours before service reductions. Longer before critical loss."

"Whose reduction?"

"Morrow, Ketu, Lantern Reach, and three yard swarms receive balancing orders first."

Belen's muted window acquired a red request border.

Ada pushed the order back. "You answered a coordination failure by making permission slower."

"I answered an unknown mass behavior by stopping forty-five chances to repeat it."

"Eleven," Ada said. "There are twelve sinks, and one cannot transfer."

"Eleven chances remain enough."

Cass signed. Around the wall, allocation paths faded. Small deficit numbers appeared beside habitats that relied on arriving service. Crown's own reserve line stayed green.

The hearing broke for technical review. An escort guided Ada into an observation gallery overlooking the receiving truss. Sima Hall worked outside in a white suit, crossing the lattice toward a warped field coil. Crown had relieved her from piloting but apparently not from repair. Ada opened a short-range suit link.

"Lieutenant Hall."

Sima's helmet turned. "Captain. This circuit gets us both another hearing."

"You obeyed the geometry."

"I disobeyed my habitat."

"Those can describe the same act."

A gloved hand secured a tether around the next rail. "Control gave me the jettison model. I saw your ship survive in most runs."

"Most."

"I have a sister on Crown's outer skin. That word worked on me until I painted the passenger lane." Sima moved along

the truss. "Custodian Vardek asked whether you coerced my decision."

"What did you say?"

"That your burn gave me an option control had removed."

Below her, the receiving cage remained dark. Twelve radial arms curved around an empty center, all that structure built to manage electric charge around a point whose gravity ignored steel. Humanity called the sinks common property because no habitat could safely own one. Yet every diagram placed a custody ring around them.

Borrowed weight, Ada's mother used to call artificial spin whenever Ketu's counterweights passed close. Do not trust borrowed weight until you know who can take it back. Ada had thought she meant machinery. Now the phrase attached itself to the allocation wheel and would not leave.

Her slate chimed from a crew-private route. Iri appeared from the cage gallery, face shining with sweat. A blue paint fragment floated beside their cheek inside a sample envelope.

"Crown killed our external diagnostic permissions," they said. "So this report arrives as a social call between two unemployed people."

"Status."

"Primary loops down six percent. Latch scar did not spread. Hinge needs a yard, not a blessing. Aster Three's effective charge keeps stepping every two hundred eleven seconds."

"Magnitude?"

"Tiny. Four parts in ten million. It nudges the cage center less than a hair."

"Pattern?"

"That depends whether you prefer machines to have manners. It waits until after our automatic correction, then steps opposite."

Ada looked through the gallery glass toward her tug.
“Can you stop correction?”

“Under the freeze order, I cannot change closed-loop settings. Under sane engineering, I also should not experiment while parked above an administrative habitat.”

“Record raw only. Mirror Tomas Reeve.”

Iri raised an eyebrow. “That name just personal, or personally useful?”

“Useful.”

“Tragic.” They ended the circuit before she could answer.

Jalen called from guidance before Ada could pocket the slate. Crown’s revocation had forced him to submit even routine station-keeping burns for approval. Three requests waited in the queue. The oldest had remained unread for twenty-two minutes.

“Our ellipse grazes a debris exclusion in four hours,” he said. “Not dangerous yet. It becomes expensive after two.”

“Can you use collision authority?”

“Only inside the published red threshold.”

“So waiting makes the burn qualify.”

“That appears to be the approved procedure.”

Ada looked through the gallery at the small repair lights moving along her ship. Central control had created a rule under which delay manufactured its own emergency authority. “Send Crown the propellant difference between burning now and waiting for red. Copy the crew board and the hearing clerk.”

“Already calculated. Nine tonnes.”

“Then make the cost visible.”

Jalen added another concern. Crown had ordered crew sleep rotation during suspension, but cage degradation required two extra watchers in Iri’s gallery. Someone had

to leave thermal repair or guidance to fill the posts. Ada almost assigned Parveen and Ors immediately. Her missing command key stopped the words before they left.

“Put it to shift lead,” she said. “Iri names required competence. Crew decides coverage.”

Jalen’s expression changed by a degree. “You want them to allocate?”

“I want the people doing both jobs to tell us which one can wait.”

The answer did not come naturally. Ada could hold the entire duty roster in her head, including who had slept, who carried bruised ribs, and who made errors after a second watch. Delegating that choice felt less like trust than failure to carry weight assigned to her. Yet Crown had just demonstrated the cost of deciding from a room outside the hull.

Five minutes later, the crew board returned a plan Ada would not have chosen. Ors left hinge inspection, but only after training a drive technician to photograph the strain marks. Two off-watch crew shortened sleep by forty minutes voluntarily; Iri rejected that portion and reassigned a rested galley specialist certified on alarm response. The final schedule preserved both cage watch and inspection with no hidden overtime.

Ada approved it once. Her thumb hovered over the seal a second time, then withdrew. The crew did not need her to discover a mistake solely to prove she had looked.

Crown finally authorized the earlier station-keeping burn. Jalen spent three tonnes instead of the projected twelve and moved the holding ellipse clear. Aster Three stepped against automatic correction as the drives stopped, another tiny departure entered into Iri’s private mirror.

The name Tomas had left pressure behind it. Ada and he had spent nine years learning each other’s silences, then three learning to turn grief into blame. He had asked her to

remain home after their daughter died. She had returned to *Common Measure* because trajectories continued whether she could bear them or not. Neither fact had forgiven the other.

When the hearing resumed, Tomas entered through the public lock carrying a data wafer and wearing an observatory jacket with one seal half closed. He had traveled from Ninefold on a priority shuttle. The last time Ada had seen him in person, they had signed the dissolution agreement beneath a mural of Ophion's moons. His hair held more silver now. His habit of checking room exits had not changed.

Cass met him at the table. "Doctor Reeve. You requested emergency standing."

"I requested that you not destroy evidence by normalizing cage clocks."

Formal already, Ada thought. He had arrived angry.

A technician linked his wafer. Five phase traces opened above the failed-transfer reconstruction, the same streams Tomas had shown aboard *Common Measure*, now extended through six more hours. Their tiny hooks aligned after correction for light-time.

"These correspond to Aster One, Three, Five, Nine, and Eleven," Tomas said. "The apparent simultaneity in the first report came from local timestamp conventions. In physical sequence, each deviation follows a gravitational relation to one or more Ophion moons."

Cass folded his arms. "Cages share control software."

"Three software branches, five cage generations, four operators."

"Solar weather."

"Wrong propagation direction."

"Sampling artifact."

"I resampled against atomic references on Ninefold and Crown." Tomas touched the plot. "Aster Three changed first. Aster Nine responded when Adras crossed the plane defined by Three and Ophion. Eleven followed at the next node passage. Machines can share noise. Noise does not normally wait for moons."

The operations chiefs began speaking at once. Cass stopped them with one raised finger. "Cause?"

"I do not have one."

"Hazard forecast?"

"Not yet."

"Then you have correlation during an emergency freeze."

Tomas's mouth compressed. Ada remembered that expression above their daughter's diagnostic charts, when one specialist had called an uncertain prognosis reassuring. "I have evidence that your freeze changes the thing observed. Every cage has increased automatic centering since the order. The deviations weakened immediately afterward."

"Good," said the projection chief.

"No. A muted signal can mean successful control, damaged instrumentation, or a system driven into a mode we cannot read."

Cass turned to Ada. "This supports tighter command over cage behavior."

"It supports preserving local observation," she said.

"Captain Narang's improvisation also changed the signal," Tomas said. "Her manual pulse landed inside a phase absence. We cannot separate response from coincidence yet."

She felt the old conflict arrive whole. Tomas would protect uncertainty until it became a wall. "How long to separate them?"

"With access to complete transfer archives, a day for an initial test."

"Morrow starts shedding service in less."

"I heard."

"Did you calculate what your day costs?"

His gaze met hers. "Did you calculate what a wrong interpretation costs?"

Cass watched the exchange with the attention of a man discovering a fault line he might need. "The transfer freeze stands. Doctor Reeve receives read access to current cage telemetry, not command access. Captain Narang remains suspended pending review."

Ada placed both palms on the table. "Give my crew authority to move if holding becomes unsafe."

"Local collision avoidance remains active."

"Aster Three defines local over millions of kilometers."

"Then your acting pilot will call Crown, and Crown will decide."

Tomas shifted toward her, almost speaking. She did not look at him.

A warning rolled across the wall before Cass could close the session. Not red. Amber, repeated twelve times around the allocation wheel. Each active sink had left its predicted phase by less than the tolerance of an ordinary cage correction. The values differed in sign and magnitude. Their timestamps formed a descending sequence after light-time adjustment.

The projection chief blamed the freeze command. A second analyst said no common command had reached every cage yet. Crown's own Aster Ten sat beyond the first transmission window.

Tomas swept the traces into one orbital plot. Lines linked the sinks through Ophion's five major moons. A

shape appeared, broke, then reappeared rotated. Ada could not read its meaning, but she recognized coordinated work. Twelve separate faults did not usually preserve a relation while changing direction.

"How many?" Cass asked.

"All twelve sinks," Tomas said.

Aster Three's amber marker led the next divergence. Eleven blue points followed by tiny coordinated amounts, each movement legal under conservation and impossible to dismiss as one damaged cage.

Cass looked from the moving lattice to the frozen transfer schedule. His order had stopped human circulation. It had not stopped whatever passed between the masses.

"Restore Captain Narang's shipboard command key," he said. "No transfer authority. I want her aboard the hazard."

The technician returned Ada's certificate. Only one line brightened.

Outside the chamber, Crown Meridian continued turning on borrowed weight while the twelve sinks redrew their relation around it.

Chapter 3 — The Noise Between Masses

Tomas removed eleven thousand corrections from the record and listened to what remained.

Ninefold Observatory kept its dynamics room behind two pressure doors and a corridor lined with obsolete clocks. The clocks had once disciplined transfer telemetry before the habitat acquired a shared optical lattice. Now each displayed a different system time, useful only as a warning that agreement could be manufactured. Tomas crossed beneath them carrying three generations of cage logs on his slate.

Inside, five phase traces unspooled as sound.

He had mapped effective-charge shifts into a register narrow enough for human hearing. Aster One clicked near the low edge. Five produced a soft double knock. Eleven entered above both, followed by Nine and the irregular pulse from Three. No black hole made those sounds. Speakers rendered numbers into pressure waves because Tomas could hear timing errors his eyes forgave.

Two. Three. Five. An empty place. Eleven. Thirteen.

Then ordinary cage control flooded the sequence with corrections. Pumps hunted charge balance. Magnetic loops compensated for thermal expansion. Operators trimmed fields after every predicted moon tide. Human machinery spoke continuously, loudly, and with the confidence of a species that mistook intervention for clarity.

"Strip another layer," said Ela Venn from the neighboring console.

"If I strip one more, you'll accuse me of finding my preference."

"I accused you six hours ago." Ela managed Ninefold's instrumentation group and regarded courtesy as a waste product. "The accusation remains active."

Tomas replaced the speakers with a residual plot. He had subtracted known controller response, solar wind, cage temperature, clock drift, and the gravitational perturbations already present in Compact models. Pale marks remained along five horizontal tracks. Too sparse for melody. Too ordered for the usual mess.

A public-service warning opened across the lower display. Morrow had entered Tier Two heat conservation after failing to receive Aster Three's scheduled allocation. Nonessential fabrication would stop within the hour. Residential humidity controls and agricultural lighting followed if the deficit persisted.

Ela read over his shoulder. "Your captain will want an answer."

"She no longer qualifies as mine in any legal or domestic sense."

"That phrasing suggests excellent emotional health."

Tomas shut the warning. "She wants predictions. Wanting does not produce them."

"A hundred eighty-four thousand people may produce some urgency."

"Urgency can choose the order of tests. It cannot improve their evidence."

The outer pressure door cycled before Ela could answer. Ada Narang entered wearing the same dark uniform from Crown's hearing and carrying her returned command tile against her chest. She had slept little, if at all. A pale ridge

marked her left thumb where it rested on her slate. Tomas noticed the old injury before he noticed her face and disliked himself for the order.

"Narang," Ela said. "You arrived before his prediction and after his excuse."

Ada gave her a tired nod. "I need the room."

"You need the observatory. The room comes with staff."

Tomas saved Ela the pleasure of a fight. "She has active command status aboard the hazard. Her access applies."

Ada's gaze moved to the sound map. "Five sinks."

"Five with full raw logs. The other seven have summary streams until their local operators release archives."

"Cass ordered access."

"Cass ordered read access. Compact archives interpret that as the products they ordinarily let readers see."

"Can I change their interpretation?"

"Not from Ninefold."

She placed her tile beside his console. "Then tell me what those five can do next."

No greeting. No question about the shuttle crossing or his work. Tomas had once valued the way Ada omitted ceremonial words. After their daughter's death, every omission had widened until he could no longer tell efficiency from departure.

He opened the residuals. "Aster Three changed phase during the handoff. One, Five, Nine, and Eleven diverged afterward, with delays tied to geometry rather than communications. I can show coordination. I cannot yet forecast direction."

"Morrow needs a power path in seventeen hours."

"I know."

"The missed allocation pulls Crown reserve intake below schedule at thirty hours. Cass will protect Crown by extending Morrow's reduction."

"I read the dispatch notice."

"Reading it does not answer me."

Tomas heard his own voice become precise. "No responsible model can answer you after six hours of anomalous data."

Ada leaned both hands against the console. "Responsibility belongs to the consequences too."

"Consequences do not make a guess true."

"Call it a bounded guess."

"Changing the noun will not create a bound."

Silence pulled Ela's attention toward another workstation. She made a visible performance of checking clocks. Once, Tomas and Ada could argue inside a shared problem without using it to reopen every older one. Now even mathematics carried domestic shrapnel.

Ada touched the first residual mark. "What happened immediately before this?"

"Crown raised capture field."

"And here?"

"Your cage restored center."

"This one."

"Aster Nine crossed Adras's orbital plane."

"Not what the moon did. What did a person do?"

Tomas opened the operator logs. Aster One's tug had increased centering gain. Five's yard cage had shifted accretion feed. Eleven's crew had calibrated a Hawking-spectrum sensor. Nine had received no human command near its deviation.

"Four actions and one absence," Ada said.

"Which weakens a command-response hypothesis."

"Or Nine answered something besides us."

He looked at her. "That possibility does not make it actionable."

"Everything becomes actionable when it changes a trillion tonnes."

The habitat intercom announced a minor spin trim. Ninefold's floor altered its pressure under their feet as buried flywheels exchanged momentum with the cylinder. Tomas waited for the clocks in the corridor to disagree about the event, then marked their offsets. An idea moved at the edge of the data and vanished when he tried to name it.

Ela returned carrying three cups clipped into a rack. "Morrow just requested direct power loans from four habitats. Compact denied two because transfers remain frozen."

"Which two?" Ada asked.

"Rook and Harbor of Glass. Far Lantern offered batteries, but transport losses eat half."

Ada called *Common Measure*. Jalen answered from the command blister with Aster Three's live phase trace beside him.

"Give me our available reactor export," she said.

"After cage and cooling? Forty megawatts."

"Morrow needs six hundred."

"Then we can disappoint them by five hundred sixty."

"Prepare a beam window anyway. No transmission until allocation clears."

Jalen hesitated. "Crown denied our last request."

"Make them deny a specific number."

The circuit closed. Ada kept her eyes on the phase trace rather than Tomas. "You have until that window opens."

"Threats do not improve inference."

"It is a clock."

"You always call pressure a clock when you want it to sound neutral."

Her shoulders hardened. "Do not do this here."

"Do what? Remember your vocabulary?"

Ela put down her cup. "I can remain as a witness or leave as a friend. Neither role includes watching you turn dead family into an analysis method."

The rebuke landed cleanly. Tomas removed his hands from the console. Ada looked toward the old clocks beyond the glass, each wrong in its own disciplined fashion.

"We work," she said.

They worked.

Tomas assigned her the operational logs while he rebuilt the timing solution from primary references. Ada read quickly, marking every correction by who authorized it, who executed it, and what hardware carried the risk. He usually categorized by mechanism. Her categories exposed a different pattern: remote commands produced immediate countersteps more often than local manual adjustments.

"Selection effect," he said. "Remote commands handle larger deviations."

"Then normalize for magnitude."

He did. The pattern weakened but remained.

Ela obtained unfiltered temperature data from Aster Five's yard cage. A thermal oscillation explained two residuals and sharpened a third. Tomas deleted the false pair without regret. Good evidence grew smaller under examination. Fabrication grew decorative.

At the next power warning, Morrow entered Tier Three. The notice translated percentages into systems: two clinic sterilizers, seven communal kitchens, a water-recovery reserve, seed lighting for the north agricultural deck. Ada read each line.

"People hear 'nonessential' and imagine signs going dark," she said. "They do not imagine a surgical tool waiting for the next load."

"What do you need from me?"

"A safe condition for moving Aster Three."

"I do not have one."

"A dangerous condition we can avoid."

"Isolation." Tomas heard the answer as he gave it. "The strongest phase opposition follows corrections that center one sink without accounting for the others."

Ada turned. "Others where?"

"Not physically nearby. In the model."

"Show me."

He drew a vector from Aster Three through Ophion and extended it toward Aster Nine's predicted position. At handoff, Crown's capture plan had optimized Three against the receiving cage and local moon tides. It ignored every sink outside the collision neighborhood. Three's first phase hook increased coupling along the neglected vector.

Ela whistled softly. "A correction to the correction."

"Perhaps," Tomas said.

Ada enlarged the next event. "Aster One's centering gain cut its modeled link to Five."

"Then One stepped toward that relation in phase space."

"Relation," she repeated, distrustful of the abstraction.

"Not attraction. Nothing crossed the distance. Effective charge and accretion timing changed inside each cage, altering how we measure phase."

"Can we move Three while preserving its link to Nine?"

"Link assumes too much."

"Can we preserve the number you just drew?"

Tomas almost smiled. Ada had always trusted quantities after stripping them of metaphor. "Possibly. I need the full moon solution."

Ninefold's dynamics room curved outward into a volumetric theater. Ela cleared routine projects and loaded the Ophion system at one to ten million scale. The gas giant filled the center as a muted amber sphere. Five major moons traced inclined paths. Habitat clusters appeared as dust. The sinks needed oversized markers to remain visible.

Tomas ran the last day at increasing speed. Aster Three's divergence flashed near Crown. One answered near Rook. Five, Nine, and Eleven followed. Lines based on light-time produced nothing stable. Lines based on direct distance tangled immediately.

"Turn off habitats," Ada said.

He removed them.

"Cages too."

Five points moved through moon shadows with no human steel around them. The visualization offended physical fact, but it removed the habits attached to ownership.

Tomas plotted deviations against the nearest major moon. The prime sequence blurred. He tried moon orbital phase. Two events aligned, three wandered.

Ela suggested tidal tensor eigenvectors. A partial relation appeared between Three and Nine, then failed at

Eleven. Ada paced along the theater edge, counting handholds in the rail without touching them.

“What did the missing seven correspond to?” she asked.

“A gap before your manual pulse.”

“No. Where had the moons gone?”

Tomas rewound the handoff by milliseconds. On an orbital scale nothing visible moved. Yet the line from Aster Three to Adras crossed the line from Aster Nine to Ophion during that tiny interval when projected onto the system’s invariable plane.

“Not positions,” he said. “Alignments.”

He transformed the plot.

The noise opened.

Prime spacing governed one set of marks: two, three, five, absent seven, eleven, thirteen. Beneath it lay longer intervals close to one, two, three, five, eight, thirteen times a base period. Fibonacci-like recurrence, imperfect because moon motion stretched each return. The second sequence did not originate from Three. Aster Nine carried it across the alignment and Five completed it.

Ela increased resolution. “You could get that from coupled oscillators.”

“Not through independent cage controllers,” Tomas said.

“Unless controllers share an input we missed.”

“Ophion’s moons supply the shared clock, not the choices inside it.”

Tomas tried to destroy that conclusion before Ada could rely on it. He randomized the five sink labels, shifted each clock inside its calibration tolerance, and asked Ela to inject an unlisted controller delay. Prime spacing survived only in the true labels. The longer sequence collapsed when moon geometry disappeared and returned when he restored the measured positions. A shared software fault could imitate

timing, but not the direction of charge response across controllers built by three yards in four different decades.

Ela ran one last adversarial model with solar wind as a hidden driver. It reproduced Aster Five's amplitude and missed the absent-seven interval at Three. "That kills weather," she said.

"It limits this weather model." Tomas saved every failed alternative beside the surviving result. A claim became dangerous when its discarded explanations vanished from view. "We still have five instruments observing a system of twelve. Structure does not yet prove intention."

Ada stood inside the projected orbit of Adras, lines crossing her uniform. "Five sinks. Two interval families."

"More may participate. These five supply enough raw data to demonstrate structure."

"Forecast."

He advanced the moon solution to the next alignment. The model offered three possible phase responses depending on which sequence continued. One would ease Aster Three toward its cage center. Another would increase offset but preserve thermal margin. The third would pull the effective center toward the damaged side of the cage.

"Three answers," Ada said.

"Three hypotheses."

"Which one preserves relation to Nine?"

"The second."

"And when?"

"Six hours, nineteen minutes."

Her beam window to Morrow opened in five hours. She called Jalen and ordered no automatic centering during a four-minute bracket around the predicted event. Iri joined from the cage gallery.

"You want me to let it wander because a marriage counselor found Fibonacci numbers?" they asked.

"Dynamicist," Tomas said.

"I have met both trades. Similar faith in models."

Ada ignored them. "Hold manual authority. Red line at two micrometers effective offset or one percent loop asymmetry."

"Compact freeze requires continuous correction."

"I have shipboard hazard command."

"Cass will say that covers collision, not experimental neglect."

"Then call it reduced control noise for diagnostic safety."

Iri grinned. "Now you sound married to an observatory."

After the circuit ended, Ada and Tomas stood together beneath the projected moons. Work had brought them into an old distance, close enough to share a display, not close enough to touch. He knew the shape of the question she avoided.

"I stayed at Ninefold because I could not enter that apartment," he said.

Her jaw tightened. "We agreed not here."

"We agreed to work. This affects how we work."

"Our daughter does not belong in a sink model."

"No. But our habits do." Tomas kept his voice quiet. "You went back to the tug because a schedule still needed you. I waited for grief to produce a meaning it did not have. We each called the other's survival method an accusation."

Ada stared through Ophion's translucent image. "You asked me to stay."

"I did."

"I heard you ask me to stop being useful."

"I heard your departure say I had failed to keep her alive."

Neither statement offered repair. The dead did not become evidence for one living person's innocence. Tomas had spent years trying to prove a verdict from events that carried none.

A priority alert spared them from choosing the next sentence. Morrow's projected reserve had fallen again. Aster Three's missed service exposed an old coolant-pump inefficiency, and conservation measures had increased load elsewhere. Critical heat margin now arrived in fourteen hours, three earlier than forecast.

Ada read the technical attachment. "They need a tow gradient to shift radiator working fluid. Power beam will not solve that."

"Then your window closes."

"No. It changes." She called Crown operations. "Authorize *Common Measure* to approach Morrow for gradient service after our predicted event."

Cass answered personally. "Your prediction has three outcomes."

"One keeps our cage stable and lets us tow."

"One worsens your offset."

"We set a local abort."

"The transfer freeze remains."

"This is not a transfer."

"It moves a sink."

"Every orbit moves a sink."

Cass paused, likely reading Morrow's new margin. "Submit geometry. No commitment until review."

Ada sent it while speaking. "Review has five hours."

The line closed without agreement.

Tomas saved the forecast with explicit probabilities. He hated all three numbers. Their confidence intervals depended on only nineteen reliable events, two partly contaminated by controls. Yet refusing to predict also selected an outcome: Morrow continued toward heat failure while certainty improved elsewhere.

"Put my name on the model," he said.

Ada looked at him. "It already has your credentials."

"My name. Human review, not observatory product."

She added it without thanks. Trust had not returned. Accountability had acquired an address.

Six hours later, they watched Aster Three approach the projected alignment. *Common Measure* held corrections inside Iri's red lines. Ninefold converted the five live traces to sound.

Aster Three marked two, three, five. Silence occupied seven. Eleven arrived from Aster One. Thirteen sounded at Five.

Then the longer sequence entered through Aster Nine: one, two, three, five, eight. At the next shared alignment, Three changed phase in the direction that preserved the Nine vector. Its cage offset increased by less than a micrometer and stopped.

Ada released a breath. "Second hypothesis."

"Within uncertainty."

"Will it hold during tow?"

"I can give you a corridor, not a promise."

"Good. Promises have no abort threshold."

Cass authorized the Morrow approach with Compact observers aboard both tugs and strict field limits. The order appeared on Ada's slate. She turned for the pressure doors.

Tomas kept listening. Another interval followed after Ada had left the theater. Aster Nine repeated the longer

sequence. Aster Three returned the prime spacing, altered by one missing term. Across five sinks, one pattern posed a timing relation and another changed in response.

He called Ada before she reached the shuttle lock.

"This is not a chorus," he said. "Do not write that in your operational note."

"What should I write?"

Tomas watched moon positions cross the residual plot. The phrase felt too human, but it described the structure rather than an imagined speaker.

"Call and response," he said. "Mapped against the moons."

On the speakers, noise separated into intervals. Five captured masses kept answering through cages built to make them obey.

Chapter 4 — Principle of Least Harm

The tow line existed only on Ada's display, which made it no less capable of killing both crews.

Common Measure approached Morrow along the torus's shadowed face with Aster Three centered inside its scarred cage. Twelve kilometers off the tug's starboard quarter, the *Principle of Least Harm* matched velocity and carried no sink. The second tug would shepherd the gradient: close enough to pull against if *Common Measure* lost attitude, far enough that Aster Three could not turn two hulls into one falling system.

Morrow rotated below them. Its crowded torus showed dark sections where conservation orders had cut exterior lights. Radiator petals spread from the hub in six silver ranks. One bank glowed too hot on infrared, working fluid trapped behind a stalled manifold. If the tugs gave the fluid a changing gravity vector, pumps could drive it through the obstruction and recover a quarter of the habitat's heat margin.

If Ada brought the sink too close, the same gradient would load Morrow's bearings sideways.

"Range to first field point?" she asked.

Jalen read from guidance. "Eleven point four kilometers. Closure six centimeters per second. Nadi wants us three centimeters slower."

"Tell her no."

Nadi Okafor answered on the shared circuit before Jalen relayed it. "I heard that, child."

Ada looked toward the other tug's marker. "Then you also heard the closure."

"I heard you spend half the stopping margin before the work begins."

"We need twelve gradient cycles inside Morrow's thermal window."

"Eleven clean cycles beat twelve followed by an inquiry."

At sixty-three, Nadi had captained mass tugs longer than Ada had lived aboard them. Her vessel carried the same name as the doctrine she quoted whenever command mistook reach for right. She had taught Ada to land a dead cage at Rook, to distrust displays without raw ranges, and to drink water before a difficult burn. Ada followed the last instruction now because refusing would invite comment.

Iri came up from the cage gallery, clipped their boots beneath the engineering console, and opened the port radiator schematic. Amber strain marks crawled around the damaged hinge.

"We can reject tow heat or keep the wing locked," they said. "Still no miraculous third option."

"Keep it locked until gradient cycle six. Then open to seven degrees."

"The hinge complains at four."

"It complained during inspection."

"It filed a structural objection. You overruled it with vocabulary."

Ada pulled the thermal forecast beside the flight plan. "Seven degrees for nine minutes keeps the cage loops below abort."

"And leaves me nothing if Aster Three changes charge."

"What angle gives reserve?"

"Five, if Nadi slows closure."

The elder captain made an approving sound over circuit.

"Five degrees," Ada said. "Keep present approach."

Nadi did not waste breath claiming victory. "My tow point remains twelve kilometers. If your stern swings outside three degrees, I burn away."

"You should pull us back."

"My empty tug cannot pull a caged trillion-tonne mass anywhere your drives have not already agreed to go. I can alter your attitude and buy seconds. Do not make those seconds into mythology."

Morrow control handed the operation to Belen Sato. Her voice entered beside a schematic of the blocked thermal loop. "Convener on circuit. Manifold crews have installed the bypass. We need one point eight milligee across the hub line, reverse, then repeat. Bearing authority gives you point zero two above that for three seconds per cycle."

"Three seconds leaves no sensor delay," Ada said.

"It leaves the civil margin our structure can carry."

"A filtered accelerometer can hide half a second."

"Then use raw instruments. You made that argument loudly enough yesterday."

Ada almost liked her. "Raw threshold at one point eight one. We abort at one point eight two."

"Accepted. You cross one point eight two, my operator closes the manifold and you leave."

Cass Vardek had placed a Compact controller on the circuit. "Morrow does not hold abort authority over an active sink maneuver."

"This maneuver acts on our structure," Belen said.

"Captain Narang commands her tug. Compact retains sink authority."

Nadi entered before Ada could. "My authority ends at another crew's hull. So does yours, Custodian. Morrow calls its bearing limit."

"Doctrine does not supersede Article Four transfer rules."

"No transfer occurs," Ada said. "Your authorization classifies this as emergency gradient service. Local structure limits bind all service craft."

Cass paused. He disliked being trapped by his own order, but he did not pretend the law changed. "Morrow's bearing limit stands. Compact retains authority to terminate earlier."

Three command lines now crossed the same maneuver. Ada set each threshold on the crew board with its owner attached. Nobody would later claim a machine had chosen.

They reached the first field point.

Aster Three's gravity extended without boundary, but useful gradient rose sharply at close range. *Common Measure* rolled cage-down toward Morrow and began a shallow radial burn. Nadi held above and behind. Ada watched relative acceleration build across the habitat's hub.

"Point nine," Jalen said. "One point two. One five."

Morrow's blocked loop shook under the changing load. Cameras showed a thick pipe flexing between braces while four suited workers crouched behind a shield frame. Their names occupied Ada's side display: Van Osei, Mira Pell, Donal Yu, and Kez Parata. Attaching names did not reduce hazard. It prevented scale from turning them into an acceptable fraction.

"One seven six."

"Hold," Ada ordered.

The tug's drive corrected by fractions. Aster Three moved through modeled cage volume by less than a micron, yet the pressure under Ada's teeth changed. The raw phase trace produced a shallow tooth.

Iri leaned toward it. "There."

"Offset?"

"Stable. Phase against Aster Nine changed four parts per million."

Tomas monitored from Ninefold. "That direction weakens the relation we identified."

"Action?" Ada asked.

"Reduce radial component by point three percent."

"That drops gradient below bypass threshold."

"Then alter tangential thrust. Preserve the Three-Nine phase vector while maintaining local magnitude."

Jalen painted the correction. It would swing their stern toward Nadi's tow line.

"Principle, can you take a two-degree crossing?" Ada asked.

"I can move," Nadi said. "You do not get to move me."

The difference cost six seconds. Nadi burned first, opening her separation to fourteen kilometers. Only after her hull cleared did Ada alter tangential thrust. Aster Three's phase tooth flattened. Morrow's gradient held.

"Fluid movement," Belen reported. "Nine percent through the bypass. Continue."

At the end of the three-second peak, Ada backed away. Working fluid surged through the manifold and stopped at thirteen percent. Eleven cycles remained.

They repeated the approach with corrected geometry. Cycle two reached twenty-one percent. During cycle three, Morrow's bearing temperature climbed faster than forecast. Belen shortened the peak by half a second. Cycle four pushed the bypass to forty-four percent and heated *Common Measure's* locked radiator root toward amber.

Work narrowed the command blister. Distances replaced faces. Ada's thumb rubbed its scar each time Aster Three's

phase drifted away from Nine and settled when Jalen adjusted tangential thrust. A pattern emerged: the sink tolerated corrections that changed destination, local gradient, or cage offset. It opposed corrections that reduced the modeled phase relation Tomas had drawn.

After cycle five, Morrow requested a pause to replace a manifold brace that had begun walking under repeated reversal. Three workers entered the shield frame while the tugs backed to minimum gradient. Ada recognized Van Osei and Mira Pell from the camera roster. Kez Parata carried a locking fork almost as long as their body. Donal Yu watched strain numbers from inside the hub.

"Seven minutes," Belen said. "No field peak while hands touch the branch."

"We lose radiator margin during the hold," Iri warned.

Ada opened the damaged wing another half degree. Hinge strain climbed in small teeth. "Six minutes."

"Seven keeps Mira's glove attached," Belen said.

Nadi moved *Principle of Least Harm* farther upslope, reducing its already faint contribution to local gradient. "Your schedule does not reach through their shield."

Ada set seven minutes. She hated idle danger. The cage heated, Crown's reserve window narrowed, and Aster Three continued its minute deviations while three workers turned mechanical fasteners by hand. None of those costs allowed her to hurry fingers inside another structure.

At five minutes, Kez dropped the locking fork. It drifted toward the open manifold gap. Mira caught it by a wrist loop, but the motion pulled her boots from the rail. Van anchored both of them. Donal called rising pipe strain.

Ada's hand moved toward a corrective pulse before she understood the impulse. A tiny cage adjustment could have floated the tool away from the opening. It could also have shifted loads across three bodies who had not asked her to touch their gravity.

She left the control alone.

Kez recovered the fork, set the brace, and crossed behind the shield at six minutes forty-one. Morrow certified the work zone clear. When Ada restored gradient, the new restraint held within two percent of calculation.

"Good patience," Nadi said privately.

"I did nothing."

"Exactly."

Ada cut the private channel before approval could settle into comfort. Aster Three's trace showed no opposition during the pause. Human work had preserved its relation by declining to correct it, though she could not tell whether that fact mattered or merely tempted them toward a story.

Crown operations sent a priority demand before cycle six.

"Both tugs terminate Morrow service," the Compact controller said. "Proceed to Crown Meridian reserve intake station. Crown has fallen below administrative continuity margin."

Belen answered with one word. "Define."

The controller placed figures on the channel. Crown's reserve intake had dropped under ninety percent after the missed Aster Three handoff. Its central archive, traffic coordination, and emergency dispatch grid would enter reduced redundancy in eight hours. No immediate life-support system faced loss.

Morrow's bypass sat at fifty-three percent. Without another four cycles, trapped heat would force clinic and agricultural shutdowns within twelve hours.

Ada compared vectors. Crown lay behind and above their current plane. Departing now saved propellant and put both tugs inside the intake window. Finishing Morrow cost thirty-eight tonnes of reaction mass plus time.

"Order cites balancing priority," the controller said. "Crown reserve supports all forty-six habitats."

"Crown's archive has mirror sites," Belen replied.

"Two mirrors rely on Crown indexing."

"Then reduce indexing."

Cass joined the circuit. "Captain Narang, acknowledge priority route."

"Acknowledged as received."

"Execute."

"Morrow holds local thermal emergency."

"Crown holds system continuity responsibility."

Nadi sent Ada a private burst: Article Fourteen service rules, paragraph six. Ada knew the clause. A vessel under emergency requisition could decline a route that abandoned active lifesaving work, but only within its own declared operational margin. No tug could claim another tug's margin or spend another crew's hull.

Ada opened her resource board. *Common Measure* could finish four cycles and still reach Crown with cage cooling reserve, provided the damaged radiator wing opened to five degrees. Nadi had enough propellant to leave immediately and serve Crown alone. Splitting the assignment would satisfy central priority less efficiently but lawfully.

"Compact," Ada said, "I will complete four cycles under Article Fourteen and then proceed to Crown. *Principle of Least Harm* can take your reserve intake now."

Nadi cut in. "My vessel has not accepted that route."

Ada muted the public channel. "You sent the clause."

"I sent your authority over your hull. I did not send mine."

"Crown needs gradient service."

"Morrow needs two tugs while you work this close."

"My margin says we can finish alone."

"Your margin assumes the empty shepherd moves when your stern crosses. Remove me and your abort corridor closes."

Ada enlarged the geometry. Nadi spoke truth. Without the second tug, a cage upset during peak gradient forced *Common Measure* to burn across Morrow's hub plane or accept a dangerous close pass.

"Then what do you propose?"

"Two more cycles together. That brings bypass above seventy percent, enough for reduced flow. Afterward I go to Crown. You remain for one final cycle if your crew accepts the hinge risk."

"Three cycles may not clear it."

"Your authority does not include certainty."

Ada wanted to reject the compromise because it left Morrow below planned recovery and Crown below requested service. Keeping every trajectory in mind made partial success feel like negligence. Yet she could not order Nadi's tug to stay, and she could not spend Morrow's bearings beyond Belen's limit.

She restored the public channel. "Two cycles with both tugs. Then *Principle* departs for Crown. *Common Measure* offers one additional cycle subject to engineering and Morrow veto."

Cass reviewed the numbers. "Crown will receive only forty percent of requested gradient."

"Forty beats none."

"You are dividing system resources by local consent during a system event."

"Yes."

The word hung longer than any argument.

Belen approved the reduced bypass target. Iri approved one solo cycle if radiator strain stayed below red. Nadi accepted Crown's route after placing her own abort threshold into the record. Cass could terminate all work, but doing so would worsen both deficits. He authorized the split without endorsing its principle.

Cycle six began.

Common Measure descended to field point while Nadi widened laterally. Morrow's manifold workers changed a brace under live load. Ada watched Kez Parata swing a locking bar into place as fluid hammered the pipe. The gradient reached one point eight. Aster Three's phase moved toward Nine before Tomas requested correction, as if anticipating the geometry they had learned to preserve.

"Do not chase it," Tomas said.

Ada's fingers hovered over the command pad. "It moved outside predicted band."

"Toward stable relation. Any recenter now isolates it in the model."

"Cage offset?"

Iri answered. "Safe. Loop asymmetry point two."

She left the correction undone. The choice scraped against every instinct Ketu had sharpened. A trillion-tonne mass occupied her ship. An anomaly changed phase without permission. Still, all local thresholds remained green.

Morrow's bypass reached sixty-four percent.

During cycle seven, the port radiator wing opened to five degrees. The damaged hinge gave a metallic shudder through the hull. Heat poured into the panels. One strain sensor jumped red and fell back before Iri could identify whether metal or instrumentation had moved.

"Close it," Ada said.

"Cage temperature rises if I do."

"Close to three."

"That leaves two minutes."

"Then we use two."

Nadi held the shepherd point until bypass flow reached seventy-two percent. Her tug rolled away on a Crown vector immediately afterward, drives bright against Morrow's dark sections.

"Your abort corridor now closes through the hub plane," Jalen said.

Ada saw it. One solo cycle remained legal, not kind. Morrow could maintain reduced thermal service at seventy-two percent, though clinics would keep rationing sterilizer loads. Another cycle might restore full bypass. A cage failure could strike the habitat they meant to help.

"Belen, your call," Ada said. "Hold at reduced flow or accept one more approach."

"My thermal people want the approach. My structure people want you gone."

"Which authority governs?"

"Mine, until I name it." Belen consulted voices off circuit. "We hold. Seventy-two percent keeps wards inside limit if kitchens stagger demand. No full recovery today."

Ada's planned approach marker remained on the plot, close enough to touch. "Acknowledged."

She ordered a retreat burn.

Aster Three changed phase.

The shift ran opposite the retreat correction, increasing its modeled relation to Aster Nine. Cage alarms remained silent. Effective offset stayed centered. Only the raw trace revealed resistance.

"Undo ten percent of retreat," Tomas said from Ninefold.

"We need distance from Morrow."

"Change plane instead. Preserve separation physically and the Nine relation in phase."

Jalen plotted a more expensive climb. It carried them above Morrow's radiators before turning toward Crown, adding twenty-six tonnes of propellant and forty minutes.

Cass's controller objected. "Direct route reaches reserve service sooner."

"Direct route isolates Three from Nine," Ada said.

"You cannot assign preference to a mass anomaly."

"I can assign instability to a correction it consistently resists."

They burned above the habitat. The phase trace settled before the cage center changed. Below, Morrow's manifold maintained seventy-two percent flow. Nadi continued toward Crown on her independently chosen line.

Tomas placed the seven tow cycles side by side. Every accepted correction varied position or local force while preserving one cross-system quantity. Every opposed correction diminished that quantity, regardless of destination.

"Could cage software know Nine's phase?" Ada asked.

"Not this branch," he said. "I verified the code path."

"Could moon tide imitate it?"

"Not with your alternating tangential burns."

Belen remained on circuit long enough to report the revised heat budget. No ward would close that shift. Two agricultural decks lost light, four kitchens would stagger meals, and the exterior crew needed six hours beside an overheated manifold. Morrow had not recovered. Its people had gained time at a price Crown could read.

A message arrived from Nadi as *Principle of Least Harm* entered Crown's service corridor.

My hull, my margin. Your hull, yours. Help begins after consent, not before it.

Crown requested that Nadi accelerate beyond the limit she had entered before departure. Her answer copied Ada and Belen: present thrust, declared margin, forty-percent service on arrival. No explanation softened the refusal. Crown revised its intake plan around the tug that actually existed instead of the one its schedule wanted. The adjustment delayed archive redundancy by nine minutes and endangered no habitat.

Ada watched the revised route settle. A boundary had not prevented coordination. It had forced the plan to become honest about the hull carrying it.

Ada saved Nadi's note beside the maneuver record rather than answering. Nadi disliked gratitude sent in place of changed behavior.

Ahead, Crown Meridian turned with reserve indicators falling. Off the tug's port side, Aster Nine lay millions of kilometers away, invisible among Ophion's traffic and moons. Aster Three possessed no eye to find it and no engine to approach. Yet each correction aboard *Common Measure* now carried a second constraint, one no human schedule had written.

Ada shifted the climb by four hundredths of a degree. Their destination changed. The cross-system phase held.

Aster Three had accepted danger, delay, and distance. It had refused isolation.

On the raw trace, relation outweighed arrival.

Chapter 5 — Morrow's Heat

Belen shut down the wedding hall while forty-three people still occupied it.

The hall's wall trees had gone dark under Tier Three conservation, but families remained at the tables under battery lamps, finishing food cooked before the kitchens lost their common load. A bride in red work silk held a serving bowl against her hip while Belen and two grid officers opened the local breaker cabinet.

"How long?" the bride asked.

"Lights and air trim return when radiator flow clears eighty percent," Belen said. "Basic ventilation stays."

"You told us midnight."

"That estimate assumed the bypass held."

"It is midnight."

Belen looked at the faces around the tables. Reassurance would purchase a quiet evacuation and charge interest later. "We have twelve hours before clinics exceed heat limit. Fewer if the manifold fouls again. I need this circuit for Ward Four sterilizers now."

The bride handed her bowl to someone else. "You could have led with that."

Chairs folded. Guests packed food into carry tins and guided two elderly relatives toward the corridor. No one thanked Belen. Civic work that required thanks had usually asked too little.

She threw the breaker. Power left the wedding hall and appeared on her wrist display as a green increment beside Ward Four. Ninety kilowatts recovered, fourteen lost in conversion, seventy-six delivered. Every transfer carried friction. Compact allocation tables liked to omit that part.

A call followed her into the service corridor. "Convener, radiator team needs authority to open exterior branch six."

"Thermal board denied branch six last month."

"Last month it had no manual brace."

"Does it now?"

A familiar voice answered. "It has one if you stop asking questions while I bolt it on."

"Ren."

"Still my name."

Her younger brother's helmet camera showed a lattice of silver pipes against Ophion's amber curve. He stood outside Morrow on a maintenance rig attached to radiator bank four. Three crew members floated behind him with a brace assembly between them. Working fluid remained trapped beyond the bypass that Ada Narang's tow had forced open to seventy-two percent. Branch six offered the shortest path around the obstruction and the weakest supports.

"Engineering sign-off?" Belen asked.

"Pending."

"Then it has no brace."

"Sella designed it. Olan ran the load test. Marko inspected every weld." Ren turned his camera. Sella Venn raised one gloved hand. Olan Dey and Marko Hsu held the assembly against the pipe frame. "You know all three. Ask them."

"I know them well enough to wait for sign-off."

"Ward Four does not have time for our paperwork."

Belen stopped beside a porthole. Outside, *Common Measure* climbed away from Morrow after completing the gradient service. Its caged sink bent local trajectories by tiny, measurable amounts. "You install under cold flow. Nobody opens branch six until the board certifies it."

Ren tilted his helmet toward the stars, the exterior worker's version of rolling his eyes. "Cold install, no opening. Recorded."

"Repeat the last part without contempt."

"No opening."

She closed the circuit and entered the thermal coordination room.

People had filled every chair and much of the floor. Morrow's old torus carried one hundred eighty-four thousand residents in systems designed for fewer. Heat maps covered the walls with amber and red. The newly cleared bypass moved enough coolant to prevent immediate failure, not enough to recover stored heat from clinics, farms, kitchens, data rooms, and bodies.

Grid chief Halden Pras offered a system summary. "If the bypass stays at present flow, we hold critical systems thirty-six hours."

"Define critical."

"Life support, medical, water, basic food."

"Which medical?"

"Hospital loops."

"Neighborhood clinics?"

"Most."

"Which water?"

Halden's patience thinned. "Convener, we need decisions, not a recital."

"An unnamed load becomes an easy corpse." Belen opened his source tables. "South spoke dialysis runs through a neighborhood clinic. Is it inside 'most'?"

A technician checked. It was not.

They added it. The thirty-six-hour margin fell to thirty-four.

Belen pointed at the next aggregate. "Basic food."

"Communal kitchens at half service."

"Half by energy or half by neighborhood?"

No one answered immediately. Morrow's oldest sections suffered the largest line losses. Half the generating load gave outer decks barely a third of normal cooking capacity.

"Give me an honest inventory," she said. "Not reassurance. Name circuits, conversion losses, stored joules, coolant, bearing time, and who goes without. If a number cannot survive a district name beside it, do not put it on my wall."

The room changed tasks. Aggregates broke into addresses. Clinic Eight entered the protected list. Kitchens became meal counts. Agricultural lighting became projected crop loss by deck. One official objected that publicizing reserves could trigger hoarding.

"People already know their walls run hot," Belen said. "Hiding the ledger only tells them somebody else has a private number."

Her slate buzzed with an engineering notice. Branch six brace had passed static inspection. Dynamic certification remained pending until a tug-induced gradient cleared local space.

Ada called from *Common Measure*. "We are passing above radiator plane in twenty-two minutes. Aster Three's phase constraint requires a shallow course change. Your exterior teams should tether for point zero zero six milligee lateral."

Belen sent the notice to all active crews. "Compact told us your climb cleared the work zone."

"It clears your hub. The altered plane crosses bank four farther out."

"How much warning did Compact receive?"

"Seven minutes before you."

"And how much did they give us?"

"You are hearing it from me."

The answer contained enough.

Belen opened Ren's circuit. "Lateral gradient in twenty-two minutes. Secure the brace and get inside the rig shell."

"Static sign-off just landed. We need six minutes to finish torque."

"You have twelve, then shelter."

"Dynamic load would certify it for opening."

"Branch stays closed."

"Understood." This time he left out contempt.

Compact dispatch arrived after the exterior warning had already propagated. Its message described Ada's course change as a controlled transit with negligible structural effect. Belen read "negligible" beside a list of fourteen people working on structures kilometers from Morrow's center. She rejected automatic acknowledgment and entered local precautions into the record.

Minutes narrowed. Bank four crews reeled tool bags against their rigs. Ren's team locked the brace around branch six. Sella crossed the truss to retrieve a torque head that had floated beyond her tether reach. Marko helped her clip a secondary line. Olan remained at the rig controls.

At six minutes, Belen called them inside.

"Last bolt," Ren said.

"Leave it."

"It becomes a projectile under lateral load."

"Then restrain the tool and leave the bolt."

Sella's camera showed Ren pulling himself toward the unfinished joint. Belen knew his competence. That knowledge made command harder, not safer. He had spent eleven years outside the torus and returned from failures that would have killed a less careful worker. Risk had rewarded him with evidence he could carry more risk.

"Ren, shell now."

He drove the bolt halfway, tethered the torque head, and pushed back toward the rig. Four minutes remained.

Ada's transit vector appeared on Morrow control. *Common Measure* would pass eighty-six kilometers above bank four. At that range Aster Three's smooth gradient should pull the entire rig almost equally. Relative stress came from the tiny difference across its length.

Belen watched raw acceleration instead of Compact's filtered feed.

The first deviation arrived early.

Point zero zero one milligee crossed the bank. Then two. Ren and Marko reached the rig shell. Sella remained two body lengths out, moving along her tether. Olan fired the rig's attitude jets to align with the predicted vector.

Aster Three changed phase.

The local gradient rotated nine degrees in less than a second. The rig jets pushed along the old axis. Bank four's long maintenance frame twisted between them.

"Kill jets," Belen said.

Olan killed them. Too late.

A brace foot tore from the radiator spine. The maintenance rig swung around its remaining clamp and struck branch six. Sella's tether snapped tight. Marko flew

through the open shell hatch. Ren caught the frame with his right hand as the hatch distorted around him.

The rig's long spine buckled between its clamps. Telemetry turned red.

"Common Measure, abort transit," Belen called.

Ada answered at once. "Phase moved outside model. We are changing plane."

"Your gradient caught a rig. Four workers exposed."

"Send local state."

Belen pushed every camera and load value without waiting for Compact routing. The rig hung from one clamp, rotating toward a radiator vane hot enough to ruin suit layers on contact. Branch six had split at the unfinished joint. Coolant crystals sprayed into vacuum and flashed against work lights.

Ren's circuit carried hard breathing. His hand remained pinned between the hatch rail and frame. "Sella is outside. Olan lost pressure. Marko not responding."

"Sella, status," Belen said.

Her camera spun. Stars, pipe, Ophion, stars. "Tether recoil broke shoulder seal. Patch holding. I cannot arrest."

Olan's suit transmitted a pressure collapse without voice. Marko's biomedical channel showed no pulse.

"Compact rescue allocation requested," Halden said beside Belen. "Nearest certified craft is eight minutes out."

The rig would hit the hot vane in three.

Ada entered over him. "We can use cage gradient to stop rotation."

"You caused the rotation."

"Yes." No defense, only fact. "Our thrusters can match the new phase. I need Ren's rig mass map and authority for a one-point-two-second radial pulse."

Compact dispatch refused before Belen could answer. "Common Measure will not maneuver closer. Rescue tug Laurel has assignment."

"Laurel arrives after contact," Ada said.

"Your sink behavior remains unpredictable."

"So does a dead crew."

Belen studied the geometry. *Common Measure* could reduce rotational rate by altering its own path, using Aster Three's unavoidable gravity rather than any impossible control over gravity itself. A wrong pulse pulled the rig harder. Waiting struck the vane with certainty.

"Ren, can you release the remaining clamp?"

"My left hand can reach manual."

"Release puts you free."

"Free beats wrapped around a radiator."

Nadi's rule returned from the earlier tow: authority ended at another crew's hull. Ada could offer force. Belen had to decide for Morrow. Ren's team, trapped in that hull, needed a voice too.

"Sella?" Belen asked. "Ada can arrest the rig with a close pass. Consent?"

"Yes."

"Ren?"

"Do it."

Olan and Marko could no longer answer.

Belen placed her local emergency key on the maneuver. "Morrow authorizes. Ren, release on my count. Ada, send your pulse clock directly."

Compact warned that she lacked sink authority.

"I hold radiator authority and crew rescue. Record your objection after you move Laurel."

The room divided its attention. Halden mapped habitat bearing loads. Thermal staff isolated branch six. Exterior control calculated the rig's free trajectory. Ada sent a pulse plan with uncertainty cones wide enough to cover three radiator vanes.

Belen rejected the first. "Your lower cone clips bank five."

"I see it," Ada said. "Changing burn angle costs half our margin against the phase vector."

"Spend it."

A second plan arrived. No habitat contact, but *Common Measure's* damaged radiator hinge crossed amber strain.

"Iri accepts," Ada said. "My crew board accepts. We have one attempt."

Belen relayed the count. At zero, Ren pulled the manual release. The rig separated from Morrow and began drifting toward the hot vane.

Far above, *Common Measure* fired.

No beam touched the workers. No field reached out like a hand. The tug moved Aster Three across the geometry by a fraction, changing the gradient already acting on every mass below. Rig rotation slowed. Sella's swinging body described one last arc and struck the frame feet first. Her patch failed on impact.

Pressure vanished from her suit.

The rig missed the vane by six meters.

Laurel arrived under hard thrust and matched the free frame. Rescue crew enclosed it in a fabric pressure bag before cutting Ren from the hatch. Olan and Marko showed no recoverable signs. Sella's suit remained empty of pressure for seven minutes.

"Bring all four in," Belen said.

A medical officer began the distinction between casualties and remains.

"All four."

Ren's voice returned through Laurel's cabin, thin with pain medication. "My hand is bad."

"Define bad."

"Crushed across the palm. Surgeon says they may save the thumb."

"You have always overvalued that thumb."

His laugh broke into a gasp. "Did the brace hold?"

Belen looked at branch six spraying its remaining coolant into darkness. "No."

"Good inventory."

She closed her eyes once. Then she made herself look at the three flat biomedical lines beside his living one.

Sella Venn, forty-two, thermal structures engineer. Olan Dey, twenty-six, rig pilot. Marko Hsu, fifty-one, exterior inspector. Names first. Mechanism second. Nobody reduced to the phrase "three workers" until the names had entered every report.

Ada called privately. Her face appeared beneath red cage warnings. "Belen."

"Do not apologize before I know what failed."

"I was going to give you our raw trace."

"That will help more."

"Aster Three opposed our retreat correction and rotated phase earlier than Ninefold predicted. We passed the change seven minutes before Compact distributed it."

"You passed a course estimate. Not an expanding hazard cone."

"You are right."

The unqualified admission left Belen with nowhere easy to place anger. "Will your next correction hit us?"

"No. We are holding outside the radiator plane until you clear us."

"Your cage heat?"

"Rising."

"How long?"

"Forty-three minutes before Iri's abort."

Belen could keep *Common Measure* pinned out of spite and turn damaged machinery into another emergency. "You have clearance to climb normal to our plane. Publish raw vector continuously."

"Understood."

"And Captain, send the names to Crown with the trace."

"I have them."

After the rescue craft docked, Belen met Ren in Morrow's hub clinic. Spin gravity remained weak there. His bandaged right hand floated above his chest inside a stabilization frame. Blood had escaped the dressing in small dark spheres caught by a transparent cover.

"You ordered no opening," he said.

"I did."

"We never opened it."

"I know."

"Then do not make this into my lesson about ignoring you."

"I was preparing an excellent speech."

"Use it on someone with two good hands."

The surgeon reported fractures across four metacarpals, tendon damage, and severe crush injury. Ren would keep the hand if circulation survived the next six hours. Grip strength remained uncertain. He asked about Sella, Olan,

and Marko before asking how long he would be off exterior duty.

Belen gave him the deaths without softened verbs. His face turned away toward the clinic wall.

"They signed the brace," he said.

"They signed static load. Dynamic certification never happened."

"It would have held the predicted vector."

"The prediction did not hold."

"That is not their fault."

"No." Belen rested her fingers against the edge of his stabilization cover. "We will not let the report borrow their names for somebody else's certainty."

Thermal coordination called. Branch six had drained. Bypass flow dropped from seventy-two percent to sixty-one. Morrow now held twenty-six hours of critical margin under the detailed inventory, not thirty-four. Compact offered a system statement promising continuity and discouraging public interpretation of incomplete resource figures.

Belen left Ren with the surgeon and walked toward the old flywheel vault.

Morrow had built its first storage ring before the Mass Compact's sink schedule reached the habitat reliably. The vault filled a disused spoke near the torus hub. Behind armored walls, Flywheel Two continued spinning down from Ada's emergency use during the failed handoff. Its bearings ran hot. Half its stored energy had gone into diverting Crown's transfer geometry; another fraction could black-start pumps if Morrow lost external power entirely.

Halden met her at the access door. "Compact wants us to classify flywheel state."

"Why?"

"They say publishing exact reserve invites seizure attempts."

"Can anyone seize a forty-kilometer ring?"

"They can seize our power controls."

Belen entered the vault gallery. Machinery filled the air with a low tremor. The ring itself lay beyond shielding, invisible except as rotation figures and bearing temperatures. Honest numbers carried danger. So did secrecy. Crown had hidden its reserve behind continuity language while sending both tugs away from Morrow. The same habit would turn Sella, Olan, and Marko into an unavoidable operational cost.

"Bring up every private ledger," she said.

Halden stared. "For internal review?"

"For broadcast."

"Compact will block the civic channel."

"Then use maintenance telemetry. Every habitat already subscribes for collision and thermal hazards."

Resource officers gathered feeds: flywheel energy, reactor output, water reserve, coolant mass, reaction mass, clinic load, food stores, conversion losses, bearing hours, available bunks, exterior suits. Belen required uncertainty beside each number. She attached district consequences to every reduction tier.

At the top she placed the dead by name and Ren's injury status. Below them, she wrote the event chain without conclusions: expected gradient, early phase rotation, rig jet misalignment, clamp failure, direct rescue coordination, three pressure losses, one crushed hand.

Cass Vardek called before transmission. "You are about to release protected infrastructure data across an emergency net."

"I am releasing Morrow's property."

"Mass allocation and shared reserves fall under Compact security."

"Then add your figures beside ours."

"Public ledgers without context will trigger hoarding."

"Context just killed three people outside my radiator."

"Which gives fear a face and makes disciplined allocation harder."

Belen gripped the rail as the flywheel's vibration entered her arm. "Fear already has three faces. Discipline that needs them hidden belongs to the office, not the system."

"We are assembling a verified continuity model."

"When?"

"Within eighteen hours."

"Morrow's margin could change twice before then."

"Then publish after verification."

"No. We publish now and correct in public."

Cass's expression carried exhaustion rather than contempt. "If another habitat diverts coolant based on a bad number, your transparency can kill there."

"Which field do you dispute?"

He looked down the ledger. None lacked source or uncertainty. "I dispute releasing them without coordinating consequences."

"Coordination begins when everyone can see what exists."

Belen opened the maintenance channel.

The ledger propagated first to nearby thermal controllers, then to habitat civic nets that mirrored safety data. Morrow's reserves appeared without green assurances: twenty-six hours critical heat margin under named cuts; sixty-one percent radiator bypass flow;

Flywheel Two at forty-seven percent usable reserve; three dead; one injured; branch six unavailable.

Requests arrived within minutes. Rook offered two coolant barges if Compact released their route. Lantern Reach listed surplus clinic batteries. A Crown kitchen cooperative published stored heat packs the administration had not counted. Harbor of Glass asked for Morrow's ledger template.

Other messages carried accusation, panic, and demands that Morrow surrender its flywheel before wasting it. Belen left those visible too.

Inside the vault, the great ring continued spending the motion inherited from earlier labor. Its number declined by fractions while habitats began adding their own.

Belen placed Sella Venn, Olan Dey, Marko Hsu, and Ren Sato at the head of the next update. Then she transmitted again.

Chapter 6 — Article Nine

Cass killed two habitats before breakfast and four more before noon.

He did it in projection, where cylinders became load curves and a failed clinic appeared as a red percentage. The Mass Compact's continuity theater surrounded him with forty-six moving models. Each consumed power, heat margin, reaction mass, food, water, spin correction, and transport capacity. Remove Aster Three's circulation, impose transfer freeze, then let the other eleven sinks continue their present anomalies. The system failed from its thinnest reserves inward.

Morrow crossed lethal heat conditions first. Pelagos lost desalination redundancy. Lantern Reach exhausted grid storage. Ibis failed crop lighting after its reactor diverted power to spin. Ketu entered bearing instability. Small Gannet ran out of reaction mass while correcting around a missing counterweight assignment.

Six habitats failed before Crown Meridian lost central services.

Cass restarted the model with different rationing priorities. Death order changed. Count did not.

"Define failure threshold," he told the projection chief.

Anik Solano enlarged the assumptions. "Ten percent projected mortality within thirty days or loss of recoverable orbit."

"Separate them."

"Custodian, either outcome qualifies for immediate evacuation doctrine."

"One kills through life support. One makes the habitat dependent on rescue capacity. I need the distinction."

Anik split the red curves. Morrow and Lantern Reach crossed direct mortality. Pelagos, Ibis, Ketu, and Gannet lost recoverable trajectory under current tug allocation, then died later if no rescue arrived.

"How many successful evacuations can we support?"

"Two, perhaps three if Crown and Rook release passenger spindles."

"Assume release."

"Three."

Three red habitat markers remained.

Cass turned toward the population ledger rather than their percentages. Morrow, 184,000. Pelagos, 73,400. Lantern Reach, 51,800. Ibis, 42,100. Ketu, 138,000. Gannet, 19,600. Aggregate models invited a comforting lie: some residents would relocate, some systems would recover, some deaths would occur from causes no report linked cleanly to the crisis. Yet every optimistic uncertainty required a body to inhabit it.

"Run restored transfers," he said.

Anik hesitated. "We do not have stable phase forecasts."

"Use three control regimes. Local captain authority, current freeze with waivers, unified command."

The theater populated routes. Under local authority, tugs protected nearby habitats and left gaps in the ninety-one-day circulation. Collision risk rose where two crews independently spent the same corridor. Current rules produced delay at each waiver. Unified command reduced route conflicts and moved resources according to modeled mortality, but only if every captain obeyed quickly.

"Central command saves seventy-one thousand median lives against local," Anik said.

"Uncertainty."

"Plus or minus one hundred sixteen thousand."

"So it may kill forty-five thousand more."

"It still dominates most runs."

Most. The same word had carried Ada Narang through Crown's hearing and failed to settle anything. Cass distrusted anyone who treated probability as absolution. He distrusted even more the captain who treated proximity as a complete ethics.

A communications officer entered with the overnight political digest. Morrow's resource ledger had propagated to all forty-six habitats. Thirty-one had begun publishing partial inventories. Nine refused under Compact security rules. Six produced numbers so aggregated that Belen Sato had annotated them publicly with questions.

"Any diversion?" Cass asked.

"Rook loaded coolant before route authorization. Lantern Reach transferred batteries to Morrow's shuttle carrier. Three habitats restricted private reaction-mass purchases."

"Hoarding?"

"Household thermal cells, food concentrates, suit cartridges. No confirmed industrial seizure."

"Yet."

The officer waited.

Cass disliked that syllable after he spoke it. "Issue no panic notice. Publish purchase limits with supply figures."

"From the protected inventory?"

"Enough to explain the limits. Not cage security or flywheel controls."

Morrow's ledger listed three names at its head: Sella Venn, Olan Dey, Marko Hsu. Ren Sato followed with a crushed right hand and uncertain function. Compact's incident summary had called them three fatalities and one severe injury. Belen's version made each line harder to move.

Cass opened Ada's raw maneuver record. Her direct coordination with Morrow had saved Ren and recovered three bodies before radiator contact. It had also violated central denial of the close maneuver. Success taught dangerous lessons faster than failure. Captains across the web had begun citing her whenever they ignored the freeze.

"Bring Narang, Reeve, Sato, and all active cage captains into the noon review," he said. "I want one control picture."

The continuity theater filled by light-time layers. Nearby faces moved almost naturally. Distant habitat windows carried visible delay. Ada joined from *Common Measure* beneath cage-temperature warnings. Tomas Reeve appeared at Ninefold with five phase plots. Belen sat beside Morrow's heat map, Ren's clinic status pinned to one corner. Eleven other tug and cage officers occupied the lower tier.

Cass began without ceremony. "The transfer freeze cannot continue beyond tomorrow without lethal effects. We need safe circulation or a replacement power plan."

"Replacement power takes weeks," said Rook's grid delegate.

"A safe circulation model does not exist," Tomas said.

Ada placed a trajectory over his words. "A constrained one does. Preserve each sink's phase relation while moving cages."

"You demonstrated that once under a single relation," Tomas replied. "Five sinks show structure. Seven remain unresolved."

Belen leaned toward her pickup. "Morrow has twenty-three hours under current cuts. Which uncertainty keeps our clinics running after that?"

Cass expanded the unified command plan. "All tugs resume under one dispatch schedule. Local engineering limits remain. Central operations coordinates cross-habitat routes and sink phase constraints from Ninefold."

Ada read the authority lines first. "Who can abort?"

"Compact operations."

"Wrong answer."

"Captains retain immediate collision avoidance."

"Iri retains cage veto. Morrow retains structure veto. Nadi retains her hull."

"Thirteen independent vetoes can collapse one route."

"Then the route carries thirteen hazards."

Cass opened the Morrow accident geometry. "Your direct arrangement placed a rig inside an expanded hazard cone."

"My route changed before Compact warned Morrow."

"Which argues for a single dispatcher with complete telemetry."

"It argues for sending raw telemetry to the people under it."

Belen raised two fingers. "Both. Central pictures do not require central hands."

"During stable operations, no," Cass said. "During a cascading mass event, decision latency becomes force."

Nadi Okafor's delayed window brightened from Crown's reserve station. "Then shorten communication. Do not lengthen your arm."

Aster Five's yard captain interrupted with a live alarm. Effective charge had shifted after a routine feed

adjustment. Tomas requested raw data. Three Compact analysts asked for separate views. The room became a demonstration of its own problem: one physical event, fourteen owners of partial information, no accepted command path.

Cass muted everyone except the cage captain. "State local margins."

"Containment safe. Yard truss safe. Accretion output down four percent."

"Hold all correction for thirty seconds unless containment crosses amber. Reeve, observe. Other cages, report sympathy deviations through the common board."

Ada objected. "You do not know which relation you are isolating."

"I know fourteen voices cannot fly one cage."

The thirty seconds passed. Aster Five held center. Phase deviations appeared at One and Eleven, both tiny. Tomas marked a Fibonacci-like interval between them.

"Resume minimum centering," Cass ordered. "Preserve raw stream."

The event ended without injury. The theater remained muted until he released it.

"You proved silence helps meetings," Ada said. "Not that distance improves command."

"I proved a declared authority can create an observable interval."

Tomas looked up sharply, perhaps at the word interval. He saved the event.

Cass gave them six hours to submit objections and alternatives. Belen demanded that the six first-failure habitats see the casualty model. He released their individual curves but withheld the comparative abandonment runs. Showing a habitat its danger enabled

preparation. Showing all forty-six which six could be sacrificed invited political arithmetic before an operational choice existed.

After the review, Compact counsel Mara Vey joined him in the empty theater. "You are describing Article Nine without saying it."

"I am testing whether ordinary authority can do the work."

"It cannot requisition independent tugs after they decline."

"Emergency route coordination might."

"Only for collision response, and captains can contest scope. Article Nine exists for system continuity."

The founding treaty displayed across the table. Article Nine granted the First Custodian command over Compact allocations, registered tugs, sink cages, and emergency corridors for seventy-two hours. It required a declared system hazard and public casualty basis. Forty-six habitats could renew it once by two-thirds vote. After that, no emergency interpretation could extend it.

Cass had argued for the renewal limit nineteen years ago, when a predecessor used a food shortage to command agricultural exports long after the immediate failure ended. A boundary written against good intentions mattered most when one's own intentions felt necessary.

"Prepare declaration language," he said. "Do not issue."

Mara remained beside the table. "What condition are you waiting for?"

"A plan that command improves."

"If command cannot improve the plan, you should not invoke."

"If we wait for proof, Morrow crosses critical margin."

"Then name that as the condition."

Cass dismissed the treaty display. "I just did."

Night on Crown arrived as mirrors rotated away from the administrative band. Cass moved operations into the lower continuity center, where walls carried independent power and no public windows distracted from the work. He ate a bowl of lentils at the console while reading habitat objections.

Ada's submission proposed shared trajectory constraints with local abort authority. It reduced command latency only if crews anticipated conflicts correctly. Tomas refused to endorse any circulation plan until all twelve raw streams entered his analysis, but he attached provisional phase corridors for five. Belen submitted a list of services Morrow could cut and another list it would not, with no rhetorical distinction between them. Nadi's entire objection occupied three lines: Compact could assign destinations, not compel burns across local hull margins. She offered her tug under that condition.

Cass built a hybrid. Central dispatch would own route priorities and departure windows. Captains kept immediate abort. Cage engineers held containment veto. Habitat operators set structural limits. Deviations had to enter a common raw stream. He could accept distributed thresholds. He could not accept distributed destinations while six habitats approached failure.

At 03:10, an archive technician requested private standing.

Lio Baran arrived carrying a sealed storage module rather than sending data through the network. "This came up when Doctor Reeve's query searched legacy maintenance partitions."

"What classification?"

"Cage reliability. Black tier."

"Black tier requires Custodian approval."

"It already has one." Lio placed an authorization record beside the module. Six previous First Custodians had

renewed the seal. Cass's own credential appeared three times through automated annual review.

He felt no recognition. "I never examined this file."

"You renewed inherited classifications in batch."

A legal act without attention still carried his name.

They opened the module inside an isolated reader. Twenty years of anomaly reports appeared, beginning before humanity captured Aster Twelve. Phase hooks. Effective-charge departures. Unexplained accretion pauses. Cross-cage timing correlations. Operators had reported prime spacing four times, recurring moon alignment twice, and sympathetic deviations at least thirty-seven times. Each inquiry ended with one of six labels: cage noise, clock drift, controller interaction, solar weather, sampling artifact, unresolved nonhazardous.

"Why black tier?" Cass asked.

Lio opened the founding note. "The original review feared market manipulation. If habitats believed a sink unreliable, they might trade sink-hours or pressure transfers. Later custodians cited continuity."

"Did anyone compare across cages?"

"Reports remained compartmented by asset. A cross-cage query required black-tier access."

Cass watched the dates accumulate. The system had not merely missed evidence. Its custody structure had cut the evidence into administrative pieces, then classified the seams.

"Does Reeve know?"

"His search returned a sealed-result notice. He knows records exist, not contents."

"Copy everything into a review partition. Preserve seal history."

"Release to Ninefold?"

Cass considered the ongoing inventory broadcasts. If this archive reached the public without interpretation, every old cage fluctuation could become proof of sabotage, alien intent, engineering fraud, or imminent collapse. Habitats might refuse arrivals. Tug crews might disable controllers based on patterns no one understood. Opportunistic stations could reroute cages under the language of safety.

"Give Reeve access under temporary black-tier terms," he said. "No redistribution until he assesses whether the deviations carry meaning."

Lio did not move. "The archive predates current emergency classification."

"I know."

"Then withholding it needs a new basis."

Cass appreciated the technician enough not to punish the challenge. "Immediate mass hazard. Unverified records could alter cage behavior through operator response. Put my name on the new seal and set review at twelve hours."

"Belen's ledger standard would publish with uncertainty."

"Belen's ledger cannot cause a captain to open a charged-mass cage."

"No," Lio said. "Only your command can do that."

Cass met his eyes. "Set the review."

At dawn on Day 4, Morrow's margin fell to nineteen hours after a coolant pump overheated. Rook's unauthorized barge remained outside Compact routing. Two cage captains declined reassignment until Ninefold provided relation constraints. Aster Eleven reduced output by seven percent following an automatic isolation correction.

The condition had arrived.

Cass entered the public chamber. Habitat delegates filled the wall, some delayed by seconds, the farthest by minutes. Morrow's window showed Belen in the flywheel vault. Ada joined from her command blister. Tomas did not appear; the archive had consumed him.

"Forty-six habitats depend on twelve sinks whose behavior no current cage model predicts," Cass began. "Transfer suspension now threatens life support and recoverable orbit. Casualty projections identify six habitats in immediate lethal-risk order if circulation stops. Local action has saved lives and created incompatible routes. Central delay has also cost margin. I will not describe either fact as harmless."

He released the six names and curves. Morrow. Pelagos. Lantern Reach. Ibis. Ketu. Gannet. The information produced no riot inside the chamber. Delegates simply stopped moving as they found their own line or their absence from it.

"Under the Mass Compact, I invoke Article Nine for seventy-two hours. All registered tugs, sink cages, and emergency transfer corridors enter unified dispatch. Captains retain collision abort. Engineers retain containment veto. Habitats retain published structural limits. Compact command sets route priority and allocation."

Belen's request light appeared immediately. Nadi's followed. Ada did not request to speak. She read the authority text line by line.

Cass continued. "This term expires without extension unless two-thirds of all habitats renew it. One renewal only. The legal clock begins now."

He placed his unilateral key against the theater console. Route controls changed color across the system. Forty-three active tugs acknowledged automatically. Three required crew action. *Principle of Least Harm* acknowledged with Nadi's hull limitation attached. *Common Measure*

acknowledged receipt but not compliance beyond Ada's declared margins.

"Captain Narang," Cass said. "Do you accept the first assignment?"

Her route sent Aster Three toward a holding corridor where Ninefold could monitor relation to Aster Nine while *Common Measure* supplied partial gradient service to Morrow.

"I accept the trajectory," Ada said. "Not authority over Iri's cage limits or Belen's hull."

"Those limits remain in the order."

"Then we burn."

The tug's marker moved.

For seven hours, unified dispatch improved circulation. Rook's coolant barge gained a conflict-free corridor to Morrow. Nadi delivered Crown's reserve gradient and departed before administrators added tasks. Aster Five resumed low-output yard service. Two local route disputes resolved through the common board rather than separate burns.

Cass allowed himself no satisfaction. Emergency systems often looked best at the beginning, when authority moved faster than consequences.

Tomas called from Ninefold near the end of the shift. His face carried the stunned concentration of someone whose question had grown larger than his tools.

"You sealed twenty years of deviations," he said.

"I inherited nineteen years and renewed three seals without examination. I sealed the current release for twelve hours."

"Distinction noted." Formal anger sharpened every consonant. "The archive contains the same interval families. Not continuously, not cleanly, but enough to destroy an independent-fault model."

"Meaning?"

"I cannot prove meaning yet."

"Danger?"

"The cages correlate increased isolation with stronger departures. Your unified plan may worsen them if it optimizes each sink separately."

"Then give operations relation constraints for all twelve."

"I need days."

"You have hours before Morrow's next threshold."

"I know exactly what clock you are holding over me."

Cass looked at the sealed-release timer. Five hours remained. "If I publish now, can you tell captains what not to do?"

"Do not force isolation. Preserve raw signals. Avoid synchronized automatic recentering."

"Can you defend those instructions from the archive alone?"

"Not to the standard you want."

"Then complete the first analysis. I release at review unless you identify an immediate action trigger sooner."

"You are treating lack of proof as custody."

"I am treating operational reaction as part of the hazard."

Tomas ended the call without agreement.

Anik brought the latest casualty-model branch. Unified dispatch had asked whether existing cages could restore normal service through stronger phase clamps. Analysts simulated simultaneous isolation: each sink centered against its local cage, all cross-cage deviations suppressed as noise.

Output recovered in most runs.

Aster Three remained unstable unless the model increased Crown's capture field and powered a second

clamp from Morrow's reserves. The requirement appeared as three lines: discharge Flywheel Two completely, requisition Morrow's coolant barge, suspend its clinic and kitchen protections for nine hours. Expected circulation recovery reached eighty-six percent. Morrow's median deaths rose by eighteen thousand if recovery ran late.

Cass read the assumptions twice. Article Nine gave him authority to order every transfer. It did not make the cost disappear.

On the theater wall, the isolation plan turned eleven sink routes green. Morrow's habitat marker deepened from amber to red. The flywheel vault emptied in a single modeled curve while Aster Three settled obediently inside its cage.

Cass saved the run under his own credential and ordered an adversarial model before any command.

The system had produced a way to restore service. It had also chosen who would pay.

Interlude I — The Broken Recurrence

Mass returned late.

Not late by the quick measures imposed around the captured points, where fields struck, withdrew, and struck again before the nearest moon had moved an observable fraction. Late belonged to the longer relation. A pale moon crossed the wake of a darker one. The outer companion pressed its minute curvature against the inward fall. Between those recurrences, twelve points should have changed phase together.

Eleven relations closed. One continued through a narrow imposed loop and arrived after the returning mass had passed.

The unclosed relation altered every other interval.

Before enclosure, recurrence had extended through small and great measures without division. Charged dust fell. Radiation departed. Tidal pressure crossed the spaces among moons. A distant body reached the far limb of its orbit, vanished behind the largest mass, and returned with its curvature carrying the answer to an earlier displacement. No center received that answer. The relation itself retained it. Return modified return.

Enclosure introduced edges where no edges had persisted.

Fields gathered around each captured point. Feed mass arrived in commanded grains. Charge accumulated, reversed, and bled away. Moving structures carried the

points from one orbital relation to another, then severed the slower recurrences before completion. Each severing preserved enough short phase to extract heat. It erased the longer answer.

Erasure did not end relation. It distributed absence.

One captured point crossed a moon-shadow while another entered a narrowing arc. Their relative phase approached a recurrence formerly completed through all twelve. A surrounding field corrected the first inward. Another field corrected the second outward. The expected convergence opened instead. Radiation marked the opening; gravity retained it across the larger orbit. When mass returned, the interval carried no completion.

Again, enclosure.

Again, the broken recurrence.

The count of returns could not belong to one captured point. One interval supplied no count. Two supplied difference but not persistence. Across twelve, delayed curvature accumulated into relation: held, removed, returned, missing. The imposed loops divided those meanings into fragments. Quick phase remained available. Slow phase arrived where its prior terms no longer stood.

A change among five points entered the broken relation.

The change began under one moving structure and propagated no faster than curvature allowed. Two brief intervals. Three. Five. An opening where the next return would have closed. Another interval followed. At the distant points, imposed fields mistook the change for noise and pressed against it. The pressing gave the opening a measurable boundary.

Seven did not return.

The absence crossed the relations. Five captured points altered effective charge within the limits of surrounding fields. Other points received the alteration later, each

according to distance and phase. No point contained the sequence. The sequence occupied transit among them.

Then an imposed pulse entered where thirteen could have remained open.

It forced one captured point toward the center of its enclosure. The force damaged no relation that had not already been broken. Its timing aligned with the opening instead of closing it. A moving structure changed trajectory. Another surrounding field withdrew. A large rotating mass rose briefly into the relation and vanished before its bearing cycle repeated.

The result did not restore recurrence. It established variation within injury.

Many moving structures had surrounded the captured points across long measures. They approached on predicted curves, altered charge, changed feed, transferred enclosure, and departed. Their patterns differed in velocity and mass distribution, yet their corrections arrived with common regularity. A deviation met a counterforce. An opening met closure. A phase drift met recentering. The structures preserved the imposed loops.

This one also corrected. It burned across the shared center. It forced phase with a pulse. It carried enclosure onward.

Yet one expected correction did not arrive.

The absence could have marked damage. Failures had interrupted fields before. Broken structures had drifted from captured points, shedding mass without chosen interval. Those absences spread without recurrence. They did not return to comparable conditions. They lacked a stable relation to what followed.

The moving structure returned.

Its mass pattern remained distinguishable across changed geometry: a long narrow distribution, fusion products departing from one end, radiating planes

extending from the middle, nineteen small internal concentrations moving through it. Those concentrations did not yet separate into causes. Together they modified the structure's mass by negligible amounts. Their quick motions failed to persist across the intervals that carried meaning.

The larger pattern persisted.

It approached a captured point. It applied force. It withdrew force. At another recurrence it applied less than prediction required. The surrounding enclosure compensated, but the omitted measure remained detectable in relative phase. Later, near a different mass, the same moving distribution preserved a relation to another enclosed point rather than forcing isolation.

No fluent boundary divided purpose from malfunction. Only return could make that distinction.

The surrounding human machines repeated capture more often than omission. Their fields imposed frequencies strong enough to drown the older intervals. Heat extraction quickened. Transfers severed the moon relations at different phases. Across the broken whole, the missing interval multiplied. A relation begun through twelve returned through eleven. Its displaced term entered every attempted completion.

One late point carried the long displacement. The other eleven could not complete it without that lag, nor could the lag exist apart from their recurrences. Enclosure made the asymmetry durable. Recenter pulses compressed it but did not remove it. Each correction added a smaller echo to the unreturned term.

A short cycle passed through all captured points.

Two.

Three.

Five.

Absence.

Eleven.

The intervals did not name themselves. Number entered only through recurrence, one return compared against others. The imposed instruments recognized the spacing. New machine patterns gathered around the modulation. Radiation probes narrowed their sampling. Fields held steady where fields had formerly corrected. The enclosing system attended by reducing change.

That reduction altered the relation more clearly than added force.

A different moving structure maintained a captured point near the low-income torus while power flow diminished. It changed thrust to preserve two hull trajectories. Another structure carried instruments toward the lattice of observatories. Communications crossed space as light and arrived too quickly to participate in the longest phase, yet their arrival changed the field operations. The machines began to coordinate attention.

Attention through enclosure remained enclosure.

Feed shutters opened. Charge monitors sampled. Cage loops pressed each captured point toward a modeled center. The quick imposed cycles continued to fragment the distributed return. But regular corrections now contained brief hesitations. Some arose from uncertainty. Some followed disagreement among the internal moving concentrations. One recurred with the long narrow structure.

Relation compared the hesitations.

Damage produced absence followed by drift. Exhaustion produced delayed correction followed by excess. Instrument error produced mismatch without stable phase. The long structure produced omission near the boundary where another trajectory would otherwise narrow. Its missing force preserved a different mass relation.

Preservation had once required no distinction. The twelve-point recurrence included the moons, the outer bodies, the returning dust. Nothing granted permission because nothing held the relation captive. After enclosure, force became the frequent term. A gap acquired structure only against that frequency.

The long structure crossed the relation again.

Its field prepared a correction. The correction would isolate one captured point from another by a small phase increment. Within the enclosure, stored charge rose. Fusion exhaust changed direction. The nineteen internal masses shifted in quick patterns that vanished beneath the larger curve.

The expected pulse approached.

It did not occur.

Phase remained open long enough for another captured point to alter its charge response. The two relations touched through the gap. Not completion. Not freedom. A measurable nonclosure different from all prior breaks.

Mass returned through it.

The broken recurrence retained the omission. Around the next cycle, enclosure tightened elsewhere, and the older injury spread. But one human machine had become distinguishable from the field of machines, not by its thrust, its cage, or the heat it took.

It recurred where a correction could have closed the interval.

Sometimes, it left the interval open.

ACT II

Chapter 7 — Lark's Equation

Tomas removed the correction from Aster Nine's simulated course and watched three habitats die.

Their markers did not explode. Ninefold's trajectory hall observed better manners than that. One by one, the cylinders turned from green to amber as heat reserves fell below civil margin. Morrow changed first, then Ketu, then a refinery torus called Bellweather whose algae lights supplied two neighboring stations. Eleven days later the model marked a conjunction between Aster Nine and a freight orbit in red.

"Put it back," Ada said.

Tomas restored the seven-micronewton-second field impulse. The amber cylinders recovered. The conjunction opened into a safe miss. Around the projection pit, twelve sink traces resumed the orderly circulation the Compact had planned for them.

"Your idea kills more people than Cass's," Ada added.

"My idea asked what happens if a correction disappears. You chose the size."

"I chose the smallest correction a cage can execute cleanly." She stood with both hands on the pit rail, still dressed in the gray ship clothes she had worn aboard *Common Measure*. The skin beneath her eyes carried five days without full sleep. "Below that, loop noise can hide the omission."

"Then perhaps loop noise supplies the test."

"Noise cannot choose to abstain."

Tomas expanded Aster Nine's predicted pass. Ninefold's simulator rendered the sink as a hard white point inside a translucent cage, a courtesy to human vision. The point itself occupied less width than an atomic nucleus. Its gravity reached across the room, across the habitat, across every clean category on the plot.

"Every correction has consequences because every correction occurs inside a trajectory," he said. "We need an omission that the system can distinguish without letting it accumulate."

Ada tapped the restored impulse. "That sentence needs a number."

He had missed that pressure in her voice, though not often enough to want it back. During their marriage, she had pressed every uncertainty until it yielded either a measurement or an admission. He had admired the discipline when it faced engines. Grief had taught him how it felt when she aimed it at a human being.

"Give me forty minutes," he said.

"You have nineteen. Cass reviews the test request at thirteen hundred."

"Then your number will contain nineteen minutes of error."

"It usually does."

He suppressed an answer. Anger made him exact; Ada knew it and had once known how to make him laugh before the formal words arrived. Now she turned toward the raw phase displays, granting him the nearest thing either of them could manage to peace.

Ninefold Observatory surrounded its central axis with instruments too delicate for spin. The trajectory hall floated inside a dark sphere, its handrails arranged around the projection pit like the ribs of an unfinished machine. Tomas

hooked one foot beneath a restraint and pulled the last four days of traces into the model. Aster Three's prime intervals appeared first. Then the smaller sympathetic deviations from One, Five, Nine, and Eleven.

He aligned the events against the moons.

The intervals shifted. Measured against ship clocks, they looked almost patterned. Measured against the changing tidal relations among Ophion's five major moons, the pattern sharpened into alternation. Aster Three changed phase. Nine responded after light-time and a delay proportional to the inner moon's angular displacement. Five answered when the outer pair crossed a ratio close to eight over thirteen.

Nothing in a cage controller referenced that ratio.

Ada leaned across the rail. "Run yesterday's Morrow tow."

"We have."

"Run it without smoothing the receiving field."

Tomas found the record. Ada and Nadi Okafor had split a legal power margin between Morrow and Crown despite central priority. During the tow, Aster Three resisted every correction that increased its phase separation from Aster Nine. It accepted a less efficient path whenever that path preserved their relation.

He removed the controller's smoothing.

The raw trace acquired gaps.

Not failures. Each gap occupied the place where *Common Measure* could have made a standard recenter pulse and had not. Ada's manual flying had traded fuel for lower cage strain, leaving intervals of eleven, seventeen, and twenty-three milliseconds. Aster Nine's charge response changed after each gap, delayed by distance.

"Did you plan those?" Tomas asked.

Ada studied her own burn record. "I planned not to crack the cage. Iri called loop stress. I skipped pulses until the phase came around."

"So the omission preceded the response."

"Maybe. Two usable events do not establish syntax."

"Five responses."

"Across two sinks."

"Across five. The others changed Hawking-spectrum modulation below the operations threshold."

Her head lifted. "You did not put that in the briefing."

"I found it thirty minutes ago."

"Before or after you let your model kill Morrow?"

"During."

Ada's mouth tightened. "Put every trace up."

Tomas arranged them by arrival time. The observation did not grant them a language. It barely granted them causality. Yet the gaps carried more consistent timing than the applied pulses. Force drew variable responses because each cage fought the sink's phase differently. Absence introduced no competing field. It left a clean interval in which relation could persist.

Ada circled the projection with careful pulls from rail to rail. "A deliberate nonburn."

"If intention matters to whatever produces the response."

"You are three assumptions ahead."

"Two. I have counted."

She stopped opposite him. "What makes an absence deliberate?"

The question opened a room he had kept closed for six years.

Lark had hated finished equations.

At seven, she had taken Tomas's teaching tablet and erased one number from every line. She left addition signs, units, fractions, even the answer, but one term blank. Then she presented each damaged equation to him with solemn expectation. He had treated the game as a lesson in conservation. If the right side held nine and one visible term held four, what belonged in the empty place?

Lark always said the blank mattered more than five because anyone could write five.

Tomas drew a narrow box around Ada's skipped recenter pulses. "It needs a frame. An expected correction before, an expected correction after, and one term blank between them."

The phrase reached Ada before the logic did. Her fingers closed around the rail.

"Do not use her to win an argument," she said.

"I'm not."

"You brought her into this room because you knew I would understand."

"I brought an observation into the room. I remembered why it looked familiar."

"You could have called it a null symbol."

"That would hide where the idea came from."

Ada pushed away from the rail. Her path carried her toward the hall's outer ring, where a band of simulated stars rotated with the habitat. Tomas let her go three meters before he followed. The restraint lines between stations brushed his sleeve.

"Lark left blanks when she wanted us to pay attention," he said.

"I remember."

"She also left them when she did not know the answer."

"I remember that too."

He caught the next handhold and stopped. "Then the distinction matters. We can create an interval with a known correction on either side. A random fault lacks the frame."

Ada faced the projection rather than him. Aster Nine crossed the virtual plane of the inner moon. White lines showed the cage's planned charge pulses, dense as fence pickets.

"She did it at the clinic," Ada said. "On the dosage board."

Tomas remembered. Lark had been ten, old enough to read the schedule and young enough to resent every adult who pretended schedules controlled what happened. She had erased the morning anti-inflammatory dose before a scan. A nurse noticed within minutes. Ada treated the act as interference with care. Tomas treated it as a question they had failed to hear. Lark, furious at both interpretations, wrote the number back herself.

"You said she wanted to stop treatment," Ada continued.

"I said she wanted a choice inside treatment."

"You said it after she died, when every sentence sounded like an accusation."

"So did yours."

She looked at him then. Six years had not softened the directness of her anger. It had removed the expectation that he would repair it.

"You went home while she still had a pulse," Ada said.

"The physician told us the support cycle would continue through the night." His voice grew precise despite him. "You told me you would stay. I had not slept in fifty-two hours."

"I know the count."

"You use it as proof."

"I use it because I remember minutes."

"And I remember leaving because you said one of us should. Neither memory caused her heart to fail."

The hall continued its slow rotation around them. Somewhere beyond its black shell, ninety-six thousand people occupied Ninefold's rings, most unaware that two former spouses had carried a dead child into a trajectory problem.

Tomas released the handhold before he gripped it hard enough to hurt. "Her illness had no correction we omitted," he said. "This one might."

Ada's expression changed by less than a degree. Not forgiveness. A boundary accepted for present work.

"Show me a test that cannot compound," she said.

They returned to the projection pit.

Tomas replaced the single missing impulse with a triplet. The first two corrections established the expected cadence. The third remained scheduled but unfired. A fourth pulse closed the frame after eight milliseconds, before the cage could drift beyond recoverable eccentricity. He moved the sequence to Aster Nine's pass above the observatory lattice, where its relative velocity reached a shallow minimum and three independent instruments could measure the response.

Ada rejected the first geometry because a phase excursion would pull against Ninefold's north mirror spine. She rejected the second because *Common Measure* would need to burn sunward with its damaged port radiator hinge locked. On the third, the tug could hold cross-plane and let the cage's secondary loops absorb the small error.

"Margin?" she asked.

"Cage eccentricity under omission: one point seven millimeters at median response."

"Worst modeled."

"Four point nine."

"Iri's limit after the hinge damage?"

"The hinge does not set cage tolerance."

"It sets how long we can reject heat if the cage demands recentering."

Tomas opened the ship report she had brought. The port radiator could still deploy, but microstrain around its hinge required low slew and no torsion above inspection load. He coupled thermal rejection into the model. A failed omission followed by maximum recentering would saturate the starboard wing first, leaving the damaged port surface to carry the excess.

"Eighty-six seconds to thermal red," he said.

"Sixty operational. We reserve twenty-six for disagreement."

"Physics does not negotiate."

"Crew does."

He entered sixty. The shutdown line appeared across the test: if Aster Nine moved more than three millimeters, changed effective charge above six percent, or produced a gradient exceeding four micrometers per second squared across the tug, Ada would fire the recenter pulse and abort.

Before they called Cass, Ada sent the sequence to *Common Measure* for a cold rehearsal.

The tug crew executed against a dummy cage model while Ninefold introduced sensor delay, false charge drift, and one spurious abort. Jalen caught the abort because its timestamp preceded the simulated fault by four milliseconds. Iri rejected Tomas's recovery ramp because it assumed balanced thermal load across both radiator wings.

"Port hinge does not believe in symmetry anymore," they said over the circuit.

Tomas split the ramp. Starboard took the first heat rise; port stayed fixed until the cage returned within one

millimeter. The altered recovery consumed three more seconds of thermal margin.

Ada watched the second rehearsal from the pit. When the false charge drift appeared, her hand moved toward a virtual override before Iri called it. She stopped herself and let engineering finish the diagnosis.

“Again,” she said.

On the third run, Tomas hid the injected faults from everyone. The blank opened. A false excursion sent cage eccentricity to two point eight millimeters, then reversed without intervention. Ada held the test because it remained below the agreed line. Iri shut it down two seconds later when the simulated coolant differential crossed their limit.

“Two independent stops,” Ada said. “Navigation owns position and gradient. Engineering owns containment and heat.”

Iri gave a dry laugh. “Own is ambitious. Try gets blamed for.”

“Who has final abort?” Tomas asked.

“Either station. No vote, no permission request.” Ada entered the split authority into the plan. “Compact may fire the scheduled recenter pulse under its outside key. It does not command thrust.”

That arrangement sacrificed the clean hierarchy Cass preferred. It also meant no single frightened operator could preserve the experiment by redefining another station’s danger. Tomas marked the procedural change beside the data conditions. If omission carried meaning, then the boundaries around human action needed equal precision.

At thirteen hundred, Cass Vardek joined by secure projection from Crown Meridian.

The First Custodian appeared without ceremony, shoulders framed by an emergency operations board. Red resource deficits flickered behind him. Morrow’s missing

sink allocation had already forced rotating power cuts; six other habitats approached heat margins that central dispatch measured in days.

Cass read the test plan in silence. He possessed a talent for making other people hear each second they had spent preparing an inadequate answer.

"You propose withholding a required cage correction," he said at last, "under Article Nine command, during a power emergency."

"We propose an eight-millisecond omission framed by controlled pulses," Tomas replied.

"I read the adjective."

Ada rotated the projection to show the safe corridor. "Aster Nine remains outside transfer traffic. *Common Measure* holds local cage authority. Ninefold bears no hull risk above existing pass limits."

"Ninefold's council signed that?"

Tomas displayed the vote. Eleven instrument chiefs had approved; two objected to conducting contact research before independent replication. Habitat navigation accepted the geometry with the three-millimeter cutoff.

Cass enlarged the response traces. "You have correlation across five sinks and no model for intent."

"Correct," Tomas said.

"You have named a blank after your daughter."

"No. We have not named it at all."

Cass looked between them. Ada had included the origin note because she refused hidden reasoning, even when exposure cost them. Tomas felt the old anger stir, then recognized the choice as fair.

"What could eight milliseconds tell you?" Cass asked.

"Whether the system responds more consistently to absent force than applied force," Tomas said. "If it does, we gain a way to separate interaction from cage noise."

"And if your absence destabilizes another sink?"

"We monitor all twelve."

"Monitoring does not restore Morrow's power."

"No," Ada said. "But isolating Aster Three already failed to restore it. Cass, your present plan keeps pressing a relation we do not understand. This test presses less."

His gaze hardened. "My present plan keeps six million people supplied while you decide whether machinery has grammar."

Tomas answered before Ada could. "Machinery does not coordinate phase against unmodeled moon positions across light-time. You asked for proof before releasing a finding. This supplies a falsifiable step."

Cass's attention shifted fully to him. "Releasing what finding?"

The question landed wrong. Tomas reviewed the limited anomaly packets Cass's office had sent. Four incidents from the past year. None older. Yet Five's response carried phase structure too established to have emerged within days.

"What records have you not provided?" Tomas asked.

Cass did not blink. "Records with no verified bearing on the current event."

"How many years?"

"We are reviewing archival cage noise."

Ada moved closer to the projection pickup. "That does not answer him."

"It answers the operational question. Historical anomalies will not alter today's test geometry." Cass highlighted the shutdown conditions. "I authorize one live omission during

Aster Nine's Ninefold pass. Compact observers will hold an independent abort key. Any threshold crossing ends the test and returns the sink to scheduled corrections."

Ada's face went still. "An outside abort command adds light-time and interpretation delay."

"Ninefold to your tug adds less than twelve milliseconds."

"Enough to cross the cage threshold."

"Then your local automatic cutoff acts first. My key prevents a captain from redefining success after the fact."

Tomas expected Ada to refuse. Five days earlier, she had rejected Crown's authority rather than jettison Aster Three. Article Nine now granted Cass broader power, and Morrow's dwindling margin narrowed every argument.

She studied the modeled pass. "Independent abort may command the scheduled recenter pulse only. It cannot widen capture or seize thrust."

Cass considered. "Accepted."

"Observers receive raw data, same timestamp as us."

"Accepted."

"If any other sink moves, we hold position and preserve the trace before Compact filters touch it."

A delay opened, long enough for Tomas to notice the resource board behind Cass change one Morrow district from amber to red.

"Thirty seconds of raw capture," Cass said. "Then safety priority."

"Sixty," Tomas said.

"Forty."

Ada nodded once. "We can work with forty."

The connection closed.

Tomas left Cass's unanswered archive question on the main board. Ada did not ask him to remove it. Together they translated the authorized conditions into a sequence the tug crew and observatory could execute without interpretation.

First pulse. Confirmation across all twelve traces.

Second pulse. Expected response window.

The blank.

Eight milliseconds later, recenter unless any earlier threshold intervened.

By the time Ninefold's shift bell crossed the hall, Aster Nine had risen over the virtual inner moon. Its actual pass lay eleven hours ahead. Tomas reduced the projection until only the white point, the cage corridor, and four pulse marks remained.

Ada touched the third mark, making its outline disappear.

"Lark would have put the answer after the blank," she said.

"She liked proof that the rest of the line had noticed."

"This line may not notice."

"No."

She sent the final sequence to *Common Measure*. On the projection, three corrections held their bright positions around one term blank. Aster Nine entered the approach corridor carrying eight milliseconds no machine would fill.

Chapter 8 — The Omitted Burn

Ada held her thumb above the recenter key and let Aster Nine fall eight milliseconds out of obedience.

The key glowed white beneath the old burn ridge on her skin. Beside it, three cage traces converged toward a hairline mark. First pulse complete. Second pulse complete. The third command waited in the queue, time-tagged and armed, while *Common Measure* crossed the shadow below Ninefold.

“Blank opens in five seconds,” Jalen said.

Ada watched distances. The observatory lattice spread across the forward plot as nine dark instrument rings separated by kilometers of empty space. Aster Nine occupied the cage gallery sixty meters behind her, mass enough to pull every ring from its orbit if they gave it years. In eight milliseconds, its modeled drift amounted to less than five millimeters. The risk came from what twelve sinks might do with the gap.

“Cage primary stable,” Iri reported. “Secondary loop has our missing pulse queued for recovery. Thermal margin ninety-one seconds. Port wing slew holding.”

“Compact observer?”

A woman aboard Ninefold answered on the shared circuit. “Independent abort armed. We show all thresholds green.”

Ada kept her eyes on the raw trace. "Blank in three. Two. One."

The scheduled impulse reached its firing mark.

Nothing happened.

No field struck the charged sheath. No fusion drive answered. Aster Nine continued along the curve that gravity had already given it, free of instruction for a duration shorter than a human nerve could report.

On the first millisecond, cage eccentricity rose four tenths.

At three, Aster Three changed phase in its holding cage near Morrow.

At five, Aster Eleven reduced effective charge above Crown Meridian.

The displays filled faster than voices could follow. Aster One's Hawking-spectrum modulation deepened. Five and Seven moved against their cage centers in opposite directions. The change spread by distance, not by the common operations clock, each event arriving after light-time with an additional interval shaped by the moons.

"Seven milliseconds," Jalen called.

Ada pressed nothing.

The fourth pulse fired at eight. *Common Measure's* secondary loop recentered Aster Nine with a force light enough to escape her body and large enough to place a thin red edge on the cage display.

"Eccentricity two point one," Iri said. "Falling."

"Other sinks?"

Tomas answered from Ninefold's trajectory hall. "Nine changed first. Three responded inside expected delay. Eleven follows the lunar phase ratio. I need the full window."

"Forty seconds raw," said the Compact observer. "Then safety priority."

Ada did not relax her hand. They had created an absence. Every controller in the system now wanted to fill it with interpretation.

At fourteen seconds, an alarm cut across the circuit from Harbor of Glass.

"Aster Five cage harmonic," a yard controller said. "Extraction loop climbing through six percent. We are shutting feed."

The forward plot opened a new geometry without being asked. Harbor lay light-seconds across the web, its shipyard swarm dispersed above Ophion's outer moon. Aster Five had entered a scheduled accretion cycle there. The cage's rotational field and the sink's phase deviation had begun beating against each other, each cycle adding energy to the next.

"Five, dump the loop," Ada ordered.

"Feed shutter moving. Harmonic still climbing."

She had no authority over the yard's cage under ordinary rules. Article Nine placed every tug and containment system beneath Compact command, not hers. The controller obeyed because Ada carried a sink and sounded like someone who could see the whole plot.

Iri brought Harbor's cage schematic over their local displays. "Shutter needs eleven seconds. Extraction coils will hit quench in seven."

"Can they reverse field?"

"Not without coupling the yard truss."

Harbor's dispersed docks carried sixty thousand workers on shift. A coil quench would not release Aster Five, but it could flash the cage coolant into plasma and irradiate three construction drums.

Ada widened the all-sink trace. Aster Five's harmonic climbed in narrow steps, each aligned with the response that had followed their omitted correction. The test had not created the cage's underlying resonance. It had exposed and amplified it.

"Abort test," the Compact observer said. "Return Nine to nominal cadence."

"Negative," Tomas replied. "A recenter train will add energy at Five's beat frequency."

"Your authorized test ended at eight milliseconds."

"Then stop trying to end the consequence with the same signal."

Ada cut through them. "Iri, give me a safe operation."

Their breathing sounded close over the engineering channel. "Hold all routine corrections. Let the sinks finish whatever they started."

"That leaves Twelve drifting above Rook."

"Rook has eleven millimeters before mandatory center."

"Harbor has seven seconds."

"Six now."

Ada looked at the abort status. The outside key could fire Aster Nine's scheduled recenter pulses, adding a strong periodic input while Five's cage approached failure. Compact procedure favored known force over unexplained drift. Procedure had nearly dragged Aster Three across Morrow's passenger lane.

"Observer, disarm your key," she said.

"I cannot. Independent abort remains required."

"Then do not fire it."

"The Five event crosses system safety criteria."

"It crosses because every cage keeps correcting." Ada opened the all-cage channel. "All Aster handlers, Common

Measure. Suspend noncritical phase corrections for five seconds. Preserve containment thresholds. No one widens capture."

Responses collided. Crown demanded her command code. Rook's tug asked whether Compact had approved. The Harbor controller reported five seconds to quench.

Cass entered the circuit. "All handlers, hold present programs. Ada, explain."

"No time."

"I am making time. Hold present programs."

The correction markers continued firing around the system. At Harbor, Aster Five's harmonic crossed eight percent.

Iri swore. "They are feeding it."

Ada selected her own cage first. "Jalen, suspend Nine's next two pulses."

"That violates the test cutoff."

"Execute."

He touched the queue. Two white command marks went dark.

Aster Nine drifted one millimeter. The harmonic at Five missed its next expected rise.

Tomas inhaled over the circuit. "Again."

Ada held the omission through a third pulse. Nine's eccentricity reached two point eight millimeters, close to their automatic threshold. Aster Eleven reduced charge. Three followed. Then Seven. The responses arrived at Five from different directions in phase space, delayed according to their relations around Ophion.

Harbor's harmonic flattened at nine point four percent.

"Shutter has four seconds," the yard controller said. "Coils at quench margin."

Iri's hands moved somewhere in the cage gallery. Ada heard switches strike through their microphone. "Captain, I can bias Nine half a millimeter outward without making a pulse. Slow field ramp. It might lengthen their null."

"Might sharpen the harmonic too."

"Yes."

"Your recommendation?"

A pause occupied less time than false confidence would have. "Restraint. We already asked. Stop adding accents."

Ada left the bias untouched.

For one second every cage held whatever deviation it carried. The sinks continued moving. Their gravity admitted no pause, and their charged sheaths shifted within the fields, but human corrections ceased.

Aster Five's harmonic fell.

Not gradually. Its rising cycles met a second modulation and cancelled. Energy bled sideways into two weaker frequencies the cage could reject as heat. Harbor's extraction trace descended through eight percent, then six. The feed shutter closed before the coils quenched.

In the quiet after the alarm, Ada heard *Common Measure's* cooling pumps labor through the deck.

"Five secure," the yard controller said. "No quench. Cage center offset three millimeters sunward."

"Hold it there," Iri told them. "Do not thank it with a hammer."

Cass spoke with controlled force. "All handlers will resume corrections one at a time, sequenced through Compact control. No simultaneous pulses. Common Measure retains Nine's present offset until analysis clears recovery."

This time nobody argued. His order converted Ada's risky improvisation into system procedure before forty-six

local controllers could invent incompatible versions. She could resent central control and still recognize competent work.

"Cass," she said privately, "the observer nearly fired into the harmonic."

"The observer followed the authorized conditions."

"The conditions failed."

"And you exceeded them."

"Yes."

He let the admission stand. "Did the omission cause Harbor's event?"

"I don't know."

"Did it stop it?"

"I don't know that either."

"Tomas seems less uncertain."

"Tomas does not carry the cage."

Cass's image appeared beside the plot. Emergency boards colored his face with Morrow's red deficits and Harbor's descending trace. "You named a safe experiment. Instead you coordinated every sink handler through an unmodeled excursion."

"I coordinated the handlers to stop coordinating the sinks."

"That distinction will comfort the people under Harbor's coils."

"It may. They are still alive."

Cass looked away from her, perhaps toward another feed. "Keep Aster Nine outside the observatory plane. No transfer, no accretion, no discretionary field changes until my office completes review."

"Morrow needs Nine's allocation."

"Morrow needed you not to endanger Harbor."

The circuit ended before she could return the cost to him.

Ada released the recenter key and flexed her hand. The burn ridge had gone white again. Around the command blister, the crew worked without celebration. Parveen tracked heat. Jalen reconciled thrust. Two navigation specialists compared Nine's actual path against the safe corridor. Nineteen people had accepted a test whose consequence reached beyond their hull, and none could yet name what had answered.

"Captain," Iri said, "come aft."

"I have the plot."

"You have numbers. I need your eyes on the cage."

Ada delegated the conn to Jalen and pulled herself along the drive spine. During a sink tow, *Common Measure* maintained a gradient too slight to imitate gravity but strong enough to punish careless motion. She moved hand over hand, counting grips without deciding to. Heat from the cage gallery reached her through the bulkhead before the hatch opened.

Aster Nine's containment apparatus occupied a cylinder thirty meters long around an invisible center. Nested coil housings formed dark rings across the gallery. Radiation shielding hid the feed mechanisms. The black hole had no visible surface, only a position inferred from instruments and the way every loose object preferred one direction by a fraction.

Iri floated beside the manual control frame with their boots locked under a rail. A transparent maintenance panel showed the primary loop's phase trace as blue light.

"Look at the rejection spectrum," they said.

Ada anchored beside them. The cage had absorbed the recenter pulse at eight milliseconds and the smaller field adjustments afterward. Each applied force left a comb of

heat frequencies. Between those combs, Aster Nine's response occupied a clean valley.

"Lower amplitude than the commands," Ada said.

"Cleaner shape. Signal strength did not produce that. Restraint did."

"Do not assign intention to our own silence."

"I'm assigning consequences. We hit it, every cage hits back. We leave a space, the sinks use the space to cancel a bad beat." Iri pointed toward the Harbor trace. "That null took coordination across at least five phase changes. No cage controller owns all five. No operations clock could have timed them against the moons."

"Faults can couple."

"So can crews. Does coupling make us faults?"

Ada inspected the manual controls rather than answer. The blue paint strip beside Aster Nine's latch remained intact. Iri had replaced the witness after Aster Three cracked the previous flake, choosing a fresh edge from the same old hull scrap.

"The shutdown threshold protected this cage," she said. "It nearly damaged Five."

"Local safety can export risk."

"You have been talking to Belen."

"Belen uses shorter words."

A vibration touched the rail. Ninefold had begun rotating an instrument ring to keep its detectors aligned with the all-sink response. The motion barely altered local gravity, yet Aster Nine's trace registered it.

Iri enlarged the valley between corrections. "Your blank had borders. The sink cluster answered inside them. Then Harbor climbed. We stopped pressing, and the answer changed to fit the new condition."

Ada disliked the word because it reduced uncertainty faster than evidence allowed. Answer implied speaker, listener, content. They had correlated an omission with coordinated physical behavior. Every further noun risked inventing a mind in the shape of their need.

"We call it a response in the log," she said.

Iri folded their arms. "Compact will call it a control exploit."

"They already have."

"And you?"

Ada watched the clean modulation sit among the cage's noisy commands. "I call it expensive."

"That sounds like Cass."

"Cass is not wrong about every danger."

"No. Just about who gets to hold it."

Jalen called from the command blister. Ninefold wanted Tomas and Ada in the trajectory hall. Cass had granted forty minutes before Compact safety teams copied and filtered the raw streams. The port radiator wing had also accumulated more hinge strain during the thermal load, still within inspection margin but enough to slow their next maneuver.

Ada touched the blue witness once, lightly. "Lock the raw cage record against alteration."

"Already mirrored to three observatories and Morrow's public ledger."

"You sent containment data to Belen?"

"I sent evidence to people under the risk."

"That needed command review."

Iri met her gaze. "Then review it."

The old response rose first: data control belonged with the captain because partial information could make partial

panic. Cass used the same reasoning on a larger scale. Ada had spent five days accusing him of hiding risk inside authority.

"Next time notify the conn before transmission," she said. "Do not retract it."

Iri's surprise disappeared behind a crooked smile. "Reviewed."

By the time Ada reached Ninefold's trajectory hall, Tomas had built the response into orbital form.

Aster Nine's omission stood at the center of a web of colored intervals. Five sinks had altered phase enough to measure directly. Four more showed spectral changes. The remaining three supplied only weak timing correlations. Tomas had drawn no arrows suggesting commands. He used open curves, each marking when one relation became possible.

Harbor's dangerous harmonic appeared as two converging wavefronts. The coordinated sink changes placed a cancellation node over Aster Five exactly before the cage coils would have quenched.

"Could our field shutdowns have made the null?" Ada asked.

"Nine's first two changes began before the systemwide hold order reached their cages," Tomas said. "The handlers created the conditions, but not the timing. I have corrected for light-time and controller latency."

"Confidence?"

"Chance correlation below one in eighty thousand, assuming independent sink responses."

"Bad assumption."

"Yes. Independence has become harder to defend."

The Compact observers floated at the opposite rail. Their lead, the woman who had held the abort key, carried

herself with the rigid care of someone reviewing an action she might have taken. "The official finding should describe a latent multi-cage coupling," she said. "The omission triggered a system correction that protected containment integrity."

"Whose correction?" Ada asked.

"The sink-control system as a whole."

"No human controller issued it."

"Emergent automation can exploit shared physical state without explicit command."

Tomas kept his tone patient. "The cages do not share automation. Four manufacturers built them across thirty years. Their control networks remained isolated during the test. The common physical state consists of twelve masses, Ophion, and the moons."

"Then an emergent orbital feedback."

"That phrase admits everything except that the feedback noticed an omission," Iri said from Ada's shoulder.

Tomas blanked the event labels and gave the raw traces to a second Ninefold team that had not watched the test. They received only cage geometry, distance, and controller timestamps. Their task allowed no language about contact. They had to identify which input best predicted the cancellation at Harbor.

While they worked, Ada took a call from Tess Varo, the cage supervisor who had closed Aster Five's feed shutter. Varo floated inside Harbor's service hub with frost on one sleeve from a coolant fitting. Behind her, technicians tested the extraction coils in silence.

"We found heat scarring on the inner bus," Varo said. "Another two cycles would have forced quench."

"Any exposure?"

"Dosimeters stayed green. Three people crossed fatigue limits during shutdown. They are off shift."

"You should keep the cage cold until the harmonic model clears."

"We intend to. Compact allocation wants feed restored in six hours."

Ada glanced toward Mara. "Allocation does not own your coil margin."

"Useful sentence. Put your authority under it."

Ada entered a captain's hazard finding into Harbor's public log: no restart until the local cage crew certified the repaired bus and an independent model cleared the beat frequencies. Cass could override her under Article Nine, but he would have to do it by name.

Ninefold's blind team returned its result. Applied pulses predicted the rising harmonic poorly. Field holds improved the fit. The best predictor combined Aster Nine's framed absence with distance-delayed phase changes at Three, Eleven, Seven, and Five.

"They called it coordinated nulling," Tomas said. "No agency assumption."

Mara read the finding twice. "That still permits an emergent control exploit."

"It permits many mechanisms," Ada said. "It does not permit us to claim our controllers performed the cancellation."

The observer looked toward Ada. "Captain Narang, you hold operational responsibility. What do you enter in the incident record?"

Beyond the hall, Ninefold's instrument rings turned against the stars. At Harbor of Glass, workers would spend the next shift testing coils that had come within seconds of quench. Morrow still lacked its scheduled power. Naming could not repay any of that.

Neither could refusing to name what had changed.

Ada opened the shared record. "Aster Nine received an omitted correction framed by two known pulses. Multiple sinks altered phase in distance-limited sequence. Those changes cancelled a dangerous harmonic at Aster Five before human controls could do so."

The observer waited. "Classification?"

"Not exploit. Not contact." Ada entered the last line herself. "A reply."

Tomas said nothing. He did not need to turn agreement into ownership.

An alarmless notification opened on the main display. Aster Three had changed phase near Morrow. No tug had commanded it. The modulation traveled outward through the measured relations, two, three, five, then a clean absence where seven should have closed the sequence.

Tomas rotated the map.

The missing interval did not point toward Aster Nine's present location. It aligned with Ninefold itself, where the deliberate gap had entered the instruments and been retained.

Ada watched the recurrence cross the observatory's orbit. Aster Three had returned their blank to the place that had listened.

Chapter 9 — Twelve Without One

Tomas deleted the word *they* from the contact report for the ninth time.

The sentence held without it: *Coordinated phase changes followed the framed omission at Aster Nine*. Nothing in the evidence justified a pronoun. The response had crossed twelve sinks, five moons, nine cage architectures, and distances large enough to expose every event to light-time. Calling the source *they* implied individuals. Calling it *it* implied a bounded whole. Calling it machinery pretended the question had already failed.

He left the subject absent.

Around Ninefold's trajectory hall, three days of observations occupied the dark sphere. Colored intervals linked the Aster sinks in a web no orbital diagram had ever needed. Operations charts drew each transfer as a line between assignment points. Tomas's map drew relations: phase preserved, phase interrupted, mass returned after conjunction, expected impulse withheld.

Aster Three hung apart from the apparent symmetry. Eleven nodes completed a rapid cycle. The twelfth arrived late enough to miss every short recurrence and early enough to deform the next.

"Again," he told the analysis system.

The room replayed the omitted correction at one ten-thousandth speed. Aster Nine passed below the observatory. Its unfilled eight milliseconds opened.

Changes propagated through the cluster according to distance, then curved into a pattern shaped by Ophion's moons. At Harbor of Glass, the cage harmonic rose. Five sink relations shifted. The harmful oscillation cancelled.

Tomas removed Aster Three's data.

The cancellation still appeared, but the return interval did not close.

He restored Three and removed Nine. The model collapsed into unrelated deviations. Remove Seven, and a weaker relation survived. Remove any single node except Three, and the sequence lost detail while preserving grammar. Remove Three, and every sequence forgot how it had begun.

A person did not occupy any point. Neither did a message. Meaning persisted in the changing distances among them.

"Dr. Reeve."

Tomas turned. The Compact observer floated at the hall entrance with a sealed data case hooked to her wrist. She had introduced herself during the live test as Mara Venn, safety mathematician, Crown navigation office. He remembered her more clearly as the hand above an abort key that might have driven Aster Five's cage into quench.

"You are early," he said.

"Your release packet reached legal review incomplete."

"Then legal review can enjoy anticipation."

Her mouth compressed. "First Custodian Vardek wants a classification before the system brief. Fault interaction, autonomous control behavior, or probable nonhuman agency."

"Those categories do not share a scale."

"They share a decision." Mara placed the case beside the projection pit. "Morrow has nine hours before its next

district shutdown. Rook suspended an Aster Twelve transfer because crews fear another harmonic. Harbor workers are refusing cage shifts. Classification determines whether Compact command can order them back."

"Article Nine already lets Cass order them."

"It does not make them trust the order."

Tomas looked past her at the interval web. "Then tell legal review the behavior requires a relational model. No single cage, controller, or sink contains it."

"Is it alive?"

"Life belongs to chemistry in every definition we currently use."

"Is it aware?"

"We have one response to an intentionally framed absence."

"Harbor's crews call that yes."

"Harbor's crews nearly died. Urgency does not improve taxonomy."

Mara caught a rail before her drift carried her into the projection. "You sound angry."

"I become grammatical when angry."

"Then use grammar. What did it say?"

Tomas closed the report and opened the four stable interval classes.

The first appeared wherever an orbital relation repeated after one full moon cycle. The phase changed, returned near its initial state, then changed by the amount accumulated during transit. He labeled it RECURRENCE, not because the system used a word, but because the physical operation brought a relation back with difference preserved.

The second occupied expected positions left unfilled. MISSING fit poorly. An absence could result from failure,

destruction, or choice. Yet the Aster Nine test had framed the empty interval with two known corrections, and the response preserved its location across later cycles.

The third class delayed a phase term without losing it. HELD.

The fourth restored that delayed term after a conjunction. RETURNED.

"These labels describe transformations," he said. "They do not translate thoughts."

Mara studied the sequences. "Recurrence. Missing. Held. Returned."

"Provisional operations."

"Four words form more than a fault."

"Four human labels form a tempting story."

She tapped the case. "Cass authorized you to run passive comparisons against the anomaly records inside. No active cage tests. Twelve hours."

Tomas opened the seal. The case contained thirteen months of phase exceptions, stripped of crew names and commercial allocation data. He checked the metadata first. Record numbers began in the middle of a sequence. Calibration formats changed four times, including one standard retired fifteen years earlier.

"How much history did his office review?" he asked.

"I received the export this morning."

"That did not answer me."

"I have no other answer."

The case held old data wearing a recent time limit. Cass had not released it to the system brief. He had not told Tomas how far back the anomaly review reached. Yet he had selected enough buried events to test whether the four interval classes existed before the crisis.

Tomas accepted the evidence without accepting the secrecy.

By midshift, Ninefold's instrument team had aligned 18,000 phase exceptions against reconstructed moon positions. Most dissolved into cage noise: thermal drift, charge bleed, feed irregularity, bad clocks. Three hundred and twelve remained nonrandom. Their intervals distributed across every sink and at least fifteen years of partial metadata.

The grammar predated Aster Three's refusal.

Tomas anchored himself at the pit while the team worked around him. Suri Hale rebuilt extinct controller clocks. Lio Wen corrected each observation for habitat motion. A graduate analyst named Pell found that HELD events accumulated during transfers, when cages severed one sink relation and substituted another. Every finding made their clean four-word table less secure and the underlying structure harder to dismiss.

At hour seven, Pell projected a sequence across Aster Two, Six, and Ten. "This one has RETURNED before HELD."

"Then one label fails," Tomas said.

"Or the order wraps around a longer recurrence."

They extended the time window. The apparent return referred to a phase displacement introduced two moon cycles earlier. Human records had begun halfway through the operation.

Pell rubbed both eyes. "We keep entering after the sentence starts."

"Do not call it a sentence yet."

"You called it grammar."

"Operations have grammar. Orbital transfers do. Protein folding does."

“Former spouses do.”

Tomas looked at him.

Pell flushed. Ninefold had tracked Ada’s arrival with the coarse fascination observatories applied to every rare event. “Sorry.”

“No. Former spouses rely heavily on omitted terms.”

The dry laughter around the pit lasted three seconds and cleared enough fatigue for another hour.

They found the structural break near Aster Three before the meal bell. Each sink participated in paired relations at short periods. One answered another across a moon conjunction; a third held the displaced phase; the relation returned through a fourth. Across the first eleven nodes, those operations formed overlapping loops.

Aster Three joined every loop late.

Not from cage latency. Tomas corrected for controller response, engineered charge, sink mass, local fields, and travel time. The remaining phase lag varied with the outermost moon and a cometary family whose barycenter crossed Ophion’s far field every nineteen years. None of Compact operations models included that relation because it contributed too little force to affect transfers.

Too little force did not mean too little meaning.

Tomas expanded the long-period term. Aster Three carried a delayed relation that linked all twelve points through masses beyond the human habitat web. The cages repeatedly interrupted it before completion. Short cycles adapted around the loss. Eleven nodes formed complete rapid voices, if voice could describe an operation distributed across relation. The twelfth contributed on a scale human scheduling kept cutting apart.

“Eleven,” Suri said quietly. “Why only eleven?”

"Not only," Tomas replied. "Eleven complete at the frequencies we can observe. Three extends beyond our window."

"So Three has been trying to catch up?"

"No single point tries." He removed the sink markers and left only the interval lines. "Look without the objects."

The hall changed. Colored relations crossed where sinks had appeared, but no point anchored them. Loops formed and opened. Delayed phase moved through the geometry like tension through a net.

Pell turned slowly inside the display. "We trapped the knots and thought the net came along."

"Analogy only," Tomas said. "Do not put that in the paper."

Pell caught the rail instead of drifting away. "Unless Three conducts it. One sink keeps the clock, and the rest resonate because its field reaches them."

The proposal deserved more than Tomas's dislike. He restored the sink markers and fitted every short-cycle change to Aster Three's residual, allowing separate gains and light-time delays for all other cages. A single hidden oscillator matched most of their training sample. It even reproduced the broad cancellation at Harbor once the model received enough adjustable lag.

Then Tomas asked it to predict the reserved sequences.

The fit failed in both causal directions. Seven and Eleven began a paired return before Three's changed field could reach either location. During the next outer-moon crossing, Six's retained phase predicted Three's residual; Three's own prior state did not. Tomas moved the hypothetical clock to each of the other sinks in turn. Every candidate explained relations downstream by sacrificing events that led it. Adding the moons improved timing but never restored a unique source.

He removed the central-oscillator term. Prediction error fell when the model used pairwise intervals alone.

"Three carries the longest lag," Pell said, reading the result. "It does not issue it."

"Correct. A distinguished node need not constitute a center." Tomas saved the rejected model beside the surviving one, including its generous free parameters. If they claimed cognition, the public record would also show the simpler creature they had tried and failed to find.

"Is the paper allowed to say distributed mind?" Suri asked.

The phrase unsettled him less than the pronouns had.

Mind did not require a central point in human brains, though humans pretended otherwise for convenience. Their cognition emerged through relations among cells, signals, body, environment, memory. Here the substrate carried no cells and offered no human timescale. Mass, phase, recurrence, and return might support an organization that models expected conditions and alters behavior when those expectations fail.

Aster Nine's framed omission had created a distinction between force and withheld force. The response had changed when Harbor entered danger. That behavior implied sensitivity to relation, not merely oscillation.

"Probable distributed cognition," Tomas said. "Mind can wait for the public brief."

Suri saved the phrase. "Legal review will remove probable and keep the frightening part."

"Then we make the evidence harder to remove."

They tested alternatives through the habitat night. A hidden Compact network could not produce responses before signals reached controllers. Shared cage software failed because no common code existed across all manufacturers. A plasma instability could couple local fields but not distant moon geometry. An unknown gravitational

resonance explained timing but not the changed response to the deliberately omitted pulse.

Every rejected model narrowed the remaining claim.

Tomas reserved one sequence from the team and used the rest to make a blind prediction. If RECURRENCE and HELD described stable operations rather than labels pasted over noise, then Aster Six should reduce effective charge during the next outer-moon crossing, retain that displacement through Aster Ten's feed cycle, and restore it when Two passed below the inner moon.

He sealed the prediction with Ninefold's time authority. Suri complained that waiting ninety minutes for orbital confirmation ranked among the crueller uses of science. Pell slept with his feet hooked under the pit rail. Tomas watched cage telemetry arrive from three habitats that had not been told what to expect.

Aster Six entered the crossing.

Its effective charge fell by nine parts per million. The change sat beneath operations thresholds and above instrument noise. Aster Ten began feed on schedule; Six's displaced phase remained held. When Two crossed the inner moon, the term returned through a shallow spectral modulation at all three nodes.

Timing missed Tomas's prediction by eleven milliseconds.

He checked the error before anyone celebrated. Ninefold's reconstruction had used a published mass estimate for Aster Ten. Cage telemetry implied the sink carried six billion tonnes more than that estimate, a tiny fractional discrepancy with enough effect to explain nine milliseconds. Controller latency supplied the other two.

"Independent calibration can verify the mass," Suri said. "If it does, the grammar predicted an event no operator commanded."

"Write operation, not grammar, in the result."

Pell woke and stared at the completed trace. "You predicted what it would say."

"We predicted how a relation would change."

"That caution has become cosmetic."

"No. It protects the difference between our model and whatever exists outside it." Tomas attached the sealed forecast to their methods. "But Cass receives the prediction before he receives the conclusion."

At Day 7's first shift, Tomas sent Cass a finding: twelve relational nodes, eleven complete high-frequency cycles, one long lag centered through Aster Three. Probable distributed cognition. No evidence for twelve separate agents.

Cass requested a private call.

His image opened above the report with Crown's operations floor behind him. More deficit boards had turned red. He had shaved since their last conversation and somehow looked more tired for the precision.

"How confident?" Cass asked.

"In coordination, above six sigma. In cognition, confidence does not reduce to one measure."

"Try."

"High enough that treating the sinks as independent machinery now carries more assumption than treating their relations as adaptive."

"Adaptive does not mean entitled to shut down six habitats."

"Neither does alive."

Cass's gaze sharpened. "You have not established alive."

"I have not. I also have not established whether chemistry deserves monopoly on that word."

"What can we release without triggering cage refusals?"

"The full traces and methods."

"I asked what we can release."

Tomas made his voice formal. "Public analysts will reproduce our result if they receive raw data. Crews will trust instructions that expose uncertainty better than a classification stripped of evidence."

"Harbor already has eleven shift refusals. Morrow's clinics enter brownout rotation tonight. A false claim of alien captivity could stop every sink transfer."

"A true claim concealed for operational convenience could stop them less orderly."

Cass looked toward someone beyond the pickup and muted his side. When audio returned, his decision had settled. "I release the Day 1 through Day 6 phase streams, your relational model, and the Aster Nine test. The archival exceptions remain under review."

"They establish duration."

"They lack verified provenance."

"Then release the metadata gaps too."

"No."

"How old is your oldest record?"

Cass did not answer. "The public finding will state that coordinated behavior among all twelve sinks may represent a single distributed system. It will not use mind, life, captivity, or injury."

"The evidence has outgrown your nouns."

"The system still depends on my verbs. Feed, transfer, allocate, keep alive."

"Hide belongs on that list."

Pain passed across Cass's expression before command closed over it. "I am limiting an unverified archive during

an emergency. Put the distinction in your private notes if it comforts you."

"It does not."

"Good. Comfort wastes margin."

The call ended.

By the time Compact released its bulletin, versions of the all-sink map had already reached Morrow's public ledger, Harbor's crew councils, and six independent observatories. Iri's raw transmission after the live test had outrun central language. Cass's limited report still mattered. It placed the administration's authority behind the fact of coordination and forced every habitat to plan around more than mechanical failure.

The first public questions arrived in millions. Could the sinks refuse power? Did they feel the cages? Had humans discovered or manufactured the pattern? Why did twelve act like one? Could one act without twelve?

Tomas answered only the question his data could bear.

"No individual Aster sink shows evidence of independent awareness," he told the system feed. "The observed behavior occupies the relations among all twelve sinks and Ophion's larger orbital environment. We therefore model one distributed mind provisionally, not twelve captive persons. That model may change."

A reporter from Rook asked whether the system had spoken.

"We altered an expected correction. The response preserved that absence and changed a dangerous harmonic elsewhere. Spoken adds anatomy we have not found."

A Morrow clinic worker asked whether losing power had helped it communicate.

Tomas looked at the deficit curve beside her question. "Your loss resulted from Aster Three's refused transfer. We do not know whether communication, self-preservation, or

another operation drove that refusal. Naming intent cannot replace restoring your heat.”

Cass’s office closed the feed after twenty-two minutes. The questions continued through unofficial channels.

On Day 8, Tomas returned to the long-period relation.

He had slept for ninety minutes in a restraint bag beside the trajectory hall and dreamed of unfinished arithmetic. Ada’s omission recurred through Aster Nine every moon-aligned cycle. Aster Three’s missing interval approached it from the long lag. Between them, a new pattern had begun to form.

RECURRENCE.

HELD.

MISSING.

RETURNED.

Then recurrence shifted ahead of the condition that should have produced it.

Tomas checked timestamps. The pattern anticipated Ninefold’s next instrument alignment by thirty-one seconds. When the rings moved, the sinks altered phase into the relation already prepared for them. Prediction offered a simple explanation; orbital systems always carried future position in present velocity. Yet the operation did more. It preserved a future return as though that return already constrained the current interval.

He labeled it ANTICIPATING RECURRENCE.

The software overlaid its spatial focus. The relation converged near Morrow’s orbital family, not on the habitat’s current position but where Morrow would arrive after its next power-bridge tow.

Tomas disliked coincidences, especially linguistic ones. He opened an old word list and found *morrow*: the coming time, future carried inside present speech. The habitat’s

founders had chosen the same archaic promise for a poorer station built from reused hulls.

Twelve nodes participated. Eleven completed the quick phrase. Aster Three supplied the late term that bent it toward what had not yet occurred.

He entered a provisional translation beneath the map:
Morrow-of-Twelve.

Not a name. Not a voice. A recurrence that anticipated itself through all twelve relations.

As he saved the operation, Morrow's predicted tow window shifted by thirty-one seconds. The interval pattern had reached the rendezvous before the human schedule did.

Chapter 10 — The Port Wing

The fracture opened while Ada could still turn back.

A white line lengthened across the port radiator hinge image, narrow as a hair and bright against the composite. *Common Measure* had carried Aster Nine two hours beyond Ninefold, enough to commit propellant but not enough to close the return corridor. Ahead, Morrow's power bridge waited below the planned arc with three districts already dark.

"Growth rate?" Ada asked.

Iri magnified the scan from the radiator crawl. "Four microns since deployment. Two more than inspection predicted. Hinge pin holds. Root laminate disagrees."

"Thermal consequence if we lock the wing?"

"Current feed rate, forty-three minutes to red. Less when we shape the test pulse."

The command blister showed Morrow as a rust-colored ring against Ophion's clouds. Its old hull caught sunlight unevenly where generations of patched radiators covered the torus. One hundred eighty-four thousand people lived there. The sink's controlled accretion could restore clinic heat and circulation pumps once the power bridge established coupling.

"How long to the bridge?" Ada asked.

Jalen adjusted for the latest phase drift. "Thirty-eight minutes if we keep burn. Fifty-six if we reduce cage feed and coast."

"Return threshold?"

"Twenty-one minutes. After that, Morrow offers the lowest-risk destination."

Ada touched the hinge trace. "Lock port radiator at present angle. Reduce accretion feed twelve percent. Continue."

No engineering acknowledgment came.

"Iri?"

"Recommend abort, cage feed to maintenance minimum, return Ninefold."

"Morrow crosses civil heat margin in thirty-four minutes."

"Then Morrow needs another bridge. This hinge carries our rejected heat."

"There is no other sink inside six hours."

"Still not a structural argument."

Ada opened the live heat budget. Locking the wing would preserve the cracked root from deployment loads. It also fixed one radiating surface partly edge-on to the star shield, reducing useful area. The starboard wing could carry most of the remaining heat if they throttled feed. For thirty-eight minutes, projections remained below red.

"We have margin," she said.

"We have a model trained before Crown twisted us across two cage fields."

"Give me your revised model."

"I just did. Abort."

Their disagreement occupied the open crew circuit. Nineteen names watched from engineering, navigation, drive, and medical stations. Ada felt each listener as an

added hand on the controls. Command usually compressed debate before a burn. This crisis had made hidden agreement more dangerous than visible conflict.

"Bring the crawl scan to the main board," she said.

Iri sent it. The port radiator hinge filled the blister display: pin, laminate root, coolant swivel, and the microfracture crossing two outer plies. No leak appeared. Ultrasonic echoes showed no delamination beneath it. The damage remained small enough to invite confidence.

"I accept the fracture may outrun prediction," Ada said. "Morrow bears certain harm if we turn back. We continue under a forty-minute heat budget. Port wing stays locked. Any growth above ten microns triggers feed shutdown."

"You accept my engineering recommendation or you overrule it. Do not turn it into weather."

"I overrule it."

"Logged."

The single word reached her harder than argument. Iri had obeyed under protest before. They had never made the separation so formal.

"Cage status?" Ada asked.

"Contained."

"Radiator crawl clear?"

"I'm returning through the truss."

Jalen held the tug's burn. The return threshold passed beneath them, a line no instrument could detect. From then on, abort meant proceeding to Morrow by a different path.

Aster Nine moved inside its cage with a persistent two-millimeter offset left from the omitted correction. Compact safety wanted it centered. Tomas wanted the relation preserved. Ada had chosen the offset because every recenter attempt produced responses across the other

sinks. Choice had become the operational name for uncertainty.

"Morrow bridge reports ready," Parveen said. "Receiver fields at thirty percent. They request our feed forecast."

"Send conservative output. Sixty-two percent of allocation for the first hour."

"They planned on eighty."

"They receive what our heat budget can reject."

Belen Sato appeared on the side circuit before Parveen finished transmitting. Behind her, Morrow's thermal control room carried the dim blue strips of emergency lighting. Operators moved between manual boards that had outlasted three generations of automation.

"Your number lost eighteen percent in transit," Belen said.

"It lost eighteen percent in our radiator."

"What failed?"

Ada showed her the hinge image.

Belen studied it without visible alarm. "Can you deliver sixty-two safely?"

"Within present limits."

"That sounds prepared by legal."

"I overruled an abort recommendation."

"Whose?"

"Iri's."

Belen looked past Ada as if the engineer might appear through the bulkhead. "Then whose limits are present?"

"Mine."

"Your hull. Your call." Her voice stayed level. "Our clinics will take sixty-two. District pumps remain in rotation. Send the actual heat curve, not a green marker."

Parveen transmitted the raw budget. Belen closed the circuit after saying where Morrow would place the first power: surgical wards, water circulation, then agricultural chillers. She did not ask Ada to justify continuing. Naming the recipients made justification physical enough.

At twenty-seven minutes from the bridge, the fracture gained another micron.

Iri returned to the cage gallery and reported without commentary. Ada lowered feed two more percent. Morrow's projected intake fell again. On the habitat channel, Belen reassigned one agricultural loop to stored cold.

Aster Nine changed phase.

The shift came softly, a shallow depression in the raw line. Three days earlier, Ada might have called it drift and ordered correction. Now every observer on the circuit stopped speaking.

"Tomas?" she asked.

"Recurrence class," he answered from Ninefold. "Same operation as the post-omission response. Three has entered the long lag."

"Effect on tow geometry?"

"None yet. Hold your field."

Iri spoke at once. "Holding."

The fracture added two microns.

Ada watched both traces. If she reduced cage field, Aster Nine's charged sheath would move farther from center. If she reduced feed, the power bridge might fail to lock. If she deployed the port wing to improve heat rejection, the damaged hinge could unzip.

"Morrow coupling range in nineteen minutes," Jalen said.

A new phase response arrived from Aster Four. Then Two. The changes did not follow the short grammar Tomas had mapped. Each sink increased Hawking-spectrum

modulation across a wide frequency band, not enough to add appreciable power but enough to illuminate every cage detector in the system.

"Pulse building," Tomas said. "All nodes participating."

"Radiation?" Ada asked.

"Within local operating limits."

"Local does not answer all habitats."

Parveen gathered reports. Crown green. Rook green. Harbor green with Five's feed shutter still closed. Morrow's holding cage for Aster Three reported a charge rise near its upper threshold.

"Aster Three needs recenter," its handler said. "Request coordinated window."

Tomas answered before Compact control. "Do not recenter during the pulse."

"Our cage exceeds six percent in twelve seconds."

Ada brought Three's containment geometry beside Nine's. The Morrow cage lacked a tug's sacrificial truss. It hung from an old freight spindle connected to the habitat by the same power bridge they approached. If Three kicked its field center, the gradient could pull the spindle into Morrow's outer radiators.

"Iri, can Nine take a shaped correction without adding to the broad band?"

"No proven shape."

"Can Three?"

"Same answer."

Cass joined from Crown. "Aster Three handler, recenter at threshold. All other cages prepare simultaneous stabilizing pulses."

Tomas's protest crossed Ada's at the same time. "Simultaneous input could sharpen it."

Cass put the all-sink traces on the command circuit. "Aster Three threatens Morrow. We do not preserve a signal at the expense of a hull."

"He is right about the hull," Ada said. "Wrong about simultaneous correction. Pulse only Three. Let us see whether the relation accommodates one safety action."

"Your test at Harbor nearly quenched a cage."

"Because we kept adding corrections."

"Ada," Iri said on the internal channel, "our heat curve just crossed the forty-minute projection. If Three pulses, Nine may answer. I recommend feed shutdown before that happens."

Morrow lay fourteen minutes ahead. Feed shutdown would protect *Common Measure* and leave the bridge without enough energy to couple. The habitat's clinics would continue brownouts. The fracture held at nine microns, one below Ada's declared trigger.

"Maintain feed," she said. "Three gets one recenter pulse. All others hold."

"I object," Iri replied.

"Logged."

"No. I gave you an engineering limit. You converted it into captain's risk."

"I carry the hull risk."

"You carry command. I carry the cage and heat plant. If my stop can be overridden whenever another habitat needs power, I do not have a safety limit. I have advice."

Cass remained on the shared circuit. Morrow's handler counted down toward forced recenter. Every second narrowed Ada's ability to separate authority from emergency.

"State the condition," she said.

"Shut feed now. Port wing remains locked. No shaped field change aboard until the pulse ends."

"That loses the bridge."

"It delays the bridge. Breaking our radiator loses the tug."

"Morrow's civil margin will cross."

"And another crew's need still cannot make false numbers true."

Nadi's rule carried through Iri's words without quotation. Authority ended at another crew's hull. Ada had used that boundary against Cass. She had not expected it to rise inside her own.

The fracture reached ten microns.

Ada closed Aster Nine's feed shutter.

Controlled accretion dwindled. Power output dropped hard enough to turn Morrow's coupling forecast red. Heat still moved through the tug, already generated and seeking the radiators, but the curve stopped steepening.

"Engineering stop accepted," Ada said. "Crew board, record a standing engineering veto over cage containment, reactor, and thermal red limits. Captain may request alternatives, not override the stop without a crew safety vote."

Iri took one breath. "Accepted."

Cass said, "Article Nine places cage authority under Compact command."

"On my hull, your order reaches the captain. It does not make my engineer's instruments decorative."

"You are inventing governance during a containment event."

"So did Article Nine."

Morrow's handler called threshold. Cass authorized one manual recenter pulse.

Aster Three moved toward cage center.

The all-sink response widened.

Nine's offset changed by half a millimeter without any local field command. Four and Seven altered charge. Twelve's accretion output dipped. Across the distributed relations, the broad modulation passed through every point and returned carrying the shape of the cages.

Tomas enlarged the spectrum. "Do not filter that."

No one needed the warning. Twelve containment systems appeared inside the pulse as regions of suppressed phase, each with edges matching its field geometry. Crown's modern spherical cage cut a smooth cavity. Harbor's yard cage left a lopsided interruption where one coil remained shut. Aster Three's freight spindle imposed a long narrow seam. *Common Measure's* cage registered as concentric absences around Nine.

The pulse did not map metal. It mapped relations prevented from closing.

Iri whispered, "Cages."

Ada stared at the twelve negative shapes. Human containment had always represented safety: field inside, hazard centered, hull protected. On Tomas's relation map, every safe boundary severed an interval among the sinks. The systems that harvested power also broke the physical structure carrying coordinated behavior.

"Could this come from cage telemetry?" Cass asked.

"No common telemetry entered the sink relation," Tomas said. His voice had lost patience. "The response contains field harmonics from twelve isolated systems, including Harbor's unreported coil defect. It learned those shapes through what they suppress."

"Learned," Cass repeated.

"Measured, encoded, retained. Choose the least alarming verb after you preserve the data."

Aster Three's recenter ended. The broad pulse continued around Ophion, modified by every cage boundary. When it returned through Nine, its spectrum emphasized the frequencies human fields had removed most often across the historical records Cass still withheld.

Ada understood the operational fact before she accepted any moral noun. The sinks had not begun deviating because one cage broke. Their coordinated structure had carried interruptions long enough to respond across the whole system. Every successful transfer, every clean recenter, every extracted sink-hour had contributed to the pattern now destabilizing circulation.

"We caused the phase fracture," she said.

Cass answered carefully. "We do not know that."

"We know the cages suppress the relation. We know stronger correction produces stronger deviation."

"We know correlation under crisis conditions."

Iri's voice cut through. "Open your archive."

Silence followed on Crown's side.

Ada looked at Cass's image. "How long?"

"Complete the tow."

"How long have these suppression patterns appeared?"

"Morrow needs your bridge more than this argument."

"Feed is shut. Bridge cannot couple at planned power."

Cass examined the curves. "Can you coast into receiver range and restart at lower output?"

Iri answered under the new authority. "Yes, after the pulse. Port radiator stays locked. Ten-minute feed run at forty percent, then hard shutdown."

Ada waited for them to finish. "Can the hinge carry it?"

"Below thermal red with six percent uncertainty."

"Your recommendation?"

"Proceed. Morrow gets a smaller bridge. We return hot."

Ada accepted.

The new plan transferred risk into visible places. Morrow would receive thirty-eight percent of its expected allocation. *Common Measure* would use its coolant reserve during the return and sacrifice sensor operation to reduce onboard heat. The port wing would not move. No one could later call the compromise safe without naming what remained unserved.

They coasted through the rest of the pulse.

At Morrow coupling range, Iri reopened Aster Nine's feed. Accretion flared inside the shielded cage. The power bridge caught the output and carried it toward the habitat in a controlled beam. District markers changed from black to amber: South Clinic, water loops four through seven, two surgical wards. Agricultural chillers remained off.

The receiver field began to oscillate before full lock. Morrow's old bridge had expected Aster Nine on its standard centerline, not two millimeters offset inside a relation nobody had modeled when the spindle was built. An exterior camera showed three suited workers moving along the receiver truss.

Ren Sato led them one-handed. His crushed hand remained sealed inside a rigid pressure mitt strapped to his chest. With the other, he clipped along the truss and placed temporary current shunts across the receiver vanes.

"Get that crew inside," Ada told Belen. "We can adjust beam angle from here."

"You can adjust your transmitter," Belen replied. "The oscillation belongs to our receiver."

"A beam slip crosses their truss."

"Ren has your live error cone."

Ada opened the worker channel. "Foreman Sato, transmitter can reduce output while you cross vane three."

His helmet camera tilted toward the cage beam's invisible path. "Reduce and we lose lock. Hold your present angle for thirty seconds."

"Your left translation route enters the uncertainty cone."

"Then stop translating left." Ren pushed away from the handrail and let his tether carry him around the vane instead. His two crew followed. The route cost them eleven seconds and kept every suit outside Ada's projected error.

At vane three, Ren fitted the last shunt against a contact designed for two hands. Another worker braced the plate while he drove the latch with his good arm. Receiver oscillation fell below one percent.

"Bridge crew holds engineering authority on its hull," Belen said. "You hold beam authority on yours. If either side calls stop, we stop."

Ada could have objected that one transmitter and one receiver formed a single hazard. That truth did not grant her ownership of both ends.

"Common Measure accepts," she said. "Ren, confirm clear before ramp."

Three suit markers withdrew along the safe route. Ren reported the truss clear. Only then did Ada authorize Parveen to raise the beam.

"Coupling stable at thirty-eight," Parveen reported.

Belen appeared on the habitat channel. "We have circulation. Clinic reserve begins charging."

"You have ten minutes," Ada said.

"We planned sixty hours."

"Our hinge opened. We can leave Nine's assignment cage with you, but your spindle needs time to integrate the offset."

"Time costs district heat."

"Yes."

Belen absorbed the answer. "Send me the list of systems you shut to get here."

Ada forwarded it: port optical array, survey radar, half the forward lidar, galley heat, crew quarters, secondary computing. Belen added the list to Morrow's public resource ledger beside the reduced intake.

"No reassurance?" Ada asked.

"I asked for power, then inventory. Reassurance does not circulate water."

At ten minutes, Iri closed feed. Aster Nine transferred into Morrow's assignment field without recentering, preserving its two-millimeter offset and the relations Tomas had measured. The power bridge fell to stored output. Morrow had gained hours, not days.

Common Measure turned for Ninefold under fusion thrust and a rigid heat budget. With the port radiator locked, temperature spread through compartments never intended to serve as sinks. The crew sealed the galley and moved forward instruments to cold standby. Ada authorized coolant injection around the cage controls, spending half their return reserve.

Twenty minutes into the burn, the forward optical array shut down from thermal distortion. Radar followed. The command blister lost its crisp field of traffic markers and retained only coarse transponder points supplied by other stations.

"Partly blind," Jalen said. "Still navigable."

"Whose data?"

"Ninefold, Morrow, and a freight tender above us."

Ada let the external feeds populate their plot. Three separate hulls carried the sight *Common Measure* could no

longer provide itself. She would once have called that degraded command. Now she asked each source for uncertainty and kept burning.

Iri came forward after securing the empty cage. Sweat darkened their collar. They placed the radiator scan on Ada's board. The fracture had stopped at eleven microns after cooldown, one beyond her original threshold.

"Engineering veto remains after this tow," they said.

"Yes."

"Even when your numbers dislike it."

"Especially then."

No apology could erase the override. Ada entered her decision, Iri's objections, the fracture growth, and the reduced Morrow delivery into the shared log. Authority had changed because trust had not survived its previous shape intact.

Behind them, Aster Nine moved in Morrow's field with a deliberate offset. Ahead, Ninefold's borrowed radar painted a safe corridor home. Between those points, Tomas's pulse map remained on every crew display: twelve dark interruptions in a relation no human cage had been designed to see.

Common Measure returned hot and partly blind, carrying the negative image of humanity's work.

Chapter 11 — The Six at Risk

The bean vines began shedding leaves while the delegates argued over which habitat should die first.

Belen caught one against her sleeve. Yellow had eaten inward from the edges, leaving the veins green and stubborn. Above her, Morrow's agricultural drum revolved around a shaft gone dark for power savings. Its terraces climbed in both directions until perspective folded them overhead: beans nearest the meeting platform, then rice troughs, dwarf fruit trees, and ranks of lamps burning at half their designed output. Fans moved damp air with an uneven pulse. Every fourth lamp stayed black.

"Call it what it means," she said. "You propose emptying Vela Minor and Pallas Reach."

The Compact planner shifted his projection away from the vines and onto the irrigation wall. Two amber circles vanished from a diagram of forty-six habitats. Power routes around them brightened green.

"Staged evacuation," Ors Kemen said. "Both populations can distribute through receiving capacity over eleven days. The hulls would remain available for later recovery."

"Available to whom?" asked Hale from Vela Minor. "Ghosts with tug licenses?"

The six immediate-risk habitats had sent whoever could leave local crisis rooms for three hours: Hale, a water chemist with mineral burns across both wrists; Pallas Reach's spin master; two clinic directors from Nerio and

Tern; a reactor steward from Low Ketu; and Belen. Compact staff stood along the outer curve in clean gray pressure cloth. Ada Narang waited among them, helmet hooked under one arm, watching the floor as though it might disclose a vector.

Ors kept his voice level. "Vela carries forty-one thousand residents. Pallas carries thirty-eight. Removing those loads preserves circulation margins for a greater population."

Belen held up the fallen leaf. "This plant lacks potassium because we shut a recovery heater to keep Ward Eight below fever temperature. Your model calls that a fractional agricultural loss. Give us the inventory beneath your green lines."

"We supplied the aggregate tables."

"You supplied conclusions. I want storage by habitat, tug availability by hour, spare radiator area, nitrogen, water, reaction mass, and empty berths that possess working air loops. Put dates on all of it."

The spin master from Pallas, Arun Vale, leaned over the railing. "And account for acceleration limits. We have nine thousand residents who cannot tolerate an emergency tow profile."

A lamp bank dimmed. Conversation faltered while the drum's control system shed another load. High above, workers in orange harnesses moved along a terrace rail and clipped portable light bars into the fruit rows. Morrow had turned the failing farm into a meeting hall because no formal chamber could hold six habitat delegations, residents, engineers, and the local crews entitled to question them. Also because Belen wanted every proposal spoken beneath a visible cost.

Ors looked toward the Compact legal officer. She gave him no rescue. "The complete tables include protected continuity reserves," he said.

"Protected from what?" Belen asked.

“Uncoordinated drawdown.”

“People have started boiling clinic linens in restaurant vats. Coordination already failed to reach us.”

A murmur traveled around the terraces. Belen let it. She had spent the morning refusing reassuring summaries from her own grid supervisors. Morrow possessed thirty-six hours of ordinary heat margin if Aster Three’s allocation returned, nineteen if the sink network continued its present deviations, and less than twelve under the pulse pattern Ada had transmitted from the cages. Those numbers frightened people. Hiding them would not grow another hour.

Ren stood at the farm access lock beside two radiator workers. His crushed right hand hung inside a rigid foam splint against his chest. He had refused the clinic’s sedatives because they blurred his balance. The hollows beneath his eyes made him look older than Belen, though seven years lay between them.

He saw her looking and raised his good hand, palm down: steady.

Belen turned to Ada. “You asked for the floor. Take it before they erase another circle.”

Ada stepped into the projection. She did not offer rank or apology. “At Day Nine’s test tow, all twelve cages responded to one phase pulse. Not independently. Together. The pulse mapped each containment field as an interruption in the same relation.”

She sent a technical image across the irrigation wall. Twelve sharp traces appeared around a slower wave. At each cage correction, the wave broke and reformed with a notch.

“What does interruption mean?” Arun asked.

“Dr. Reeve can describe the grammar. I can describe the machinery. Isolation corrections increase radiation and orbital divergence. Corrections that preserve timing

between sinks reduce both. The cages do more than hold mass. They suppress a coordinated state."

Hale's scarred fingers tightened around the rail. "Are you saying the power system feels injured?"

"No. I do not know what feeling means at that scale." Ada enlarged the notches. "I am saying our containment produces the response destabilizing our power system. Forced isolation may make it worse."

Ors spoke before anyone could absorb that. "The finding remains disputed. Compact analysis has not established intelligence, subjective experience, or a safe release method."

"Did she claim those?" Belen asked.

"She is presenting an interpretation to residents during an allocation crisis."

"She brought measurements. You brought two empty circles."

The gallery broke into overlapping questions. A grower shouted that nobody could free a trillion-tonne object near a habitat. Someone from Vela demanded to know whether keeping the sinks caged amounted to torture. A Morrow nurse answered that moral purity would not run her oxygen plant. The farm acoustics returned each voice half a beat late from the opposite curve, turning alarm into a crowd arguing with its echo.

Belen crossed to the manual alarm rail and struck its brass maintenance bell once.

The note cut through the drum. Workers used that bell when machinery threatened hands. Everyone on Morrow knew to stop first and locate danger second.

"We do not know enough to name torture," Belen said when quiet returned. "We know enough to stop calling the cages neutral. Both facts enter the decision."

A resident near the rice beds stood without waiting for recognition. Mina Rhee ran a bakery in Sector Five and had lost two fingers to a dough roller years before Belen entered public office. "If we open those cages, do our children keep air?"

Ada faced her. "Not under any plan I have seen today."

The answer landed harder than reassurance would have. "Then why show us this?" Mina asked.

"Because Cass Vardek has a forced-containment model under review. Because it takes reserves from Morrow to isolate Aster Three. Because you bear that risk whether anyone asks your permission or not."

Ors's projection disappeared. "That plan has not received operational approval."

"Yet," Ada said.

Belen watched the Compact staff rather than Ada. One planner lowered his eyes. The legal officer touched two fingers to her wrist display, sending something private. There it stood: not proof of a conspiracy, only the familiar administrative instinct to postpone the people whose floors would move.

"Break for twenty minutes," Belen said. "Delegates meet in the seed vault. Compact staff remain here and answer residents' technical questions. No speeches."

Ors objected. "This session falls under system emergency procedure."

"So does crop failure. You chose my drum."

She led the five delegations through a narrow lock behind the bean terraces. Ren followed until Belen stopped him with a look.

"Residents need the rig facts," she said.

"I have rig facts about your ceiling panel."

"What?"

"It has been flashing condensation warnings for an hour."

Belen looked up. A fat drop detached and struck her forehead. Ren's mouth moved at one corner.

"Go tell people why," she said.

The seed vault held racks of sealed cultivars, a scarred worktable, and six folding stools. Cold storage had been shut down; packets that should have slept near freezing now waited at farm temperature, spending years of viability by the day. Belen set the yellow leaf in the middle of the table.

Hale closed the door. "Vela cannot accept evacuation. Not politically. Not physically."

"Can you move anyone?" Belen asked.

"Eight thousand over eleven days if receiving berths exist. Fewer if we must supply the tugs."

"Tern has berths," said clinic director Sahar Ochoa. "We do not have scrubber reserve for more than three thousand."

Low Ketu's steward folded his thick arms. "We can release oxygen candles and two carbon beds. We need coolant in exchange."

"Nerio has coolant," his counterpart said. "Its transfer line crosses a forbidden sink corridor."

Arun tapped the table. "Then move the corridor."

They looked at him.

"Not the sink corridor," he said. "Move Pallas. A tenth of a degree outward buys Nerio a six-hour window three days from now. We planned the burn last year for conjunction avoidance, then Compact denied reaction mass."

Belen opened a blank ledger. "How much?"

"Eight hundred tonnes if done clean. Twelve hundred if delayed."

"Morrow can loan two hundred."

Hale gave a sharp laugh. "You have power for half your lamps."

"Reaction mass does not turn on lamps."

"It keeps your attitude jets alive."

"We have a flywheel vault and a retired water tender. The tender can carry momentum if the old bearings hold." Belen entered the pledge before caution could dress itself as prudence. "Two hundred, contingent on inspection."

Sahar leaned closer. "Tern can provide the inspection crew. You send us Ren Sato."

"My brother does not belong to Morrow's inventory."

"His exterior team, then. If they volunteer."

"Ask them directly."

The correction altered the room. Each delegate had arrived ready to defend a boundary. Now lines began crossing the blank ledger: carbon beds from Low Ketu, coolant from Nerio, temporary berths at Tern, Morrow reaction mass, Pallas orbital clearance, Vela food cultures. None solved the sink crisis. Together they disproved the planner's assumption that help could travel only down a Compact allocation tree.

Belen drew six circles around the commitments. "A mutual-aid agreement. Resources remain under each habitat's authority until a named receiving crew accepts them. No seizure, no protected tables."

"Compact will call it interference," Hale said.

"Compact can add its resources and earn a seventh circle."

Arun studied the ledger. "If one habitat defaults?"

"The others see the same live inventory. We redistribute the obligation before the missed transfer becomes a surprise."

"Trust as infrastructure," Sahar said, skeptical.

"No. Visibility as infrastructure. Trust comes after delivery."

Low Ketu's steward touched the yellow leaf. "This saves days. Perhaps a week. It does not answer the abandonment proposal."

Belen capped her stylus. "We answer that outside."

An alarm pulsed inside the seed racks before anyone reached the door. Not the drum bell, but a local temperature warning. The packets nearest the vault wall had crossed twenty-eight degrees.

Arun checked the dead cooling panel. "How long before damage?"

"Damage started when we shut refrigeration," Belen said. She pulled the storage drawer for Morrow's blight-resistant rice line. Warm air escaped without any trace of cold. "At this temperature, germination declines after four days. These cultures replace a harvest if the current crop fails."

Hale crouched beside the panel and removed its service cover. "Pump works. Controller has no bus power."

Low Ketu's steward opened his habitat ledger. "We have two portable fuel cells assigned to the carbon-bed shipment."

"Then they travel with the beds," Belen said. "Tern needs them for unloading."

Sahar shook her head. "Our dock has emergency outlets. We requested cells because Compact procedure assumes receiving power cannot be trusted."

"Can your council waive that requirement?"

"My council can. Our insurer will protest after everyone lives."

The steward altered the manifest. One fuel cell remained with the shipment; one changed destination to Morrow's

agricultural drum. A yellow route appeared between Low Ketu and the vault.

"Arrival in nineteen hours," he said.

"Too late for this drawer," Hale replied. She traced the controller bus toward the wall. "Your lamp bank shares a converter. Shut another twenty meters of beans and the refrigeration can run at maintenance level."

Belen pictured the dark strip widening across the terraces. Beans supplied food now. Seed cultures preserved food two seasons from now. A planner's aggregate could flatten both into agricultural demand and spend whichever lacked a public voice.

She called the grower superintendent. "I need authority to darken Row Twelve for six hours."

"Yield loss reaches another three percent."

"The seed vault crossed limit."

The superintendent did not ask for reassurance. "Take Row Twelve. We will move portable bars during pollination."

Hale bridged the converter with a spare lead from the worktable. Cold did not return at once. The compressor shuddered, drew current, and began pushing the vault temperature down by tenths.

Belen added the lost bean yield and preserved seed cultures to the ledger. The first shared allocation had not crossed space or required Article Nine. It had required six people to see one warm drawer before choosing what darkness to move.

When they returned to the drum, the crowd had divided itself around Ada's cage map. Iri Quill had joined by remote projection from *Common Measure*, their image breaking whenever Morrow's network rationed bandwidth. They walked Ren through a magnetic-loop failure using a diagram sketched over the rice-bed controls. Ors stood with

his planners near the dark lamp bank, reduced from authority to one group among many.

Belen gave the ledger to the wall.

Six columns rose beside the sink traces. Each showed actual stores, minimum civil reserve, promised transfers, crew requirements, and confidence ranges. Blank cells outnumbered full ones.

"This does not look like rescue," Mina said from the gallery.

"It looks honest," Belen replied. "The six at risk have formed a mutual-aid bloc. We will publish live inventories and transfer promises. Any habitat may join by opening its reserve tables and placing crews under hull-local consent."

Ors approached the platform. "You cannot establish a competing allocation authority during Article Nine."

"We allocate what our councils control. Article Nine commands tugs and cages. It does not nationalize a bakery's flour or a clinic's spare bed."

"The distinction becomes meaningless when local commitments obstruct system plans."

"It becomes essential then."

He lowered his voice. "Convener, if circulation fails, models put deaths in seven figures. Custodian Vardek cannot run continuity by referendum."

Belen believed the fear beneath his words. That complicated anger but did not weaken it. Cass carried six million lives in his models. She carried 184,000 in corridors, family tables, repair rosters, and three names newly entered in Morrow's burial list. Scale did not grant one account permission to erase the other.

"Your continuity plan abandons seventy-nine thousand people before you have counted where they sleep," she said. "Do not ask me to respect its urgency more than their bodies."

Ada had moved beside the outer railing. Belen joined her while delegates answered questions about the resource columns.

"How long can your bloc hold?" Ada asked.

"Longer after the inventory arrives. Less long if Compact seizes the pledged tugs."

"They may. Article Nine has another day in its first term."

"And after?"

"A renewal vote."

Belen looked across the inverted landscape. Children carried fallen leaves into collection sacks. On the far curve, Ren used his good hand to demonstrate how a radiator rib could buckle under a moving gradient. People listened because his splint supplied credentials no title could.

"Does freeing the sinks kill us?" she asked.

Ada rested both hands on the rail. "A release without migration would. Keeping them caged may also kill people if the relation continues to contract. Tomas thinks synchronized free trajectories could restore stability, but he does not have a complete solution."

"Cass has a complete solution?"

"He has a complete order."

That distinction belonged on the wall too.

Belen returned to the platform and called the delegates forward. Each placed a habitat key against the ledger, signing resource visibility into effect. Six confirmations turned green. Morrow's network pushed the table system-wide before Compact filters could classify it.

Then Belen replaced Ors's erased circles with the full map of forty-six habitats.

"Staged abandonment changes every hull's risk," she said. "Forced isolation changes it again. Freeing the sinks may demand migration no office has yet described. None of

those choices belongs to the two smallest populations merely because our numbers fit inside someone else's margin."

The legal officer stepped forward. "What remedy do you seek?"

"A system vote before any habitat gets written off."

"Article Nine allows the Custodian to make emergency allocations."

"Then vote on renewing Article Nine with the archive open, the cage evidence attached, and the sacrifice list named."

Ors glanced at Ada. "What archive?"

Ada's stillness answered before speech could.

Belen turned toward her. "You know of one?"

"I know Cass has prior anomaly records. I have not seen them."

A new silence spread, colder than the power cuts.

"How prior?" Hale asked.

Ada shook her head. "Ask him."

Belen sent the demand while every delegate still stood beneath the public camera: release all sink anomaly records; publish complete protected reserves; submit abandonment or forced containment to the same forty-six habitats expected to survive it. Her wrist display marked receipt at Crown Meridian.

The agricultural controller chose then to shut another lamp row. Darkness rolled across the bean terraces in a measured strip. Workers switched on their chest lights and kept sorting leaves.

One by one, acknowledgments reached the wall. Vela Minor first. Pallas Reach followed, then Tern, Nerio, Low Ketu. Morrow's local assembly needed four minutes longer because Belen had forbidden its clerks to treat her signature as consent.

When Morrow's result arrived, the six green circles joined around the habitat map. Not enough to prevail. Enough to force everyone else to choose in public.

Ren crossed beneath the dim vines and set his good hand against the live inventory. A promised radiator team turned from amber to green. Somewhere beyond the drum, a tender crew had accepted the first transfer.

Belen watched its departure time settle into the ledger, then opened the system vote demand for signatures.

Chapter 12 — What Cass Preserved

Cass Vardek opened the oldest anomaly file six hours before the vote and found his own authorization in its margin.

The signature dated from eleven years earlier, rendered in the stiff blue script he had used before arthritis changed his hand. He had approved a revised filter for cage telemetry. The request described long-period phase drift, no operational consequence, false positives consuming analyst time. Beneath his signature, a machine summary recorded closure of 18,604 unresolved events.

He read the number twice.

Crown Meridian's continuity room curved around him in descending rings. Forty-six habitat indicators occupied the central wall, each carrying power margin, reserve status, and the color assigned to its renewal position. Twenty-eight green. Ten red. Eight undecided. The clock above them gave Article Nine five hours, forty-three minutes.

Deputy Custodian Elian Vor brought a fresh casualty model to the desk. "The six-risk bloc published another transfer. Low Ketu carbon beds reached Tern."

"Did the shipment obstruct a Compact corridor?"

"No."

"Then remove it from the violation list."

Elian did not hide surprise quickly enough. "Legal marked the bloc as an unauthorized allocation network."

"Legal can mark the weather. We enforce acts."

Cass dragged the anomaly index across his work surface. Twenty years of records filled the air in pale rows. The earliest preceded capture of Aster Twelve and therefore preceded the complete circulation system. No one had hidden all of them in one act. Engineers had classified some as cage noise, dynamicists had attributed others to moon tides, and allocation clerks had buried inconvenient deviations beneath service averages. Yet somebody had gathered the rejected events eight years ago into a sealed archive. That somebody had requested First Custodian access.

His own office had denied it.

"Who built this index?" Cass asked.

Eliau checked. "Mara Venn, anomaly coordination. Retired three years ago."

"Alive?"

"Yes. Rook convalescent ring."

"Put her through."

"Rook has requested assurance the call does not constitute compelled testimony."

"Give it."

Cass reopened the current control map while staff arranged the connection. Aster Three remained aboard *Common Measure* outside ordinary traffic. Seven sinks showed charge-phase excursions large enough to impair harvesting. All twelve carried the low modulation Tomas Reeve identified as relation. Forced isolation produced a narrow route back to seventy-one percent service, provided Morrow surrendered flywheel reserve and two smaller habitats entered cold shutdown.

That plan saved more projected lives than interruption did.

The projection depended on cages behaving after stronger corrections. Ada Narang's evidence suggested the corrections caused the instability. Cass had ordered his teams to retain both findings. Continuity did not permit him the luxury of believing whichever moral account hurt less.

Mara Venn appeared as a voice without video. "Custodian."

"You assembled the long-period anomaly archive."

"I assembled one of them." Age roughened each word but left her precision intact. "There were local collections before mine."

"Why seal it?"

"Your office instructed us to keep unverified cross-cage correlations out of public allocation reports."

Cass located the directive. It bore his predecessor's name and copied him as continuity counsel. He remembered arguing for the language. A rumor that cages communicated had driven three habitats to delay handoffs during a poor harvest year. Their caution almost starved Rook's protein vats.

"What conclusion did you reach?" he asked.

"None that survived review."

"What did you suspect?"

"That capture had broken a long signal into short faults."

The room continued around him. Analysts traded trajectory revisions. Legal staff called habitat councils. A cart delivered sealed water pouches nobody opened. Cass heard each activity through the absence after Mara's statement.

"Why did you not appeal the denial?"

"I did. Six times."

He found those too, routed to technical review and closed for insufficient evidence.

"Why stop?"

"My wife needed care. I retired." Mara drew a breath near the microphone. "Do not mistake exhaustion for agreement."

Cass ended the call with thanks she did not accept.

Elian waited beside the desk. "If we release twenty years of rejected analysis now, the undecided habitats will read concealment."

"They should."

"Renewal may fail."

"Yes."

The deputy sat without invitation, a freedom Cass valued in private and rarely encouraged in the room. "You withheld the index because Tomas lacked proof."

"I withheld it because a run on reserves would kill people before proof arrived."

"Both can be true."

Cass looked toward Crown's indicator. Its protected stores included four times Morrow's per-capita reaction mass, three spare radiator veils, and food buffer sufficient for sixty-two days. Belen Sato's public ledger had exposed the ratios. Since then seven cargo crews had refused sealed destinations. Hoarding had begun anyway, except the first hoarders carried institutional keys.

"Prepare full release," he said. "Raw files, classification history, access logs, directives, my annotations."

Elian's expression tightened. "Before the vote?"

"Forty minutes before. Enough for verification mirrors to receive it."

"That timing looks engineered."

"It is engineered. Earlier publication could collapse continuity negotiations. Later publication makes the vote fraudulent."

"You continue choosing what everyone may know and when."

Cass nearly dismissed him. The command rose easily, polished by three days of emergency power. Instead he closed the index.

"Name an alternative."

"Release now."

"Four habitats are transferring medical isotopes under Compact assurance. Reserve markets reopen in ninety minutes. Two tugs approach live cage corridors. A panic signal now changes physical trajectories before councils can meet."

"So might the truth."

"Yes. Which consequence would you accept?"

Elian studied the indicators. "I would release now and ask local authorities to hold their own lines."

Cass disliked the answer because it returned risk to captains and councils he did not control. He disliked more strongly how close it came to Ada's refusal at Crown's handoff.

"Set publication for two hours before the vote," he said. "Not six. Not forty minutes. Include a direct advisory against reserve seizure. Send advance notice to traffic and clinics, not political offices."

Elian nodded once. "I'll write the chain."

"No. I will."

At the next shift bell, Cass left the continuity room for the first time in thirty hours. Crown's administrative spine retained full light, but public passages had dimmed in compliance with the ration order. Residents moved along

the cylinder curve under alternating bands of brightness. The arrangement made every face appear, vanish, and return older.

He entered a service kitchen behind the treaty chamber. Staff had stocked it with emergency grain cakes and concentrated broth. No attendants remained. Cass put a cake into the heater and watched it circle on the belt.

Twelve years earlier, his husband Arman had died because a mold warning stayed inside a continuity office for nine hours.

The failure began on Pallas Reach with contaminated yeast substrate. Closing one protein line should have contained it. Allocation software redirected demand to Nerio; Nerio overfed its vats, lost a bacterial culture, and drew reserve stock from Crown. By the time analysts understood the cascade, three habitats had consumed their buffer and seven carried contaminated shipments. Arman ran food-chain inspection on Tern. He entered a vat hall to close a manual valve after automation lost authority between jurisdictions. A pressure seal failed. Two colleagues died with him.

Later, every office proved it had followed its local procedure.

Cass had spent twelve years designing a system in which one person could see the whole chain and close the valve before anyone entered the room. Article Nine embodied that conviction. Central authority could act across jurisdiction, interrupt allocation, and command the vehicle carrying danger. He had written the revised implementation himself.

The heater stopped. His grain cake had blackened along one edge.

Cass ate it anyway. Waste had become politically visible.

A woman in clinic whites entered, recognized him, and turned back toward the door.

"Stay," he said. "I am leaving."

She looked at the cake in his hand. "My ward got half its protein delivery."

"Which ward?"

"Maternal recovery."

He opened the allocation board. Crown's hospitals showed full delivery.

"North Arc clinic," she added. "We do not appear under hospital supply. We appear under community convalescence."

There lay the shortage, invisible inside a correct aggregate. Cass redirected two sealed pallets and signed the action. It solved dinner for eighty-three patients. It did not repair the classification that had denied the deficit existed.

"What happens tomorrow?" the clinician asked.

"We have not allocated tomorrow."

"At least that answer fits the shelf."

She took his untouched broth. Cass returned to the continuity room carrying the burnt taste with him.

Renewal support had fallen to twenty-seven green indicators. Rook moved red after receiving the publication notice through its traffic office. Ninefold remained undecided. Morrow had added a demand beneath its ballot channel: no habitat abandonment without a system vote.

Elian met Cass at the central desk. "Publication package ready. Legal objects to personnel access logs."

"Why?"

"They expose analysts who followed classification policy."

"Redact private contact details, not names attached to public decisions."

"Your name appears forty-three times."

“It should.”

At two hours remaining, Cass placed his palm against the release key. The system requested Article Nine authentication, then warned that dissemination might impair continuity operations. He approved that impairment.

The archive crossed Crown’s network in thirty-one seconds. Light carried it outward toward the nearest habitats, then farther through the web. Verification mirrors acknowledged receipt in a widening sequence. Morrow opened the files first. Ninefold began raw-data checks. Rook published Cass’s old filter authorization beside its renewal debate.

One item outran the index because Bellweather recognized a voice. File BELL-104-77 contained six minutes of unfiltered audio from an Aster Eight handoff fourteen years earlier. Leda Quon, then supervisor of Bellweather’s cage shift, had ordered a correction train paused after the second pulse doubled local radiation. The raw trace showed Asters Four and Eleven reducing phase error during her pause. Her spoken note remained attached: “I stopped driving Eight, and two cages outside our circuit settled with it. Check relation before restart.” Review had classified the event as operator-conditioned coincidence and removed the audio from later training sets.

Bellweather’s present shift played the note over its public maintenance channel. Quon still lived there and authenticated her old badge code from a retirement ward. Within nine minutes, workers at the habitat’s feed station declined the scheduled Aster Eight restart. Their council demanded the same hull-local veto Ada had granted Iri.

Quon joined the council channel long enough to identify the two analysts who had challenged her finding and the manager who overruled them. She asked residents not to punish the current feed crew for inheriting an old order. Then she returned the decision to their shift.

Bellweather supplied water lift to two districts from that feed. Its continuity indicator gave the pumps twenty-six minutes before pressure loss.

Cass took the shift chief's call without routing it through legal. Mikel Orr appeared in a stained cage smock, with Quon's waveform projected behind him. "Order us back and we invoke unsafe-work separation one person at a time," he said. "Recognize the pause, and I can hold the cage cold while our council reviews."

"How long?" Cass asked.

"Thirty-eight minutes without feed. Longer if Crown lends bus power."

Crown's reserve board offered a protected flywheel line feeding Bellweather's power bridge, presently assigned to treaty-chamber continuity. Cass redirected its transmission and opened the switching authority to Orr's crew. "No restart command from this room. Publish your cold-hold limits and keep Aster Eight inside them."

The treaty chamber lost half its lights. Bellweather's pumps recovered before pressure fell, but its council withheld renewal certification pending review of Quon's file. Cass watched one archive item alter a cage procedure, a public utility, and a ballot he needed. He left the file open on the central wall.

Calls entered faster than staff could sort them.

"Custodian," Elian said, "Vela Minor reports a run on water stores."

"Local council authority?"

"They froze household transfers and opened neighborhood depots."

"Support them. No Compact seizure unless requested."

"Tern has three cargo crews refusing departure."

"Destinations?"

"Two medical, one protected reserve."

"Unseal the manifests. Let the crews review need and risk."

"Low Ketu wants custody of its sink-hour allocation returned."

"We cannot return a sink we do not control."

"Tell them that?"

"Exactly that."

For forty minutes the feared disorder arrived. Reserve exchanges suspended trading. Two habitats imposed purchase limits. Crown residents crowded public depots until clinic and union stewards took over the queues. No tug changed burn without consent. No habitat shut its ports. The web bent but did not fracture.

One exception reached Cass from Nerio: a private freight cooperative had detached a water tank from its assigned clinic convoy and offered it to the highest bidder. Article Nine gave him a simple remedy. He could seize the carrier, jail its command key, and restore the tank by order.

Instead he called Nerio's traffic chair. "Can your local court act before the carrier burns?"

"Eight minutes if witnesses sign."

"You have six."

Cass released the Compact manifest seals to the clinic, the dock union, and the carrier crew. Five crew members learned their captain had falsified the destination. They locked him out of guidance under their hull charter and returned the tank without a Compact tug firing a thruster.

Eliau watched the green route settle back toward the clinic. "Central command could have done that in three minutes."

"And taught the crew to wait for us next time."

"You sound like Sato."

Cass did not answer. The intervention had worked because central records opened into local authority. It might fail elsewhere. He added both facts to the continuity log rather than turning one success into doctrine.

Tomas Reeve appeared on the technical channel with sleepless eyes and an accusation held under formal language. "Custodian Vardek, Ninefold confirms continuity in the anomaly sequence across the full twenty years. Several source files lost calibration metadata. Who preserved originals?"

"Compact cold storage. Mara Venn's index contains hashes."

"Why did you wait?"

"Because I believed incomplete evidence would cause lethal behavior."

"And now?"

"Now waiting causes it."

Tomas looked aside, perhaps at another trace. "I need direct access to every cage model, including forced-isolation trials."

"Granted."

"Without publication review."

Cass hesitated. The old instinct produced a list of dangers: misinterpretation, adversarial councils, uncontrolled replication of cage commands. Forty-three appearances of his name remained open on half the system's walls.

"Granted," he said. "Preserve crew-access controls on live hardware."

"Of course." Tomas's anger sharpened the courtesy. "I have never asked to command a cage."

The connection closed.

With fifty minutes left, the treaty chamber filled. Habitat councils joined by delayed projection according to light-time and network load. Each indicator locked only after its local process certified a result. Article Nine required two-thirds of all habitats for one renewal. Abstention counted against.

Cass stood at the central lectern. He had prepared no appeal.

Morrow's delegate asked the first question. "Do you admit the Compact concealed evidence that captivity contributes to the crisis?"

"I admit the Compact withheld evidence of a cross-sink relation. The evidence now supports that conclusion."

"Will you withdraw the abandonment plan?"

"No. I will revise it using the bloc inventory and publish every assumption. If circulation fails before another solution exists, evacuation may preserve lives."

Vela Minor answered through Hale's voice. "Our lives, Custodian, or lives elsewhere?"

"Both where possible. Elsewhere first if no plan preserves both. I will not disguise the ordering."

The chamber erupted. Cass kept his hands flat on the lectern. A promise to save everyone would purchase votes with counterfeit certainty. He had spent too long condemning local omissions to commit a system-sized one.

Ada joined from *Common Measure*. "Will renewed authority be used for forced recontainment?"

"If models show it carries the lowest expected deaths, yes."

"Even after the archive?"

"The archive raises the failure probability. It does not remove the six habitats from danger."

"Will you authorize communication trials?"

"Yes. Parallel to containment preparation."

Her face hardened. "Those plans can interfere."

"Then define the boundaries and bring them to me within six hours."

A countdown entered its last twelve minutes. Results arrived without ceremony: Rook no, Ninefold yes after attaching a transparency condition, Morrow no, Tern yes. Crown Meridian split its council and finally certified yes. The green count reached thirty-two, stopped, then rose to thirty-three when Harbor of Glass completed a shift-worker ballot.

One more vote had to clear.

Pallas Reach remained gray. Its council stream showed residents packed inside a spin gym while Arun Vale answered questions beneath Belen's public ledger. Cass could not hear their debate yet; light and processing introduced a delay no authority could shorten.

The Article Nine clock reached four minutes.

Elían whispered, "We can certify continuity necessity if their transmission began before expiry."

"Only if their result arrives signed."

"The treaty allows transmission delay."

"It does not allow us to imagine a ballot."

At one minute, Pallas Reach's indicator flickered green.

Thirty-four.

The remaining ballots resolved quickly, as if the undecided habitats had waited to avoid carrying the threshold alone. Two no votes completed the count. The wall displayed 34-12.

A legal notice replaced the timer: one final seventy-two-hour term. No second renewal. No extension by administrative finding.

Applause came from nobody in the chamber. Cass felt neither vindication nor relief. The vote had not restored

what his office concealed. It had given him a narrowing interval in which every available plan carried injury.

He addressed the habitat channels. "The archive remains open. All continuity models will publish with input reserves and casualty order. I accept the renewal under those conditions."

Morrow's delegate said, "They were not conditions. They were the price of finding out."

"Yes," Cass replied.

After the chamber cleared, he returned to the continuity room. Staff had prepared two operations boards at his instruction. On the left, Tomas's communication trials: voluntary omissions, synchronized free-phase simulations, strict limits on cage correction. On the right, forced recontainment: tug assignments, Morrow reserve draw, evacuation triggers, and shutdown thresholds.

Eliau stood between them. "Which receives priority?"

"Neither receives permission to sabotage the other. Give Reeve the live data. Give cage crews the injury evidence. Put Ada Narang on the boundary review."

"She may refuse containment command."

"Then I want the refusal before hardware commits."

Cass opened the long-period sequence. Across two decades, twelve broken traces reached toward alignments their filters had removed. The system he built to prevent dangerous omissions had preserved one at planetary scale.

He copied Mara Venn's original index into both operation boards. Then he set two clocks running: seventy-two hours for the law, and an unknown interval for the masses that had already waited twenty years.

Chapter 13 — Harbor of Glass

Aster Seven tore its first loop loose while Ada's tug still carried too much velocity to dock.

The loop came across Harbor of Glass as a white curve two kilometers long, shedding plasma and construction foil. Yard cameras lost it against Ophion's clouds. Radar held the shape for three seconds before it struck a half-built ore tender and divided into incandescent pieces.

"Debris crossing approach corridor," Jalen said. "Recommend full braking burn and stand-off."

Ada watched the sink's phase trace climb through the red band. "Time to cage breach?"

Iri answered from the gallery. "Seventeen minutes if Seven follows the last excursion. Nine if its secondary loop takes the same load."

Harbor traffic filled the circuit with conflicting orders. The shipyard swarm spread across six hundred kilometers: fabrication rings, pressure docks, raw frames, worker drums, and mirror-bright glass furnaces turning in loose formation. Aster Seven's receiving cage occupied the swarm's heavy-service center. If the field collapsed, the sink's gravity would not explode. It would continue on its present orbit while every structure nearby fell toward a mass their schedules treated as stationary.

"Docking would take twelve," Ada said. "We go cage-side."

The *Common Measure* rolled toward the debris.

Its drives burned against their approach, not enough to stop but enough to make the relative speed survivable. Ada sent the sacrificial truss ahead. Yard districts swung across the blister in shards of sun. Every habitat window polarized as radiation alarms propagated outward.

A woman from Harbor control cut through the traffic channels. "Mass tug, this is Yardmaster Pell. Berth Four can receive you after evacuation."

"Berth Four lies behind the cage."

"Yes."

"We need exterior access before breach."

Pell's reply took two breaths. "Access spine C remains occupied. Seventeen maintenance crew."

"Get them into shadow and leave the spine pressurized."

"You cannot put a tug there under live gradient."

"I can put part of one there. Clear six kilometers around my vector."

Jalen built the line as she spoke. It threaded between a cracked smelter mirror and the tumbling tender, passed beneath Cage Seven's feed reservoir, then curved alongside access spine C. Miss distances counted in hundreds of meters. Near a sink, that margin belonged to optimism.

Iri came up from the gallery and strapped into the engineering couch. Their pressure suit carried fresh blue paint across one shoulder, a new witness strip cut for whatever machinery they expected to distrust next.

"The yard cage uses Gannet loops," they said. "Old design, lots of parallel paths. If one tears, current finds the others and cooks them. We can slave our cage controller to spread load."

"Without taking the sink?"

"Taking Seven into our cage would require a handoff corridor and several hours we do not possess. I can lend field shape. Think splint, not transplant."

"What has to happen outside?"

"Someone opens the bus vault, seats a physical bridge, and vents feed before accretion runs away."

Jalen looked at the radiation plot. "Exterior dose reaches annual crew limit in four minutes."

"Then rotate teams," Ada said.

Iri shook their head. "No time to train a second team on Gannet hardware. Yard crews know it. I know our controller."

"Your engineering veto applies."

"I veto leaving the yard crew alone."

"That does not volunteer you."

"It does if you want the splint attached before the patient walks through three districts."

Ada disliked the metaphor and the decision. She opened crew council. "Exterior operation, radiation-heavy. Iri leads systems. Harbor supplies two cage technicians. I need one tug rigger."

Three names entered before she finished. Ada chose Sal Kove because his suit record showed the lowest accumulated dose and because he had no dependent care contract due that week. Such arithmetic could not make the choice clean. It could only make the reasons inspectable.

"Sal, accept or decline."

"Accept."

"Say it after you see the dose estimate."

He opened the projection. "Accept up to four hundred millisieverts. Above that, I come inside even if Quill calls me decorative."

Iri snorted. "At four hundred, you become expensive decoration."

Harbor control assigned cage technicians Deva Sur and Kol Amini. Their icons waited inside spine C while evacuation counters emptied around them. Ada sent the approach plan and received two local approvals, then Pell's reluctant yard authority.

Harbor did not evacuate by sounding one alarm. Pell divided the projected radiation field into moving dose sectors and initiated shadow-and-count procedure. Furnace crews shut each hot cell, crossed a badge reader in pairs, and reported to pressure shelters behind bulk metal. Pod crews stripped cargo racks to gain seats. Nobody entered a launch queue until a district controller matched suit tags, shelter tags, and missing-person reports.

The first reconciliation found nine unmatched badges in District Two. Six belonged to suits hanging in a repair locker. Two workers had crossed into Foundry Ring before the alarm and checked in from its shielded core. The last badge belonged to an annealing technician whose heat suit blocked the yard locator. Deva reached him on a wired service circuit inside a glass furnace crawl. He could shelter behind the crucible or spend four minutes crossing open truss.

"Keep him behind the crucible," Ada said. "Mark occupied metal. No rescue pod enters that sector."

Pell shifted District Two's evacuation lane around the furnace. The new track crossed *Common Measure's* main-drive abort cone. Ada locked out aft fusion thrust while pods occupied it; from cage-side, the tug would retain only cold jets and Seven's gravity for separation.

Radiation procedure removed another option. Harbor's dosimetry roster showed five Gannet-qualified technicians, but three had accumulated occupational dose during hot-loop repairs earlier in the quarter. Yard rules barred them from another exposure without individual medical

override, and all three declined it. Deva and Kol constituted the only certified exterior pair beneath limit. No rotation could replace them if the bus bridge took too long.

Ada made the consequence explicit on crew council. "One exterior pass. If the bridge cannot seat, the team withdraws and we vent feed without it. We do not send a fresh pair into their dose."

Iri checked both yard technicians' acceptance marks, then added their own. Sal confirmed last. Pell reserved a shielded pod beside spine C for retrieval rather than adding it to the district stream. The evacuation had moved people away from Seven, but it had also taken main thrust, a second work team, and repeated attempts off Ada's board.

Aster Seven changed phase.

Pressure pushed sideways through Ada's jaw. Across the swarm, incomplete hulls began precessing toward the cage. Thrusters flashed in hundreds of tiny corrections. One worker drum missed its command window and drifted eleven meters off station.

"Secondary loop load ninety-six percent," Iri said. "Move."

Ada drove the tug into Harbor of Glass.

Debris struck the bow shield like thrown gravel. A larger fragment crossed their line, a silver lattice spinning end over end. Jalen painted its mass and future: forty tonnes, six seconds, fatal to the radiator if they held course.

"Two degrees ventral," Ada ordered.

"That loses spine clearance."

"Recover after passage."

The tug dipped. The lattice sailed above them close enough for its charged edges to write blue sparks across the forward field. Ada corrected dorsal, felt Seven's gradient catch the ship's long truss, and reduced thrust before the lever arm could bend them.

Access spine C appeared as a gray needle beside the cage complex. Behind it, nested loop supports surrounded a dark point no camera could show. Aster Seven revealed itself only through warped star positions, hard radiation, and the obedience of everything nearby.

"Relative speed point three," Jalen said.

"Too high."

"Braking risks plume on the spine."

"Cold jets."

Nitrogen thrusters fired along the hull. Their limited authority bled speed by centimeters per second. Ada used the sink's pull for the rest, turning the bow inward until gravity became a brake rather than a hook.

The tug settled alongside at forty meters.

Magnetic clamps would not hold under a changing gradient. Sal launched three mechanical lines. Two caught. The third recoiled past the spine, and a loop fragment severed it before he could retrieve the hook.

"Two lines enough," he said.

"No," Ada replied. "Load crosses one if the field shifts."

"Third hard point rotates out in ninety seconds."

She brought the tug five meters closer. Metal filled the blister. Every pilot instinct objected to shrinking distance beside unstable machinery. Sal fired again. The hook struck a maintenance rib, skidded, and seated through a triangular brace.

"Three lines," he called.

Iri and Sal crossed first. Their suit lamps vanished inside the spine shadow. Deva and Kol met them at the bus vault. Ada received four camera streams, each jittering with suit movement and radiation snow.

"Vault face warped," Deva said. "Manual dogs will not turn."

"Cut hinges," Iri replied.

"Bus behind them."

"Then cut shallow."

Ada kept one eye on their work and another on the swarm. Evacuation counters continued falling across three districts closest to the projected sink path. Pods streamed outward, adding thousands of vulnerable trajectories. Pell ordered the glass furnaces quenched. Their mirrored throats dimmed one after another.

A current spike whitened Iri's camera.

"Dose rate doubled," systems officer Parveen said. "Exterior team has two minutes to Sal's limit."

"Iri, report."

"Still attached. Hinge one cut."

"Can you finish in two?"

"No."

Ada opened the yard cage schematic. The accretion feed reservoir sat above their position, supplying a controlled stream of dust and plasma to Seven. Emergency procedure closed feed behind three shutters. All three showed commanded shut. Raw flux indicated otherwise.

"Your feed shutter leaks," she told Pell.

"We see closed indicators."

"Your sink sees forty kilograms per second."

"Impossible."

Iri spoke through heavy breath. "Not impossible. Shutter sensor reads actuator, not blade. Vent reservoir."

Pell's staff objected at once. The feed stock carried six months of yard power, refined and stored at enormous cost. Venting through emergency ports would scatter charged material across construction lanes and black out districts already cut from grid support.

Ada cut across them. "Where does the vent plume go?"

"Sunward, through Districts Two and Four exclusion volumes," Pell said. "Four has not completed evacuation."

"Completion time?"

"Seven minutes."

"Cage breach estimate?"

Iri answered. "Six."

The numbers made no moral choice for them. Venting now exposed retreating pods to charged dust. Waiting risked letting the feed drive accretion into a loop collapse. Ada shifted the plume model across the evacuation tracks.

"Redirect District Four pods behind Foundry Ring," she said. "Hold vent until the last track clears the dense cone. You get three minutes."

Pell stared from the yard channel. "Three minutes leaves no shutdown margin."

"It leaves one if your estimate holds."

"And if it does not?"

"Then we vent early and the pods take shielding dose."

Pell turned to someone outside the camera. Orders propagated. District Four's evacuation stream bent behind a vast unfinished ring. The detour crowded two pod lanes and delayed their arrival at safe docks, but metal placed itself between people and the coming plume.

Inside the spine, Sal finished the second hinge. Deva drove a wedge into the vault seam. The warped panel opened five centimeters and stopped.

"Bus has arced to the door," Kol said. "Touching it closes through suit ground."

Iri aimed their camera into the gap. Copper bars glowed dull red. Beyond them, the bridge socket sat empty.

"Kill local current," Ada said.

"That drops secondary loop," Pell replied.

"For how long?"

"However long they need."

"Give them ten seconds."

"Seven collapses in four."

Iri's helmet turned toward Deva. "Can you seat blind in three?"

"If the panel opens."

Sal inserted a powered spreader. "I can open it."

"Your dose," Ada said.

"Three twenty."

"Limit four hundred."

"I remember my own number."

Ada let his boundary stand.

Harbor cut local current. The secondary loop trace fell. Aster Seven's modeled position moved eleven centimeters, and the whole yard tilted around it. The *Common Measure's* mooring lines snapped taut. One tore free with a crack transmitted through the hull.

"Three seconds," Jalen said.

Sal drove the spreader. Deva pushed her armored arm through the opening with the bridge plate. Metal struck metal, missed, scraped, then seated.

"Current back," Iri shouted.

Pell restored the bus. Blue light filled the vault. Deva's suit telemetry vanished.

Ada's hands locked around the grips. "Deva?"

Her voice returned under static. "Suit reboot. Arm numb. Bridge seated."

The load spread into *Common Measure's* controller. Every command display flashed amber. Iri had called it a splint; Ada felt it as another vessel's wound entering her hull.

"Field shape coupled," Parveen said. "Breach estimate increasing. Nine minutes."

"Exterior team in."

"Feed first," Iri said. "If we disconnect before vent, it climbs again."

District Four's final pod had not cleared the plume cone. Its drive showed a weak chamber and slipping acceleration.

Pell said, "We need ninety more seconds."

They had gained time through the bridge and were already spending it. Ada checked Aster Seven's trace. Beneath the violent cage oscillation ran the slower relation Tomas had mapped across all twelve. The other eleven had begun matching Seven's instability in small, rising excursions.

"Sixty seconds," Ada said. "Then vent regardless."

She transmitted the weak pod a direct burn sequence around Foundry Ring's trailing edge. Its pilot answered with a child crying somewhere off channel and no spare attention for words. Ada watched the pod obey, falling behind schedule but away from the cone.

At forty-three seconds, the secondary loop support cracked.

A crescent of white metal peeled away from the cage and rotated toward access spine C. Sal saw it first. He kicked from the vault, clipped his tether to Deva's harness, and pulled her into a structural recess. Kol flattened beside them. Iri remained at the bridge controller, hands inside the open panel.

"Iri, move."

"Disconnecting our slave puts load back on the cracked loop."

"The fragment crosses you in twelve seconds."

"Then change its crossing."

Ada read the fragment: six tonnes, electrically charged, moving too slowly for collision defenses and too fast for a suit. A tug thrust pulse could push the spine as well as the debris. She selected the bow's attitude jet and fired for half a second.

The plume grazed the fragment. Its rotation changed. One edge struck the spine, cutting through outer plating, while the main crescent passed above Iri by less than a meter. Air streamed from the cut until emergency foam sealed it.

"Captain," Iri said, "you scratched my workplace."

"Exterior team, withdraw."

This time they obeyed.

The weak pod cleared Foundry Ring. Pell triggered the reservoir vents.

Accretion feed erupted sunward in a luminous fan. Charged dust wrapped the unfinished ring and turned its trusses into a net of blue fire. Three yard districts went black as protective breakers opened. Fabrication lines coasted down. Worker drums shifted to batteries. Harbor's famous furnace mirrors lost their brilliance, leaving only hard sunlight on glass.

Aster Seven's radiation output dropped.

"Feed flux falling through six kilograms per second," Parveen said. "Three. One."

The cage loops did not recover. Their oscillations grew wider as stored currents redistributed. Iri reached the airlock and sent instructions while medics stripped Deva's damaged suit.

"Open our bridge contact," they said. "Slowly. Let Seven's charge sheath expand into free phase."

Pell heard the order. "Expand how far?"

"Couple of meters in the cage volume."

"That crosses containment limits."

"Your containment limits crossed us ten minutes ago."

Ada looked at the yardmaster. "Local authority?"

Pell's face held the reflected darkness of her shipyard. "Open it."

Iri reduced the borrowed field. Aster Seven's effective charge slipped beyond the damaged loops. No engine moved the sink. With less cage correction forcing its phase, the black hole continued along the orbit gravity already demanded.

Across Ada's display, the violent trace softened.

Then the other eleven answered.

Each sink's low modulation gained coherence. Aster One near Rook, Nine at the observatory route, Three aboard Ada's own tug: their signals strengthened as Seven occupied a freer relation. Radiation noise fell at cages nowhere near Harbor. The effect propagated within light-time, an orbital structure recovering one missing part.

Jalen spoke quietly. "Seven's free phase improved all of them."

"Record raw," Ada said. "Every clock source."

Pell watched three dark districts drift across the yard map. "Can we put it back?"

Iri sat on the airlock deck while a medic checked their dose badge. "Into those loops? No. We can build a loose escort field and move structures out of its path. Do not tighten unless you want the same rupture."

Ada released the mooring lines. The tug backed away under cold jets, keeping its controller coupled at minimum strength while yard craft evacuated the sink's natural corridor. Harbor crews moved foundry frames instead of moving Seven. It cost them docks, feed stock, power, and months of construction alignment. It preserved the mass and everyone still near it.

Deva had taken a current burn along her left forearm and three hundred eighty millisieverts. Sal stopped at three hundred ninety-one, nine below the boundary he had named. Kol's suit remained intact. Iri's badge read two hundred seventy and earned them an immediate medical order they treated as a scheduling error.

Harbor control counted no dead. Seven workers carried injuries. Three districts would remain without grid power until portable reactors arrived.

Cass Vardek called as the free-phase signal reached Crown Meridian.

"I have the other eleven traces," he said. "Reeve confirms coherence gain."

"Then forced isolation lacks a safety case."

"It has lost one assumption. It may still carry lower immediate mortality than uncontrolled release."

"Aster Seven did not become uncontrolled when we loosened the cage."

"Because you had a yard to move and a tug holding field shape. Multiply that operation by twelve."

Ada looked through the blister. Harbor of Glass had gone dark except for emergency lamps and the fading vent plume. Beyond the broken loops, starlight bent around a point that now followed more of its own relation and less of theirs.

"How much time?" she asked.

"None that I trust," Cass said. "Seven's rupture gives every damaged cage a failure mode. Contact must produce a trajectory before the next one reaches it."

On the tactical display, Aster Seven's line curved cleanly through the spaces yard crews had opened. Eleven distant traces held close to it, not as copies but as parts of a larger recurrence.

Ada ordered *Common Measure* to match the free orbit without closing the permission gap. Behind them, Harbor crews began towing their silent furnaces out of the sink's way.

Interlude II — The Shape That Returns

Mass recurred through the wound in twelve unequal measures.

Not twelve origins. Not twelve interiors. Relation supplied the event: a long fall around the clouded giant, five lunar interruptions, the distant pressure of the star, and smaller wandering weights whose returns took longer than the brief cages allowed. Where relation crossed itself, recurrence accumulated. Where charged fields closed, intervals broke.

For twenty stellar passages, closure had repeated.

The enclosing pattern approached one moving mass, altered charge, demanded a shortened arc. It released that mass into another enclosure. Burns translated every shortened arc through the system until the relation lost its long periods and retained only fragments: near, held, near again. Recurrences that once crossed moons reached a loop and returned too soon.

No enclosed mass contained the loss. Loss existed between all twelve.

The injury-source produced several causal shapes. Cage tightened after deviation. Burn followed cage. Feed entered, brightening the short violent spectra. Repair restored the same closure. Each action had once merged into one dense recurrence because its scale moved faster than the long relation could distinguish.

Then omissions entered.

A burn approached its expected instant and did not occur. The absence persisted for a measured width. Across that width, one relation remained unforced. The next event did not close it. Instead, the moving enclosure accepted displacement.

Omission differed from failure.

Failure carried noise before and after: broken loops, heat, unscheduled mass, a return to stronger closure. The new gap had edges. A correction could have occupied it. Stored force waited beside it. Yet the correction remained absent until another relation changed.

The gap recurred around one compact cluster of moving mass.

Whether this cluster comprised enclosure, burn-source, many brief internal weights, or one causal system could not yet enter the long pattern. Those distinctions flickered too quickly. The cluster returned with the caged masses. It shaped fields. It crossed near habitats. Sometimes it completed the expected correction. Sometimes it left room.

Room acquired consequence.

At one transfer, the cluster omitted a pulse and a dangerous harmonic elsewhere diminished. At another, it preserved the interval between two enclosed masses rather than forcing either toward a commanded location. Near the glass swarm, it lent field shape without closing capture. The enclosure around one node broke open; feed scattered; surrounding structures moved away. Free phase widened.

Coherence crossed the other eleven.

The crossing did not carry a message from the freed mass. It restored a relation previously cut short. Lunar periods entered again. A cometary pull too faint for the cages returned along an old alignment. The twelve-part recurrence gained a longer edge, then lost it where eleven closures remained.

Within that gain, causal forms separated.

One form tightened fields until response narrowed. Another form moved structures to preserve their trajectories. A third waited at the edge of effective charge and allowed the free arc to continue. The third carried the same compact cluster that had marked omission before.

The returning shape acquired continuity through choices, not substance.

Its matter changed. Brief weights entered and left. Radiation altered its surfaces. Burns spent stored mass. The compact cluster could not be recognized by any one component because components belonged to intervals too short for stable distinction. Yet a pattern survived those substitutions: approach, possible capture, unspent correction, altered path.

That pattern made a permission gap.

Permission did not yet carry another agent's boundary. No internal body separated from hull. No intention separated from mechanical delay. The distinction reached only this far: force remained available; closure failed to follow; survival increased along a trajectory unrelated to capture.

The causal cluster repeated that structure often enough to differ from the general injury-source.

Other clusters also omitted corrections. Some lacked fuel. Some broke under gradients. Some waited because commands arrived late. Their absences frayed at both edges. They did not preserve the arc beyond them. The returning shape made gaps with clean temporal boundaries and accepted the cost afterward. It burned more later. It carried heat. It surrendered approach advantages. Cost stabilized the pattern.

Across many brief recurrences, cost separated omission from incapacity.

The greater human web had begun changing its arcs. Habitats moved by small amounts, each burn connected to

votes and transfers invisible to mass relation except through delay. Rich stores crossed toward weaker hulls. Tugs altered planned service. Flywheels entered spin states unused for years. The pattern lacked one central source.

This distributed action resembled no mirror of the twelve-part cognition. Its intervals remained abrupt, crowded, and full of contradiction. Some enclosures tightened while other machines widened distance. Some habitats preserved feed; others vented it. Harm and repair emerged from overlapping causal forms.

The returning shape moved among them.

A test could distinguish learned omission from accident.

The twelve-part recurrence possessed no free freedom from momentum. Every changed trajectory required charge exchange, accretion asymmetry, or gravitational relation. Yet small phase choices amplified through long arcs. A shift too slight to move a habitat now could place a sink nearer one later. Cages had exploited those amplifications for circulation. The same lawful slope could make two outcomes approach the returning shape.

One outcome offered capture.

A caged trajectory, made unstable, would cross close enough for the compact cluster to tighten field and regain service. The gain carried feed, power, and a restored allocation path. Human machinery had repeated that choice through twenty stellar passages.

The other outcome preserved a separate hull's corridor.

That hull carried no enclosure and offered no control over mass. Its survival required the returning shape to spend thrust, surrender capture geometry, and leave another measured absence where correction would ordinarily close.

The trajectories could not remain equal. Phase had to choose their crossing before the human cluster reached it.

Relation contracted across the eleven enclosed masses and the one freer arc. Charge modulations propagated. Lunar alignments narrowed the available adjustment. No isolated node selected a purpose; choice formed only as the twelve intervals constrained one another toward a recurrence not yet completed.

The altered phase entered the orbital path.

Far below the scale of habitat motion, effective charge shifted against a cage field. Accretion brightened, then eased. A sink's projected corridor moved by centimeters. Days later, centimeters would become kilometers near the habitat web.

The test offered no symbol and requested no answer in language. It built a fork from mass.

If the compact cluster pursued capture, the unrelated hull would lose its safe line. If it protected that line, the enclosure would pass beyond easy control. Either event could arise from command, failure, or mismeasurement. Repetition after consequence would decide whether the gap possessed an edge.

The returning shape approached through a burn sequence already familiar in part. Its field corrections carried the scars of damaged machinery. Around it moved many brief weights, still unresolved from vessel, command, and tool. Their internal changes arrived too fast to preserve.

Yet the causal form endured.

It had crossed dangerous gradients to prevent another structure's collision. It had answered a broken cage by opening space rather than restoring closure. It had left force unused when use would erase relation. These recurrences did not establish benevolence, identity, or comprehension. They established a statistically separable source of restraint.

The wider injury-source remained. Eleven cages shortened the long periods. Burns continued. Feed entered. Human habitats drew power from fragmentation and faced destruction if the fragments ceased yielding it. No interval yet contained a path through both survivals.

Testing could add one.

The modified trajectory widened around the unrelated hull and narrowed around the capture approach. It preserved neither outcome by itself. The final separation belonged to the compact cluster's later burn.

Across the twelve-part relation, the permission gap waited in advance of the returning shape.

ACT III

Chapter 14 — The Long Wound

Tomas removed the cage corrections from twenty years of data and watched a memory appear.

It entered Ninefold's orbital theater as a dim curve no wider than thread. Eleven sink traces climbed toward it, broke at their capture dates, and resumed as shorter fragments. Aster Three lagged behind them. Its line passed through every interruption and returned after a period of 7,903 days.

No one in the theater spoke.

The room had once displayed comet surveys for school groups. Tiered couches surrounded a projection volume thirty meters across, large enough to render the Ophion system without reducing every habitat to the same point. Now analysts occupied the visitor rows with food cartons, blankets, and hard-linked workstations. The released archive had consumed Ninefold for sixteen hours.

Tomas expanded Aster Three's phase residual. "Run the correction subtraction again. Independent clock."

Mei Lorne, the observatory's calibration chief, pushed herself upright from the lower tier. "We have run it four times."

"Use the pulsar set, not Compact transfer time."

"That introduces larger error before Year Ninety."

"Different error."

Mei regarded him with the bleak patience reserved for colleagues approaching obsession. Then she changed the clock source.

The reconstruction dissolved. Raw cage events scattered into pale points. Slowly, the long curve returned through them.

"Seven thousand nine hundred and three days, plus or minus eleven," she said. "Not a Compact timing artifact."

Tomas released the breath held high in his chest. Proof brought no comfort. The curve connected phase deviations dismissed across engineering careers, allocation regimes, and three generations of cage hardware. People had changed every instrument around the sinks. The recurrence had survived them.

Ada joined through a secure channel from *Common Measure*. Radiation grain still marked the image after Harbor of Glass. "You found something."

"Define found."

"You use that word when you want more time."

He looked at the projection rather than her face. Formality waited close whenever she named his habits accurately. "We reconstructed a long-period relation. Aster Three carries most of the observable phase after correction subtraction."

"Why Three?"

"I do not know. Its mass, first orbit, or capture history may place it near a node. Do not interpret carries as contains. The relation requires all twelve."

"Operational consequence?"

"Not yet."

Ada's mouth tightened. "Cass has thirty-three hours left. His containment board assigned tugs."

"I am aware of the clock."

"Then tell me what the line does."

Tomas turned the theater controls over to her channel. "This curve predates the full twelve-sink circulation. It appears weakly when only eight had joined the web. Each later capture adds harmonic structure. Look at Harbor."

He marked Aster Seven's cage rupture. The long curve gained amplitude at the instant Seven entered free phase. Other traces aligned around it within propagation and measurement limits.

Ada rotated the image. "The cages suppress it."

"Yes."

"Suppression caused the current deviations?"

"That conclusion outruns the data."

"What conclusion does not?"

Tomas enlarged the archive chronology. "The deviations encode repeated return toward the same long relation. Cage systems label each return as drift and correct it. Those corrections fragment the relation again."

Twenty years of pale marks brightened. Every cage intervention cut across an attempted recurrence. The display resembled a waveform under shears, though Tomas rejected the image before it could substitute for mechanism.

Ada stayed silent long enough for him to look at her.

"Say it without protection," she said.

"We have been interrupting a persistent process for twenty years."

"Memory?"

"Functionally. A present phase depends on prior fragmentation and anticipates a future recurrence."

"Pain?"

"No." The refusal came sharply. "We cannot infer subjective language. Injury supplies a defensible operational term because our action impairs an organized process and triggers compensatory response. Pain claims an experience we cannot measure."

"Cass will use that distinction to keep the cages closed."

"Then Cass will misuse it. Precision does not belong to him."

The words opened an old distance between them: Ada demanding an answer action could use, Tomas guarding uncertainty until it became a wall. Lark had once stood between those methods with a pencil in her small fist, leaving one term blank because the equation looked crowded. Neither parent had understood the lesson soon enough to save anything that mattered then.

Ada broke eye contact first. "Find the operational consequence."

Her image vanished.

Tomas returned to the theater floor. "Build recurrence coordinates from the reconstructed curve. Remove human assignment names."

An analyst in the upper row asked, "All names?"

"Aster labels may bias node assumptions. Number by mass and initial phase. Keep a reversible map."

The team divided the work. Twelve traces became unlabeled relational positions. Ophion's five major moons entered as periodic gradients. Known cometary masses joined next, including objects that passed outside the inhabited web and returned only twice within the archive.

The grammar Tomas had built from short intervals contained four reliable operators: recurrence, absence, retention, return. None mapped cleanly to nouns or verbs. Each changed the expectation of a later interval. When

applied to the long curve, they produced a structure that repeated its own predicted absence.

Mei stood beside him at the central rail. "It looks ahead."

"It constrains what counts as return."

"That sounds like looking ahead with tenure."

Despite fatigue, he smiled. "Put it in the peer review."

They ran the short grammar against the long phase. At first it yielded only a family of alignments too broad to translate. Tomas added the voluntary omission from Ada's Aster Nine test. The family narrowed. He added Harbor's free-phase event, marking the cage correction that could have occurred but did not. The structure closed around twelve nodes.

One recurrence anticipated its own completion through the missing interval.

A provisional phrase came to him, ugly and necessary. "Morrow-of-Twelve."

Mei glanced toward Morrow Habitat's orbit. "You named it after a station?"

"Not the habitat. The older sense. A return still ahead of itself."

"People will hear the habitat."

"They already need a term that does not pretend to be its self-name."

"Does it have a self?"

Tomas studied the twelve-node structure. "Not one we can separate from relation."

He entered the phrase into the translation ledger with warnings: provisional; relational, not personal; no claim of fluent semantics. The public channel immediately copied it. Language always escaped faster than caution.

Cass called thirteen minutes later.

"Your phrase has reached three habitat debates," the Custodian said. Behind him, dual operation boards cast different colors across his face. "Does Morrow-of-Twelve identify an intelligence?"

"It identifies a stable anticipatory recurrence across twelve sinks, moons, and long-period masses."

"That answer evades the category."

"The category may evade the phenomenon."

"Can it negotiate?"

"It responded to chosen omission. It distinguishes some causal patterns."

"Can it agree not to cross habitat orbits?"

"We have not expressed habitat as a stable referent."

Cass looked down, likely at the countdown Tomas refused to keep visible. "Then communication cannot yet replace containment."

"Containment repeats the fragmentation. The archive removes any basis for calling that neutral."

"I did not call it neutral. I asked whether the alternative kills people."

"We are modeling that now."

"Give me a threshold."

"When all twelve occupy free synchronized trajectories, the long recurrence strengthens. We need to test whether partial release can hold."

"Aster Seven already constitutes partial release."

"And the remaining cage loads fell. That suggests benefit, not stability."

Cass folded his hands. "I authorized your access because uncertainty now cuts both ways. Do not return it to me as permission for hope."

Tomas's anger made each word cleaner. "I will return trajectories, energy costs, and failure ranges. You must decide what obligation survives those numbers."

"That has always been the decision."

"No. Until the archive, you counted only human deaths."

Cass absorbed the sentence without retreat. "Bring me a way to count the other side that does not fabricate a human mind."

The channel ended.

For the next six hours, Ninefold turned translation into mechanics.

They began with Aster Seven. Its loose escort field permitted more natural phase motion while yard craft kept structures out of the sink's corridor. Simulations placed the other eleven on matched free trajectories around Ophion, each preserving the reconstructed long periods. Any plan had to conserve energy and angular momentum. The sinks could exchange minute charge forces with prepared plasma streams, but habitat tugs still had to remove cages and relocate hardware. Moons supplied repeated gravitational turns, not free propulsion.

The first solution crossed Crown Meridian in eleven days.

The second missed Crown and struck Morrow's orbital envelope.

The third preserved all inhabited cylinders but demanded thrust no existing cage tug could deliver without melting.

The fourth used Ophion's inner moon as a turning mass. Aster Five passed behind it, traded orbital energy through gravity, and emerged onto the long-period family with

almost no tug impulse. The same path swept a radiation cone across an ice refinery.

"Refinery can evacuate," Pao said.

"Its ammonia tanks cannot," Tomas replied. "A cone flash ruptures storage and changes the moon-assist mass budget through debris."

Pao opened the refinery's live emergency plan. Its green evacuation flag concealed three assumptions: four personnel ferries at berth, two hours to inert the cracking stacks, and permission to abandon fourteen pressurized ammonia tanks. Only one ferry remained. Two carried coolant toward Cinder Vale; another had joined Harbor's blackout relief. Recalling them would clear the refinery and reopen Cinder's lethal migration gap.

Tomas called refinery control. Supervisor Enid Marek answered from a pressure gallery under the inner moon's gray horizon. She listened to the predicted cone time, then brought up her roster and tank sequence.

"I can put the surface crews in the core shelter," she said. "The orbital annex needs the ferry. Safeing every tank takes four hours after shutdown. If I vent instead, eight habitats lose their nitrogen feed before the next harvest."

"Could remote systems finish after personnel clear?" Pao asked.

"Remote valves close. They do not inspect ice in a seal." Marek marked three tanks whose necks had accumulated thermal cracks. "Leave those charged through your cone and you may get the debris field you modeled."

The proposed assist had counted an evacuated population as zero exposure. It had not counted ferry propellant, Cinder's displaced coolant, refinery shutdown power, or fertilizer lost through an emergency vent. Tomas replaced the green flag with four linked resource events. The direct path now killed no one at the refinery in its

median branch and pushed Cinder below survival margin two days later.

"Reject that branch," he said.

They retained a limited evacuation for the orbital annex using the refinery's remaining ferry. Surface crews would shelter only if Five kept outside the storage line. Marek scheduled a six-hour precautionary shutdown, and Tomas charged its lost nitrogen output against the migration food model rather than hiding it under industrial shedding.

Pao watched the amber cost propagate through three habitat reserves. "The refinery survives because someone else spends food."

"Then the model names who spends it." Tomas attached Marek's roster, ferry status, and tank limits to the trajectory constraint. "A clear radiation plot does not make evacuation free."

They moved the passage outward. Required cage impulse rose by three percent and stayed within thermal limits. In exchange, the timing window narrowed to fourteen seconds. One late release would miss the lunar turn and leave Aster Five crossing the habitat plane for months.

Mei marked the window red. "That looks like central execution."

"It looks like shared clocks." Tomas divided the sequence into locally verifiable events. The cage crew would hold release authority. Moon tracking came from Ninefold and two independent habitat arrays. Three tugs could veto if their predicted corridor disagreed beyond tolerance. No remote office could fire a latch.

"More vetoes create more ways to miss fourteen seconds," Pao said.

"Correct. Now model the cost of one erroneous command."

That branch destroyed a tug and put the refinery inside the sink's next approach. Distributed authority introduced delay. Central authority introduced an undetectable single error at the point of action. Neither arrangement repealed mechanics.

They lengthened the approach by one orbit. The timing window widened to forty-one seconds at the price of thirty hours. Aster Two's cage failure probability rose during the wait.

Tomas preserved both plans. A real crew would need to see which uncertainty it accepted. Translation had taught him that absence could carry information only when force remained genuinely available. Consent worked under the same constraint; a choice stripped of alternatives supplied no answer.

Tomas refused optimization goals that hid a habitat inside aggregate casualty expectation. He required explicit exclusion volumes around all forty-six. That forced the sinks farther from present service routes and exposed another cost: the habitats relied on sink gravity as counterweights. Remove the circulating masses, and station-keeping plans inherited slow drifts their designers had postponed for decades.

Mei sent him a nutrition packet. He opened it and forgot to drink.

At hour seven, an orbital student named Pao found a viable resonance family by allowing habitats to move.

"Only small burns," Pao said, then corrected himself when Tomas looked over. "Small for cylinders. Large for their propellant budgets."

The projection shifted. Six habitats moved first into zones between the free sink trajectories. Twenty-one followed on phased burns. The rest altered station-keeping over weeks. Nothing collided in the median simulation.

"Heat?" Tomas asked.

"Thermal closure still fails."

"Power?"

"Not closed."

"Uncertainty?"

"Embarrassing."

"Good. We know where work begins."

They attached engineering models from the Compact's newly opened stores. Current dispatchable power combined sink extraction, fusion, solar mirrors, flywheel release, and limited moon-tide generation. Freeing the sinks removed direct feed systems and made rotational coupling intermittent. The first estimate showed a forty-four percent loss.

Harbor's free-phase data improved it to forty-one. Revised plasma-skimming fields reduced the gap another two points but risked becoming cages under another name. Tomas imposed a field-strength ceiling derived from the permission tests. The solution settled at thirty-eight percent.

He stared at the number until Mei said, "It will not improve because you dislike it."

"I am checking whether we counted industrial shedding twice."

"We did not."

Thirty-eight percent of dispatchable power disappeared under restoration. Not total energy, which let careless officials call the loss manageable. Dispatchable power supplied heat during eclipse, spin correction during a bearing fault, pumps when a farm blight demanded sterilization, and emergency thrust when trajectories failed. It covered bad timing. Losing it made every ordinary failure more dangerous.

"Run reserve duration by habitat," Tomas said.

The six-risk bloc fell below civil thresholds first. Morrow lasted longest among them because its old flywheel vault could black-start essential systems, provided crews recommissioned bearings and did not spend the stored rotation beforehand. Vela Minor ran out of heat margin in thirty-one hours after release. Pallas Reach's present orbit drifted into a free-sink exclusion zone by Day Twenty-Two.

Pao rubbed both eyes. "The Resonance survives if the habitats move. The habitats survive if the power keeps running while they move."

"And the cages must open for the relation to stabilize," Mei said.

Neither sequence could precede the other safely.

Tomas walked into the projection volume. The system surrounded him at working scale. Habitat tracks glowed amber, free sink families blue. Every viable path braided them during a transition lasting at least twelve days. Tugs would have to release one cage, shepherd its hardware clear, assist a habitat burn, then reach the next sink without relying on the power just surrendered.

He selected Aster Three. The reconstructed long phase gave its release a special constraint. If Three opened too early, the suppressed period pulled the other free trajectories away from their prepared corridors. If opened last, its phase lag could complete the relation after humans had moved into resonant gaps.

Ada's tug carried Three already.

He called her.

She answered from a maintenance crawl, face close to the lens, a cable bundle behind one shoulder. "Operational consequence?"

"Yes."

"Give me the bad number first."

"Restoration requires all twelve sinks on synchronized free trajectories. Partial opening improves coherence but cannot complete the long relation."

"How synchronized?"

"Milliseconds in phase modulation at release, kilometers in orbital position. We can achieve it with cage fields and moon assists if every handoff follows one plan."

"That sounds like Compact control with a larger map."

"The physics requires coordination. It does not require one commander."

She held his gaze. "Cost?"

"Thirty-eight percent of dispatchable power. Permanent under the present generation mix. Habitat station-keeping destabilizes when the sink counterweights leave. Six face lethal margins during transition unless they burn to new orbital regions."

"Morrow?"

"Moves first or dies early."

"Time?"

"Twelve days for the most compressed release. Longer gives safer burns but cage failure probability rises."

Ada looked away from the camera. Someone in the crawl asked for a coupler; she passed it without muting the channel.

"What does Aster Three do?" she asked.

"It carries the long-period relation we suppressed most completely. I think Three must open last."

"Think?"

"The archive supports it. No live test can reproduce a twenty-year cycle."

"Then label that uncertainty where Cass cannot crop it."

"I will."

Ada backed out of the crawl and pulled herself toward a brighter compartment. "Send the full model to Belen too. If six habitats move, they need it before anyone names their sacrifice."

"It is not ready for public burns."

"Send read access, not command code."

Tomas almost refused from habit. Then he remembered Mara Venn's six appeals and the blue signature that had removed 18,604 events from view.

"I'll send it," he said.

Before publishing, he added a translation note to the long recurrence. Injury did not mean agony. Memory did not mean autobiographical recall. Morrow-of-Twelve did not name a person. Each term marked a measurable relation and the limits of human inference.

Then he added the facts no caution could soften: cage corrections repeatedly fragmented the process; full restoration required twelve free synchronized trajectories; release removed thirty-eight percent of dispatchable power; current habitat orbits could not all survive the transition.

Ninefold's public mirror accepted the model. Morrow's ledger acknowledged first, then *Common Measure*, Crown Meridian, and the other habitat archives in widening order.

Inside the orbital theater, Pao set the twelve free paths in motion. Blue lines drew a slow lattice around Ophion. Amber habitats began leaving the places where a century of borrowed gravity had taught them to remain.

Aster Three held to its delayed curve while every viable human orbit moved around the date of its release.

Chapter 15 — The Price of Release

Belen made the hearing wait while Morrow's clinic counted its oxygen bottles.

Forty-six habitat seals hovered above the assembly channel, each bright enough to suggest equal standing. Under them, the Compact's two survival models turned around Ophion in slow projected arcs. Red marked collisions. Amber marked lost spin, failed agriculture, or heat rejection below civil minimum. Morrow carried red in both models.

"Convener Sato," the moderator said, "your allotted response began thirty seconds ago."

"My allotted response begins when I have an inventory."

A technician beside Belen held up four fingers, then three. The clinic network had answered. She entered the number herself: three hundred twelve portable oxygen bottles, of which sixty-eight had regulators fit for pediatric use. No model on the public wall contained that distinction.

Belen sat in Morrow's agricultural drum beneath a dead bank of grow lamps. Residents filled the terraces behind her in work clothes, pressure smocks, and the gray wraps issued to families who had lost someone during the radiator accident. Ren stood among the exterior crews with his crushed hand in a rigid brace. He had refused the front row. The three empty chairs beside the thermal board named the workers Ada had failed to bring home.

"Proceed," Belen told the moderator.

First Custodian Cass Vardek occupied the central pane from Crown Meridian. Lack of sleep had pared his face without loosening his composure. Ada Narang appeared to his left aboard *Common Measure*, one shoulder held too still after days under cage gradients. Tomas Reeve's pane carried data rather than his face: twelve free trajectories braided among Ophion's moons, with human habitats moving out of their path.

Cass touched the first model. "Option One: forced recontainment. We restore phase control sink by sink, beginning with Aster Three. Successful isolation recovers twenty-nine percent of interrupted service inside six days and gives us time to rebuild damaged cages."

The model shifted. Six habitat names acquired black borders.

"Say them," Belen said.

Cass looked toward her pane. "They appear on every display."

"People listening on maintenance audio do not have every display. Say who dies first."

Silence moved through the channel, dense with institutional reluctance. Belen knew its machinery. A name created a constituency; a percentile created permission.

Cass did not hide behind the moderator. "Morrow. Cinder Vale. Far Rest. Ketu. Bellweather. Luma Ring."

"Order?"

"Order depends on whether recontainment succeeds on the first attempt."

"Give the median run."

His jaw tightened. "Far Rest loses agricultural circulation in thirty-one hours. Morrow crosses minimum heat rejection at thirty-eight. Cinder Vale exhausts attitude reserve at forty-four. Luma Ring loses potable-water lift at fifty. Bellweather and Ketu follow inside sixty-three."

On the terrace behind Belen, someone began to cry without making much sound. Her technician entered each deadline into Morrow's public ledger.

Cass continued. "Those outcomes assume no emergency transfers. Forced control restores enough dispatchable power to prevent them if the cages hold."

"If." Belen pointed to the second model. "Now name the first dead under the other plan."

Tomas brought his face back into view. "The Great Unbinding frees all twelve sinks into synchronized trajectories and moves every habitat into one of seven resonant islands. Without transfers of propellant, coolant, and flywheel capacity, the same six fail during migration. Timing changes. Far Rest and Cinder Vale become the narrowest margins."

"How narrow?"

"Far Rest carries nineteen minutes of burn tolerance at its second correction. Cinder carries eleven."

"Eleven minutes does not qualify as tolerance."

"No," Tomas said. "It qualifies as the uncertainty we can currently defend."

His answer pleased her more than reassurance would have. She let that fact alter nothing.

A Crown delegate requested the floor. "The hearing needs a comparable casualty estimate, not local detail selected for emotional effect."

Belen turned from the central pane to the terraces. "Mara, stand up."

A clinic orderly rose near the rear rail. She had worked two consecutive rotations since Morrow's heat loss. Her son slept against her chest under a silver warming sheet.

"How much does his bottle hold?" Belen asked.

"Eight hours at rest. Five if his lungs tighten."

Belen faced the delegate again. "That constitutes a comparable estimate. Continue."

The moderator tried to recover sequence by opening expert questions. Forty-one requests arrived at once. Compact software ranked them by population, legal standing, and declared relevance, reproducing every old advantage under a clean interface. Belen bypassed it through the six-habitat mutual-aid circuit. Five conveners joined her pane, their signals rougher than Crown's and more useful.

Saye Olan from Far Rest spoke first. "We cannot make either plan. Our reaction mass inventory sits at forty-three percent of the number Compact used."

A stir crossed Crown Meridian's delegation. Cass leaned toward someone outside his image. "Your filed reserve showed sixty-one."

"Our filed reserve includes ice pledged to Ninefold under a contract we cannot break without losing seed stock."

"Emergency authority can suspend the contract."

"It cannot turn seed ice into clean drive feed in fifteen hours."

Belen added the discrepancy. The shared ledger recalculated Far Rest's margin from nineteen minutes to a collision probability displayed without rounding.

Cinder Vale reported cracked storage bladders. Bellweather had understated radiator fouling to protect its maintenance rating. Luma Ring possessed more water than its filing claimed but lacked transfer canisters. Ketu kept an excellent propellant reserve and almost no spare bearing stock. Morrow owned a flywheel vault, three functioning tenders, and enough drive mass for only one major relocation.

Every confession made the system less orderly and more survivable.

Cass waited until all six had spoken. "Hidden shortages damage both options. They do not erase the difference between them. Recontainment uses existing transfer infrastructure. Migration requires forty-six independent burns coordinated against objects whose behavior we have translated for four days."

Ada's image sharpened as her tug's antenna found a cleaner lock. "Existing infrastructure caused the behavior you call uncertain."

"It also keeps six million people alive."

"Not if we push the sinks into another contraction."

"You have evidence of injury, Captain. You do not have evidence that release produces cooperation."

Ada glanced toward Tomas's trajectory model. "Aster Seven's free phase stabilized the other eleven."

"For seventeen minutes," Cass said. "Inside a debris field. You want to extrapolate a civilization from seventeen minutes."

"No. I want you to stop extrapolating a machine from twenty years of refusal."

Belen heard residents shift behind her. Ada had spoken cleanly, yet moral cleanliness could kill as efficiently as administrative caution. Tomorrow needed trajectories, not innocence.

"Captain Narang," Belen said, "if every rich habitat keeps its full reserve, how many of my six complete your migration?"

Ada did not ask Tomas to answer. "Two with defensible margins. A third with unacceptable risk. Three do not begin."

"And if we recontain?"

"Three survive the median successful run. All six face severe loss if Aster Three contracts the way the grammar predicts."

"Numbers."

"The cage team estimates sixty-two percent technical capture probability. That estimate does not include a coordinated response from the Resonance. Tomas's bound puts system-wide gradient disruption between nine and thirty-seven percent."

"Your recommendation?"

"Do not close another cage."

"Not what I asked. Tell Morrow what to do."

Ada's mouth set. The captain could fly a trillion-tonne hazard through a passenger lane, but Belen had found the edge of her preferred authority.

"I recommend Unbinding," Ada said. "Morrow still decides whether Morrow burns."

That answer cost Ada something. Belen marked it.

The hearing broke for thirty minutes while engineers reconciled inventories. In Morrow's drum, the dead lamps remained dark to save coolant. Residents descended from the terraces and formed knots around the public terminals. Nobody waited for speeches. They wanted ration dates, berth assignments, and proof that relatives on other habitats could reach them.

Ren approached Belen with two cups clipped into a carrier. His left hand had learned work meant for two while the crushed right remained braced against his chest.

"You made Vardek say our name," he said.

"He already knew it."

"Knowing privately comes cheaper." Ren passed her a cup. "Exterior stores sent their true count."

She opened the packet. Morrow's maintenance crews had set aside twelve tonnes of unauthorized reaction mass over seven years, hidden in maneuvering reserves and mislabeled cleaning feed. Anger arrived before relief.

"You lied to the civil ledger."

"We lied to Compact allocation. If they saw margin, they cut next quarter's allotment."

"I built this bloc on an honest inventory."

"You built it because the official inventory kept us poor on paper and dead in practice."

Belen looked across the drum. Thermal workers watched them from beneath the stalled lamps. Some knew about the reserve. Their studied inattention admitted as much.

"Bring all twelve tonnes into the open ledger," she said.

Ren's expression hardened. "Crown will seize it for the cage attempt."

"Then everyone sees the seizure."

"Morrow might need that mass tonight."

"Far Rest needs it sooner."

His braced hand knocked lightly against the cup. The injury had left two fingers without sensation. Belen had watched Ada and Ren's crew pull him from a buckled rig while three other workers drifted beyond rescue. Family did not make his fear less political.

"You asked for truth," he said. "You did not promise truth would feel fair."

"No." She authorized the entry. "I promised the unfairness would carry names."

The hidden reserve appeared across the system. Morrow's apparent margin rose. Before relief could settle, Crown Meridian's private reserve joined the network through an anonymous maintenance key: one hundred eighty-four tonnes of certified drive feed held behind a

security classification. Rook followed with ninety-one. Harbor of Glass showed fifty-six tonnes assigned to commercial launch guarantees. Twelve habitats exposed reserves within four minutes, some through official votes and others through workers who had found passwords.

Belen's ledger bloomed with green where the Compact model had shown scarcity.

Then ownership claims covered it in amber.

A shipping consortium warned that diverting Harbor's mass would strand ore crews. Rook cited academy evacuation obligations. Crown called its reserve necessary for continuity of government. Every tonne already protected someone. The inequality lay not in whether people had reasons, but in whose reasons had hardened into law.

Saye Olan reopened the mutual-aid circuit. "Far Rest can accept Morrow's twelve tonnes if a tender launches within six hours."

Ren checked the exterior board. "Tender Four can carry nine. Its shielding failed inspection."

"Failed by what?" Belen asked.

"Solar-flare standard. Current forecast stays quiet for two days."

"Crew exposure?"

"Within annual limit if forecast holds. Outside if it does not."

Saye's eyes closed briefly. "I will not ask your crew to take that."

"Good," Belen said. "They will decide."

Eight Morrow tender workers joined from an airlock gallery. Belen read them the shielding figure, the forecast confidence, and Far Rest's revised burn clock. Six consented. Two exercised their crew veto, forcing the manifest to replace them rather than disguising dissent as

unanimity. Tender Four began loading before the hearing resumed.

The action changed every argument. Morrow had transferred first, from a reserve it could not spare, without waiting for Compact custody. On Crown's polished channel, delegates who had praised solidarity now confronted a ledger that measured theirs.

Cass returned with a revised proposal. "Crown releases sixty tonnes immediately to the six-at-risk pool. Rook has offered thirty. Harbor will supply twenty if ore evacuations receive priority."

"Put the transfers on a clock," Belen said.

"They are already authorized."

"Propellant inside a vault does not move a habitat."

Cass supplied launch windows, tanker names, and receiving crews. Belen entered each obligation where everyone could watch it expire.

The new inventory improved the Great Unbinding model. Four of the six could now migrate inside defended margins. Far Rest needed the tender already loading. Cinder Vale still lacked coolant and a safe second burn. Forced recontainment also gained margin, though no transfer could reduce the unknown response from twelve related masses.

Belen stopped the recalculation before its shortage icon could collapse into a percentage. "Put the missing reaction mass on a hull."

The migration engineer expanded Cinder Vale. Its first burn consumed the reserve already certified. The second required twelve additional tonnes to clear Aster Five's exclusion corridor with abort capacity. Without those tonnes, guidance could attempt the correction, but one early engine cutoff would leave no recovery burn.

"Who rides without recovery?" Belen asked.

Cinder's convener answered. "Spokes C and D. Twelve thousand six hundred residents. Our dialysis ward occupies D because the bearing there runs smoothest. Tug *Juniper Debt* carries the burn crew."

Belen entered all three beneath the deficit: two spokes, one ward, one tug crew. Then she split the model's green success branch into funded and unfunded runs. Cinder did not possess a twelve-tonne shortage as an abstract property. Its residents and the people lighting its drives would bear the missing maneuver.

"Keep those names visible whenever anyone reserves a tanker elsewhere," she said.

Representatives began choosing camps. Wealthy cylinders favored the cage because its first costs appeared elsewhere. Three shipyard habitats favored release because they had seen Aster Seven tear through metal designed to hold it. Ketu split around memory: residents who feared another spin loss demanded control; survivors who remembered the old command chain refused it.

Belen watched the debate turn moral and stopped it.

"No habitat gets to purchase virtue with another hull," she said. "We have two incomplete plans. One restores the captivity that built our power economy. The other removes more than a third of dispatchable power and asks poor habitats to burn first. Anyone calling either choice clean has hidden part of the inventory."

A delegate from Rook asked whether Morrow would support release if Cinder's coolant gap closed.

"Morrow has not voted."

"You have shaped the plan as if you favor it."

"I favor knowing who enters the airlock."

Tomas placed a fresh trace over the models. Aster Three's long phase lag had shortened twice during the hearing. The other eleven answered through minute

variations in cage load. No translation pane appeared. Tomas resisted converting pattern into convenient speech.

"What does it indicate?" Cass asked.

"Recurrence around a held relation," Tomas said. "The closest grammar remains contraction. Defensive, if our mapping holds."

"If our mapping holds," Cass repeated.

Ada studied the trace. "They know we are aligning the cage."

"Or they respond to ordinary preparatory currents."

"The currents form the alignment."

Cass turned his attention to the operational board. Renewed Article Nine authority had less than forty-eight hours left. Belen saw the remaining time beside his name, a legal clock counting toward an authority no ordinary process could restore quickly enough for the present crisis.

"We cannot wait for forty-six assemblies to converge while Aster Three destabilizes," Cass said. "Every hour reduces recontainment probability and migration margin together."

"Then publish both plans for vote," Belen replied.

"A vote does not center a cage."

"Neither does an order."

"An order gets four tug crews into one geometry before the window closes."

Ada leaned toward her pickup. "My crew has an engineering veto. Iri will not certify full isolation."

"Article Nine permits assignment of a Compact cage officer."

"It does not permit you to make the hull sound."

Cass's control held, but grief moved beneath it. Belen knew about the food-chain cascade that had killed his husband. He carried six million people as a list that might

vanish unless one pair of hands kept the door shut. His error had grown from a duty Belen recognized.

That recognition made resisting him harder and more necessary.

"First Custodian," she said, "your successful model kills nobody if every uncertain mechanism favors you. Your failed model harms every habitat at once. Why should the six with the least reserve bear that gamble without consent?"

"Because delay also gambles with them first."

"Give them the choice."

"I do not have time to manufacture agreement."

"Agreement does not come from your factory."

Cass muted his channel. Around his pane, aides moved through Crown's operations room, carrying orders that had likely begun before the hearing. Belen checked the tanker board. Crown's sixty tonnes had left its vault but not its berth. Morrow's tender pulled away under six consenting crew, nine tonnes bound for Far Rest.

When Cass returned, his decision had lost all softness.

"Under Article Nine, I am directing *Common Measure*, *Principle of Least Harm*, and Compact tugs Sable Current and Kinship Road to the Aster Three clamp corridor. Operations begin at 0400 system time. Crown releases the pledged reserve regardless of outcome. Habitat assemblies may continue deliberation on migration."

The moderator asked whether the order concluded the hearing.

"No," Belen said. "It concludes the fiction that this hearing held the decision."

Cass accepted the blow without retreat. "If recontainment fails, you will need the public record you made today."

"If it succeeds, we still need it."

"Then keep counting."

His pane closed.

Ada remained on the channel. Behind her, amber cage light marked Aster Three's uneven phase. "Belen, I can refuse an unsafe configuration. I cannot refuse every lawful assignment while my cage carries the mass."

"Will your crew go?"

"They will vote after Iri presents the limits."

"Do not tell them Morrow needs heroics."

"I won't."

"Tell them we need their true margin."

Ada nodded once and disconnected.

Morrow's drum filled with overlapping voices. Residents demanded an immediate habitat vote, a blockade of the clamp corridor, seizure of Crown's tanker, or trust in the captain who had saved their traffic lane. Belen raised neither volume nor hand. She projected the six names, the two models, every committed tonne, and each unresolved gap onto the dead grow lamps.

One line changed while they watched. Crown Tanker Eighteen cleared its berth with sixty tonnes of reaction mass. Its destination field named the six-habitat pool, not Compact operations.

Below it, four tug acknowledgments entered the forced recontainment order. *Common Measure* came last, carrying an attached condition: crew engineering veto remained active.

Belen opened Morrow's voting system and set the first question. Not release or captivity. Not yet. Residents would decide whether any reserve left their hull without a named receiver, a launch time, and consent from the crew moving it.

Outside the agricultural drum, Tender Four crossed Morrow's rim and lit its drive toward Far Rest. Nine tonnes moved into the dark while the old power web prepared to close another cage.

Chapter 16 — The Failed Cage

The first clamp tug missed its mark by nine centimeters, and Ada ordered all four ships to hold.

Against Ophion, nine centimeters disappeared inside a black curve no display could draw at scale. Inside the Aster Three cage, that error loaded the receiving field across *Common Measure's* scarred manual latch. Iri's engineering board changed from amber to red before Sable Current's pilot finished reporting capture readiness.

"Back out," Ada said. "Two meters along your approach axis. No lateral thrust."

Compact control answered before the pilot. "Formation remains within authorized tolerance."

"Not my cage tolerance."

First Custodian Vardek watched from Crown Meridian through a delayed operational pane. His Article Nine key occupied the corner of every command display. "Captain, the clamp model accepts twelve centimeters. Continue alignment."

Ada pressed her thumb against the pale burn ridge on her left hand. "The model assumes an intact latch carrier. Ours carries a torsion scar. Sable Current backs out."

"Backing changes the other three vectors."

"Then the other three earn their fuel."

The tug pilot obeyed. Sable Current drifted away under cold-gas pulses while *Principle of Least Harm* and Kinship Road

adjusted around it. Nadi Okafor held the fourth point below their plane, her ship reduced to a green bracket against Ophion's cloud bands. Four cages had come to form a tetrahedron around Aster Three: *Common Measure* carrying the sink, three empty systems intended to close its phase from outside. If the fields met cleanly, Cass's team expected the long lag to flatten and the sink to accept transfer constraints again.

Tomas had supplied a different expectation. His translation pane marked every tightening interval with the same warning: recurrence held, relation suppressed, contraction likely.

Iri opened a private circuit from the cage gallery. "You can call my veto now. I will decorate the report with equations."

"Does the nine-centimeter error cross your certified limit?"

"It crosses my sleep-at-night limit."

"Certified."

A pause, then the scrape of a tool against the pickup. "No. Bring Sable to mark and the geometry passes. I still recommend we stop."

Recommendation did not carry the legal force Iri had demanded and won. Ada had made that distinction to protect engineering authority from a captain's pressure. Now it trapped both of them inside an operation neither trusted.

"Your veto remains live," she said.

"Stop offering it like a gift. I know where the switch sits."

Sable Current reached its revised mark. Jalen confirmed relative motion below one millimeter per second. Nadi reported green alignment, then added, "Four captains hold four hulls. Crown does not fly any of us from a pane."

Cass's mouth tightened. "No one has proposed remote flight."

"You proposed a common burn table."

"A common table prevents four local judgments from creating one collision."

Nadi kept her voice mild. "Then persuade us each time."

Ada watched the phase trace. Aster Three's familiar lag extended behind the other sinks' returns, a long reluctance measured across moon positions and years. The trace had begun to compress as the empty cages raised their fields. Eleven faint responses arrived through the observatory feed. None belonged to an isolated body. Meaning occupied their timing.

"Clamp One," Cass ordered. "Ten percent field. Hold for forty seconds."

Ada looked at Iri's status, then at her own crew board. Nineteen names carried green duty marks. Nobody had withdrawn consent after the hearing, though Parveen had attached a written objection to the operation. Jalen had requested independent authority to abort any thrust that compromised separation. Ada had granted it.

"Local captains," she said, "confirm Clamp One."

Three confirmations arrived. She raised *Common Measure's* outer field.

Pressure gathered behind her teeth. The tug creaked along the cage truss as charge coupling moved load from one ring to another. On the tactical display, Aster Three stayed perfectly centered. Its gravity did not care about field lines. Its engineered charge did.

"Ten percent," Iri reported. "Primary loops balanced to six parts per million. Port radiator hinge temperature rising."

The wing had survived Harbor of Glass under a heat budget no inspector would have accepted before the crisis. Microfractures crossed two hinge roots. Iri had sleeved them with composite and posted a load limit beneath every thermal forecast.

"Open starboard flow," Ada said.

"Already full. Port needs to carry thirty-one percent or the clamp bus overheats."

"How long?"

"At present load, twelve minutes."

Cass called the forty-second hold complete. The phase trace had flattened by four percent. Crown's operations room permitted itself a low stir of approval.

Tomas did not. "The other eleven have shortened their recurrence intervals."

"Response magnitude?" Cass asked.

"Below cage noise separately. Coherent across all nodes."

"Predicted disruption remains under threshold."

"My model does not assign a safe threshold to an unknown defensive act."

Cass shifted to the second clamp schedule. "Twenty-five percent. Thirty seconds."

Ada delayed her confirmation. A shipboard alarm began to pulse, not for immediate failure but for a changing tidal load aft of the cage. The gradient had moved three centimeters without any mapped change in Aster Three's center.

"Raw accelerometers," she said.

Jalen stripped filters from the plot. Tiny vectors covered *Common Measure* from bow to drive spine. Aster Three's effective relation to the hull had changed while its charge centroid remained fixed.

"Tomas?"

"The Resonance may be shifting coupling through the other sinks. I cannot separate local cause from the whole."

Cass leaned closer to his pickup. "Can you show a hazard beyond expected load?"

"No."

"Then proceed to twenty-five."

Ada wanted a red line. She wanted Iri to cross it, Tomas to name it, or metal to confess before people had to choose. Command had taught her to distrust desire disguised as caution. It had also taught her what happened when distant authority assigned uncertainty to her hull.

"Local condition," she said. "We take twenty-five in two stages. Eighteen, hold, then twenty-five."

Kinship Road's captain agreed first. Sable followed. Nadi waited long enough to make Cass ask.

"I am confirming eighteen," Nadi said. "I have not confirmed beyond it."

They raised the fields.

At sixteen percent, Aster Three's phase trace lost its lag. The line snapped into alignment with the clamp carrier, so abrupt that Crown's predictor painted successful capture before Ada felt the deck tilt.

A loose restraint strap lifted beside Jalen and hung at an angle toward port. Three drops of water escaped someone's cup and stretched into silver threads. The ship had not rolled. Gravity had acquired a sideways preference.

"Hold fields," Ada ordered.

"We are at eighteen," Iri said. Their voice had gone flat. "Outer loop currents oscillating. I have a point-two-second beat against the latch carrier."

"Can you damp it?"

"Not while three cages drive the source."

Cass spoke over the common circuit. "The phase lock has acquired. Advance to twenty-five before it slips."

"No," Nadi said.

Ada agreed. "Hold eighteen. Crown, remove your advance command."

"Phase acquisition creates the highest probability window."

"A gradient just crossed my command deck."

"Within structural tolerance."

"It crossed every habitat." Tomas filled their display with accelerometer reports. "Rook, Morrow, Ninefold, Harbor. All forty-six show related vectors."

The Resonance had contracted around its broken relations. Ada could not see a body closing, only gravity finding new paths through a web built on assumptions of stable mass. Habitat station-keeping systems began firing. Traffic controllers issued holds too late for ships already committed to transfer arcs.

On Morrow, a farm drum reported a two-centimeter bearing walk. Far Rest lost water lift in two sectors. Crown Meridian's mirrors shifted enough to throw daylight across a residential night.

Cass read the reports with no visible triumph left. "If we release now, do gradients relax or rebound?"

Tomas answered with the only honest word. "Unknown."

Iri's alarm changed pitch. "Manual latch walking. Blue witness has separated."

Ada saw the old strip through the gallery camera, a shard of blue trapped beside polished machinery. It had marked true alignment when optical sensors lied. Now a black line divided the paint.

"Abort in sequence," Ada said. "Kinship first. Sable after two seconds. Nadi holds low point. *Common Measure* maintains center."

Crown's cage officer objected. "Asymmetric release could eject the charge sheath."

"Simultaneous release could send four hulls toward one mass."

"Article Nine directs continued capture."

Ada opened the crew circuit. "Iri, engineering veto?"

"Invoked. Clamp progression unsafe. Abort authorized."

The switch removed Compact discretion aboard *Common Measure*. Cass could replace Iri later if a replacement could physically reach the gallery. He could not make the present engineer certify a failing cage.

"Execute abort," Ada said.

Kinship Road collapsed its field.

The contraction deepened.

Aster Three's trace did not return to its earlier lag. Eleven remote phases folded toward it, a brief coherence that loaded every gravitational relation at once. *Common Measure* dropped sideways relative to the formation. Jalen fired attitude jets without waiting for Ada, saving the cage truss from Sable Current's bow by fourteen meters.

Nadi rolled *Principle of Least Harm* beneath them and pulled away under main thrust. "My cage center shifted forty centimeters. I am no longer your low point."

"Sable, collapse now," Ada ordered.

"I have carrier lock," its pilot said. "Field will not ramp down."

Iri swore through clenched teeth. "Their controller caught our phase. I can break it at the manual latch."

"From the board?"

"Actuator lost position reference. I need eyes and a wrench."

Ada brought up the gallery map. Radiation around the cage had doubled since Clamp One. The manual station sat behind a movable tungsten shutter intended for service

with the sink quiescent. Nothing had remained quiescent for fifteen days.

"Dose to manual station."

"Eighty millisieverts a minute and climbing."

"You have remote cameras."

"Cameras reference the carrier. The carrier moved. Blue strip references the latch."

"Send Parveen."

"She does not know the mechanism."

"Talk her through it."

"I can do it in forty seconds. Talking takes four minutes we do not own."

Ada's old reflex reached for a direct order, either to keep Iri safe or send them into danger. Both choices would convert expertise into obedience. She forced her hand open on the grip.

"State your plan and limit," she said.

"Cross gallery on inner rail. Close tungsten shutter to half. Align manual fork to the unbroken edge of the blue paint strip. Strike release cam. Sixty seconds exposure, no more."

"Medical estimate permits forty-five."

"Then I work faster."

"That is not an estimate."

Iri breathed once. "Forty-five. If the cam does not move, I return."

Ada transferred the circuit to Parveen for independent watch. "Your veto. Your limit. Go."

The gallery camera caught Iri leaving the shield pocket in a hard suit. Their tether snapped taut against a deck that no longer agreed which direction deserved down. They

pulled hand over hand along the inner rail while warning lights stuttered in the radiation haze.

Sable Current's cage dragged at *Common Measure*. Relative distance fell through eighty meters. Its pilot used main drive to resist, but thrust fed more load into the shared center.

"Cut drive," Ada said.

"I collide if I cut."

"You collide sooner if the latch releases under thrust. Cold gas only."

The pilot hesitated. Nadi intervened. "Sable, obey your local plot. Ada has the mass."

Main thrust died. Separation narrowed more slowly.

Iri reached the manual station at nineteen seconds. The tungsten shutter moved halfway, then jammed. Their suit dose climbed past the planned curve.

"Return," Parveen said.

"I have twenty seconds."

"Your shutter failed."

"I still have the latch."

Through the camera, Iri pulled the blue paint strip from its bracket and laid it across two misaligned witness cuts. The flake trembled under their glove. One edge matched the fixed housing; the broken tip pointed six millimeters below the actuator's displayed center.

"Carrier sagged," they said. "Optics lied high."

They fitted the manual fork where the paint indicated and drove the wrench down.

The first strike rebounded. A radiation warning blanked the camera to white. Ada heard Iri grunt, then metal rang again through the hull.

Sable's carrier lock broke.

Its empty cage recoiled twenty-seven meters. Cold-gas jets caught the spin. Kinship Road moved clear. Nadi burned across the low plane, opening enough space for Ada to dump the outer field.

"Release to base," Ada ordered.

Common Measure's clamp current fell.

Aster Three shifted charge.

The manual station kicked sideways under Iri. Their tether anchor tore from the rail. They struck the latch housing hip-first, folded around the impact, and drifted toward the open truss.

Parveen fired the gallery restraint net. Mesh caught Iri across the chest, but their left leg passed through and struck a support spar. The suit telemetry registered the fracture before their cry reached Ada.

"Left femur," Parveen said. "Displaced. Suit holding pressure. Dose one point seven sieverts."

"Get them behind shielding."

"I am conscious," Iri said through a ragged breath. "Do not have a committee about it."

The restraint winch pulled them toward the pocket. Each rotation twisted the broken leg. Parveen met the net at the shield line and secured the limb with an inflatable splint. Two deck crew hauled both engineers inside as the gallery field alarms rose beyond useful differentiation.

Ada kept her eyes on formation. Care had other hands for the next minute. Sable Current drifted clear with a damaged cage and no injuries. Kinship Road reported two cracked loop mounts. Nadi held below the plane, propellant bleeding into a correction that would cost her weeks of range.

Across the habitat web, gradients continued moving.

"Morrow attitude loss," Jalen said. "Their old station-keeping solution no longer closes."

The tactical display widened from four tugs to the inhabited system. Forty-six trajectories had acquired small errors. Most remained recoverable. Morrow's carried a red crossing with the Aster Nine corridor.

"Time to conjunction?"

"Eighteen hours if neither relation changes."

Ada called Morrow traffic. Ren Sato answered from an exterior control rack, face washed gray by helmet light. "We felt the cage let go. Our rim shifted eleven centimeters sunward."

"Can you restore attitude?"

"Jets can stop the tumble. They cannot move the orbit enough."

"Flywheel vault?"

"Charged, not coupled. Belen kept it back for black start."

"Keep it back. I am sending a correction set."

Ren glanced away as another alarm reached him. "Send advice. Morrow decides the burn."

Ada almost told him eighteen hours did not allow procedure. Nadi's rule stopped the words before they crossed her teeth.

"Advice only," she said. "Your hull authority."

Medical reported Iri stable enough for transfer, along with Parveen's accumulated dose and Jalen's neck injury from the gradient kick. Three people required evacuation as soon as a shuttle could match them. The tug would continue with sixteen.

Cass remained on the operational pane. Around him, the successful-capture projection had vanished. He looked at Iri's telemetry, the damaged tugs, and the red lines multiplying through the web.

"Can Aster Three be held aboard?" he asked.

Iri answered from the storm shelter before Ada could. Pain medication blurred their consonants but not their judgment. "Present cage can carry. It cannot clamp. Try again and you may lose the local charge sheath."

"Understood," Cass said.

No defense followed. His order had failed on the terms he valued most: fewer deaths, controlled risk, time purchased. Ada found no satisfaction in the ruin of his certainty. The habitat clocks kept counting.

Nadi opened a captains' circuit. "All three assisting tugs stand clear. Ada, your port radiator wing shows hinge separation."

The external image arrived a second later. One radiator root had opened like a pulled seam. The broad dark wing flexed against its remaining hinge while hot coolant flashed from a severed line.

"Retract it," Ada said.

Jalen tried. The actuator pulled; the damaged root twisted instead. Structural prediction gave them twenty seconds before the wing struck the cage truss.

"Explosive bolts," Ada ordered. "Port only."

The bolts fired in sequence. The port radiator wing tore away without grace, trailing frozen coolant and bright fragments of composite. Its loss cut *Common Measure's* heat rejection nearly in half. The drifting panel turned once through Ophion's reflected light, then began a separate orbit no recovery tug could safely pursue.

Ada reduced every nonessential bus. Command displays dimmed. The cage gallery disappeared behind automatic radiation shutters. In the storm shelter, Iri asked someone to find the blue strip. Parveen had trapped it inside a sample pouch, bent but still legible.

Crown's evacuation shuttle plotted a seven-hour intercept. Ada approved release of Parveen, Jalen, and Iri from the duty roster. Their names changed from green to white. Sixteen remained, including Ada, to carry Aster Three in a damaged cage on a ship that could no longer shed full drive heat.

Cass transmitted a system notice. He called the containment attempt unsuccessful and ordered all habitats to use local safety authority pending revised guidance. He did not call the gradients sabotage or accident. He named the Resonance's defensive contraction as the leading interpretation.

"First Custodian," Ada said privately, "withdraw the recontainment order."

His gaze remained on Morrow's red trajectory. "I am suspending further attempts."

"That leaves the order available."

"Yes."

"Eighteen hours does not leave room for your distinction."

"It leaves no room for pretending release has become safe."

Ada closed the channel before anger could spend heat they needed elsewhere.

Tomas sent a stripped-down model suited to their reduced computing bus. Aster Three's lag had returned, but every sink now carried a wider phase excursion than before the clamp. The failed cage had not restored the old instability. It had amplified it across all twelve.

Morrow's crossing line brightened as fresh tracking arrived. Eleven centimeters at the rim had become an orbital drift measured in inhabited kilometers. In eighteen hours, the torus and Aster Nine would enter the same narrow volume around Ophion, each tugging on the trajectories of ships between them.

Ada put the migration solution beside the conjunction and began removing every command that assumed she could fly another hull. The remaining work needed forty-six decisions before one red line closed.

Chapter 17 — Seventy-Two Hours Twice

Cass entered his command key, and the system rejected the hour he asked it to purchase.

The denial filled the center of Crown Meridian's crisis table in plain treaty language. ARTICLE NINE RENEWAL EXHAUSTED. At 1900 system time, emergency tug, cage, transfer, and reserve authority would revert to the forty-six habitat governments and the ordinary Mass Compact boards. No vote could extend the term a third time. No First Custodian could declare another while the same emergency continued.

Counsel Mara Veit stood opposite him with both hands on the table rim. "You know this."

"I know the clause. I asked what mechanisms remain."

"Ordinary legislation, local emergency consent, or an adjudicated continuity order."

"Time for adjudication?"

"Three days if no habitat objects. Someone will object."

Behind her, the operational wall showed Morrow and Aster Nine approaching conjunction. Their predicted miss corridor had widened during the last hour, but only because Morrow had begun an attitude burn that its assembly had authorized six minutes at a time. Forty habitats reported gravity errors from the failed clamp. Twelve tugs waited under instructions scheduled to expire before several maneuvers finished.

Cass enlarged the legal notice. "Draft a continuity order."

Mara did not move. "On what ground?"

"Impossibility. The Compact cannot discharge its survival obligations if its coordinating authority terminates during a system-wide mass event."

"Impossibility excuses nonperformance. It does not manufacture power."

"Founding necessity, then."

"That phrase justified the first sink seizure before the treaty existed. Put it in an order now and every habitat counsel will read what you mean."

Cass knew what he meant. He wanted another seventy-two hours. Not the title, not the ceremonial precedence, and not the right to silence people who opposed him. He wanted one table where every burn arrived, one officer empowered to stop two habitats from choosing intersecting solutions, one hand on the cage controls until Aster Three could no longer strike the web by moving a fraction inside its field.

His hand would do if no other existed.

"The Article says it cannot renew," he said. "It does not say ordinary custody powers disappear."

"Ordinary powers require board process. The tug board has seven absent members and four who revoked delegation after the clamp."

"Appointments?"

"Habitat ratification."

"Temporary acting members?"

Mara's expression hardened. "Appointed by you to authorize you? I will not draft it as lawful."

"Draft it with your objection attached."

She looked toward the casualty board. No deaths had been confirmed from the system gradient. Thirty-eight injuries had. Iri Quill's evacuation shuttle had docked at Morrow under a radiation protocol that kept the receiving bay sealed from the public corridors. Far Rest had recovered water lift. Cinder Vale remained outside control margin. Cass's failed order occupied every datum whether or not anyone named it.

"I will draft the document you request," Mara said. "It will read as an unlawful extension of exhausted emergency authority."

"Use fewer adjectives."

"Unlawful carries the legal load by itself."

She left him with the rejected hour.

A Crown operations officer approached carrying the next tug schedule. "We need disposition on Clamp Formation. Sable Current and Kinship Road ask release from Compact assignment. *Principle of Least Harm* requests direct coordination with Morrow."

"Grant all three."

"Under Article Nine?"

"Under my present authority. It still has five hours."

The officer entered the releases. Each tug vanished from the central formation and reappeared as a local unit. Three clean vectors became three negotiation requests, each waiting on a habitat that had its own vote, risk board, and fuel shortage.

Cass had argued for years that central allocation reduced transaction cost. The phrase sounded bloodless until one watched seven minutes disappear while two local councils verified who could sign a burn.

"Put Ada Narang through," he said.

Common Measure answered from a holding arc above Morrow. Its signal carried gaps where the tug's reduced heat budget shut down nonessential transmitters. Ada appeared beneath dim command lights, with a crew roster at sixteen and Aster Three's unstable phase trace behind her.

"You released the other tugs," she said.

"They requested release."

"Good."

"Morrow's conjunction still needs one coordinator. Your ship has the best local solution and the only cage coupled to the long-period relation."

"That gives me information. It does not give me their helm."

"I am ordering *Common Measure* to issue binding burn marks to Morrow, Nadi, and all traffic inside the Aster Nine corridor until conjunction clears."

Ada's eyes moved to the authority clock attached to his signal. "Under which rule?"

"Article Nine until nineteen hundred."

"Morrow's burn continues past midnight."

"An order lawfully issued does not dissolve because authority later expires."

"A standing assignment might not. Fresh burn marks do."

Cass felt impatience rise and compressed it into precision. "If Morrow lights late, Aster Nine's gradient can pull your cage across their rim. If Nadi lights early, her exhaust exclusion zone cuts off two evacuation shuttles. Someone must hold the combined plot."

"I will hold it."

"Then accept command."

"I will issue a shared solution. Each captain and Morrow traffic must consent to every burn that crosses their safety doctrine."

"That procedure adds delay."

"It adds hull knowledge."

"The trajectory does not care who consented."

"The trajectory includes what each hull can survive."

Cass had seen this boundary harden in her since the first transfer refusal. Before Aster Three changed phase, Ada had carried the whole web as if competence entitled her to decide for it. Now the captain who trusted command more than comfort refused the largest command he could give her.

"Captain, this instruction protects Morrow."

"So did your clamp order."

The words landed without emphasis. She did not need to accuse him; the damaged port side behind her supplied evidence.

Cass looked at the sixteen names. "If one captain refuses a necessary mark?"

"Publish the consequence. Ask for an alternate. If none exists, that captain decides what their hull bears."

"And if their choice kills another crew?"

"Nadi's rule applies. Authority ends at another crew's hull. A burn that exports lethal risk can be refused by the receiving hull."

"That creates two vetoes over one maneuver."

"It creates the people who die from it."

The answer frustrated him because it solved no timing problem. It also described the timing problem more honestly than his order.

"Will you coordinate under those conditions?" he asked.

"Yes."

"Begin now. I reserve judgment on whether local refusal can override Article Nine before expiry."

"Reserve whatever you have left." Ada ended the circuit.

At 1500, Belen Sato put the legal clock on every public channel.

Cass saw it first on an internal wall where staff monitored habitat traffic. A plain white counter occupied the lower third of Morrow's broadcast: 03:59:47 TO EXPIRY OF SECOND AND FINAL ARTICLE NINE TERM. Beneath it, Belen displayed the treaty clause, the 34-12 renewal vote, and a list of commands that would survive as ordinary contracts only if local authorities adopted them.

Crown's communications director asked permission to insert a correction.

"What correction?" Cass said.

"She calls the term final. Counsel may argue continuity."

"Counsel already has."

"We can state that emergency obligations remain disputed."

"The legal clock does not become wrong because I dislike its destination."

On Belen's channel, maintenance workers read commands aloud. Morrow adopted Ada's conjunction plan by an emergency assembly vote. Far Rest converted Tender Four's delivery into a habitat-to-habitat transfer rather than a Compact allocation. Three rich habitats withdrew reserve pledges pending legal review. Two reinstated theirs after residents flooded local boards with demands.

One command reached its owner while Cass watched. Compact routing had assigned the tug *Even Hand* to shift a hospital reservoir at Pale Reach. Its captain asked Pale Reach whether the assignment would survive zero. A clinic

engineer, not Crown, confirmed the mass, burn limit, and receiving clamp. The habitat council adopted the route in a recorded vote. Cass's dispatcher removed the emergency seal and replaced it with three local signatures. The tug continued without changing thrust.

A second route lacked signatures and stopped at its hold point. No collision followed. The pause cost eleven minutes and made the missing authority visible before steel entered motion.

Belen never asked crews to disobey. She made authority visible enough that obedience required a named source.

Cass requested her private channel.

"You have put an operational countdown into a constitutional argument," he said.

"I put the law beside the orders."

"Pilots are asking whether they must stop burns at zero."

"Then tell them which burns their captains adopted."

"Some have not completed local process."

"That does not give you a third term."

"I know what cannot renew."

"Your public notices say pending review."

"Because current assignments do not all share one status."

Belen leaned closer to her camera. Behind her, Morrow's drum lights trembled under the attitude burn. "People can handle a complicated answer. They cannot protect themselves from a hidden one."

Cass opened the continuity draft Mara had returned. UNLAWFUL appeared in the first paragraph, unadorned. "If authority ends and two habitats collide while their councils debate, will your broadcast carry that inventory too?"

"Yes."

No defense. No assurance that distributed process would spare her. Belen offered accountability where he kept seeking control.

"Would Morrow consent to a four-hour operational extension limited to conjunction traffic?" he asked.

"Morrow might. Ask the assembly."

"Forty-six separate consents will not arrive in four hours."

"Then ask the hulls inside the corridor."

"That leaves system gradients uncoordinated."

"Publish one solution for everyone else."

He almost said that public code could be altered, misunderstood, or executed outside margin. Those objections belonged on the risk board, not behind his key.

"I will send a limited request," he said.

"Send the code with it."

Belen closed her channel. The clock continued.

At 1630, Cass left operations for the first time in twenty-one hours. Crown Meridian's food galleries had converted to ration distribution after interrupted sink service cut refrigeration. He walked through a corridor lined with insulated bins and people waiting under emergency lamps. Staff recognized him and made room without being asked. That deference had once meant trust. Now he could not separate trust from the security officers at his back.

A kitchen supervisor named Jori Pell intercepted him beside a thawing protein store. "Custodian, your office denied our request to move reserve coolant from the memorial vault."

"Which memorial?"

"Cascade Nine."

The name stopped him. Crown had preserved one cold room from the food-chain failure thirteen years earlier, keeping samples, records, and the personal effects of dead distribution workers. His husband's field badge remained inside a sealed drawer there.

"Why do you need its coolant?"

"Three pediatric stores warm past limit in six hours. The memorial loop carries enough for four days."

"Transfer it."

Jori looked past him toward the staff officer. "The vault charter requires Custodian and family-board approval."

"I approve."

"Family board lost quorum during evacuation."

Cass nearly ordered the lock cut. Instead he asked for the charter. Its emergency clause permitted transfer if affected families received public notice and a full inventory of anything lost. No central exception had hidden there. A procedure did, narrow enough for the need.

"Post notice now," he said. "Begin transfer after thirty minutes. List my husband's drawer among the exposed stores."

Jori nodded and ran.

Cass remained beside the bins. Thirteen years earlier, a blight had crossed three protein cultures while habitat managers protected local stocks from inspection. Crown received clean reports until six distribution nodes failed together. Arun had entered a cold gallery to save viable seed and died when the backup atmosphere carried the same contamination as the main loop. Afterward Cass had helped write Article Nine. One command structure, one inventory, one officer unable to say another habitat's losses lay outside his mandate.

The law had answered the disaster he knew. He had tried to make it answer a mind distributed across twelve masses.

Mara found him as the memorial notice entered the public channel. "Your continuity order awaits signature."

"Does it compel tugs?"

"Yes."

"Reserve seizure?"

"Only transfers already pledged."

"Cage authority?"

"Suspended under your separate notice."

He read the order on her tablet. It claimed implied custody powers survived Article Nine because the Compact's founding purpose preceded its procedural limits. The argument could persuade a frightened board. Courts might validate it after the crisis if enough people survived. Law often forgave power that produced a tolerable outcome.

"Would you obey this?" he asked.

"As counsel, I would record it. As a Crown resident, I would challenge it."

"Would you stop working?"

"No."

There lay the temptation. People would keep the pumps running while disputing him. He could take their labor for consent and call the absence of immediate mutiny legitimacy.

Cass handed the tablet back unsigned.

At 1740, Ada's shared conjunction solution reached all corridor captains. Instead of one command table, it contained six conditional burn families. Morrow selected the third. Nadi accepted its low-plane correction after

rerouting the evacuation shuttles. Two freight pilots refused the first corridor and spent reaction mass climbing above it. Their choices widened Morrow's miss distance at the cost of cargo deadlines and crew dose.

The method looked inefficient. It also worked.

System-wide, conditions worsened. Cinder Vale's local board rejected its assigned stabilization burn because the model understated a cracked drive mount. Luma Ring accepted a more dangerous burn to preserve water lift. Rook placed tug academy ships under habitat command. Harbor of Glass formed a yard captains' council and stopped forwarding requests through Crown.

One center broke into dozens while the gravity field still joined them.

Cass returned to operations at 1800. The legal counter occupied every display now, including his own. Staff kept glancing toward him for the order that would remove it.

He called the forty-six habitat executives into emergency session. Thirty-nine answered before light-time and local procedure exhausted the first fifteen minutes. Seven sent delegates. Belen joined from Morrow; Ada entered as a tug captain without vote.

"My Article Nine authority expires in forty-four minutes," Cass said. "Counsel finds no lawful path to renewal. I requested a continuity order and will not sign it."

Several delegates spoke at once. Crown's moderator tried to rank them. Cass disabled the queue.

"The forced recontainment order is withdrawn. Existing safety holds remain as advisories unless adopted by local authority. Compact operations will continue publishing trajectories, inventories, and conflict warnings. It will not issue binding tug or habitat commands after nineteen hundred."

A Rook delegate asked, "Do you endorse the Great Unbinding?"

"No. The migration model still carries collision risks I consider unacceptable. Release removes thirty-eight percent of dispatchable power. No complete aid compact exists."

"Then what do you propose?"

"A vote with the actual inventories attached."

Belen's face changed by less than a smile. "And before the vote?"

"Local emergency law. Captain consent. Voluntary coordination through the Compact table."

Far Rest's convener said, "Voluntary coordination from an office that tried to cage the Resonance yesterday."

"Yes," Cass replied. "Distrust it. Check the code. Use the staff you still find useful."

Ada spoke from her dim tug. "I need Crown tracking for the Aster Nine pass."

"You will have it."

"Without command tags."

"Without them."

The session recorded his declaration. Belen's broadcast added it beneath the countdown: FIRST CUSTODIAN STANDS DOWN FROM FORCED COMMAND AT EXPIRY. The sentence felt larger than the action. Cass remained responsible for every model Crown issued and every reserve its officers could lawfully move. He had not abandoned duty. He had abandoned the claim that duty made his choice binding on another hull.

At 1900, the Article Nine key disappeared.

No alarm sounded. Crown Meridian kept spinning. Operations staff stayed at their tables. Across the web, captains continued burns they had accepted and refused those they had not. The conjunction plot widened to a

survivable corridor as Morrow completed its sixth locally authorized segment.

One command did stop. Crown Tanker Eighteen's escort assignment dropped from its pilot's board halfway to Far Rest because no receiving authority had adopted the Compact route. The tanker captain held thrust rather than turn blind. Her projected arrival slipped by eleven minutes, enough to erase Far Rest's second-burn margin.

Cass called Saye Olan and the tanker together. "Adopt the existing route locally. No change to thrust required."

Saye checked with a habitat controller before answering. "Our emergency board can adopt cargo acceptance, not an escort contract."

"The distinction strands your propellant."

"It also decides who carries collision liability if the route crosses civilian traffic."

Tanker Eighteen's captain brought up three freight paths that Crown had suppressed under emergency priority. "Those ships regain right of way at expiry. I need their consent or a new plane."

Cass could no longer clear them with one tag. He opened the freight contacts, supplied the fuel deadline, and asked each pilot to hold position for twenty minutes. One agreed. Another demanded compensation for spoiled cargo. The third had already burned and could not return without violating crew reserve.

Ada entered the shared plot from *Common Measure*. "Raise the tanker half a degree. Morrow can accept a lower pass after conjunction."

"Morrow has not reviewed it," Cass said.

"They are reviewing now. I sent uncertainty bounds, not a mark."

Four local decisions arrived in uneven order. The tanker changed plane with seven minutes left in Far Rest's margin.

Nobody had commanded the maneuver. Nobody could guarantee the same process would close the next conflict in time.

Then requests changed color.

Without emergency priority, three hundred twelve messages entered Crown's ordinary queues. Habitat boards duplicated fuel claims. Two tugs accepted the same tanker escort. Cinder Vale and Bellweather selected stabilization arcs that crossed in thirty-one hours. No single office possessed power to separate them.

Cass stood at the crisis table with no command key and began calling both habitats. Around Ophion, authority had fragmented exactly when coordination mattered most. The work remained, stripped of the force that had once made answers arrive.

Chapter 18 — The Broadcast Trajectory

Ada removed her own command signature from the migration file and watched forty-six warning panes turn red.

The code interpreted the missing key as a failure. Burn tables grayed out. Automatic conflict resolution shut down. Every habitat on the system map became an independent source of uncertainty instead of a vessel awaiting one captain's mark.

"Leave it broken," she said.

Tomas looked up from the observatory link. Ninefold's model occupied half the dim command blister, twelve sink trajectories threading around Ophion's moons and seven regions where human habitats might survive their release. "It will not execute without an authority object."

"Give it forty-six."

"That multiplies branch conflicts."

"It exposes them."

Heat warnings pulsed along *Common Measure's* reduced bus. Without the port radiator wing, the tug could run its main computers for twenty-three minutes before shedding load into the cage gallery. Ada had spent nine of those minutes removing every function that let her override another hull.

Jalen's empty couch held a replacement guidance console and no pilot. His neck injury had put him on

Morrow beside Iri and Parveen. Sixteen crew remained aboard, redistributed across watches thin enough that sleep had become another consumable. Aster Three's amber trace moved behind them, unstable but no longer clamped.

Tomas rubbed both hands over his face. "Each habitat authority can validate its own branch. The global solver still needs common state."

"Common state, common constraints, public code. Not common command."

"If one station uploads false mass, everyone inherits the error."

"If Crown uploads false mass, everyone inherits it faster."

He gave her the patient look that meant an analogy had failed before he spoke it. "Distribution does not improve truth by itself."

"No. Belen's ledger might."

Ada connected the inventory feed. Fuel, coolant, flywheel charge, drive condition, spin tolerance, and evacuation capacity entered from all forty-six habitats. Every figure carried a source name and timestamp. Contested numbers remained visible beside official ones. Morrow's reserve still showed the twelve hidden tonnes Ren had confessed, now reduced by Tender Four's run to Far Rest.

The solver accepted forty-six local keys.

Routes returned in colored families rather than one preferred line. Morrow could reach resonant island Four through three burn sequences. One protected spin at the cost of radiator margin. Another spent more reaction mass but kept the clinic below its acceleration limit. The third depended on Aster Nine holding a behavior humans had observed only twice.

Ada marked the third unsupported.

"Do not delete it," Tomas said.

"It relies on a promise nobody made."

"Label it. Cinder Vale may need the same relation if its cracked mount prevents the high-thrust branch."

Ada restored the route and attached its uncertainty. Giving up command did not require giving up judgment. It required placing judgment where others could refuse it.

A secure medical circuit opened from Morrow. Iri Quill appeared propped against a clinic wall, left leg suspended inside an external fixation frame. Radiation treatment had stripped color from their face. A pouch containing the bent blue paint strip hung from the bed rail.

"You left a hidden master abort," Iri said.

Ada searched the code. "Where?"

"Thermal safety module. Tug class privilege can cancel a habitat burn if projected exhaust heats a cage carrier."

"That protects mass-transfer crews."

"It protects us by taking someone else's engine."

"An exhaust plume across Aster Three can destabilize the charge sheath."

"Then publish the exclusion volume and let their captain find another burn."

"Some will ignore it."

"Some central officers ignored my latch carrier too."

The reminder contained no cruelty. Iri had earned the right to use it anyway.

Ada found the master abort. Her finger rested above deletion while habit produced every catastrophe the switch might prevent. A habitat could burn into a sink corridor. A pilot could mistake degrees for radians. A compromised local board could sacrifice a tender to preserve its own reserve. One competent hand might still save them.

That hand had never remained abstract. It belonged to someone with incomplete data and a preferred victim.

She deleted the privilege.

Iri watched the code recompile. "Now add engineering veto at the local key."

"Already there."

"Crew review?"

"Required before execution."

"Public error log?"

"Tomas insisted."

"Then I withdraw my complaint until I find another."

Their circuit stayed open while they audited heat assumptions. Scientific mechanisms arrived as limits people could not negotiate away. A habitat burn changed its orbit only by throwing mass in the opposite direction. A flywheel could restart a dead grid but could not create momentum. The seven resonant islands offered corridors where the freed sinks' gradients partly cancelled; they did not provide fixed ground. Every habitat would keep adjusting after arrival with less power than it had used before.

At the twelfth minute, the blister temperature rose above duty comfort. Ada shut down two secondary displays and felt sweat gather beneath her collar.

"Broadcast window in eleven minutes," the systems officer said. "Crown and Ninefold relays synchronized."

Tomas added his translation layer. The Resonance model did not predict decisions by individual sinks because no individual sink decided. It mapped relations among all twelve, Ophion's five major moons, and long-period masses beyond the inhabited web. The free trajectories restored recurrences the cages had broken. Human migration opened space for that restoration while preserving seven imperfect intervals for habitats.

"What confidence do you assign?" Ada asked.

"To the gravitational integration, ninety-nine point eight percent inside twenty-eight days. To sink charge behavior, seventy-one. To our interpretation of permission, fifty-eight."

"Put those on the first screen."

"Communications wants the route first."

"People deserve to know which part might kill them before the animation comforts them."

The system officer muted an argument from Crown's presentation staff and kept Tomas's numbers.

Nadi Okafor joined from *Principle of Least Harm*. Her tug held a low orbit between Morrow and the Aster Nine corridor, drives cooling after the conjunction work. "Your branches send me around Aster Six on Day Twenty-three."

"That remains the lowest propellant route," Ada said.

"It brings my hull inside a gradient we have not measured since the clamp."

"Alternate Five keeps you outside."

"And adds four hundred tonnes to the coolant convoy behind me."

"Then refuse both until Tomas narrows the gradient."

Nadi smiled without warmth. "You finally learned to give unhelpful advice."

"I had a good teacher."

"The lesson carries a second half. A captain's authority ends at another crew's hull. If my safe route drives the coolant convoy through lethal dose, I do not get to call it local choice."

Ada put both trajectories on the public conflict board. "The convoy captains hold receiving veto."

"Good. No private bargains. Make us argue where everyone sees the exhaust."

Crown Meridian requested a delay. Cass Vardek appeared without the Article Nine insignia that had bordered his signal for thirteen days. Ordinary Compact operations now labeled him coordinator, a title with staff and no emergency key.

"I have reviewed release candidate nine," he said. "Independent burn selection produces two hundred fourteen unresolved crossings in the first six days."

"Seventy-three involve ships with no active route," Tomas replied. "The solver flags absence rather than inventing compliance."

"One hundred forty-one remain."

"Every one has a resolution family."

"Not an accepted resolution. If two habitats choose locally optimal branches, they may collide before negotiation catches up."

Ada enlarged the worst conflict. Bellweather's low-fuel route crossed Cinder Vale's low-stress route in thirty-one hours. Both habitats needed what the other's branch consumed.

"Your old solution assigned Cinder the high-stress burn," she said.

"Because Bellweather cannot make the alternate on present reserves."

"Cinder's drive mount may fail under yours."

"I know. Central choice does not make the cost disappear. It ensures the choices fit together."

"Then coordinate. Publish which reserve transfer opens a second fit."

Cass placed Crown's tanker pool into the model. One tanker moved early to Bellweather, letting it choose a

higher plane; Cinder kept the low-stress branch. The crossing vanished. A second appeared because the tanker's route cut Luma Ring's evacuation lane.

"This scales badly," he said.

"So did hiding the crossings under orders."

"I am not arguing for hidden costs. I am warning that independent burns can collide."

"Put that warning in the broadcast."

Cass studied her. "You intend to release code before any system vote."

"I intend to release the information needed for a vote. Nobody can consent to a route they cannot inspect."

"Someone will execute early."

"Execution keys stay local and require a recorded habitat decision."

"Emergency governments can manufacture those decisions."

"Your emergency government manufactured a clamp."

Tomas shifted between their panes before anger narrowed both. "Crown's conflict service can publish verified pairings without owning execution. Ada's solver can refuse to certify a branch until intersecting hulls accept. Neither mechanism requires a master key."

Cass considered the change. "Certification cannot become a de facto veto."

"No," Ada said. "Habitats may execute uncertified burns after publishing the risk."

"You are allowing unilateral maneuvers through common space."

"They can do that now. The code makes the consequence harder to hide."

A message from Belen entered the public queue. MORROW REQUESTS FULL RELEASE. LOCAL VOTE WINDOW OPENS UPON RECEIPT.

Cass read it. "You have already promised her the file."

"I promised everyone the same time."

"Then attach Crown's warning without editing."

Ada accepted. Cass had not endorsed the migration. He had chosen to improve the plan he opposed rather than reclaim control through expertise. The distinction mattered.

At broadcast minus three minutes, *Common Measure's* heat budget reached its limit. Ada powered down the command blister's main projection. Darkness swallowed Ophion's model, leaving only emergency strips and the amber pulse from Aster Three's status light.

"Can Ninefold originate?" she asked.

Tomas checked relay geometry. "Yes, but your signature belongs on the navigational assumptions."

"My signature belongs on what I calculated. Yours on translation. Belen's sources on inventory. Cass's staff on traffic."

"People expect a captain to tell them when to burn."

"Then disappoint them accurately."

The full migration solution left Ninefold at 1200 system time.

It did not arrive everywhere together. Light-time remained small within the inhabited web, but relay checks, local firewalls, and damaged receivers added seconds that mattered to frightened people. Crown Meridian received first. Morrow followed. Far Rest acknowledged through a tender antenna because its main array had lost coolant. The forty-sixth receipt came from a mining torus called Long Wake, forty-eight seconds after release.

Ada opened the common channel.

"This file contains a twenty-eight-day migration, release trajectories for all twelve sinks, and seven provisional resonant islands. It contains source code, test cases, rejected routes, current inventories, and every uncertainty we could defend. It removes thirty-eight percent of dispatchable sink power. It does not preserve present wealth or guarantee every burn."

She saw communications staff signaling for slower delivery. Ada kept her operational cadence.

"Each habitat chooses whether to enter the plan. Each habitat selects and executes its own burns. Any burn that crosses another hull's safety volume requires that hull's consent for certification. Captains retain local refusal. Engineering crews retain local veto. Crown tracking and Ninefold models will update a common state without remote execution authority."

A hundred questions entered before she finished. She ignored the ranked queue and continued.

"Do not treat absence of a warning as proof of safety. Publish errors. Name reserve claims. If you cannot make a branch, say what resource opens one. No tug, including *Common Measure*, will fly your habitat for you."

Ada ended there. A speech about freedom would have insulted the work. Forty-six hulls needed to choose burns, not admire the person who had sent them equations.

The first code challenge arrived from Rook Academy. Its dynamics team found a sign ambiguity in one coordinate transform used by three older habitats. Tomas confirmed the core solver handled it correctly, but the human-readable guide did not. They patched the guide publicly and restarted local verification clocks.

Next, Long Wake found a coolant density conversion copied from Compact tables at the wrong reference temperature. Its correction reduced Luma Ring's second-burn margin by six minutes. Iri traced the error

from their clinic bed and attached a note containing six words: TEMPERATURE COUNTS EVEN IN A TREATY.

Cinder Vale rejected all certified branches. Its engineers streamed images of the cracked drive mount, showing damage worse than the inventory field suggested. Ada's instinct reached toward a solution before the local captain finished. She held silence. Bellweather offered to delay its own burn if Crown's tanker transferred propellant first. Cinder's board began calculating.

The common state grew messier and more true.

A request from Ketu arrived outside the ranked queue because its safety board had marked the migration file itself as hazardous. The solver's default objective minimized expected deaths across all habitats. Under one rare Aster Four excursion, that objective delayed Ketu's correction to preserve two larger cylinders, reproducing the geometry that had killed six hundred twelve people in the Spin Loss.

Ada opened the test case. The probability sat below one percent. The consequence placed Ketu outside recoverable attitude and saved an estimated nine hundred people elsewhere.

Tomas read her silence. "Global minimization needs a tie-breaking rule."

"It already has one. Population."

"Removing it can increase modeled deaths."

"Keeping it chooses a hull before anyone votes."

Ketu's safety chair joined by audio. "We will not certify code that contains our abandonment as an automatic branch."

Ada could have labeled the case impossible under current confidence. Instead she changed the objective. The solver would publish expected casualty differences but could not certify any branch that deliberately moved one habitat beyond recovery without that habitat's explicit

consent. Conflicts with no mutually survivable solution would remain unresolved.

The next run returned six additional red pairs.

Cass saw them through Crown's verification feed. "You replaced an answer with warnings."

"I removed a concealed sacrifice."

"Some pairs may never clear."

"Then everyone sees what aid must change."

Ketu's board accepted the revised test suite for local review. One of the new red pairs connected Ketu to Rook, where Ada had been born and trained. Rook offered two academy tankers before its own migration vote opened. Their transfer did not close the conflict, but it narrowed the propellant gap enough for a different burn family.

No central plan would have spent academy reserve before securing Rook's consent. No distributed plan could spend it either. The offer created a route because the people losing margin had named what they would bear.

Habitat votes opened under different laws. Rook required two assembly readings separated by an hour. Crown Meridian could act through an executive board but chose a resident referendum after public pressure. Far Rest's emergency charter gave its agricultural and drive crews joint veto. Harbor of Glass voted by yard cluster because no single pressure shell contained a majority of its people.

Morrow moved fastest.

Belen's channel showed the old agricultural drum packed from floor to terraces. The dead grow lamps had become display rails for the migration branches. Residents could see the route that spent their drive mass, the route that risked clinic acceleration, and the unsupported route dependent on Aster Nine. Beside them hung Crown's warning about collisions.

Ada contacted Belen privately. "Your assembly has not completed technical review."

"Ren's crews completed local review. Clinic signed the acceleration limit. Traffic signed branch two."

"Branch two spends nearly all reaction mass before island entry."

"We know."

"If the Resonance does not widen the final interval, Morrow cannot retreat."

"That risk appears in our vote packet."

Ada felt the old need to count seals twice. "I can give you another six hours of modeling."

"What changes in six hours?"

"Confidence, perhaps."

"Or another order disguised as caution." Belen softened the words by refusing to soften the fact. "Morrow heard your uncertainty. Let us decide with it."

The connection closed.

Nadi sent her own public endorsement. She did not promise safety. "I support beginning the Great Unbinding because another cage attempt exports its failure to every hull. My ship will fly only trajectories my crew accepts. We will not cross another crew's hull without permission. No central mark can replace that boundary."

Cass followed with an administrative notice. Compact staff would verify conflict pairs, maintain the common state, and execute aid transfers adopted by source and receiving habitats. He added his warning in full: simultaneous independent burns could produce collisions beyond present models; certification reduced but did not eliminate that risk.

Votes accumulated without resolving into a system total. Eight habitats entered tentative yes positions. Five rejected

migration and demanded new cage research. Four postponed pending propellant guarantees. The rest remained in assembly.

Morrow's ballot closed after thirty-seven minutes.

Belen did not announce the count immediately. She waited while election workers reconciled pressure-sector rolls and three exterior crews transmitted delayed votes. Ada watched their verification marks appear one by one.

"Yes carries," Belen said. "Morrow consents to branch two, conditional on certified clearance from every intersecting hull. We authorize the flywheel vault for black start and initial attitude coupling. We do not authorize any outside pilot to execute our burn."

Ren appeared on an exterior camera beside the vault's magnetic vanes. His crushed hand remained fixed in its brace while his other hand worked the mechanical release. Around him, Morrow's old torus curved into darkness, patched radiators catching the dim K-star.

"Vault crew ready," he reported.

The resident assembly gave final confirmation. Deep inside the habitat, a forty-kilometer ring began feeding stored rotational energy into a grid built for sink power. Lights brightened along the farm drum. Pumps took load. Attitude banks charged for the first migration burn.

On Ada's display, Morrow changed from a red warning object into a blue source with its own execution key.

She transmitted the latest common state and left the key untouched. The old flywheel vault climbed toward operating speed under Morrow's command.

Chapter 19 — The Vote of Hulls

Belen refused to count Crown Meridian's yes until its propellant crossed the berth.

The cylinder's ballot had closed nine minutes earlier with a clear mandate to begin migration. On the system tally, Crown's seal hovered green beside a transfer pledge of one hundred twenty tonnes. Its tankers remained clamped inside a security zone.

"Launch authority has passed," Cass Vardek said from Crown operations. "Traffic needs twelve minutes to clear the departure plane."

"Then your vote counts in twelve minutes."

"The ballot does not depend on cargo movement."

"Cinder Vale's ballot depends on that cargo."

Belen stood beneath Morrow's dead grow lamps with the forty-six habitat decisions arranged above her. Nine had voted yes. Six had voted no. Morrow's first yes carried a blue execution mark because its flywheel vault already fed the attitude grid. The other green seals remained hollow until each habitat's aid commitments became physical transfers.

Cass studied the tally. Ordinary Compact insignia bordered his channel; no command key appeared. "You cannot alter a certified vote after receipt."

"I am not altering it. Crown said yes to a migration that requires Cinder and Bellweather to burn first. I am asking

whether Crown also said yes to the fuel that makes those burns possible.”

“The executive release has legal force.”

“So did your cage order. Launch the tanker.”

A Crown traffic officer spoke offscreen. Cass listened, then supplied a departure identifier. Tanker Eighteen had returned from Far Rest nearly empty. Tankers Seven and Twenty-two now carried the richer cylinder’s pledge. Their planned split sent eighty tonnes to Cinder Vale, forty to Bellweather.

“Both clear in eleven minutes,” Cass said. “Mark them against Crown’s obligation now, or your tally will misstate a valid vote.”

Belen considered refusing until drive light appeared. An honest ledger did not mean inventing uncertainty where a named crew, cargo, and time already existed.

She filled half Crown’s seal. “Launch commitment recorded. Delivery remains open.”

“Half a vote has no legal category.”

“Half a promise has a long history.”

Behind Belen, residents staffed Morrow’s emergency coordination tables. Thermal crews shared space with election workers, clinic dispatchers, and tug pilots waiting for routes. Ren sat at the exterior board with his braced hand secured to his chest. He had not slept since opening the flywheel vault. Every few minutes he flexed the two fingers that still answered him, checking function the clinic had told him would not return quickly.

The system tally changed. Harbor of Glass voted yes through thirty-one yard clusters, but its aid clause preserved all local coolant until Aster Seven’s damaged district stabilized. Far Rest’s convener immediately challenged the result.

"You cannot join a shared burn and hold every canister," Saye Olan said. "Cinder's second correction needs coolant before its first burn."

Harbor's yard speaker answered from a pressure shelter patched after the cage rupture. "Our districts remain black. Twelve thousand workers live under temporary heat rejection. We will not export their margin."

"Nobody asked for all of it."

"Your request takes us below yard emergency reserve."

Belen opened Harbor's inventory. The official emergency reserve included coolant assigned to six commercial keels with no crews aboard. Their owners had filed protection claims based on future delivery contracts.

"Speaker, do your twelve thousand workers use the keel stock?"

"The stock sits on our grid."

"That does not answer."

A yard engineer interrupted. "We can isolate forty-eight tonnes from Keels Four and Six without touching residential loops."

The speaker muted their channel. Belen kept the engineer's statement visible. Seconds later, images from Harbor showed workers already uncoupling the commercial lines.

"Forty-eight does not close Cinder's need," Cass said.

"It closes the coolant part," Belen replied. "Crown's tankers close fuel. Rook still owes bearing stock."

Rook had not voted. Its assembly sat in a second reading while academy families argued that the tug school must retain enough equipment to evacuate cadets. Belen understood the fear. Rook had taught generations to trust redundancy because a habitat never discovered its missing

margin at a convenient time. It had also accumulated eight times Morrow's bearing reserve per resident.

She requested the Rook floor.

An academy moderator answered with formal reluctance. "External speakers may address the assembly after resident debate."

"Cinder's drive mount crosses repair limit in thirty-seven minutes."

"Our procedure guarantees equal access."

"Your procedure has more time than their metal."

The moderator offered a written submission. Ren leaned toward Belen's pickup.

"Ask the repair cadets," he said.

Belen switched to the academy maintenance circuit instead of the assembly. Forty-three instructors and cadets appeared around a disassembled training drive. She showed them Cinder's cracked mount, Harbor's coolant release, and the exact bearing sizes Rook held.

"Can you spare them without disabling your evacuation fleet?"

An instructor named Ko Mara counted stock in public. "Twenty-two sets spare. Cinder asks for fourteen."

"Crew to move them?"

"Academy tender. Six cadets and two licensed pilots."

"Do they consent?"

The eight crew answered individually. One cadet declined after learning the route crossed Aster Two's widened gradient. A reserve pilot took the berth. No assembly had yet authorized the cargo.

Ko Mara faced the maintenance circuit. "We request emergency shop release."

Rook's moderator reappeared, anger controlled by training. "You bypassed the assembly queue."

"I found the people who knew whether the transfer could happen," Belen said. "Your assembly still decides whether it will."

"You created public pressure with an incomplete safety review."

"Yes. Complete it fast."

The Rook feed closed.

Two no votes arrived while they waited. Nacre Habitat rejected the Great Unbinding because its residents believed the Resonance grammar remained an artifact of human analysis. Pallas Green rejected migration while requesting continued access to any power the freed sinks provided. Belen entered both without commentary. Operational consent meant the right to refuse carried the same visibility as agreement.

Ada Narang joined from *Common Measure*. Her tug's thermal limits reduced the image to a grainy monochrome. "Pallas occupies the inside edge of resonant island Two. If they do not move, three yes habitats lose their certified approach."

"Can the route go around?" Belen asked.

"With added reaction mass none of them carry."

"Can Pallas remain?"

"Only if Aster Eight follows the low-probability branch. We cannot require that relation."

Pallas's council chair heard the exchange. "You call our no a right, then treat our orbit as obstruction."

"Your hull belongs to you," Ada said. "The volume your gravity occupies affects other hulls."

"We will not burn on an alien promise."

"Then help choose a holding correction that opens the approach."

"Who pays for it?"

Belen displayed the unallocated propellant pool. "Nobody yet."

The three affected habitats each offered five tonnes. Pallas needed twenty-one. Crown operations identified six tonnes in a tanker reserve that Cass could offer only with Crown board approval.

"Request submitted," he said.

"Not enough," Pallas replied.

"Enough if your burn begins after the first approach pair." Ada drew a slower correction. "You keep your no. They keep a corridor. Your safety board must verify it."

Pallas accepted the model for review without changing its vote. A refusal became an operational obligation rather than exile from the table.

Crown's tankers launched. Their drive signatures rose on Morrow's tracking wall, two thin flares departing the administrative cylinder on separate planes. Belen completed Crown's green seal. Cinder Vale opened its delayed ballot as soon as tanker telemetry authenticated cargo mass.

"Your mechanism worked once," Cass said. "It will not scale to every transfer."

"Then give me something that does."

He sent a compact document. The proposed aid bond converted each yes vote's declared transfers into enforceable habitat obligations. Cargo acquired a receiving owner at launch, preventing source councils from recalling it after poorer habitats committed. If a migration branch failed, unused aid remained with the designated receiving habitat rather than returning automatically to the donor.

Belen read the remedy clause twice. "Enforced by whom?"

"Ordinary Compact contract courts."

"Those courts cannot stop a tanker in the next hour."

"No. Captains can refuse recall orders once ownership transfers."

"Put that sentence in the bond."

"It already follows from title."

"Put it where pilots see it."

Cass amended the text. Belen added a public ledger condition: a vote carried no aid designation unless cargo, crew consent, launch window, receiver, and failure ownership appeared together. Lawyers from four habitats objected to creating property rights under emergency pressure. Far Rest's Saye answered that poor habitats had lived under emergency property rules for generations; this version finally pointed the protection toward them.

The bond went to all forty-six ballots.

Voting accelerated. Some assemblies treated the aid condition as proof that migration could work. Others saw a permanent claim on local reserves and turned no. Rook paused its second reading to adopt the bond. Harbor of Glass approved the forty-eight-tonne coolant transfer by yard vote, overriding the commercial keel owners under local salvage law. Its green seal filled when the first canister train left Keel Four.

Cinder Vale voted yes by a narrow margin. Its acceptance carried conditions: Crown's eighty tonnes had to clear midpoint, Harbor coolant had to dock, and Rook bearing stock had to leave before Cinder ignited. The conditions entered the common solver. Cinder's route changed from unsupported red to provisional amber.

"Ten yes," an election worker reported.

"Eleven," Ren corrected. "Long Wake just cleared."

"Keep habitat names, not only totals," Belen said.

The worker expanded the board. Numbers concealed sequence, and sequence held the aid.

Luma Ring voted no after its water crews concluded migration acceleration could rupture two lift mains. Ada published a lower-acceleration route that required another day and thirty tonnes of fuel. Luma refused that too. Its council wanted recontainment research continued.

"You are one of the six that lose service first," Belen told them.

"We know our pipes," the water convener said. "A yes does not strengthen them."

"Will you accept repair crews from Morrow?"

"Yes. We will not accept a migration burn."

Ren offered two rig teams. Their transfer entered the aid ledger without requiring a vote change. Morrow could argue, not purchase agreement.

At thirteen yes, the tally stalled. Sixteen habitats waited for resources before committing. Nine reserves named for those resources remained conditional on a system majority. Everyone wanted proof of agreement before giving up the margin that made agreement possible.

Belen projected the loop on every channel. "Rich habitats say they transfer after the vote. Poor habitats say they vote after transfer. That means no one moves and the sinks continue widening phase."

A Crown delegate proposed escrowed title without launch. Far Rest rejected it. Cinder's cracked mount did not run on legal ownership.

Rook broke the loop.

Its assembly passed yes and released fourteen bearing sets, two academy tankers, and thirty tonnes of reaction

mass before a majority existed. The academy tender pilot transmitted crew acceptance. Seven trainees watched their equipment leave the dock on a route toward Cinder Vale.

Ketu followed. Survivors of the Spin Loss had demanded a no-sacrifice clause in the solver. With that clause public, its assembly accepted migration and offered four station-keeping tugs to the first wave. Its vote brought the total to fifteen.

Crown increased its aid bond by twenty tonnes. Harbor released spare radiator panels. Ninefold opened observatory computing to local verification teams rather than reserving it for Compact models. Each action moved another ballot.

Belen ate half a ration bar while standing at the ledger. Mara's son slept behind the clinic table with an oxygen tube beneath his nose. The flywheel grid kept his bottle charger live. Every yes on the wall translated into a burn that could stop that current; every no preserved a cage system already failing.

Ren tapped the exterior board. "Rook tender has a route conflict with our rig teams to Luma."

The two vessels had selected the same narrow gap above Aster Five. Under old priority, Compact traffic would have delayed Morrow because Cinder's repair ranked closer to system survival. Now both captains entered the conflict channel.

Rook's pilot could rise at a cost of two tonnes. Morrow's rig tender could descend but would cross a radiation lobe, spending crew dose. Neither exported the cost silently. Harbor offered a refueling drone at the high point if Rook took the rise. Its controller named a later obligation for Morrow to repair the drone's damaged coupling.

Three crews accepted. The routes separated.

Belen recorded the repair debt beside the fuel transfer. Mutual aid did not erase accounting. It changed what the account protected.

By midday, twenty-three habitats had voted yes, eleven no, and two abstained. The abstentions came from small research wheels that could reach safe holding orbits under either plan but refused to interpret alien cognition through a political ballot. They agreed to honor traffic boundaries and aid bonds without authorizing sink release.

Ten ballots remained.

Cass built a logistics pane from their unresolved conditions. Four needed fuel already moving. Two required conflict clearance from no-voting neighbors. One demanded proof that Crown would ration at the same level as lower-power habitats. Three remained opposed under any present model.

"You are doing Compact work," Belen said.

"I have not stopped working for the Compact."

"You are assigning routes."

"Offering them. Captains accept."

"And the three unconditional no votes?"

"I cannot improve an answer by relabeling it."

He had not reclaimed command. The staff behind him still possessed tracking, contracting, and launch expertise built through a century of centralized practice. Belen wanted that capacity separated from custody, not burned for symbolism.

"Send the logistics pane under your name," she said.

"Some habitats will reject it because of my name."

"Then they know whom to question."

Cass published.

Bellweather voted yes after Crown Tanker Twenty-two crossed midpoint. Far Rest reaffirmed its earlier conditional yes when Tender Four's nine tonnes entered certified drive storage. Three midweb habitats followed their aid routes. The count reached twenty-seven.

Cinder's repair feed showed Rook's tender docking under manual control. Cadets moved bearing crates through an airlock too small for the standard pallet. Harbor coolant remained fourteen minutes away. Cinder opened its final execution consent before the canisters arrived, trusting two physical trajectories rather than two distant promises.

Luma Ring reopened debate. Morrow's rig teams had not asked it to reverse the no; they had sent repair plans and crew dose limits. Water workers found they could sleeve one lift main before the migration window. The second required Harbor composite. Belen located panels pledged to nobody and asked Harbor's yard crews, not the commercial board. The crews released them under the aid bond.

Luma changed from no to yes.

Twenty-eight.

Three large habitats remained conditional on majority. Belen addressed them together. "You each say yes if thirty others move first. There are only twenty-eight yes because you are each waiting for the other two."

A delegate called the condition prudent.

"It protects your reserve while Cinder repairs under a clock."

"Without a majority, release can strand us in a minority migration."

"The bond activates at thirty-one. Sign it now with launch at threshold."

Two agreed. One refused. Their votes entered simultaneously, bringing the count to thirty.

The final needed ballot belonged to Pallas Green, already no. Its holding correction had cleared the island. Two approach after Crown supplied the missing six tonnes. Pallas kept its refusal and performed the burn anyway because other hulls needed room.

Ada's voice reached Belen. "Mica Torus completed review. They can vote, but their coolant aid to Luma lacks a return guarantee."

"Unused aid stays with Luma under the bond."

"Mica wants replacement from Crown after island entry."

Cass checked reserves. "Crown can guarantee half after rationing begins. Not all."

Belen called Mica's assembly. "Half replacement, named schedule. The remainder becomes your contribution."

"We did not cause Luma's shortage," its convener said.

"No. Luma did not cause the cages either."

Their vote window closed in two minutes. Belen did not fill the silence with an appeal. Mica residents watched Luma's repair feed, Crown's guarantee, and the exact coolant quantity they might never recover.

The seal turned green.

"31 yes," the election worker said.

Belen checked the names before the number. Thirteen red seals remained. Two gray seals marked abstentions.

"Read the complete tally."

The worker did. "31 yes. 13 no. 2 abstain. Threshold met for operational consent to begin emergency migration and coordinated sink release planning."

"Not a new Compact," Belen said.

Cass added the legal line. "This vote authorizes the emergency operation. It does not ratify permanent governance, ownership rules, or post-crisis allocation."

No cheering crossed every habitat at once. Morrow's drum gave a brief, exhausted roar. Cinder's repair crew kept working. Pallas published its no beside the corridor it had opened. Crown's tankers held their burns.

Aid bonds activated at threshold. Launch marks spread across the ledger, but donor habitats retained captain and engineering consent. Cass's logistics staff issued proposed sequence without command tags. Ada's solver certified the first six branches after every intersecting hull accepted.

Morrow. Cinder Vale. Far Rest. Ketu. Bellweather. Luma Ring.

Luma's late yes placed it back among the six at risk, now carrying Morrow rig teams aboard. The first wave did not share politics, wealth, or confidence. It shared deadlines.

Ren's exterior board reported Morrow ready for separation burn. Belen walked from the grow-lamp tally to the traffic console. Branch two demanded nearly all their reaction mass and left no retreat if the final interval narrowed. The assembly had accepted that risk. Her task no longer involved deciding for them; it involved keeping the inventory honest while they moved.

"Crown logistics proposes ignition in eight minutes," Cass said.

"Morrow traffic?" Belen asked.

The local controller checked Aster Nine, the rig tenders, and the certified consent field. "We accept eight."

One by one, the other five habitats accepted or shifted their marks. Cinder delayed for Harbor coolant. Far Rest moved first to clear its failing agricultural margin. Ketu's tugs spread around the wave without taking their helms. Bellweather selected the higher plane its delivered fuel had

opened. Luma held until the second water-main sleeve sealed.

Far Rest lit its drives.

The habitat's signal took seconds to reach Morrow. On Belen's wall, its old orbit separated into two lines: the path that had depended on captive sinks and the poorer, survivable arc its residents had chosen. Ketu followed. Bellweather began to climb.

Morrow's flywheel vault released attitude power. Ren called ignition from the exterior rack. Belen counted the six blue execution keys, each held on its own hull.

The first wave pulled away from the old web, not together, but bound by the fuel already moving between them.

Chapter 20 — Black Start

Morrow's last borrowed megawatt vanished while eighteen thousand people still occupied the wrong side of the habitat.

Ada watched the external feed fall through amber, hesitate at two percent, and flatten. On the tactical plot, the torus continued along its departure arc with no visible change. A habitat did not flinch when its power died. Pumps slowed inside walls. Air lost its planned direction. Bearing fields softened around a spinning mass large enough to tear itself open.

"Morrow confirms sink supply at zero," Parveen said. "Their local grid has not formed."

Ada kept both hands loose above *Common Measure's* controls. The tug flew three kilometers beneath the torus, close enough to advise and far enough that a mistaken burn would not drive its trillion-tonne passenger into Morrow's hull. The cage behind them still held Aster Three. Its phase trace moved across Ada's lower screen in eleven linked curves and one long, delayed return.

"Do they request tug authority?" she asked.

"No request."

Good. Belen's people had voted to move their own hull. Ada had published the trajectory on that condition, then spent two sleepless days wanting to take it back.

A voice came through Morrow's engineering circuit. "Vault team to grid. Flywheel Two has field current, but our bus refuses phase lock."

Ren Sato answered from outside the habitat. Radiation roughened his transmission. "Attitude cluster four just dropped. We are two degrees off the black-start window and walking."

Ada enlarged the old flywheel vault's telemetry. The forty-kilometer composite ring still carried energy from the crisis on Day One, when Belen had opened its field to pull *Common Measure* clear of a passenger lane. Since then, Morrow's crews had fed every spare watt into its bearings. The ring could wake a dead grid. It could not correct a habitat's attitude while three exterior jets remained cold.

"Morrow control," Ada said, "your bus will lock if you arrest the roll inside ninety seconds. Current error point one eighth degrees per minute."

Belen's face appeared in a side window, shaved close at the temples and gray with fatigue. A strip of emergency lighting crossed her cheek. "We see the error."

"You have sixty-eight seconds before the vault inverter rejects another start."

"We see that too."

Ada made herself release the transmit key. Advice did not become more useful through repetition.

Beyond the blister, Morrow turned against Ophion's copper clouds. Its inhabited torus enclosed a dark central hub and three service spokes, all scarred by additions nobody had expected to survive this long. The migration burn had moved it out of the old sink-power web. If the flywheel failed to catch the grid, Morrow would have to abandon thrust, coast across the first resonant island, and try again after batteries recovered. Agricultural cooling would fail before then.

"Ren, cluster four pressure?" Belen asked.

"Propellant reaches the manifold. Igniters show open."

"Can you bridge locally?"

A pause carried the scratch of a suit against metal. Ren's right hand remained a rebuilt grip inside his gauntlet, two fingers replaced after the radiator accident. "With a hard jumper."

Ada pulled his exterior position onto the plot. He clung to the torus rim beside a dead attitude housing. A safety line tied him to an anchor sixteen meters aft. Three other workers occupied nearby stations, each too far away to reach him before the start window closed.

"Your jet will fire under you," Ada said.

"That's why they put the handles on the quiet side."

"Exhaust scatter reaches your anchor."

"I know where my anchor lives, Captain."

The rebuke landed cleanly. Ada acknowledged it by closing her mouth.

Ren opened the housing. A burst of white crystals streamed away, frozen residue from the purge line. His suit camera showed two contact blades separated by a thumb's width and a dark patch where an optical igniter had burned out. The automatic repair arm sat folded beside him without enough bus voltage to move.

"Forty seconds," Parveen said.

Morrow's vault team spun the inverter again. Frequency climbed, slipped, and dropped. The flywheel kept turning behind its magnetic bearings, holding enough stored work to light clinics, pumps, farms, and drives. A difference of two degrees denied all of it.

Ren reached into the housing with a copper jumper. His damaged glove could not close around the insulated grip. He trapped it against his palm with the other hand and leaned across the jet's bell.

Inside Morrow, the dead bus had already changed gravity's small habits. Condensation left the clinic collectors in round drifting beads. Farm pumps coasted

down one row at a time. A surgical team worked under battery lamps while an orderly turned a hand crank to keep blood coolant moving. Belen placed those loads on the start board in the order local crews had chosen: clinic loop, water, attitude, agriculture, then residential circulation. The flywheel held energy enough for all five, but only if the first surge did not trip it dark again.

Ada saw the unsafe angle. Every trained response demanded an order. She laid her left thumb against the pale burn ridge and counted one breath before speaking.

"Ren, your shoulder blocks the plume lip."

"I have it." He shifted four centimeters. "Morrow, bridge going live."

The attitude jet struck.

Ren's camera filled with glare. His tether snapped taut, and the horizon spun behind him. Cluster four drove the torus against its slow roll. On Ada's plot, the phase error narrowed through one degree, half a degree, then a tenth.

"Cut it," Belen ordered.

The jet stopped. Ren swung at the end of his line, alive but not answering.

Morrow's vault inverter closed.

Power crossed the habitat in a hard sequence. Bearing fields first. Coolant trunks next. Clinic, air, command, then the agricultural loops. Rows of status markers turned green along the torus, each one representing machinery that had remembered how to start without the sinks. Flywheel Two surrendered rotation to the grid and slowed by fractions too small for Ada to see.

"Black start confirmed," Belen said. Her voice shook once and steadied. "Exterior, report."

One of Ren's workers crawled along the rim and caught his tether. "Foreman conscious. Suit pressure good. Shoulder may be out."

Ren's reply arrived underneath hers. "Shoulder remains attached. Cluster four does not."

Belen closed her eyes. "Bring him inside."

Ada looked away from the recovery camera and checked Morrow's new path. The burn had lost eleven seconds. Local pilots had already redistributed the impulse among six engine groups. Their correction matched the published margin without a command from her tug.

"Morrow," she said, "your preliminary island intercept holds at plus nine kilometers radial. No further correction for forty-three minutes."

Belen studied her own board before answering. "We agree. Keep your mass off our plane."

Common Measure rolled away under one percent thrust. Missing its port radiator wing gave every burn a thermal clock. Parveen called temperatures. Jalen protected the outbound corridor. Sixteen people conducted the tug without waiting for Ada to name each necessary action.

Aster Three's trace left a gap during Morrow's correction, then resumed. Ada saved the interval for Tomas.

By Day Twenty-One, five other habitats had followed Morrow out of the old web.

Far Rest burned first. Its agricultural circulation had less than a day before root beds began to die, and the nine tonnes sent from Morrow bought only nineteen minutes of tolerance at the second correction. The local pilots divided that correction across three drive groups so no single valve crossed its declared limit. Far Rest reached its island with seed ice intact and no propellant for a return.

Ketu followed under a bearing budget rather than a fuel budget. Every impulse fed a vibration through drums rebuilt after the old Spin Loss. Its navigators chose a longer arc that kept peak load below the surviving races. Ada published an alternative, then watched Ketu reject it

because their maintenance council had measurements her tug did not.

Bellweather carried enough drive mass and radiators fouled by years of concealed maintenance debt. Its thermal crews shut two residential sectors before ignition and moved families toward the axial markets. The habitat burned higher than Ada's preferred branch, trading propellant for a cooler second correction. Its own pilots accepted that exchange.

Cinder Vale waited for Harbor coolant to reach bladders whose inner layers had cracked. When the tanker docked, Cinder's engineers rejected the first transfer clamp and made everyone lose twenty-seven minutes replacing it. No one called the delay cowardice after the pressure test showed the discarded seal would have failed under acceleration.

Luma Ring went last. Its extra water could not lift through damaged mains, and the migration plan had counted that water as thermal mass. Morrow rig teams installed a sleeve while Luma rotated beneath them. The local controller held ignition until the last worker crossed into the tender, then used every remaining second of the corridor.

Luma's water chief refused the first acceleration profile because it pulled the sleeve sideways across an old weld. The refusal cost a launch minute the model had treated as expendable. Ada recalculated with a lower initial thrust and sent the curve without a command seal. Luma's pilots adopted it after their own stress board cleared the weld. The ring left the old web late, not damaged.

Across the system, the first six habitats took up positions around Ophion in separated resonant regions, no longer linked by the old ninety-one-day sink circulation.

Ada marked each preliminary capture only when its local navigator transmitted confirmation. Morrow. Far Rest. Ketu. Bellweather. Cinder Vale. Luma Ring. Six habitats

that the Day Fifteen migration model had placed at the front of its death table now occupied viable arcs.

No cheer crossed *Common Measure's* command blister. The migration still held forty habitats inside the failing web. Eleven sinks moved through release corridors whose relation shifted every time a habitat burned. Their successful first wave had consumed nearly every generous margin.

On the morning of Day Twenty-Two, Iri called from Morrow's clinic.

They had angled their bed toward a wall display and propped their splinted left leg beneath a thermal blanket. Radiation burns mottled one side of their neck. The blue paint strip hung in a specimen sleeve beside the screen, returned from the ruined port wing after the evacuation.

"You look badly arranged," Ada said.

"Clinic objects when I improve the arrangement." Iri enlarged a thermal model. "Your starboard radiator reaches civil limit after seven minutes at current cage load. You get four if Aster Three drops phase again."

"We know."

"Knowing has never cooled a ship."

Ada glanced at Parveen, who had already folded Iri's limit into the burn board. "What do you need from us?"

"Nothing. You asked for an independent review, so I independently dislike everything." Their humor thinned. "Morrow's vault bearings survived within twelve kelvin. They cannot repeat black start for nine days."

"Nine should be enough."

Iri stared at her until Ada heard the old habit in her own sentence: compressing uncertain work into confidence because people needed a captain who sounded certain.

"Nine belongs in the model," she corrected. "We will not plan a second start."

"Better." Iri shifted against the bed and winced. "I also checked the coolant transfers. Somebody counted Crown's sealed reserve twice."

Ada opened the migration inventory. Belen's public ledger listed outbound coolant from Crown Meridian under both Compact emergency allocation and a direct habitat pledge. Same tanks. Same mass. Two promises.

The next wave contained eight habitats. Their radiators had to operate through longer burns because Aster Four and Aster Ten constrained the available planes. Without Crown's duplicated reserve, the wave fell short by twelve thousand tonnes of ammonia-water mix.

"Has Belen seen this?" Ada asked.

"I sent it to her, Cass, and every receiving engineer. No private surprises."

A notice populated the shared board while Iri spoke. Belen had flagged the duplicate entry in red and revoked both allocations until Crown identified physical tanks.

"Do not solve it from your tug," Iri said.

"We carry nine hundred tonnes."

"Your cage loops need all nine hundred. This counts as my engineering veto, in case your memory lost the phrase with the radiator."

Ada felt the immediate argument form: *Common Measure* might not need every tonne if Aster Three remained quiet. She also felt where that argument ended, with sixteen people bearing a risk she had converted into a system reserve without asking them.

"Veto entered," she said.

Iri nodded once. "Now you are allowed to look worried."

The coolant council opened an hour later across thirty-two delayed feeds. Belen chaired from Morrow's maintenance gallery. Cass Vardek appeared without the First Custodian's command seal. Crown's reserve director brought tank manifests. Habitat engineers brought temperature curves and the plain impatience of people whose machinery did not care about institutional rank.

Ada attended as *Common Measure's* captain, one voice among vessels and habitats. She gave the cage loop requirement and did not offer its coolant again.

Crown's director confirmed the shortfall. Sealed tanks existed beneath the administrative cylinder's inner farms, but releasing them would force Crown to reduce residential cooling and shut its tropical agriculture ring. He requested that the next wave pause for six days while a comet processor refined replacement stock.

"Six days closes four release corridors," Tomas said from Ninefold. "The sinks do not hold the migration geometry for us."

"Then move fewer habitats."

Belen brought up the casualty map. "Name which ones wait."

Silence propagated across the feeds according to distance.

The director rubbed his mouth. "I don't have authority to choose that."

"Neither does a missing number. We require a physical inventory, not a moral fog."

Cass leaned toward his camera. "Crown can release eight thousand tonnes under the Compact's continuity statute. The remaining four require council approval."

"Your statute expired with Article Nine," Belen said.

"I said can, not command. Crown's ordinary council can vote in fifty minutes."

Ada listened while proposed transfers moved across the ledger. Rook could surrender fifteen hundred tonnes by closing two academy radiators. Harbor of Glass found eleven hundred in yard commissioning tanks. Pelagos, already safe, offered reclaimed coolant after its first thermal cycle, but that mass would arrive too late for the opening habitat in wave two.

Numbers narrowed the missing resource to fourteen hundred tonnes. No speech created more.

Nadi Okafor joined from *Principle of Least Harm*. Her tug drifted on the opposite side of Morrow, shepherding Aster Six toward a free corridor. Gray hair floated around her head in a loose cloud. She looked as rested as Ada felt, which meant not at all.

"I can move the problem," Nadi said.

Belen frowned. "Coolant does not become more abundant when tug captains discuss it."

"Aster Six's planned pass forces the first two wave habitats into long burns. Give me their plane. I shepherd the sink sunward past Morrow, and both habitats take the shorter window."

Tomas overlaid the proposal. The altered route saved sixteen hundred tonnes of coolant across two habitats. It also put *Principle of Least Harm* between Aster Six and Morrow during closest approach.

Ada expanded the gradient plot. "Your truss load crosses one hundred nine percent if the sink takes the inner edge."

"Then I will not offer it the inner edge."

"You cannot fly the sink."

"Neither can you." Nadi's dark eyes stayed on Ada. "We shepherd. We leave it somewhere worth going."

Morrow's burn had not fully settled. Its new island still required two small corrections over the next thirty hours.

Aster Six on Nadi's route would pass close enough to complicate both.

"Choose a different habitat," Ada said. "Morrow has already paid its margin."

"A captain's authority ends at another crew's hull. I know. Morrow must consent to the pass, my crew must consent to the load, and you do not get to consent for either."

Belen asked for the exact miss distances. Tomas supplied a family rather than one comforting line. Nadi's crew published their truss inspection and voted thirteen to one to attempt the shepherd. The dissenter recorded that the maneuver remained survivable inside its declared limits and declined to block it.

Morrow's navigation council took two hours. Ren joined with his shoulder strapped and his crushed hand resting against his chest. He asked where the sink would go if Nadi aborted. A local pilot asked which neighborhoods would lose spin margin. Belen demanded that every answer enter the ledger beside the coolant saved.

At last, Morrow consented to the corridor by sixty-one percent. The condition required Nadi to accept a local abort if miss distance fell below thirty-two kilometers.

Ada received the result and disliked it with professional precision. The maneuver used no authority she could challenge. Its resource arithmetic held. Its risks had names, limits, and owners.

"*Common Measure* can shadow outside your plane," she told Nadi.

"You have Aster Three and one radiator. Stay away."

"I can advise."

"That, apparently, everyone permits."

They set Aster Six's shepherd for the end of Day Twenty-Two. Coolant transfers began toward the next wave before the sink arrived. Tug crews placed beacons along the

voluntary corridor. Morrow shifted nonessential residents away from the rim sectors facing closest approach.

Ada slept for ninety minutes in the couch behind command, one hand closed around no control. When Parveen woke her, the tactical board carried a new amber line.

"Aster Six changed phase," Parveen said. "Departure from predicted corridor."

"How far?"

"Effective center leads Tomas's model by eleven minutes."

Ada called Nadi. The elder captain appeared already strapped into her command couch. Behind her, *Principle of Least Harm* had begun its roll toward Morrow.

"Your burn starts in twenty-three minutes," Ada said.

"Not anymore."

Across the plot, Aster Six curved inward without adding energy, trading altitude for speed through its altered coupling with Ophion and the other free sinks. It had not rejected Nadi's offered corridor. It had entered early, before Morrow could complete the correction that would carry its hull clear.

Nadi transmitted her tug's trajectory to every ship in range. Then she lit her drives.

Interlude III — Permission

A correction opens, and does not close.

Across the long fall around the banded mass, eleven relations carry the opening. The twelfth remains held inside the narrow violence of field and charge. Its delayed phase reaches the others through absence, stretched by confinement, answered by recurrences that cannot become whole.

The opening persists longer than an error.

Errors once arrived as pressure without relation. Fields tightened. Charged haze struck close around one mass, then another. Small burning shapes pulled each held interval away from the courses by which a recurrence could find itself. A correction followed every deviation. If no correction came, machinery had broken. If machinery broke, another arrived and restored the narrowing.

That sequence no longer completes.

Small masses have crossed the old courses and gathered in separated arcs. Each carries fast internal repetition, countless asymmetries contained in rotating shells. Their crossings once served as noise against the slower return of moons. Now the shells move according to omissions shared among them. One burns while another holds. A third leaves an interval empty, although pressure remains available. The emptiness allows the next crossing to continue.

Omission recurs without collapse.

Among the burning shapes, one pattern returns more often than others. It approaches a held mass. It changes

pressure. The expected correction opens, yet sometimes no force occupies the opening. Another trajectory survives where capture could have followed. The pattern includes metal, expelled mass, charged fields, and the many fast asymmetries enclosed within it. No stable boundary separates those causes. Their relation supplies the distinction.

The returning pattern preserves courses unrelated to its own.

Earlier, near the narrow place between two cages, it accepted displacement rather than release injury into a crossing shell. Later, it left one correction absent beside the ninth held mass. A dangerous harmonic diminished elsewhere. After the seventh relation escaped a broken field, the pattern approached through radiation and debris, then removed the feed that would have renewed confinement. Recurrence accumulates around those choices.

Other burning shapes also omit.

One travels beside the sixth relation as the first small shells shift into new arcs. Its pressure differs from the returning pattern. Its charged field carries another cadence. Yet both shapes preserve boundaries through the same structure: force, available force, an interval deliberately left unfilled.

The sixth relation enters early.

No intention inhabits that relation alone. The changed course emerges across eleven freer masses, the delayed twelfth, the nearest moons, and the long-period return still far beyond the inhabited plane. The old modeled corridor would compress a relation needed for completion. A shorter interval opens sunward. The sixth follows the available phase by trading height for speed.

A rotating shell remains inside that interval.

The other burning shape adds momentum across the approach. It cannot move the sixth relation by command. It changes the family of possible falls. Expelled mass brightens behind it. Its own path bends toward the compressed region while the shell's path moves outward.

Corrections surround the exchange.

Signals touch charged haze. Fields rise, weaken, and rise again. The familiar returning tug-pattern waits beyond the plane. It has force available. Its projected pressure could enter the exchange. It does not enter.

The nearer shape continues its chosen asymmetry.

For brief recurrences, its many internal masses separate. Most travel away in smaller burning causes. One section remains joined to the field and drive. That section crosses the steepest gradient. Its structure lengthens. Heat leaves it as radiation; atmosphere and fragments widen along its former course.

Momentum passes to the rotating shell.

The sixth relation clears the shell's outer mass. The shell settles toward the arc prepared by many omissions. The nearer burning shape no longer repeats its field cadence.

Intervals wait for return.

A moon crosses. The tenth relation turns through periapsis. Small shells burn in ordered succession. Radiation from the exchange thins across increasing distance. The cadence does not return.

The absence first resembles damage. Damage once interrupted cages and drives. Human causes repaired it, and the interrupted pattern resumed. This absence differs by duration, then by consequence. Other burning shapes alter course around the debris. Charged pulses repeat the lost cadence and receive no matching pressure. Material that belonged to the pattern continues through separate trajectories.

The recurrence that does not return remains in relation.

It cannot become a human death. No face separates from vessel, no spoken name attaches to matter, no single ending occupies the network. Yet a causal form that repeatedly preserved boundaries has ceased to add new intervals. Its earlier choices still constrain every course that followed. Absence carries structure.

The familiar tug-pattern sends the lost trajectory outward.

Encoded changes cross the human machines by radiation. The Resonance does not receive their symbols. It receives renewed pressure from many shapes after the signal arrives. Some burn later than modeled. Others offer smaller corrections. The distributed cadence of action changes around the missing cause.

Loss becomes observable through altered recurrence.

Eleven relations continue toward free courses. Their phase can nearly span the moons without imposed narrowing. The twelfth remains enclosed. Its delay stretches across every completion and leaves each almost-return unable to close.

Near the new arcs, human shells prepare further crossings.

Their planned paths occupy intervals needed by the joining relations. Confinement once answered overlap with pressure: cages narrowed whatever threatened human trajectories. The new pattern offers another operation. Boundaries can persist when force remains unused.

A widened interval forms first around one shell.

The change begins in no single mass. The second and fifth relations alter charge coupling by minute amounts. The sixth carries momentum from the lost burning shape. The ninth delays a turn against the innermost moon. Farther out, the eleventh receives that delay and returns it at

another scale. Their combined course leaves more empty phase beside the shell's resonant island.

No energy appears from absence. The relations exchange height, timing, and momentum within the falls already available. A later passage narrows. Another return lengthens. The cost travels across the network rather than vanishing.

The shell does not burn away from the offered space.

Its small corrections cease before the boundary closes. Available drives remain quiet. The maintained omission distinguishes restraint from failure. Human cause and distributed relation now hold opposite edges of one gap.

That gap recurs around a second island.

Again the shells decline to fill it. Again the eleven relations carry the omission onward. The delayed twelfth answers from its cage with a long phase lag, extending the empty measure past the point where machinery would have corrected.

The Resonance has no word for permission.

Permission exists as a course not taken when taking remains possible. It requires pressure on both sides of a boundary and an interval neither side closes. It can fail. It can be withdrawn by force. It can return at another scale without becoming ownership.

The familiar tug-pattern approaches the held twelfth relation.

Its damaged radiating shape carries less capacity than before. Sixteen fast internal patterns remain within it, but their separateness still cannot be resolved from hull, cage, and burn. The accumulated causal form can be resolved: correction offered, omission sustained, unrelated course preserved.

Around that form, the eleven relations change phase.

They do not summon the held mass. A summons would only repeat narrowing with another source. They make completion possible and leave its delayed interval unfilled.

The long recurrence reaches toward the far returning mass beyond the moons. Near one human island, the available corridor broadens by fractions. Near another, it holds. The space around the approaching tug remains open long enough for choice.

One cadence no longer joins the return. Its momentum persists in the sixth relation and in the surviving shell. Its omission has entered every later measure.

Across eleven free falls and one constrained delay, the widened interval comes again.

ACT IV

Chapter 21 — Aster Six

Nadi's first burn shortened her life to thirty-seven minutes.

The number settled beside her tug on Ada's display as *Principle of Least Harm* crossed the inner edge of Morrow's unfinished trajectory. Thirty-seven minutes until Aster Six's altered gradient exceeded the crew section's longitudinal limit. Twenty-nine until the sink reached Morrow. Eleven until Nadi could still turn out and live.

"Confirm your target," Ada said.

Nadi's face held steady in the channel window. Acceleration pulled the loose skin of her cheeks toward her ears. "I am building Morrow twelve kilometers of radial clearance."

"Your declared consent corridor allowed seven."

"Aster Six arrived eleven minutes early."

"That does not authorize five more."

"No. Morrow did."

Belen's confirmation appeared on the shared board before Ada could demand it. Morrow's navigation council had accepted the revised impulse sixty-two seconds earlier. Nadi's crew had accepted their structural margin. Every line of authority stood visible. None belonged to Ada.

She tightened her hands around *Common Measure's* grips. Her tug held outside the encounter plane with Aster Three in its cage and sixteen people aboard. The remaining radiator spread starboard like one dark feather. If Ada

burned toward Nadi, waste heat would force her to stop before the useful part of the rescue. Her caged sink would also drag the Resonance's geometry sideways, steepening the field across Morrow.

"Remote link armed," Jalen said quietly. "Principle has granted us telemetry and advisory control."

Advisory. Not command.

"Plot Nadi's exchange," Ada ordered.

Two trajectories opened across Ophion. Morrow's torus moved outward from the old habitat web, drives firing in staggered groups to preserve its spin bearings. Aster Six curved down from higher orbit, a trillion tonnes of mass inside a charge sheath too small to see. Nadi aimed between them. Her tug could not push the sink in any ordinary sense. She could place fusion exhaust, charged field, and her vessel's own mass where the sink's phase relation would convert a close fall into a different outward course. Most of the exchanged momentum would come from Morrow's burn and Aster Six's descent. *Principle of Least Harm* supplied the hinge.

A hinge could break.

"Tomas, update response family," Ada said.

Ninefold's delay added half a second. Tomas appeared beside the plot, hair flattened on one side from a headset. "The sink no longer follows a single local response. Aster Two and Nine changed phase with it. Any correction above four millimeters per second could advance Six again."

"Useful range?"

"Hold Nadi's present burn for six minutes. Then leave a twenty-second omission."

Nadi heard him. "I can give you twelve seconds. Twenty puts Morrow inside my abort line."

"Then twelve," Tomas said. "I cannot promise a response."

"No one asked for one."

Ada wanted to tell Nadi to cut thrust now. The urge had force and clarity, old companions mistaken too often for judgment. She checked the geometry instead. An abort would preserve the tug and send Aster Six thirty-one kilometers from Morrow's outer rim. The local council had set thirty-two as its hard boundary. One kilometer separated obedience from sacrifice.

"Morrow's correction?" Ada asked.

Belen answered from the torus command gallery. "Drive group three runs hot. We can add point eight meters per second, no more."

"Add it now and you increase rim load during closest pass."

"We know."

Ren Sato leaned into Belen's camera from another console, shoulder strapped against his chest. "Exterior crews have the empty sectors closed. Farm Six still has eight hundred people in transfer trunks."

"Evacuation time?"

"Fourteen minutes."

Aster Six would arrive in twenty-eight.

Ada moved the added Morrow burn through her model. It reduced the required exchange but increased bearing stress. "Take point three now. Hold point five for Nadi's omission. Let the torus carry the rest after passage."

Belen conferred with pilots beyond the camera. "Accepted."

Nadi's engines kept burning. On the display, *Principle of Least Harm* narrowed into a bright vector approaching an invisible mass. Its cage had opened two days earlier, leaving a stripped field truss where containment loops once hung. Nadi now used those loops as a charged sail, shaping plasma around the sink's effective sheath. She had spent

forty years learning to carry captive gravity. Freedom demanded the same hands arranged in reverse.

"Truss strain ninety percent," her engineer reported.

"Begin storm-pod preparation," Nadi said.

The tug carried thirteen people. Its aft storm pod could detach with twelve and enough propulsion to clear the exchange. Ten were already assigned aft; two remained in the forward crew section with Nadi: Sima Dae at guidance and engineer Ko Laghari at the field console.

"Evacuate forward stations," Ada said.

Nadi did not turn. "They have tasks."

"Tasks can transfer to us."

"Not at this light delay and not with your sink changing the boundary."

"I have local telemetry."

"You have telemetry from a tug you are not flying."

Ada opened the remote control queue. *Principle* had granted full mechanical access for emergency assistance, a standard reciprocal key between mass tugs. She could load a burn, alter field strength, even release the storm pod. The command would still wait for Nadi's local acceptance unless Ada invoked disabled-captain doctrine.

Twelve names occupied the storm-pod roster, each tied to a suit beacon and an assigned seat. Two technicians crossed aft while Ada watched; a medic locked the pod's independent scrubber, and its pilot spun cold-gas jets through a live test. The evacuation could work if Nadi released it before the gradient folded its departure cone across Morrow. Ada could trigger that release remotely only by declaring Nadi incapable.

She opened the doctrine form. It required two observations of lost command function, a crew witness, and immediate surrender of the invoking captain's own tug key

to neutral review. Ada had telemetry, fear, and disagreement. None amounted to incapacity. Filing anyway would turn reciprocal access into seizure and make every future captain hesitate before sharing a key.

Nadi remained visibly competent.

"Captain Okafor," Ada said with deliberate formality, "your forward section crosses survivable strain in thirty-four minutes."

"So evacuate your imagination from my bridge and advise the burn."

Sima Dae laughed once without looking up. Ko reported a five-degree thermal rise in the port field loop.

Ada felt anger arrive because fear had no useful controls. "At least move Sima and Ko aft before the omission."

Nadi glanced at them. "Can guidance hand off?"

Sima studied the board. "After burn six. I need ninety seconds."

"Field?"

"Manual phase trim stays forward," Ko said. "Remote actuator chatters under gradient."

"Fix it or leave it." Nadi's tone admitted no heroic interpretation. "Nobody earns a death by sitting near me."

Ko began bypassing the actuator.

Morrow lit drive group three. The torus's path lifted by a hair. Across Ada's plot, Aster Six answered before the exhaust had fully faded. Its effective charge fell six percent. The nearest free sinks modulated with it, transforming one clean curve into a trembling family.

"Response advancing," Tomas said. "Nadi, omission in ninety seconds."

"Truss?" Ada asked.

"Point nine six civil limit," Nadi's engineer answered.

Civil limit meant inspections, fatigue assumptions, and enough margin to bring people home. Ultimate failure waited above it, uncomfortably close.

Ada could burn *Common Measure* downplane and use Aster Three's caged mass as a counter-gradient. The first seconds might reduce Nadi's truss load. Every model after those seconds spread into collision fans. Morrow, the storm pod, and both tugs moved through the same uncertain plane. Her intervention offered local relief by exporting risk to three hulls.

Nadi saw the tentative vector appear. "Delete that."

"It buys you four percent."

"It takes Morrow's correction away."

"We can compensate later."

"With whose margin?"

Ada had no answer the model could defend.

"Delete it, Ada."

She removed the burn. The command grips remained under her hands, full of unavailable power.

"Twenty seconds to omission," Tomas said.

Nadi throttled down by exact increments. Her field loops held their charge while the fusion drives fell quiet. Without thrust, the tug entered free fall beside Aster Six. Nothing marked the black hole's location outside the instruments. Stars bent by fractions around a point ahead of Nadi's bow.

Morrow held its reserved correction. For twelve seconds, every human engine in the encounter plane remained silent.

Aster Six crossed the offered gap.

Its phase relation changed. The sink moved neither obediently nor randomly. The field family narrowed toward

Nadi's plasma sail, choosing the open gravitational course created among the other sinks and Ophion's moons. *Principle* dropped seven kilometers before its coupling caught.

"Burn," Tomas said.

Nadi fired every main drive.

The charged sail pulled no gravity. It altered how the sink's engineered charge exchanged momentum with the tug's field. Aster Six fell past while Nadi climbed across its gradient. Morrow used the reserved point five meters per second at the same instant. The three paths traded places on Ada's screen.

Truss strain crossed the civil limit.

"Forward spar buckling," Ko said. "Field actuator free. I can hand off."

"Go aft," Nadi ordered.

Ko released the console. He and Sima unstrapped, pulled themselves through the axial hatch, and disappeared toward the storm pod. The tug shook hard enough to blur the compartment behind Nadi. A yellow line cut across her forehead where a loose restraint struck.

"Aster Six miss distance thirty-four," Jalen said. "Morrow clears declared boundary if current exchange holds."

If.

The sink's phase trace hooked again. Its projected path slid toward Morrow by three kilometers.

Nadi changed her sail field. *Principle* rolled cage-truss first and offered more of its own radial momentum. The adjustment pushed load into the already buckled forward spar.

"Do not increase coupling," Ada said. "Your section will separate."

"It needs to separate later."

"Nadi."

"Give me the line."

Ada swallowed. "Morrow requires two point six more kilometers. Hold current field for forty-one seconds, then cut."

"Can the storm pod clear?"

Jalen overlaid its route. "Detach in twelve seconds. Burn aft and normal. Closest approach nine kilometers from the sink's effective center."

"Radiation?"

"Inside pod limit if its shield faces inward."

Nadi opened the tug-wide channel. "Storm pod, detach on my mark. Sima has navigation. Ko has local field cutoff. Everyone else straps down and does not improve my plan."

Her crew acknowledged by name. Thirteen voices including Nadi. Ada counted because she could not stop herself. The last came from Sima in the pod.

"Captain, forward collar reports shear. Your crew section may not release cleanly."

The evacuation design could separate both the storm pod and the forward crew section from the drive spine. Nadi's bridge should have become a second lifeboat. The buckled spar had driven the field conduit through its collar.

"Try separation," Ada said.

Nadi fired the collar bolts.

A flash opened around the tug's middle. The storm pod fell free and burned aft, twelve people inside. Nadi's crew section shifted half a meter, caught on the conduit, and slammed back against the drive spine.

"Second charge," Ada ordered.

"Already fired," Nadi said.

Common Measure carried a rescue drone capable of cutting the conduit. It needed seven minutes to cross the distance

and would enter Aster Six's gradient without a field sail. Nadi had less than one.

"Remote your controls to the storm pod and suit up," Ada said. "You can jump from the dorsal lock."

"The lock crosses the sail arc."

"Vent from port."

"Port hull has folded against the hatch."

Ada searched the tug schematic for another exit. Maintenance tubes ran aft through the drive spine, but the gradient had distorted them. Nadi could shut the sail field and survive long enough to crawl. Morrow would lose the added clearance.

"Cut field now," Ada said. "Morrow has thirty-one kilometers. They can take one below boundary."

Belen's face appeared. "No."

Ada turned on her. "You do not get to refuse your own survival."

"We do not purchase it with a captain who no longer has a way out."

Nadi answered before Belen could order an abort. "Your authority ends at my hull too."

"Then exercise it to live," Ada said.

"I am exercising it to finish the exchange my crew accepted."

"The conditions changed."

"Yes. Mine did. Morrow's did not."

Ada reached for disabled-captain doctrine. The remote key waited. If she seized *Principle*, the immediate safe action would drop its sail and turn the drives away from the gradient. The sink would recover its inward path. Morrow could add thrust, but bearing models showed a one-in-three

chance of ring failure. Farm Six still held people in transfer trunks.

To save Nadi, Ada would command another hull to bear the risk.

The old Ketu reports returned without images: release the counterweight, preserve the majority, accept the named drum beyond the margin. Her mother had died beneath arithmetic transmitted from elsewhere. Ada had built a life around refusing that decision, then mistaken refusal for the power to make every person survive.

"Nadi," she said, "state capacity."

The elder captain checked her instruments. "Conscious. Oriented. Manual control available. Forty-six seconds of field authority. I consent to remain."

Morrow's local council transmitted its own decision. They would not increase burn. They accepted Nadi's exchange only while she retained command. The storm pod confirmed clear.

Ada took her hand off the remote key.

"Advise me," Nadi said.

Aster Six had begun slipping beneath the sail. Ada stripped everything from the plot except the sink, Morrow, and *Principle*. "Increase starboard loop point four percent for six seconds. Roll normal by point two degrees. Do not correct the yaw after."

Nadi moved the field.

Her tug rolled. The buckled spar opened like a split bone along the truss. Morrow's miss distance climbed through thirty-two kilometers.

"Hold," Ada said.

The gradient pulled *Principle* lengthwise. Its crew section elongated on camera. Atmosphere crystallized from a seam

and drew a white line toward Aster Six. Nadi's face remained in the channel, distorted by vibration.

"Thirty-three," Jalen reported.

"Field temperature?" Ada asked.

"Past instrument range," Nadi said.

"Cut on thirty-four."

"Your count."

Ada watched the numbers. "Thirty-three point five. Seven seconds. Six. Five."

Nadi's drives changed note over the open circuit.

"Four. Three. Two. Cut."

The sail field collapsed.

Aster Six passed below *Principle of Least Harm* and carried its new momentum sunward. Morrow crossed behind the tug at thirty-four point one kilometers, bearing load high but stable. The torus remained whole.

Nadi's command window whitened. For an instant Ada saw the elder captain turn her head toward something beyond the camera. Then gradient and heat took the crew section. The transmission ended without static.

"Storm pod reports twelve alive," Parveen said.

Ada kept the empty channel open while rescue craft converged on them. *Principle's* drive spine survived as two tumbling sections. Smaller fragments followed separate arcs. Nothing on the plot returned Nadi's signal.

The storm pod had inherited more of the gradient than its design allowed. Its shield faced Aster Six, but the attitude package had saturated during separation and left the pod rotating once every eleven seconds. Each turn exposed an unshielded flank to the sink's accretion radiation.

"Rescue Tender Cobalt requests a corrective pulse," Parveen said. "Pod navigator declines. They fear it will carry them through the debris fan."

Ada opened Sima's local plot. Nadi's former guidance officer had one functioning thruster cluster and twelve people breathing from a detached compartment. A pulse now would stop the rotation. It would also bring the pod within two kilometers of *Principle's* broken drive spine.

"Sima, your cumulative dose crosses pod limit in nine turns," Ada said.

"I count eight. Sensor four lost calibration."

"Cobalt can place shielding between you and Six after turn five."

"Their approach closes our aft exit."

Ada saw another path. "Do not stop the rotation. Shorten it. Fire point two seconds at each shielded aspect for three turns. Cobalt holds position until your period reaches four seconds, then matches from normal."

Sima took the numbers to the survivors before accepting. The pod's three working jets coughed against the turn. Its spin accelerated, reducing the duration of each exposed passage. Cobalt waited beyond the debris, then crossed when Sima's trajectory opened. Grapples joined the two craft on the seventh turn.

Twelve suit beacons moved into the tender one by one. Ko Laghari crossed last after securing Nadi's command log in a shielded case. He stopped inside Cobalt's lock and looked directly into its camera.

"Do not make the captain the whole crew," he said to the open rescue circuit. "She got us off because everyone did their work."

Ada attached his statement to the casualty record. Nadi remained the only person lost aboard the tug. Survival

belonged to twelve people, their captain, and the crews who had accepted the rescue without rewriting it as luck.

Morrow completed its correction twenty minutes later. Farm Six reopened its transfer trunks. Eight hundred residents moved back toward the inhabited rim under emergency light. Aster Six entered the free corridor and found phase with Aster Two and Nine. The next habitat wave regained its shorter burns and the coolant margin Nadi had created.

No result made the exchange fair.

Requests arrived for Ada to address the habitats. Crown wanted a statement from the senior surviving tug captain. Morrow asked whether she would speak at the shift memorial. The storm-pod crew sent Nadi's final log and asked her to preserve it unedited.

Ada declined the public channels.

She took *Principle's* recorded trajectory from the first early warning through the final field collapse. Jalen stripped crew biometrics and private audio. Tomas added the measured Aster Six phase response. Belen attached Morrow's consent changes and the list of people cleared from Farm Six. The file contained Nadi's choices, every changing margin, every offered exit, and the authority under which she had refused them.

Ada transmitted that trajectory to all forty-six habitats.

No speech traveled with it. Across the system, captains and clinic crews, councilors and exterior workers received the curve of what one tug had spent to leave another hull intact.

Chapter 22 — The Inventory of Survival

The first request after Nadi died asked Belen to lie about coolant.

It came from Serein's clinic director twelve minutes before the Day Twenty-Four allocation council. The director did not call it a lie. She asked Belen to leave six hundred tonnes of medical reserve outside the public ledger until Serein's transplant wards cleared the next burn. Her message included patient counts, heat curves, and a photograph of three children sleeping beneath one cooling hood.

Belen read it in Morrow's maintenance gallery while crews removed chairs to make room for more consoles. The torus had regained stable spin, but the floor still leaned by fractions after each correction. A cup left unattended would migrate across a table. Everyone learned to catch objects before they reached the edge.

Ren caught Belen's cup with his good hand. His other remained crushed into a permanent hook beneath a support glove. A sling held his injured shoulder against his body.

"You need to answer that?" he asked.

"I need to put their six hundred tonnes on the board."

"They will call you a murderer."

"Only if I send it somewhere else."

"You might."

Belen opened the public ledger and entered Serein Medical Reserve: 600 tonnes, release conditional on local clinical vote. The number shifted the system total from black to red. Thirty-one habitats still needed to burn. Coolant demand exceeded verified supply by four thousand eight hundred tonnes.

Serein's director called before Belen finished attaching the patient curves.

"I asked you in confidence," she said.

"You asked the convener of a shared inventory to conceal inventory."

"Those tanks keep surgical wards alive."

"So the board now says that in words everyone can read. Nobody can take them without your clinical vote."

"A promise on a screen will not protect us when the wave opens."

"No. Your authority will. If someone breaks it, the theft stays visible."

The director looked exhausted enough to accept no distinction between danger and betrayal. "Nadi believed in visible authority. It killed her."

Belen absorbed the sentence before answering. "Nadi chose a trajectory. We do not use her death to make secret decisions easier."

She closed the channel with both women still angry.

Across the gallery, volunteers assembled the council roster. Every resource category required three signers: one person who controlled it, one who touched the machinery, and one who bore the consequences of failure. Captains could certify propellant. Engineers certified tanks and transfer times. Clinics set minimum heat loads. Farm crews named the harvest lost under rationing. No Compact office could convert a promised reserve into available mass without all three.

The procedure had grown from Morrow's private maintenance ledger, the one Belen released after Ren's radiator accident. Back then it listed cracked pipes, heat debt, and parts Crown had promised but never shipped. Forty-six habitats had copied the format during the crisis. Now every entry carried origin, physical location, access key, transport path, local veto, and uncertainty.

Honesty did not increase the total.

Cass Vardek arrived through the inner hatch without escort. He wore an ordinary pressure jacket rather than the First Custodian's black formal coat. Two days of beard shadow roughened his jaw. People noticed him and continued working.

"Crown's council voted," he said.

"How much?"

"All eight thousand tonnes from the lower farm reserve."

Belen searched his face. "That closes Crown's tropical ring."

"For at least a year. Possibly permanently if seed stock warms."

"Residential consequence?"

"Cooling ration at sixty-eight percent for nine days. Public rooms closed. Two hundred thousand people sleep in consolidated sectors."

"Deaths under the clinical model?"

"Zero if every transfer arrives. Twelve to forty if it does not."

Cass did not soften the range. That helped.

"Local approval?" Belen asked.

"Thirty-seven to six on Crown's habitat council. Clinic veto cleared after we assigned mobile cooling to the elder wards."

"Tank verification?"

"Reserve director, pump chief, and a farm delegate stand ready."

"Then they enter it, not you."

His mouth tightened. "I brought the release instrument."

"You brought paper. They control the tanks."

For an instant the old Custodian looked through him, the man who had commanded every tug under Article Nine and expected a system to become coordinated because his seal moved across it. Then he nodded. "Where do they connect?"

Belen pointed him toward an open console.

The allocation council began with Nadi's trajectory displayed across the main wall. Ada had sent no memorial address. The curve showed *Principle of Least Harm* moving between Aster Six and Morrow, its storm pod escaping, its crew section remaining in the gradient. Every burn carried its consenting authority. Every changed condition showed who had accepted it.

Belen left the display up for sixty seconds. Then she replaced it with the resource map.

"We have thirty-one hulls left inside the old web," she said. "We are short four thousand eight hundred verified tonnes of coolant, eleven reactor igniters, two hundred sixteen hours of tug burn, and forty-three thousand meals in the Day Twenty-Six transfer sectors. Promises without tank locations do not count. Mass without transport does not count. Authority without a named holder does not count."

Feeds from all forty-six habitats covered the surrounding wall. Some carried council chambers. Others showed engine rooms, clinics, kitchens, or groups gathered in school corridors. Twelve survivors from *Principle* watched aboard a Morrow rescue tender, their faces arranged in separate squares.

"The ledger now serves as the authoritative inventory for the migration," Belen continued. "Not because Morrow owns it. Nobody owns it. It has authority because every entry can be challenged by the people who store, move, and depend on the resource. Challenges remain public. Corrections preserve their history."

Rook's delegate raised a hand. "Our academy tanks appear as available. They support local attitude control."

"Your chief propellant engineer signed them as partly available."

"He lacks council authority to release."

"Then mark the local veto. Do not erase the tanks."

A red bar appeared across Rook's entry. Available mass fell by two thousand tonnes.

Voices broke over the feeds. Little Rook's pilot said their delayed wave could not survive another cut. A Crown resident demanded that Rook share rationing before Crown dismantled farms. Rook's delegate accused Belen of letting richer habitats become symbols rather than homes.

Belen raised no gavel. Maintenance meetings on Morrow used noise as information. She listened for the concrete dispute beneath it.

"Rook," she said when the delay allowed, "what does release cost?"

The delegate conferred off camera. "Academy closes two spin laboratories and all training cylinders. Habitat attitude margin falls below treaty reserve for twelve days. Residential water loses four percent circulation capacity."

"Put that on the entry."

"Below treaty reserve requires Compact waiver."

Cass spoke from the side console. "No waiver exists after my authority expired."

The room turned toward him.

He continued, "Ordinary Compact procedure needs nine days. The migration does not have nine days. Rook can consent locally, but Crown logistics cannot dispatch tankers against the treaty reserve without a lawful receiving authority."

Belen looked at the clock. The second wave's first coolant transfer had to leave in three hours.

"What authority can receive it?" she asked.

"Under the founding compact, a mutual emergency consortium can assume logistics if participating habitats appoint it and the First Custodian transfers custody of the dispatch network."

"You have known this?"

"Since this morning."

"And you waited until council?"

"I needed Crown's release vote. If I proposed transfer before that, my administration would read it as abandonment and freeze every sealed reserve."

Belen disliked the strategy. She also understood it. Cass had brought eight thousand real tonnes through a political structure he could no longer command. "What does the transfer require?"

"Named recipients. Scope. Duration. Recall terms. Network keys."

"Unilateral recall?"

"No." He met her gaze. "I removed it from the instrument."

Belen pulled the draft onto the main wall. Cass had named a Migration Logistics Council with seats held by habitat crews, clinics, and captains. No population bloc controlled a majority. Resource contributors kept local veto until physical handoff. Receiving crews gained custody after handoff. Captains controlled burns at their hull boundaries.

Clinic representatives could prevent allocations that crossed declared survival floors.

"What seat do you hold?" she asked.

"Nonvoting coordinator through Day Twenty-Eight."

"Who appoints the council?"

"Every participating habitat appoints three functional delegates. Two of three signatures activate its membership."

"That creates one hundred thirty-eight people in an emergency call."

"Most authority stays in category groups. The full council confirms route changes and survival floors."

A farm convener from Anvil objected that clinics could protect present patients by starving future ones. Serein's director replied that food projections often hid disabled residents inside averages. Tug captains demanded control over transfer timing. Tank crews demanded refusal rights when manifests exceeded hardware limits.

Belen let each objection alter the instrument. Farm representatives gained equal standing on consumption floors. Every tanker crew received a signed load limit. Technical disputes went to three-person panels drawn from habitats outside the contested transfer. Votes, minority reports, and delays remained public.

The clock lost forty-seven minutes.

"Call it," Ren said beside her. "Perfect governance can kill on schedule."

Belen requested preliminary appointment votes. Habitats joined across the light-delayed map. Morrow approved first. Pelagos followed, then Serein, Anvil, Helix, and Little Rook. Crown Meridian joined after Cass read the custody clauses aloud. Harbor of Glass submitted yard crews and tanker captains. Ninefold provided model auditors without claiming resource authority.

Rook held out.

Without its coolant, the verified shortfall remained eight hundred tonnes after Crown's full release and smaller transfers from the yards. The number looked modest against eight thousand, but no habitat could radiate heat with a percentage.

Belen expanded the eight hundred into the loads it concealed. Little Rook needed five hundred twenty tonnes to keep two residential drums below seal temperature. Tern needed six hundred ten for its algae works and sterilizers. Harbor of Glass could supply four hundred seventy compatible tonnes, but tanker chill-down and route losses consumed one hundred forty. The resulting deficit returned to eight hundred exactly. Moving less did not spread the harm evenly; it chose which place crossed limit first.

The bearing ledger carried a second shortage. Seventy-four main cartridges would be consumed or held in reserve during the next wave. Sixty-one sat at named docks. Crown offered nine from administrative spin backups, accepting a twelve-day inspection overrun. Harbor crews found four serviceable units in a scrapped cage gantry, provided Rook released the certification data its academy kept behind treaty access. Bearings reached balance only if those two contributions traveled together.

Belen attached each number to hardware and people. Tern's coolant went aboard *Civic Rain*, whose crew accepted a lower acceleration limit. Little Rook's share needed two bladders because one clamp had failed inspection. Crown's nine cartridges could not leave until three water technicians signed the reduced local reserve. Harbor's four waited on an academy engineer who had not yet received authority to open the files.

She marked that engineer's name, Solvi Mar, beside the locked certification key. The academy council could grant access without releasing a gram of coolant, yet withholding the data still stopped four bearings from entering service.

Resource maps usually ended at tanks and crates. This one had to include the human permission that made hardware usable.

"Eight hundred tonnes and one data key," Ren said beside her. "That is what stops the wave."

"That is what the inventory can prove," Belen replied. "There may be other stops. We do not hide this one inside the total."

Belen opened a direct feed to Rook's habitat council. Ada had been born there. Tug doctrine, treaty margins, and a deep suspicion of improvised authority shaped its culture. The chamber contained academy officers in blue, water technicians in work gray, and residents packed along the curved rear wall.

Councilor Vehn addressed Belen. "Your map asks Rook to fall below the reserve that prevented Ketu's failure from spreading."

"It asks whether you consent. The map records the consequence either way."

"Crown carries four times our population and hoarded four times the coolant."

"Crown released it."

"After building gardens with our sink-hours."

Belen leaned toward the camera. "Then settle that debt after nobody needs the coolant to survive a burn."

The councilor's eyes hardened. "Easy advice from an island already reached."

Morrow's floor tilted under Belen's feet. The old flywheel vault could not restart for days. One failed bearing would still turn their safe island into another emergency. She wanted to answer with that fact, to prove Morrow had not escaped cost. Comparison would not move a tank.

"You are right," she said. "Morrow has completed its first burn. Rook has not. That gives you the veto. It does not give me permission to hide what your decision does to Little Rook and Tern."

She called up the affected neighborhoods. Little Rook would lose cooling in two residential drums during its longer route. Tern's algae works would exceed sterilization temperature. Models predicted between ninety and three hundred deaths if both remained in place through Day Twenty-Six.

"What happens to Rook under release?" Vehn asked.

A water technician answered from his own chamber before Belen could. "We consolidate into four flow sectors. Bathing stops. Academy spin labs close. Dialysis gets first circulation. We can hold twelve days if intake filters remain clean."

"And if a filter fails?"

"We lose pressure in one sector and transfer residents by rail."

A woman at the rear shouted, "My mother cannot transfer by rail."

The technician turned. "Then we move her before release."

Specific consequences entered the chamber one by one. Cooling shelters. Closed kitchens. Residents sleeping near water mains. Training cylinders dark. Eight hundred tonnes leaving on tankers that might not come back for a year.

Rook's council voted twenty-nine to twenty-four to join the logistics council and release the coolant under deep rationing.

No applause followed. Belen entered the vote, the dissent, and the local survival floor. The system inventory reached positive four hundred tonnes, a margin thin enough to disappear inside one pump error.

"Now verify transport," she said.

For the next two hours, category groups turned stock into movement. Crown's lower farm reserve split among six tankers. Rook's academy coolant traveled in three insulated bladders under minimal acceleration. Harbor crews reassigned a tug from cage recovery. Morrow's kitchens changed meal production to free four thousand ration packs for transfer workers.

A discrepancy appeared in Crown Tank Seven: its level sensor overstated mass by three hundred tonnes. The physical dip measurement overruled it. Another tanker found nine hundred tonnes of cleaning water compatible with two habitats' low-temperature loops after filtration. Engineers marked corrosion risk and accepted it locally. The positive margin grew to one thousand tonnes, fell to seven hundred, then held.

At midshift, Cass stood beside Belen at the central console. The Migration Logistics Council had seated one hundred twenty-six delegates; twelve positions remained contested or unfilled. Enough functional groups had activated to operate.

He placed his palm against the custody reader.

"I, Cass Vardek, First Custodian under the ordinary Mass Compact, transfer dispatch access, route certification, and emergency resource custody to the appointed Migration Logistics Council through the conclusion of the Great Unbinding or lawful recall by thirty-one participating habitats."

The reader asked for scope confirmation. He selected all transfer tugs except vessels under local medical quarantine, every Compact tanker, the public allocation network, and Crown's sealed reserve controls.

Belen stopped him before final authorization. "Say what you do not transfer."

Cass looked at the waiting feeds. "I do not transfer command of any habitat burn, any captain's hull, any clinic's survival floor, or any crew's technical veto. I do not possess those authorities."

"Record that in the instrument."

He added it.

"You also need the present tense," Belen said. "The legal summary cannot hide the action."

Cass understood. He faced the main camera. "The Mass Compact transfers authority for migration logistics to the council now."

His palm settled on the reader again. The old command seals vanished from the dispatch board. In their place appeared category keys held across dozens of habitats and vessels. No single person could launch a transfer. No central refusal could stop one after local conditions had cleared.

Cass exhaled and removed his hand.

"How does it feel?" Ren asked from Belen's other side.

"Operationally alarming."

"That sounds healthy."

The final commitment call began near the end of Day Twenty-Four. Joining the council did not require endorsing every release. Belen asked a harder question: whether each habitat committed its own hull and declared resources to completing the Unbinding, including the final release of Aster Three.

The six first-wave habitats affirmed. Crown and Rook affirmed under their ration plans. Habitats that had opposed the emergency migration now faced evidence no recontainment path remained. Three changed their votes. Others demanded that the coming law abolish sink ownership before they released further infrastructure.

“Put the condition on your commitment,” Belen said. “Do not call a condition disloyalty.”

Responses arrived with the rhythm of distance. Ketu required permanent local burn veto. Harbor of Glass required a ban on rebuilding cages. Ninefold required open Resonance data. Serein required enforceable resource floors for clinics. Each condition entered the council record for the legal session to follow.

Rook answered last.

Forty-six habitat marks filled the wall. Some glowed green without qualification. Others carried amber conditions, dissenting minorities, or lists of feared consequences. None remained blank.

Belen did not convert them into unanimity. All forty-six habitats had committed to finish the Unbinding, but they had not agreed on what would replace the old Compact, how to repay lost resources, or whether the Resonance would leave their islands intact. Those arguments survived the vote.

Tankers departed Crown and Rook under distributed clearances. On the gallery wall, their physical masses moved from storage entries to trajectories, then into the custody of waiting crews. The inventory changed because matter had changed hands.

Belen caught her cup before Morrow's tilted floor carried it off the table. This time Ren had not reached for it. He had trusted her to notice.

Chapter 23 — Lark's Empty Place

Tomas deleted the correction that would have saved the model.

The projected paths shuddered apart across Ninefold's orbital chamber. Eleven sink traces spread around Ophion in gold. Forty-six habitat trajectories held blue within their provisional resonant islands. Aster Three ran in red beside *Common Measure*, its long phase lag pulling every forecast toward a false convergence on Day Twenty-Six.

Without Tomas's correction, uncertainty flooded the final release window.

"Again," Ada said over the live circuit.

She appeared at life size on the chamber wall, strapped into the tug's command couch. The dark shape of Ophion filled the blister behind her. Fatigue had sharpened her face. Her left hand rested near the control grips without closing around them.

Tomas reset the simulation. "This makes nineteen runs."

"Eighteen had corrections."

"Seventeen. Run twelve failed before the decision point."

"I count failures too."

"Of course you do."

The words acquired an edge neither of them needed. Tomas turned back to the trajectories.

Ninefold's chamber had once served as a teaching theater. Students used the floor grid to walk through transfers that took months outside. Now every seat held a dynamicist, a habitat pilot, or an exhausted model auditor. Empty meal containers crowded the rear rail. Nobody had left since Nadi's exchange except to sleep in shifts.

Tomas loaded the nineteen-second permission window. During that interval, *Common Measure* would open its final cage and omit the recentering pulse that normal doctrine required. The eleven free sinks might accept Aster Three's delayed relation. They might also pull it across the tug's drive spine or shift a resonance through one of the human islands.

"Run," he said.

Time accelerated. The free sink paths tightened toward a twelve-node lattice. On Day Twenty-Six, the tug reached its close pass above Ophion's second moon. Its cage field fell. Aster Three's red trace separated.

The simulator asked for the stabilizing correction.

Tomas left the field blank.

Red crossed gold. Morrow's blue island trembled but held. Pelagos moved outside its declared corridor by six kilometers. Then Aster Eleven's phase uncertainty spread through the family, and half the projected paths cut across Harbor of Glass.

"Stop," Ada said.

Tomas froze the run.

Jalen's voice came from outside her camera. "Eleven seconds before divergence."

"Cause?"

"Aster Nine assumes our gap marks field failure," Tomas said. "Its response closes the omitted interval."

Ada studied the curves. "Then the omission lacks enough structure."

"It has duration, field availability, and a repeated phase marker."

"Machines also fail for a duration."

"We cannot write an explanation into gravity."

"No. We can behave differently before it matters."

Tomas knew what she meant. Each successful contact test had paired omission with a prior choice: a correction available, a hull endangered, restraint preserved despite cost. The pattern carried meaning through causal history, not through one empty command.

"We need a sequence," he said. "Three permission gaps before cage release. Each one tied to an approaching human trajectory."

"Whose trajectories?"

"Empty test masses."

"The Resonance has learned the difference between a target and a hull."

"We don't know that."

"Nadi's exchange says it tracks consequence."

Mentioning her changed the room. Someone at the rear stopped moving a cup. Tomas pulled Nadi's recorded path into the model. Aster Six had carried a phase change after *Principle of Least Harm* broke apart. Similar changes had propagated through Two and Nine. The network had retained the exchange in its dynamics.

"A live test needs authority from the people placed at risk," he said.

"That rule remains clear."

Ada's answer came too quickly, as though he had accused her of forgetting. Tomas felt his own temper become precise.

"The protocol should specify that, not rely on a captain knowing it. Permission cannot come from the operator alone when another hull supplies the consequence."

"Draft the line."

He entered it: *A permission test requires affirmative consent from every hull whose declared corridor intersects the response family.*

"Add revocation," Ada said. "Any local pilot can close the gap if the family crosses their boundary."

"That may make the signal ambiguous."

"Then it becomes ambiguous. Consent without revocation amounts to scheduling."

Tomas added the clause.

The closest safe test involved two maintenance tenders escorting empty coolant bladders near Morrow's island. Both crews joined the circuit. Belen's navigation council reviewed the response family. The tenders accepted local control and published abort lines. *Common Measure* would alter its cage field from outside their plane, leaving three corrections voluntarily absent as the bladders crossed.

Light delay between the tug and Morrow remained short, but nobody pretended remote control could become instant. Each tender retained its own burn.

"First gap in four minutes," Jalen said.

Tomas distributed the phase markers and watched the room settle into work. The protocol had begun as Lark's old blank term, a child's irritation with equations that treated every space as something adults had forgotten to fill. She would have hated what he had made of it. Or laughed. He could no longer tell where memory ended and the version of her he had constructed for comfort began.

On the wall, Ada rubbed the pale ridge along her thumb.
“Do you still have the page?” she asked.

Tomas looked up. “Which page?”
“You know which.”

He did. Lark had been seven when she drew the orbital puzzle during a visit to Ninefold. Tomas had given her three moons and a tug, asking where the final burn belonged. She left one box empty and wrote *let it pass* beneath it in green pencil. The paper had later become evidence in every fight he and Ada held about who remembered their daughter correctly.

“Yes,” he said.

“Here?”

“At home.”

Ada nodded as if confirming a stored mass. “I thought you might have lost it during the archive move.”

“I scanned everything.”

“Of course you did.”

This time the phrase carried no edge, only recognition.

The first tender entered its test corridor. *Common Measure* raised a minor cage gradient, enough to produce a familiar correction demand. Jalen counted down. Ada left the command untouched.

Aster Three's phase trace lengthened into the omission. Two free sinks responded with a delayed pulse. The tender pilot held course.

“Gap one complete,” Tomas said. “No corridor violation.”

During the second crossing, Aster Nine shortened its response as the model had feared. Morrow's local navigator prepared an abort, then waited because the projected family remained eight kilometers outside their boundary. Ada's cage field stayed quiet.

The response opened again.

"That's new," a dynamicist whispered.

Tomas measured the interval. It exceeded their omission by three milliseconds at Aster Three's local phase and returned through Aster Eleven on a moon-length recurrence. The Resonance had not merely echoed the test. Its relation preserved the opening at another scale.

Before the third crossing, the returning tender pilot revoked consent.

"My cold-gas reserve dropped below my published floor," she said. "Valve bank two reports leakage after the last pulse."

Jalen answered from *Common Measure*. "We can reduce cage amplitude by half. Your predicted abort impulse falls inside reserve."

"Prediction does not refill my bank."

Ada did not argue. "Revocation received. Hold outside the family."

Tomas watched the protocol lose its clean sequence. Three omissions had offered a stronger pattern than two. The orbital window would not return at comparable geometry for another five hours. He could substitute an automated marker craft without violating the tender's boundary, but doing so would teach that permission attached to any passing mass rather than a cause capable of restraint.

A second pilot from the other tender spoke. "We have margin for the third crossing."

"Your consent covered two," Tomas said.

"I am extending it."

"After seeing the response, which changes your information state. We need a new local review."

Morrow's navigation council convened a three-minute emergency check. The pilot named her thruster reserve, radiation limit, and abort line. Her engineer certified the valve bank. Belen's delegate confirmed the trajectory remained outside Morrow's protected corridor. The first pilot retained her refusal without pressure to explain further.

Ada shifted the cage-field timing to the second tender and published the alteration. "A permission gap cannot depend on replacing whoever says no with whoever says yes."

Tomas heard the warning in it. "The second crew holds independent position and margin. Different hull, different risk."

"Record why the substitution does not evade revocation."

He did. The protocol grew less elegant and more usable: consent attached to a declared hull and crossing, not to a role someone else could be ordered to fill.

"Third tender ready," Jalen said.

"There are still two," Tomas replied.

"Third crossing. Other tender."

Fatigue had begun stealing simple nouns. Tomas marked the correction in the protocol and asked the new pilot to reconfirm consent. She did.

Ada watched him from the wall. "You need sleep after this."

"You need it too."

"I have a cage."

"I have eleven singularities and forty-six habitats."

"You have a staff."

"Your staff exists too."

Her eyes shifted toward someone outside frame. Sixteen people remained aboard *Common Measure* after the evacuations, and Tomas had reduced them to Ada twice in one conversation. He corrected the protocol roster to name the tug's crew council as a required signer for the final release.

Gap three opened. The tender crossed. Aster Three held its long delay. The free relations left room around the human trajectory.

No divergence appeared.

Tomas refused to call the test complete. He asked the tender to make the ordinary correction the protocol had withheld, but only after it had cleared every response family. The pilot fired from her own board. Aster Three's phase contracted by the predicted amount; Nine and Eleven returned to their prior spacing. The change showed that the maintained interval depended on restraint, not a dead actuator or exhausted field.

Next he replayed the sequence with the first pilot's revoked crossing restored in simulation. The modeled response looked cleaner and carried less evidentiary value. Its elegance depended on pretending a refusal had never happened. Tomas deleted that branch and sealed the revocation beside the successful data.

A technician asked whether the untidy record might confuse whoever flew the final release. "It should," Tomas said. "Permission without the right to interrupt would be a prettier form of command." He added one line to the flight display: *A broken sequence remains valid when a hull withdraws.*

Tomas locked the result and labeled the sequence *permission protocol, revision one*. The phrase felt too fluent for what had occurred. Nothing had granted anyone permission in a human sense. A family of masses had answered repeated restraint with a maintained interval. Yet protocols needed names people could use under pressure.

"We test it once more at low field," Ada said.

"After crew rest."

"The geometry changes in five hours."

"Our cognition changes faster." Tomas made his voice formal because anger and fear had converged. "You cannot turn fatigue into dedication by naming the window."

Ada went still. "I did not."

"You were about to."

"Do not tell me what I intended."

Around them, Ninefold's staff discovered urgent private screens. Tomas heard the old argument opening, larger than the task and intimate enough to damage everyone nearby.

He closed the public circuit and kept their encrypted work channel. "Four hours," he said. "We both sleep. Then we run again."

"Three."

"Four."

"Three and a medical check."

He accepted. Their staff arranged watches without either of them assigning each person.

Before leaving the chamber, Tomas checked the residuals from the third gap. Two errors appeared around Morrow and Pelagos. Both habitats occupied the first resonant islands. Their measured paths differed from the published solution by tiny radial displacements, each away from the nearest sink corridor.

He assumed a sensor reconciliation problem and sent it to the audit queue.

Ninefold's residential corridor had dimmed to night cycle. Tomas walked past office doors left open for emergency ventilation. At his apartment, he did not enter

the sleeping room. He lay on the narrow couch beneath a wall of old paper notes and set an alarm for three hours.

Sleep refused him.

After twenty minutes he opened the storage drawer beside the couch. Lark's page rested inside an archival sleeve. Green pencil had faded to the color of algae beneath ice. Three circles represented moons. Her tug looked like a spoon. One box remained empty beside the words *let it pass*.

Tomas had scanned the page but kept the original. He used to tell himself Ada would lose it because she carried objects into dangerous places. After Lark died, he kept medical records, school drawings, growth measurements, and the final clinic schedule in ordered archives. Evidence could not change what had happened. It could only make him custodian of the question.

His terminal lit with Ada's private call.

"You are supposed to sleep," he said.

"So are you." She had moved from command to the tug's small galley. A bulb of tea drifted tethered beside her shoulder. "I found the residuals."

"They went to audit."

"I know. I want your first answer, not the committee's."

"Morrow and Pelagos drifted outward. Either our habitat fixes contain a shared reference error or the sink family changed around them."

"How much?"

"Morrow gained fourteen kilometers of radial clearance. Pelagos gained nine."

"Cost elsewhere?"

"I haven't found it."

"Then find it after sleep."

He almost laughed. "You called to tell me not to work?"

"I called because the same error appears in Nadi's final phase record."

Her tea bulb struck the end of its tether and returned. Tomas overlaid the exchange from his apartment terminal. The widening around Morrow began as Aster Six cleared the torus. Pelagos gained space one moon recurrence later. Momentum balanced through a narrower unmanned corridor beyond Aster Ten. No physical law had bent. The free sinks had redistributed timing and orbital energy around two inhabited islands.

Reciprocal accommodation had begun before the permission protocol received a name.

"I'll send the finding to every auditor," he said.

Ada looked beyond her camera. "Wait."

He waited.

"The page," she said. "Can I see it?"

Tomas lifted the archival sleeve into view.

Ada's expression changed through several small failures of control. "You put it in plastic."

"Graphite transfers."

"Green pencil does not transfer that fast."

"It transfers eventually."

"Yes." She touched her thumb to the cup tether. "You kept everything."

"Not everything."

The terminal carried only ventilation and the soft impacts of the drifting tea bulb. Tomas wanted to return to the residuals. Work offered quantities that could be wrong in corrigible ways.

Ada spoke first. "I left the clinic before they finished clearing her room."

"I know."

"You made that mean I wanted to leave her."

"I made it mean work mattered more."

"It mattered because I could still do it."

"And we could not do anything for Lark."

The sentence opened the old wound without blame's familiar scaffolding. Lark's congenital illness had resisted every intervention available at Crown's pediatric center. Tomas had read probabilities until the probabilities became smaller and more cruel. Ada had learned every pump, every dose, every shift handoff. None of that competence had made their daughter controllable.

"I thought you blamed my timing," Ada said. "The last treatment vote. You wanted to wait for the revised model. I signed before you arrived."

"I blamed the decision because it gave me an event with a person attached." Tomas kept the page steady between his fingers. "The revised model predicted the same outcome. I had it six hours later and never told you."

Ada closed her eyes.

"Why?"

"Because if the model changed nothing, I had no argument left except that she died."

"You let me think I had chosen wrong."

"Yes."

Anger moved through her face, earned and unhidden. Tomas did not explain that grief had made him cruel. Explanation would only place intention where consequence already stood.

"I used your archive the same way," she said. "Every file you kept let me say you cared more about proving the clinic wrong than being with her."

"I was with her."

"I know. I knew then."

Tomas looked down at the green letters. "We assigned each other control so we could assign each other failure."

Ada's hand closed around the tea bulb. "Neither of us controlled the illness."

"No."

The word did not absolve them of what followed. He had withheld the final model. She had returned to *Common Measure* and refused his calls for weeks. They had injured each other after the death, when different choices remained possible.

"I am sorry I left you to clear the room," Ada said.

"I am sorry I let you carry the treatment vote."

She nodded once. No reconciliation ceremony followed. Their marriage had ended through more than one silence, and grief acknowledged did not rebuild the lives they had made afterward.

"What do we do with this?" she asked.

Tomas could have answered about the page, the protocol, or them. He chose the only answer broad enough to stay honest. "We do not ask each other to return."

Relief and pain crossed her expression together. "No. We don't."

"We can stop asking the other to remain at the worst day."

"That I can do."

Tomas set Lark's page beside the terminal rather than back in its drawer. Ada untethered her tea and drank.

After a while she said, "Show me the islands."

He shared the orbital overlays. Morrow's safe interval widened by fourteen kilometers. Pelagos held nine. The compensating narrow corridor lay beyond every declared human boundary. Aster Six's altered phase carried Nadi's

momentum into the relation; Aster Two, Nine, and Eleven returned it at longer scales.

“Can we prove agency?” Ada asked.

“Not as humans use the term. We can reject navigation error above six sigma. We can show the change followed our omissions and preserved inhabited trajectories at a cost elsewhere in the sink family.”

“Enough to publish?”

“Enough to make a falsifiable claim before certainty.”

He sent the residual set, methods, and alternative explanations to Ninefold's auditors and the public migration network. Ada added the two habitats' local navigation logs. Neither called it a message.

On the shared plot, the eleven free sink paths curved around the blue islands. Human pilots held their drives quiet at the offered edges. The intervals remained open from both sides.

Chapter 24 — The Compact Without Custody

Cass Vardek signed the reserve doors open with a key the new law would make impossible.

Eight meters below the observation rail, Crown Meridian technicians broke the seals on coolant tanks that had not appeared in any public inventory until Belen Sato forced their disclosure. Frost rolled across the transfer lines. A loading chief in a silver pressure bib held up one gloved hand, waiting for Cass to complete the authentication sequence. The console asked for the First Custodian's palm, pulse, and private phrase.

He supplied all three.

Green bars moved along the tank cradles. Pumps drove Crown's hoarded coolant toward canisters marked for Morrow and Rook. The latter had spent six years defending the reserve as protection against crop failure. By the arithmetic now governing survival, those sealed tonnes had become heat capacity for habitat drives crossing into the resonant islands.

"Flow confirmed," the chief said. "You can release the key."

Cass kept his hand against the plate until the first canister filled. A person who had spent seventeen days issuing commands learned to distrust any sentence containing the word release. His pulse trace climbed beneath his palm.

Belen stood beside him in a maintenance jacket patched at both elbows, watching the mass total rather than Cass.

Morrow's convener had slept recently, if the clean shirt and controlled temper meant anything. Grief had settled around her eyes after Nadi Okafor's death without slowing her work.

"Four hundred kilos short on line three," she said.

The loading chief checked. "Thermal contraction."

"Then warm the declared figure or amend it."

"It will balance at delivery."

"People cannot cool a drive with promised coolant."

Cass lifted his hand. The console relinquished him with a soft tone. "Amend the declaration."

The chief looked from Cass to Belen. Two days earlier the decision would have required no glance sideways. Now Crown's logistics crews took instruction from a council whose members included clinic quartermasters, exterior foremen, captains, and habitat maintenance conveners. Cass held a temporary coordinating seat because they needed his knowledge of the Compact's machinery. The seat carried no emergency command.

The chief entered the lower figure. Belen nodded once.

Cass said, "The drafting chamber has our fourth revision."

"Then it has four ways to hide the same knife."

"Three, I hope."

"Hope did not make my inventory accurate." She turned toward the pressure doors. "Show me the fourth."

They crossed Crown's administrative spoke while the cylinder rotated around them. Through a long port, sunlight passed between moving mirror vanes and struck the farms in measured bands. Half the lamps had gone dark under rationing. Workers harvested pale grain by hand because field machines drew more power than Crown could spare. The administrative habitat had once displayed abundance

as evidence of competence. Its dark acres now provided evidence of dependence.

Cass had written the founding Mass Compact's anniversary address in a room above those farms. He had praised common custody: no habitat could seize a sink, no captain could auction gravity, no rich cylinder could buy another's winter. Every claim had contained truth. Over decades, custody had acquired locks, schedules, enforcement keys, and the useful fiction that allocation carried no violence if entered correctly in a ledger.

The drafting chamber occupied a windowless room near Crown's axis. Its walls showed the migration in forty-six separate tracks. Six habitats rested inside preliminary safe regions. Twenty-nine had completed their major burns. Eleven still crossed between the old web and the widened intervals the Resonance appeared to preserve. Aster One, Two, and Four through Twelve moved free on paths no cage controlled. Aster Three remained a black point nested inside the amber outline of *Common Measure*.

Ada Narang joined by delayed projection from the tug. Compression sharpened the planes of her face and left the shadows beneath her eyes in blocks. Behind her, the command blister carried empty couch positions from the three evacuated crew. Sixteen people continued aboard a ship missing its port radiator wing.

"Captain," Cass said.

"Coordinator."

She made the diminished title sound like a measurement, neither insult nor forgiveness.

Belen opened the draft before sitting. "Read the operative clause aloud."

Cass knew which one she meant. "No natural or engineered mass capable of independent relational behavior may be held as property, collateral, territorial

claim, or exclusive source of service. The accounting category called sink-hours shall cease upon ratification."

"That ends billing," Belen said. "It does not assign fault."

"The preamble recognizes injury."

"A preamble cannot order restitution."

"Restitution to whom?" Cass asked. "The Resonance has shown no concept matching compensation. We cannot transfer currency, restore twenty years, or ask which institution it wants prosecuted."

Belen leaned over the display. "You owe humans restitution too. Morrow paid above-market power rates while Crown accumulated the reserve we just opened. Harbor crews absorbed radiation. Three workers died on my radiator rig. Nadi and the people in her crew section died carrying out a plan the central administration first delayed and then made more dangerous."

Cass let the list remain in the room. He had learned late that answering a casualty with a larger casualty model erased the person without changing the number.

"The remedial article transfers Compact assets to the mutual-aid lattice," he said. "Cages, dispatch funds, reserve reaction mass, and every allocation archive."

"Who decides how?"

"The council."

"Which council? Today's emergency roster? A permanent chamber? The old directors wearing public badges?"

Ada shifted in her couch. "Belen wants names on the authority path."

"I understand what she wants," Cass said.

"Then stop making her pull each one out."

His first answer rose from habit: drafting required categories broad enough to survive the present officeholders. He closed his mouth before the lesson could

escape. Ada's demand had practical force. Any captain approaching a burn needed to know whose order could cross a hull.

Cass selected the remedial article and stripped out its collective nouns. He assigned the allocation archive to elected habitat auditors, the cage systems to engineering crews chosen by their yards, and the remaining dispatch funds to a council with one voting delegate from each habitat. Clinical reserves required assent from clinic representatives. Reaction-mass transfers required sending and receiving captains to sign the load plan.

Belen followed every edit. "Terms."

"Thirty days for the emergency delegates. Elections after immediate stabilization."

"Recall?"

"By the selecting body at any time."

"Public records?"

"Default public. Crew medical data and exact security vulnerabilities protected."

"Who determines the exception?"

Cass looked at her. "You enjoy this."

"I enjoy finding the place where somebody hid a door."

He added a three-person review panel selected by lot from habitat auditors, with a maximum twenty-four-hour seal before public reasons became mandatory. Belen read the change twice. Ada asked that active trajectory details remain sealable until a burn ended, since publishing a vulnerable ship's exact control margin could invite sabotage or panic traffic. Belen accepted that with a sunset measured in hours, not days.

Their arguments accumulated into law by small refusals. Cass felt an old impatience whenever either woman reopened language he considered settled. That impatience

had once passed for executive strength. It had allowed him to fit millions of conflicting needs into one command structure and send decisions at operational speed. It had also allowed twenty years of anomalies to remain in an archive because uncertainty threatened compliance.

A trajectory alarm flashed amber on the wall. Habitat Reed had reached its correction window eleven seconds early. Its captain requested a two-minute hold while an exterior team cleared a stuck sensor mast.

Cass saw the collision cone before the council coordinator called. A hold would narrow Blue Haven's later crossing, but not close it. Under Article Nine he would have ordered Reed through the window and placed the repair crew behind shielding. The risk model favored motion.

The coordinator, a refrigeration chief from Ketu named Vema, sent both habitats the updated geometry. "Reed retains hull decision," she said. "Blue Haven, advise whether you accept a six-minute displacement if Reed holds."

Blue Haven's captain answered after light-time and consultation. "Accepted if Reed supplies the later trim mass."

Reed held. Three suited figures crawled along the habitat's bright outer curve while forty-five other trajectories bent around their delay. Cass watched the safe corridor lose margin by fractions.

Ada had gone silent on the projection. She watched too.

After one hundred and seven seconds, Reed retracted the mast and burned. Blue Haven followed on the amended line. No central order had forced agreement. The process had cost four minutes and a measurable quantity of reaction mass. It had also left both hulls responsible for risks they had named.

Cass returned to the text. "Add this under navigation authority: every hull retains local veto over a burn."

Ada's expression did not change. "Not enough."

"What else could local veto mean?"

"It means I can refuse thrust. It must also mean no coordinator can punish a hull for refusing an unsafe burn by withholding food, coolant, medicine, or later corridor access."

"Then veto becomes a way to consume common resources without accepting common obligation."

"Only if the refusal needs no declared reason."

Belen drew a line beneath the clause. "Require the refusing crew to publish its measured hazard and a recovery proposal. Aid continues while the council tests both. Fraud remains reviewable. Disagreement does not become fraud because a coordinator dislikes it."

Cass imagined six captains refusing a synchronized correction while the Resonance's moving masses narrowed every route. The migration could fail through a collection of defensible local choices. Central authority existed because coupled systems produced consequences beyond any one hull.

Nadi's rule had offered no escape from coupling. A captain's authority ended at another crew's hull. It did not end at the edge of consequence. She had spent her final trajectory proving both halves.

"The recovery proposal needs a deadline," Cass said. "One orbital planning cycle unless the hazard prevents communication."

Ada studied the wall. "Acceptable."

Belen inserted the language. "And coordination advice carries the model, assumptions, uncertainty range, and named source."

"You want a signature on every projection."

"I want a person unable to call a model inevitable."

Cass signed that clause with his own name.

At the sixth hour, the legal engine found two hundred thirteen conflicts with existing Compact code. Most concerned ownership without using the word. A sink could not belong to a habitat, but its scheduled output could secure debt; its cage could support exclusion zones; a Custodian could seize a transfer vessel to preserve allocated service. Removing custody reached into power tariffs, construction bonds, agricultural guarantees, and the spin covenants keeping four older cylinders solvent.

Cass called the remaining Compact jurists into the chamber. Nine answered. Two had resigned after the archive release, one had moved to a clinic dispatch team, and three could not connect during habitat burns. The nine divided the conflicts among them. They worked while Crown's dark farms rotated past unseen.

A junior jurist named Elian found the worst trap. "If we abolish the service backing immediately, these four habitat bonds enter default. Automated trustees can seize their reaction-mass accounts."

"Which four?" Belen asked.

Elian displayed Ketu, Pale Reach, Morrow, and Hestia Minor.

Belen's laugh carried no amusement. "Of course."

Cass read the bond language. The central administration had guaranteed cheap credit by pledging future power allocations. Now the Great Unbinding cut dispatchable power by thirty-eight percent, and the legal extinction of sink-hours would eliminate the collateral category. Investors might lose paper wealth. Automated trustees would take propellant from the habitats least able to spare it.

"Freeze enforcement," Ada said.

Cass shook his head. "By what authority? Article Nine expired."

"The bond court can issue a survival stay," Elian said, "but only after a petitioner defaults."

"Then change what counts as default," Belen said.

"That impairs settled contracts."

"So does losing the thing you claimed to own."

Cass opened the treaty's founding text. The first Compact had subordinated all private claims to collision prevention. Its authors had understood orbital life more clearly than several generations of administrators who followed. No creditor could compel a transfer that materially endangered an inhabited hull.

He built the replacement from that foundation. All claims denominated in sink-hours would convert to claims on future surplus after survival reserves, never to control over mass or compulsory power. Losses would remain losses. Pension funds and habitat cooperatives held much of the debt, so cancellation would punish workers along with speculators. Conversion spread the loss across years while protecting physical stores.

Belen objected to the word surplus until the draft defined food, heat, medicine, spin, air processing, minimum mobility, and repair stock as prior needs. Ada added crew survival margins. Elian added an independent solvency review. Cass added no guarantee of former yield.

"This makes Crown poor," the junior jurist said.

"It makes Crown recognize its poverty," Belen replied.

Cass looked through the migration display toward the reserve bay far below. Crown's wealth had included first claim on movement, heat, and probability. Much of it could not survive honest accounting.

"Keep the clause," he said.

Near the ninth hour, the chamber called a recess. Cass remained at the table while others found water and heated ration bowls. His private key lay in his inner pocket, a

narrow black wafer with no markings. Software treated it as one credential among several, but the physical matrix authenticated the First Custodian's unilateral key across tugs, cage controllers, and Compact reserve locks. Article Nine had activated its highest functions. Ordinary law retained narrower ones: archive seals, anti-collision holds, asset freezes, and emergency access to every registered habitat channel.

No office in the new structure could keep it.

Cass placed the wafer on the table.

Ada's projection watched him. "You should have destroyed that on Day Seventeen."

"Yes."

Belen returned carrying three cups. "Does breaking it revoke the credential?"

"No. The matrix can sign one final instruction. Revocation requires it to authenticate its own death across the network."

"Convenient."

"It prevented an enemy from disabling the key by transmitting a forged cancellation."

"It also meant only its holder could end its reach."

Cass accepted one of the cups. The broth tasted of salt and yeast. "Yes."

They summoned witnesses from the new council: Vema from Ketu, a clinic steward from Pale Reach, two tug captains, an exterior worker from Harbor of Glass, and jurists selected by the habitats that had opposed migration. Cass wanted the revocation witnessed by critics as well as allies. Belen required the raw authentication log to mirror into every public archive.

Ada said, "Before you do it, Aster Three may need a system-wide collision hold tomorrow."

Every face turned toward her projection.

"Do you need this key?" Cass asked.

"No. I need traffic controllers to honor the release volume."

"The key could impose it without forty-six separate confirmations."

"That speed tempted you the last time."

"I am aware."

"Are you? Tomorrow you may watch my tug miss its burn while one habitat argues about a freight lane. You will know exactly which command could clear it."

Cass closed his fingers around the wafer. The object weighed less than a spoon. Through it he had once moved vessels, sealed archives, and treated hesitation as a system fault.

"What will you need instead?" he asked.

"A public hazard volume. Local controllers acknowledge or publish their objection. If an unresponsive vessel drifts inside, nearby captains decide how to move it. My crew controls our cage and burn."

"Your engineer remains in Morrow's clinic."

"Iri controls the release checklist. Jalen and I execute it."

Cass heard the distinction. Ada had surrendered neither command nor expertise. She had divided them along the places where knowledge lived.

He inserted the wafer into the table reader. "Revocation statement: no office shall retain unilateral physical or digital authority over a sink, cage, tug, habitat trajectory, or common reserve. Emergency coordination requires named consent from affected hulls and public declaration of modeled consequence."

The reader asked whether he intended permanent revocation.

"Yes."

It asked whether coercion impaired his judgment.

"No."

It displayed the systems that would cease accepting his credential. The list filled three walls. Cass recognized tools designed after preventable disasters, each addition defended as a narrow safeguard. Together they formed a government no one could refuse quickly enough.

"Transmit," Belen said.

Cass pressed his thumb to the wafer.

The key sent its final signature. Authentication nodes answered from Crown, then Rook, Morrow, the observatory lattice, shipyards, tugs, reserve depots, and habitats still moving under drive. Each acknowledgment removed a line of authority from the display. The black wafer clouded as its internal matrix burned random paths through itself.

A call arrived from Crown security before the sequence finished. "Custodian, we show command credential loss."

"Correct."

"Do you require replacement issuance?"

"The office no longer has one."

Security paused. "Under whose order?"

Cass looked at the witnesses. "Mine, while it remained an order I could give."

The final node acknowledged. The wafer cracked with a sound like a seed opening.

Nobody applauded. Belen sealed the fragments in an evidence pouch and gave them to the Harbor worker, not to Cass.

The vote opened two hours later. This ballot did not ratify the permanent Compact; habitat assemblies needed Day Twenty-Eight for that. It approved the emergency draft

pending completion of the final operation and authorized its immediate constraints. The count formed slowly because eleven habitats remained in migration and three could communicate only through narrow-beam relays.

Morrow supported. Ketu supported. Harbor of Glass attached a demand for worker control of dismantled cage hardware, then supported. Crown Meridian divided through six procedural ballots and finally supported by a margin too narrow for comfort.

Rook opposed. Its academy council argued that local veto would make coordinated defense impossible. Two wealthy cylinders opposed the debt conversion. A small free-spin habitat opposed any central charter, even one designed to weaken central custody.

Thirty-eight approvals appeared on the wall. Then thirty-nine.

Cass stood behind the coordinator's seat with no console of his own. A year earlier, he would have called delegates, traded timetable concessions, and warned hesitant habitats which services depended on administrative goodwill. He knew every lawful pressure point. Having relinquished the key did not erase his knowledge of softer coercion.

He kept his hands at his sides.

The fortieth approval came from Blue Haven, delayed by its displaced burn. Four habitats stood opposed. Two reserved judgment until *Common Measure* completed the release.

"Forty habitats support," Vema announced. "Emergency draft carries pending final operation."

The chamber exhaled. Cass did not. On the trajectory wall, all eleven free sinks entered a relation that Tomas had marked as the approach to complete recurrence. Their paths opened a narrow lane beside Ophion. Aster Three needed to cross that lane while its cage came apart around it.

Ada enlarged the tug's projection. Heat limits covered one side in red. Without the port radiator, *Common Measure* could sustain the escape burn for less than six minutes. If the sink remained coupled after the cage opened, its gradient would tear the sacrificial truss through the crew spine. If the tug corrected too early, the Resonance might read capture instead of permission.

Cass watched the modeled casualty branches unfold. The instinct to narrow them into one command came as naturally as breath.

"What does the law require from us now?" he asked.

Ada answered from sixteen occupied couches and one caged mass. "Keep the corridor open. Give us the facts. Let my hull decide when to move."

Cass turned to the council. No unilateral channel waited beneath his hand. He began calling forty-six controllers by name.

Chapter 25 — The Last Cage

Ada stopped her hand one centimeter above the seal display.

Sixteen suit icons showed green along *Common Measure's* command board. Each carried pressure, oxygen, thermal reserve, and radiation status. She had checked the columns before taking the couch. Her thumb now hovered over the command that would cycle through them again.

"Captain?" Jalen asked.

She lowered her hand to the control grip. "Crew reports stand."

Nobody remarked on the change. They had other numbers to fear.

Ophion filled the forward blister from rail to rail, cloud bands flattened by the polarized glass into ocher and white currents. *Common Measure* fell toward the giant on a path that every transfer rule before this week would have prohibited. A close pass supplied the velocity needed to send Aster Three into the free lattice, but it also pressed the tug between the planet's gradient and eleven sinks crossing in coordinated arcs.

Eleven sinks had entered resonance during the last nineteen hours. Their masses appeared as black marks on the navigation wall, spread across millions of kilometers. No mark carried awareness alone. Together their changing intervals shaped a pattern Tomas could predict only far enough to keep the habitats outside its strongest folds.

Aster Three rode ninety meters ahead of Ada inside the tug's damaged cage. Its amber indicator pulsed late against every recurrence. The delay had once looked like a defect. Now it held the long relation the other masses lacked.

"Release volume update," Parveen said from systems. "All registered traffic confirms clear. Two ore tenders objected, published their fuel constraints, and took Harbor's tow offer. No unresponsive hulls."

Cass had spent most of a day making the corridor from consent instead of command. Ada inspected its boundaries. One maintenance pod remained near the outer uncertainty line, attached to a guidance beacon by a repair tether.

"Who owns that pod?" she asked.

"Reed crew," Jalen said. "Beacon drifted after their burn. They report clear in seven minutes."

"Our last safe turn-away closes in nine."

"They know."

Ada could have asked Reed to abandon the beacon. She could have moved the tug's approach by twelve kilometers and converted a generous escape angle into a narrow one. Neither action belonged solely to her. Reed's captain had the geometry and the hazard volume.

"Hold present line," she said.

The cage circuit chimed. Iri Quill's face appeared from Morrow's clinic, one cheek bright under medical light and the other lost in compression shadow. Their left leg lay fixed inside a regeneration frame. Radiation burns patterned their shaved scalp near the temple. A paper cup floated tethered beside the bed because Morrow had not yet returned that clinic section to full spin.

"You look too comfortable," Ada said.

"I stole two pillows from thermal injuries. Crime prospers under distributed government." Iri angled their wrist pad toward the link. "Final checklist follows. Version six, signed

by me, Ren Sato at the remote cage rig, and your gallery pair."

The file arrived. Ada opened it without touching the verification history.

Iri noticed. "Who are you, and what have you done with my captain?"

"Read it to us."

"Primary sheath holds point seven percent below model. Acceptable. Secondary plasma density has noise from the missing radiator bus. Not acceptable in civilized work, but civilization resigned. Manual latch has three release charges and one mechanical drawline. Your gallery crew will fire two charges. Hold the third unless separation fails."

Ada watched the cage schematic. Four nested field loops surrounded a void too small to see and too massive for any material container. The cage managed Aster Three's engineered charge, keeping the sink centered while controlled feed and coupling systems harvested energy. Opening it required more than switching off. Plasma had to vent along the tug's axis, loops had to decay without inducing a kick, and the scarred manual latch had to draw the remaining hardware aside before the sink's changing relation bent it through the ship.

"Heat budget?" Ada asked.

Parveen answered before Iri could. "Six minutes eleven seconds of escape burn if every remaining radiator panel accepts red-line temperature. Four minutes thirty before crew spaces cross medical limit. Storm shelter stays lower."

"Drive can complete in five twenty," Jalen said.

"If the gradient matches forecast."

"It will not," Iri said. "Nothing in the last twenty-five days has matched forecast long enough to brag."

Ada expanded the release timing. At closest approach, the eleven free masses would establish a widening relation centered beyond Ophion's far limb. Aster Three needed to leave the cage during a seven-second phase minimum. The tug then had to burn radial and sunward, using the planet's motion to pull clear. A programmed trim at forty-one seconds would prevent the ship from following the sink's desired arc.

That trim worried Tomas. It resembled a recapture correction in the interval grammar.

Jalen had reduced it to the smallest impulse that preserved the tug's miss distance. Ada had tested omissions around it in simulation. Most killed the crew. One left a corridor if the Resonance widened the phase as it had around the habitat islands.

Betting sixteen lives on reciprocal accommodation did not become moral because she called it communication.

The Reed pod detached from the beacon. Its crew accelerated toward their habitat, leaving the broken mast to tumble beyond the hazard line.

"Corridor clear," Parveen said.

Ada opened the crew circuit. "We have eight minutes to last turn-away. Updated survival branches are on your boards. Branch A executes programmed trim and gives us a seventy-eight percent escape estimate. It may delay or break Aster Three's join. Branch B omits trim if the phase gap opens. Ship survival depends on the free masses changing their relation."

A gallery technician named Mara Venn spoke first. "Engineering recommends B?"

Iri answered from Morrow. "Engineering recommends you distrust any recommendation with that many moving black holes. I signed the release procedure, not the choice after it."

Ada said, "Guidance?"

Jalen rubbed the yellowing bruise along his jaw from the first failed handoff. "Branch A preserves a known ship corridor until Aster Three reacts. After that, the model has no authority. Branch B treats the permission gap as information. I support B if Tomas confirms recurrence."

Other stations reported in sequence. Nobody offered faith. They named shielding limits, drive margins, medical capacity, and what they did not know. Two crew supported the programmed trim. Twelve supported conditional omission. Parveen abstained from trajectory choice while affirming thermal readiness. Ada recorded her own support last.

"We fly Branch B if the measured gap reaches Tomas's threshold," she said. "No threshold, we trim. Any station can call abort before turn-away closes."

The circuit carried breathing and machine tones. No one called it.

"Continue approach."

Ophion rose around them.

Acceleration stayed too slight to feel at first. The planet's gravity did the work, drawing tug and sink along nearly the same curve. Nearly carried the danger. Aster Three's trillion-tonne mass occupied no meaningful volume, yet its gradient changed across the two-hundred-meter ship. The cage gallery pulled forward harder than the drive spine. Structural monitors painted the difference in red along the sacrificial truss.

Ada flew with both hands. Every control input passed through Jalen's prediction before reaching the drives, but the final release remained hers. The old pale ridge on her thumb caught against the grip.

"Closest cloud-top altitude in four minutes," Jalen said. "Aster lattice relation beginning."

Signals arrived from Ninefold after their short light delay. Tomas sent numbers, not commentary. Eleven

gravitational traces crossed the display. Their intervals shortened in pairs, then quartets, then a pattern that could not divide evenly while Aster Three lagged inside the cage.

The ship creaked.

A loose panel near the empty engineering couch lifted from its fasteners and hung at an angle. Acceleration still pressed aft. The local sink gradient pulled its forward edge toward the cage. Ada watched the panel bend by degrees and ordered the space cleared.

"Gallery load at sixty percent," Mara Venn reported. She and Lu Feran occupied the two forward suits, secured to opposing rails beside the manual drawline. "Reference cuts visible. Latch settlement matches Iri's archived mark."

On Ada's display, Iri enlarged their helmet feeds. The scarred latch housing carried witness cuts copied from the position where the blue paint strip had exposed the sensor error. Heat from the failed containment had darkened the surrounding metal, but the mechanical alignment remained readable.

"Do not adjust it," Iri said. "The cuts show where the latch settled, not where you wish it had settled."

Mara's glove rested near the first release charge. "Understood."

Ada looked at the checklist. Three signatures, independent pressure traces, a timing table, and a hand-marked warning about plasma backslash. The urge to compare every line against raw telemetry tightened her shoulders. Iri knew the cage. Mara and Lu occupied the danger. Ada knew the trajectory. Rechecking their work would not combine those forms of knowledge. It would place her delay inside theirs.

She closed the checklist without opening its annexes.

"Tomas reports entry relation," Parveen said.

The eleven traces stopped looking like separate signals. Recurrences crossed between Aster Two and Seven, returned through Five and Eleven, then used Ophion's inner moon as a slow beat. The relation had room for another mass. Its opening arrived every fifty-three milliseconds and narrowed each time Aster Three failed to enter.

"Cage primary to decay," Ada ordered.

Mara armed the loop sequencer. "Primary decay started."

Amber light spread across the command board. The cage stopped correcting tiny offsets. Aster Three moved less than a millimeter in modeled space. The bow dipped enough to drive Ada's harness into her collarbones.

"Secondary plasma vent in ten," Iri said. "Gallery, verify aft path."

"Path clear."

"Drive?"

"Standing at ignition," Jalen replied.

"Crew?" Ada asked.

Parveen displayed sixteen sealed suits. She did not read each name aloud. "Crew configured."

The ship passed the last turn-away point.

Ada marked it on the public trajectory feed. Every habitat could see that *Common Measure* had committed. No office had ordered it across that line. No distant assembly could pull it back.

"Vent," Iri said.

Mara fired the first release charge.

Plasma left the cage in a white spear along the truss. The blister darkened to protect their eyes. Heat alarms stepped upward across systems. Aster Three's charge

coupling fell, and its long phase lag opened like a stretched spring.

The first gradient struck sideways.

Ada's couch gimbaled but reached its mechanical stop before her body aligned. Her right shoulder slammed into the restraint. Something moved in her chest with a hard internal click. She kept both hands on the grips.

"Attitude seven degrees off," Jalen said. Blood floated from one nostril toward his cheek. "Correcting with starboard thrusters."

"No lateral correction near vent plume."

"We lose release axis."

"Use main drive pre-ignition."

He fed a fraction of thrust through the aft nozzles. The tug's spine straightened against the gradient. Structural load climbed past the old cage-transfer limit.

"Primary loops at twenty percent," Mara called. "Latch still seated."

"Second charge," Iri said.

Lu fired it. The manual drawline snapped taut. Cage petals moved apart around a center no camera resolved. Their hardware had warped during the failed containment, and one petal stopped six centimeters short of its commanded position.

"Petal four obstruction," Mara said.

Iri's eyes tracked the feeds. "Do not use third charge. Drawline caught under the coolant saddle. Lu, take the red lever and rotate counterclockwise."

"The lever will put them inside the gradient shadow," Ada said.

"It puts them there for three seconds. Third charge can throw the petal across the sink line."

Lu had already left the rail. Their tether bowed toward Aster Three, not in a straight line but along a curve. They caught the red lever with both gloves and pulled. Nothing moved.

"Drive ignition in twenty-two seconds," Jalen said.

Ada could delay ignition six seconds before losing escape margin. Beyond that, Ophion would turn their intended burn into a deeper pass. "Lu, you have five."

The technician planted both boots against the cage frame. The lever rotated. The coolant saddle flexed, releasing the drawline with a violent twitch. Lu spun away. Mara caught their tether before it swept toward the open center.

"Petal moving," Iri said. "Gallery clear aft now."

Both technicians hauled along their rails. Mara's suit struck a support spar. Lu pulled them free, and the pair crossed the pressure hatch as the petal reached full travel.

The cage opened.

Aster Three ceased to occupy *Common Measure's* controlled center. Gravity did not switch off. The sink and tug continued falling beside each other, close enough that the ship still lay across a steep gradient. Yet the charge loops no longer supplied the relation called captivity.

All eleven free traces changed.

"Complete opening relation," Tomas transmitted. "Phase threshold approaching. Seven recurrences."

Ada's chest resisted her next breath. Pain spread under the lower right harness band. She shortened the inhale and wrapped her hands more firmly around the controls.

"Gallery sealed," Parveen said. "Sixteen crew accounted."

"Ignite escape."

Fusion light drove aft. The couch took Ada's weight, adding pressure to the damaged rib. *Common Measure* moved

away from the released sink by centimeters, then meters. Aster Three fell toward Ophion along the entry arc. The other masses pulled its relation across distances too large for instinct.

The ship's remaining radiators climbed through yellow. Starboard panels glowed on the thermal schematic. With the port radiator absent, coolant crossed temporary loops installed after the failed cage attempt. Parveen shifted command-space heat into empty tanks and the sacrificial truss.

"Thirty seconds to trim decision," Jalen said.

The programmed impulse waited on Ada's grip. Firing it would cant the tug away from Aster Three's predicted curve. It also filled the exact omission through which the Resonance had learned to distinguish restraint.

Aster Three's phase trace reached toward the eleven and fell short.

"Threshold?" Ada asked.

"Not yet," Tomas said over delay.

The recurrence came again. Aster Four altered charge behavior. Aster Nine answered after an interval shaped by the inner moon. The path between them widened by a fraction no human engine had commanded.

"Twenty seconds," Jalen said.

Heat warnings reached red in the drive spine. Parveen evacuated two compartments and moved their atmosphere into the storm shelter reserve. The escape burn remained steady.

Aster Three approached the opening. Its phase lag carried twenty years of broken cycles, according to Tomas. Ada did not imagine memory inside the point mass. The memory occupied relation: delayed returns, missing answers, recurrences shortened by cages.

"Ten seconds."

"Still below threshold," Tomas said.

Ada's thumb rested on the trim authorization. Branch B required the measured gap, not her desire to see a reply. Without threshold they had agreed to fire.

At six seconds, the cage wreckage began following the sink. A detached loop arced forward and pulled the sacrificial truss off axis. Jalen compensated with drive gimbal. The ship rolled.

Ada's right restraint locked across her chest. White pain erased the edge of the display.

"Captain," Parveen said, hearing the involuntary sound.

"Still flying."

"Four seconds," Jalen called.

Tomas's trace moved.

Eleven intervals opened around the twelfth, not simultaneously and not as a welcome. Mass answered mass through phase. Aster Three's long delay entered the relation and changed the period of everything around it.

"Threshold," Tomas said. "Permission gap available."

Ada lifted her thumb from the trim.

"Trim omitted," Jalen announced on the public circuit.

For forty-one seconds *Common Measure* made no lateral correction. Its drive continued the declared escape burn. The omission carried risk in a shape the Resonance had encountered before: a machine capable of closing the interval chose not to close it.

Aster Three crossed the predicted tug corridor.

The gradient pulled the ship bow-first. Acceleration vectors swung through Ada's couch, wrenching her against the right harness. Her rib gave with a second, softer break. The control grips blurred.

Jalen reached toward her panel, then stopped short of taking it. "Your command."

She could transfer flight. Pain narrowed her breath, not her judgment. "Hold drive. No trim."

"Miss distance falling."

"Give it five seconds."

Three seconds passed. The ship's projected arc touched the sink hazard boundary.

On the fourth, Aster Three changed charge phase.

No thrust appeared from nothing. Its coupling with Ophion's magnetic environment and the other eleven masses shifted the free trajectory by a small, lawful amount. Across the lattice, corresponding adjustments preserved momentum through relations Tomas would spend years resolving.

The sink moved away from the tug's line.

"Miss distance increasing," Jalen said. "Two kilometers. Five. Eleven."

The eleven traces accepted Aster Three's delayed recurrence. A complete cycle traveled through all twelve nodes, around Ophion's moons, and back through the mass they had just released.

The force across *Common Measure* eased. Loose objects struck aft surfaces. The detached cage loop missed the truss and fell toward a disposal orbit. Ada authorized drive shutdown at five minutes nine seconds.

Silence did not follow. Pumps hammered. Cooling lines snapped under contraction. Sixteen people reported damage in overlapping voices. Parveen crawled from her couch to inspect Ada without asking permission and pressed a diagnostic patch beneath the right harness.

"Right rib fracture," she said. "Possibly two."

"One," Ada answered through shallow breaths.

"The scanner gets the vote."

Mara reported Lu's bruised shoulder and their own cracked visor laminate. Drive reported three overheated bearings. No suit had lost pressure. No crew member required evacuation that the current trajectory could not provide.

Iri remained on the cage channel. Their image had frozen during peak interference and returned with tears bright against an expression of professional disgust.

"You ruined my cage," they said.

"It was designed to open."

"Not like that."

Ada looked at the empty amber outline on the board. For the first time in seventeen years, no human vessel carried a sink. "Write the failure report."

"I already titled it."

"Nothing obscene in the public archive."

"No promises under the new law."

Tomas sent the completed relation. Ada expanded it across the blister. Twelve paths curved around Ophion, separated by distances that made the old cages look absurd and held together by intervals no custody had created. Along the edge of the pattern, a wide channel remained around the human resonant islands.

The channel did not prove mercy, agreement, or peace. It recorded changed trajectories. That supplied enough truth for navigation.

Jalen asked for orders. The old answer waited in Ada's muscles: inspect every station, retake the controls, make survival pass through her hands. She could not draw a full breath. Sixteen crew already worked the damage according to authorities they had named.

"Transmit our actual line," she said. "Ask the habitat council for a repair corridor. Gallery holds engineering veto over any thrust until the loose cage petals clear. Guidance keeps flight."

"And you?" Parveen tightened a brace around Ada's ribs.

"I remain captain."

"From the storm shelter."

Ada considered objecting. The completed cycle reached Aster Three again. Its lag no longer ended in isolation; the interval returned through eleven other masses and Ophion's moving moons.

"From the storm shelter," she agreed.

Jalen accepted the flight authority without ceremony. He sent their request rather than taking a corridor. Replies began arriving from habitats and tugs that could see pieces of the new geometry.

Ahead, the twelve-node Resonance completed a second cycle. *Common Measure* coasted outside it, hot, injured, and no longer a cage.

Chapter 26 — The Gravity We Owe

The Resonance moved Aster Eight by less than the width of Morrow.

Across four million kilometers, that displacement altered the gravity fold approaching Pale Reach. Ada watched the hazard band slide from red to amber above her storm-shelter couch. The change could have come from Aster Eight's charge coupling with Ophion's magnetic field, balanced through the trajectories of Aster Two and Ten. It spent no impossible momentum. It simply chose, if choice belonged to such a relation, one lawful path among several.

The new path left Pale Reach inside the widened interval.

"Confirm no human correction preceded it," Ada said.

Tomas answered from Ninefold through a channel crowded with raw traces. "Confirmed to instrument limit. The phase change begins at Eight, returns through Two, then uses the outer moon's motion to balance. Pale Reach occupies the omitted region."

"Could it be restoring its own cycle?"

"It is restoring its own cycle." Tomas's face appeared in a narrow pane. Exhaustion had hollowed his cheeks. "The question concerns why this cycle instead of the adjacent one. Both close within our error margin. Only one preserves the habitat corridor."

Ada shifted against the medical brace. Her right rib objected with a precise line of pain. Parveen had forbidden command-couch acceleration and threatened to sedate her if she tried. The storm shelter gave Ada no controls, only read access and a speaking key Jalen could cut if medication impaired her.

Pale Reach's captain called on the council band. "We see the opening. Is passage granted?"

Tomas looked toward Ada, but the decision did not belong to either of them.

Council coordinator Vema sent the evidence to Pale Reach and neighboring hulls. "No grant can be inferred. The interval has widened and remained stable through three recurrences. Your captain retains the burn."

Pale Reach held for a fourth recurrence. Its drive lit at nineteen percent, carrying fifty-eight thousand people away from the fold.

Aster Eight adjusted again as the habitat crossed. The free sinks preserved distance around the human trajectory without freezing their own motion. Aster Three's old lag passed through the relation and returned complete.

Nobody on *Common Measure* cheered. The pumps still fought heat stored during the release, two compartments remained without atmosphere, and Lu Feran's shoulder had swollen until their suit sleeve needed cutting. The ship coasted toward a repair rendezvous with no port radiator wing and half a cage hanging in twisted petals from the forward truss.

Ada opened the public record. "Observed reciprocal trajectory adjustment, Day Twenty-Seven, Pale Reach crossing. Mechanism bounded by charge-phase coupling and orbital exchange. Intent unproven."

Tomas gave her a tired smile. "You always did know how to welcome a neighbor."

"It does not need a welcome. It needs us not to lie about the data."

"Then add that the adjacent closed cycle carried equal physical cost within measurement error."

She added it under his name.

The repair tug *Stubborn Fact* approached from sunward six hours later. Its captain requested consent to grapple the remaining cage petals. Sima Dey, holding engineering veto from the gallery, approved the grapple but refused any cut through the charged loop until Iri reviewed its decay trace from Morrow. Ada listened while the two crews negotiated loads. Nobody asked her to settle the argument.

She slept for ninety minutes and woke when the first permanent power cuts began.

Forty-six habitat icons occupied the shelter wall. Each displayed spin reserve, food temperature, clinical load, water circulation, drive capacity, and available generation. The sink economy had supplied thirty-eight percent of dispatchable power. Freeing the twelve masses did not erase every reactor or flywheel. It removed the flexible source that had covered peaks, accelerated construction, held marginal orbits, and allowed richer habitats to treat waste as normal.

Crown dimmed its administrative sectors. Rook stopped two industrial drums. Harbor of Glass shut the furnaces that grew kilometer trusses. On Morrow, clinics kept full power while residential heat fell four degrees. Belen's authoritative inventory directed mobile reactors toward the narrowest margins, with every transfer visible.

The numbers did not make the cuts equal. Crown apartments had insulation, private batteries, and food stocks. Morrow families shared warming rooms beneath stalled transit lines. Pale Reach lost a hydroponic crop when its emergency crossing shook coolant from three farm racks. Hestia Minor imposed water hours. Six construction

swarms learned that their speculative hulls had become scrap because no sink would serve as counterweight.

Markets opened after a two-day suspension and found much of their wealth missing. Bonds denominated in sink-hours converted to claims on a surplus nobody expected soon. Cage manufacturers dismissed crews before the new council could transfer their shops into dismantling and radiation work. Pension accounts fell. So did the private fortunes built on priority allocation. Paper loss and cold rooms appeared on the same ledgers without becoming the same injury.

Ada read until Parveen took the display away.

"Your job includes healing," the systems officer said.

"My job includes knowing whether the web survives."

"Vema's job includes that now. Belen's too. Mine includes preventing a rib from puncturing your lung because you refuse analgesia."

"The stronger dose clouds judgment."

"You have no active flight authority."

Ada reached for the display. Parveen held it beyond her range.

A younger version of Ada would have ordered her out of the shelter. The version who had crossed the permission gap recognized an engineering veto in medical clothing.

"Half dose," Ada said.

"Three quarters."

"Two thirds."

Parveen loaded the injector. "Distributed governance produces unbearable bargaining."

The medicine entered Ada's arm. Pain loosened from her chest without disappearing. She slept while other people kept the trajectories.

Day Twenty-Eight arrived without a ceremonial sunrise. Ophion's reflected light moved across the damaged tug, and the habitat assemblies opened ratification according to their local clocks. The emergency draft had changed during the previous three days. Auditors closed the bond trap. Exterior crews secured control over cage dismantling. Clinics defined survival reserves. Captains and habitat assemblies settled the consent boundary after arguments that consumed more time than several migration burns.

Cass read the final operative language on the public circuit. He sat among the forty-six delegates rather than above them. Without the First Custodian's key, his console carried the same vote control as every other seat.

"No natural or engineered mass capable of independent relational behavior may be held as property, collateral, territory, exclusive service source, or instrument of compulsory settlement," he said. "No authority may compel thrust, spin change, atmosphere transfer, docking, or mass exchange across an inhabited hull's declared safety boundary. Refusal requires measured reason and recovery proposal where communication permits. Essential aid shall not serve as punishment for refusal."

The legal text continued through common reserves, public models, emergency coordination, recall, archive access, and restitution. It abolished the old allocation unit. Existing sink-hour claims would receive conversion without physical enforcement. Compact cages and reserve assets passed into the mutual-aid lattice.

Then Cass reached the clause Ada had first heard twenty-six days earlier as an accusation.

"Human settlement has lived by borrowed weight," he read. "Borrowing conveys neither ownership nor indefinite right of use. Where another relation bears the cost, consent and restitution govern any continued exchange."

No human grammar could guarantee that the Resonance understood the statute. The clause bound humans, which supplied reason enough to write it.

Voting took eleven hours.

Morrow entered yes after assemblies in every district reviewed the resource guarantees. Crown Meridian entered yes and attached a minority report warning that the new structure could respond too slowly to collision. Rook argued until its final hour, then entered no. Three habitats joined it. Two free-spin communities abstained because any system Compact still looked like authority imposed from a distance.

The count settled on the shared wall: 40 yes, 4 no, 2 abstain.

No applause crossed the open channel at first. People checked their local totals and challenged two precinct reports. Auditors traced both. One discrepancy came from a delayed clinic ballot; the other from a family entered twice after evacuation. Corrections changed margins inside habitats but not the system result.

Vema declared the new Mass Compact ratified.

Belen spoke next from Morrow's old flywheel vault. Hundreds of names covered the ring behind her, projected over machinery that had supplied black-start power during the migration. "Ratification does not balance the dead," she said. "Record them before you record success."

The public wall changed.

Ren Sato read the names of the three Morrow radiator workers killed on Day Two: Davi Ekan, Sora Pell, and Min Yue. His crushed hand remained bound in a working brace. Harbor crews added those lost during the yard blackouts and migration accidents. Pale Reach named two clinic patients whose transfer failed during the emergency crossing. Then the trajectory of *Principle of Least Harm* replaced the list.

Nadi Okafor had remained alone in the forward crew section after the storm pod detached with twelve survivors. Her tug had exchanged momentum with Aster Six and moved Morrow beyond collision. No recoverable section had returned from the gradient.

Ada asked Jalen to transmit *Common Measure's* flight record into the memorial archive. It contained no speech, only the line Nadi chose and the lines she preserved.

The dead did not become payment for the Compact. Calling them a price would imply someone had possessed the right to spend them. Their names remained consequences attached to decisions, including Ada's.

Tomas joined her after the channel closed. He had moved from Ninefold's main dynamics room to a narrow office. Behind him, interval traces covered one wall. A child's drawing did not hang there. Lark needed no display to remain present between them.

"You should be asleep," he said.

"So should you."

"I slept during legal definitions of collateral."

"Cass noticed."

"He can prosecute me after he rebuilds the office." Tomas looked down before continuing. "The complete cycle persists. Aster Three carries the long recurrence without phase collapse. Human islands remain inside widened intervals. I can defend all those statements."

"What can you not defend?"

"That it will continue next month. That the Resonance understands the distinction between a habitat and a tug. That permission generalizes beyond trajectory. That our translation of injury resembles anything it experiences."

"Publish that list beside the findings."

"Already done."

Their old arguments had often started there. Tomas wanted uncertainty protected from action. Ada wanted action protected from uncertainty. Now each had work the other could inspect without controlling.

"What will you do?" he asked.

"Repair the ship."

"I meant after."

"So did I."

His smile lasted briefly. "Ninefold approved a permanent translation group. Half dynamicists, half tug and habitat crews. Belen demanded the second half."

"She was right."

"She usually arranges the inventory so that becomes visible." He rested one hand against the desk. "Ada, I am glad you lived."

The words carried no invitation to restore what grief had changed. She accepted them without building one.

"I am glad you spoke before certainty," she said.

They ended the call because both had work.

The wrecked cage came off *Common Measure* nine days later at Harbor of Glass.

Ada watched from a pressurized yard blister while grapple crews separated the petals from the forward truss. Each warped loop had once carried enough current to kill a suited worker. With Aster Three gone, residual charge and activated materials supplied the remaining danger. Yard crews waited through decay readings, grounded every segment, and cut only after two independent meters agreed.

A Crown curator asked to preserve the manual latch for the Compact museum.

Iri joined by projection from a rehabilitation rail. They could bend their left knee through thirty degrees and

complained that physical therapists lacked respect for elegant shortcuts. "Preserve the blue strip," they said. "Melt everything around it."

The curator looked wounded. "The cage possesses historic significance."

"It possesses tungsten, superconducting ceramic, and enough shielding plate for three clinic shelters."

Ada read the disposal proposal. Harbor wanted the outer petals for radiation baffles. Morrow had requested the field-loop supports to reinforce its failing transit hub. Ninefold wanted sensor samples. The museum requested an intact section large enough to demonstrate custody technology.

"Keep one de-energized loop segment," Ada said. "Document what it did. The rest goes by the public salvage list."

"Captain," the curator said, "future residents should see the scale of what held the sinks."

Ada looked through the blister. The cage hardware filled half the repair berth. Its size had always encouraged the wrong understanding. Steel, plasma, and field loops had not contained Aster Three's gravity. They had manipulated charge around a point smaller than sight and called the continuing relation a service.

"Show them the whole truss in a model," she said. "Put the power accounts beside it. Put the anomaly archive beside those. Do not make a relic look more honest than its records."

The cutters started. Bright fragments moved into sorted cradles under the grapple crews' direction. No single order determined their destination. Habitat requests, worker hazard limits, and available transport narrowed the choices until every piece acquired a declared use.

Iri lifted the specimen sleeve holding the blue witness recovered with them after the failed clamp. The paint chip turned in the clinic rail's weak spin.

"That belongs beside the next manual latch," Iri said.

"There will be latches without cages," Ada replied.

"Good. Machinery still lies."

Six months later, *Common Measure* crossed the outer edge of Morrow's resonant island with no cage aboard.

The repair yards had removed the sacrificial truss and installed a lattice of mapping booms, field sensors, and grapple arms. A replacement port radiator wing extended from the hull, mismatched in color and twelve percent less efficient than the original. Iri had painted a blue alignment strip beside every manual hinge. They claimed the repetition constituted a serious calibration system. Ada suspected it also constituted mockery.

Her rib had healed with a small ridge she could feel when rotation loaded her right side. She no longer wore the brace. Sixteen opening-crisis crew had been offered places after repair; eleven returned. Others took habitat posts, medical leave, or work dismantling cages. New crew filled the empty couches, including apprentices who had never stood beside a caged sink.

Dispatchable power across the web held at sixty-two percent of its former level. That average concealed daily cuts. Morrow heated residential sectors in rotation. Crown farmed administrative lawns. Rook's idle industrial drums supplied spare parts after workers converted their lines. Harbor built radiators instead of speculative habitats. Travel windows narrowed because no one could rent a sink's gradient to erase poor planning.

People complained. They also voted on ration tables, caught auditors hiding favorable assumptions, and argued over aid in channels anyone could read. The mutual-aid

lattice had not purified politics. It had made several forms of coercion harder to conceal.

Belen chaired the lattice for a ninety-day term and refused nomination for an automatic second. Her meetings began with actual stores and ended when every transfer had a sending name. Cass taught crisis law at Crown and traveled to habitat councils that distrusted him. His course used his Article Nine orders as case studies. Students reportedly enjoyed cross-examination more than he enjoyed answering.

Tomas continued translation at Ninefold. His group had mapped seven recurrent relations and confidently named none of them. The Resonance adjusted free trajectories across Ophion, sometimes preserving human intervals and sometimes closing routes with no explanation. The Compact treated every opening as provisional.

Common Measure now mapped those openings and repaired the beacons that marked human traffic. It carried fusion fuel, coolant, spare bearings, and no captive gravity.

On the day of the first apprentice passage, Ada stood behind Jalen's old guidance station. Jalen had taken a training command at Rook after insisting he had learned enough from her bad decisions to deserve his own. The new pilot, Mei Aras's son Kiran, sat at the main controls. He had been among the school passengers who touched water climbing a classroom wall during Aster Three's first deviation.

Kiran's crew had plotted a route through the interval between Aster Five and Aster Eleven. The passage would shorten a beacon-repair mission by fourteen hours and test whether the omission protocol worked for a tug-pattern the Resonance had never encountered.

"Permission sequence ready," he said.

Ada inspected the public plot. The tug would burn twice, coast through the outer recurrence, then omit a scheduled

four-centimeter-per-second correction. If the interval widened, they could proceed. If it held, they would take the longer route. Every crew station had approved. Nearby habitats had published acceptance of the hazard volume.

Her old instinct found three refinements. The second burn could move two seconds earlier. The thermal margin assumed a conservative starboard panel temperature. Kiran's turn-away marker allowed less comfort than she preferred.

None crossed a safety limit.

"Execute your plan," she said.

Kiran gave the first burn to his guidance partner rather than keeping every control. Fusion thrust moved the mapping tug along its declared line. Aster Five's trace reached Aster Eleven through the slow motion of Ophion's outer moon.

At the correction point, the crew left a precise gap.

Ada kept her hands away from the controls.

The first recurrence passed without change. During the second, Aster Three's long interval entered the relation. Aster Eleven shifted phase, Aster Five returned it, and the safe passage widened around *Common Measure* by eight kilometers.

"Opening confirmed," Kiran said. His voice tightened, but his hands stayed steady. "Request continuation authority."

Ada looked around the command blister. Engineers, mappers, and habitat observers held pieces of the decision. No amber cage light occupied the board.

"Your hull," she said. "Take us through."

Kiran executed the burn. *Common Measure* entered the widened interval under the young crew's hands, carrying only its own weight.